

A MERGED RESEARCH REPORT

Differential Algebra and Differential Equations over the Surreal and Surcomplex Numbers

The finite-primitive criterion, the phase obstruction,
and support-certified coordinate systems

September 2026

Abstract

Eleven independently written manuscripts developed the theory of the Berarducci–Mantova derivation ∂ on the surreal class field $\mathbf{No}$, normalized by $\partial\omega = 1$, and of its unique extension to the surcomplex field $\mathbf{No}[i]$. This report is their union. Seven of them share a twelve-item spine of scalar and workspace theory; four joined later and carry the matrix, regular-singular, autonomous and transfinite second-order strata.

The organizing identity is exact: the image of the logarithmic derivative is

$$\{\partial y/y : y \in \mathbf{No}[i]^\times\} = \mathbf{No} + i\mathcal{A}, \quad \mathcal{A} := \partial\mathbf{m} = \partial\mathcal{O},$$

where $\mathcal{O}$ is the class of finite surreals and $\mathbf{m}$ its ideal of infinitesimals. Hence $\partial y = (a+ib)y$ has a nonzero surcomplex solution exactly when b has a *finite* surreal primitive. The two printed forms $\mathbf{No} + i\partial\mathbf{m}$ and $\mathbf{No} + i\partial\mathcal{O}$ agree only through a proved lemma, and $\mathcal{A}$ is a *strict* subclass of $\mathbf{m}$: the coefficient ω^{-1} is infinitesimal while its primitive $\log \omega$ is infinite. The purely infinite part of the accumulated phase is an $\mathbb{R}$ -linear surjection onto the class of purely infinite surreals, a complete gauge invariant, and the middle term of a four-term exact sequence. From it follow sharp support and iterated-logarithm thresholds, a decision procedure for rational coefficients, the pointwise maximality of the strip $\mathbf{No} + i\mathcal{O}$ as the domain of any derivation-compatible exponential, the nonexistence of a nonzero solution of $\partial y = iy$ or $\partial^2 y + y = 0$, a complete constant-coefficient and constant-matrix classification in which only real characteristic roots contribute, a triangular fundamental-matrix criterion, exact resolvents with factorially divergent Hahn solutions, differentially stable set-sized workspaces, and oscillatory Picard–Vessiot extensions with differential Galois tori — at the cost, in the real form, of the H -field ordering rule.

A later part classifies *finite matrix* systems. Every $\partial y = Ay$ with A an ordinary finite matrix over $\mathbf{No}[i]$ is gauge equivalent to $i \operatorname{diag}(\partial P_1, \dots, \partial P_n)$ with each P_j purely infinite, and the multiset $\{P_j\}$ is a complete invariant of differential equivalence; for a Hermitian H it is computed by integrating the ordinary algebraic eigenvalues of H and keeping the purely infinite parts of the primitives, with neither commutation nor a spectral gap assumed. From that follow a gap-free finite-primitive perturbation theorem, an exact kernel count, normalized non-Abelian integration, the complete cone of positive invariant metrics, a real rotation-block descent, a positive Ermakov–Pinney dichotomy, an exact Hahn expansion for the oscillatory Airy equation, and the phase-lattice Picard–Vessiot torus of an arbitrary finite system. A deliberate counterexample — a nonnormal coefficient with instantaneous eigenvalues $\pm i$ and a fundamental matrix already in the base field — marks the boundary of the eigenvalue formula, not of the finite-system classification. That part also invokes a separately imported linear-splitting and first-order surjectivity theorem, not a consequence of the scalar criterion and flagged wherever used.

Three later parts classify regular-singular systems $\partial y = -tAy$ with an ordinary residual matrix and a positive real-power Hahn perturbation, for which $\operatorname{Ph}(-tA) = \{(\operatorname{Im} \lambda_j) \log t\}$ is an explicit non-Hermitian sector reached without the import; first-order autonomous equations with constant coefficients, for *every* normalized surreal derivation, with the logarithmic-derivative image of a constant abelian variety; and second-order equations on the log-atomic tower, with a sharp threshold $1/4$ at every ordinal depth.

A deliberately separate half treats *coordinate* differential equations in fixed-common-domain Hahn rings $\mathcal{O}(U)((t^\Gamma))$: existence, uniqueness, support certificates, variation of constants, a Liouville determinant identity, monodromy valued in $\operatorname{GL}_d(K_\Gamma)$, and a polynomially nonlinear initial-value theorem, all built on strong summability and an ordered-word support lemma proved here in full. The two halves share notation and almost nothing else; several theorems are true for one operator and false for the other on the same printed equation, and the report keeps their type distinctions intact rather than flattening them.

The Berarducci–Mantova construction is imported, not reproved. Nothing here is claimed as new, refereed, or machine-checked.

Status. A merged exposition assembled from eleven independent manuscripts: seven pinned to VladimirReshetnikov/Surreal at commit e260237db9b71da8b74a0c13c8e6355119091100; the eighth, which supplies Part III, pinned to the later revision aa846271b4dcae2c055b216126a87210292ec19b; the ninth and tenth, which supply Parts V and VI, pinned to a3124af79f66b8b9c196d76b4cbc5ac3938907c4; and the eleventh, which supplies Part VII, pinned to 048b72cf7cbfc8ab246e4f73788c10460cb3f6e0. Deep structural inputs are attributed; the connecting deductions are proved in the text. No priority claim, no solution of a named published open problem, no independent refereeing, and no proof-assistant verification is asserted. Section 51 is the consolidated register of what is *not* claimed; it is part of the result, not a disclaimer appended to it.

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Part I

Foundations, and the distinctions that must survive

1 What this report is

1.1 Eleven manuscripts, one theory

This report is the union of eleven independently written manuscripts on the differential algebra of the surreal and surcomplex numbers. Seven of them form the original core: they were prepared separately, they are all pinned to the same documentation snapshot, and they prove — in seven notations — the same twelve-item spine:

- (i) the Berarducci–Mantova derivation ∂ extends uniquely to $\mathbf{No}[i] = \mathbf{No}[i]$, because differentiating $i^2 + 1 = 0$ forces $\partial i = 0$; its constant field is exactly $\mathbb{C}$;
- (ii) derivatives of finite elements are infinitesimal;
- (iii) an infinitesimal exponential and logarithm, defined as Hahn sums, with their chain rules;
- (iv) a branch-free polar decomposition with a finite angle;
- (v) a primitive normalized by a vanishing ω^0 coefficient;
- (vi) the rank-one criterion — $\partial y = (a + ib)y$ has a nonzero surcomplex solution if and only if b has a finite primitive — and the exact logarithmic-derivative image;
- (vii) the purely infinite obstruction, its exact sequence, and a gauge normal form;
- (viii) $\partial y = iy$ and $\partial^2 y + y = 0$ have only the zero solution;
- (ix) there is no globally derivation-compatible complex exponential;
- (x) the constant-coefficient classification, in which only real characteristic roots contribute, with its constant-matrix corollary;
- (xi) a set-sized differential localization;
- (xii) an oscillatory Picard–Vessiot extension with no new constants.

Every one of the seven proves items (i)–(vi) and (viii)–(ix). The remaining four are proved by most but not all: the gauge normal form of (vii) is absent from three members, its exact sequence from a fourth; the constant-coefficient classification (x) is absent from one member, which explicitly declines it, and its constant-matrix corollary from two others; and at least five prove each of (xi) and (xii). Appendix C gives the member-by-member record, and no claim below rests on a member that does not prove what it is cited for. Keeping them apart would mean printing one theory seven times, including seven proofs of $\partial i = 0$ and seven proofs that $\partial y = iy$ has no nonzero solution. Each is nevertheless the unique source of some genuinely distinct payload, and those payloads are all built on the shared spine and cannot be extracted without carrying it along. Appendix C records, member by member, what each contributed.

The eighth manuscript joined later and is different in kind. It does not reprove the spine; it cites it, names the gap this report’s own scope ledger recorded — no classification of arbitrary

nontriangular matrix systems — and closes the Hermitian half of it. Its mathematics is Part III, and it is pinned to a later revision, `aa846271b4dcae2c055b216126a87210292ec19b`, of the same repository. Several of the twelve spine items are reprovved inside it by its own routes, and where that happens the routes are recorded as agreement (Remarks 15.1 and 15.2) rather than printed twice. It also brings one *new imported input*, the linear-splitting and first-order surjectivity theorem of Theorem 15.6, which the first seven do not use and which the register of non-claims accordingly records as an import rather than as a proved consequence.

The ninth and tenth manuscripts — members 11 and 12 in the numbering of appendix C — joined next, arrived together, and are pinned to a third revision, `a3124af79f66b8b9c196d76b4cbc5ac3938907c4` [31]. Each compared itself with the repository’s root README, its catalogue and this report’s README at that revision, not with this article, and neither contradicts the report or the other. Between that revision and this merge the article changed only by the addition of Remark 19.1, and the two members’ descriptions of what the report already contained were checked against the article and hold. They share no theorem and are printed as two new parts.

- Member 12 supplies Part V: regular-singular matrix systems over real-exponent Hahn workspaces. It is the matrix version of the rank-one resonant normal form of Part IV, and over $\mathbf{No}[i]$ it computes one explicit non-Hermitian sector of the phase spectrum of Part III without that part’s imported input. It partly answers items N24 and N57 and the computable-filtration question of Warning 17.13, and it does not bear on item N41.
- Member 11 supplies Part VI: all first-order autonomous equations with constant coefficients, on curves and on abelian varieties. It lifts items N21 and N25 and the remark after Corollary 8.11 for that class, and it is proved for every normalized derivation, which required a scoped amendment of the one-derivation rule (Convention 29.2).

Their reproofs of results already here — smallness, the missing oscillator, the ordered-word lemma, the strip logarithm, the logarithmic separator, the missing logarithmic primitive — are recorded as agreements and cited, not printed twice.

The eleventh manuscript — member 13 — joined after them and is pinned to a fourth revision, `048b72cf7cbfc8ab246e4f73788c10460cb3f6e0` [32]. At that revision this article was byte-identical to the one it was merged into, and the member’s description of what the report contains — the imported derivative, the type distinctions, the summability conventions, the finite-primitive criterion and the normalized primitive — holds. It supplies Part VII: the second-order equations $\partial^2 y + (Q_\alpha + c(\ell_\alpha^\dagger)^2)y = 0$ along the ordinal log-atomic tower, with the sharp threshold $c = \frac{1}{4}$ at every ordinal depth, the failure of every set of earlier scales to decide it, and a limit-stage filtration by differential Hahn fields with a Borel Picard–Vessiot group. It amends item N21 for one explicit family of second-order equations, and it is adjacent to, without answering, member 12’s layered logarithmic continuation (Section 51.13). Its reproof of the obstruction for nonreal powers of an infinite scale is this report’s criterion, and is recorded as agreement (Corollary 37.1); its connection with the missing oscillator is made precise in Remark 37.5.

Merging carries a specific risk, which Section 5 addresses before any theorem is stated: several different objects in these manuscripts are called “phase”, and two nearly identical classes are both called the exponential domain. A merged sentence that unifies either word, or either symbol, proves false statements. This report fixes one name and one symbol per object, keeps the two classes visibly distinct, and states the distinction wherever a reader might carry one sense into another.

1.2 The documented gap

The seven audits agree on the gap and on its character. All are pinned to the same snapshot of VladimirReshetnikov/Surreal:

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That snapshot describes fifteen research packages covering surreal normal forms and constructions, surcomplex analysis and geometry, polynomial algebra, trigonometry, foundations and symbolic computation [21]. Differentiation is not absent from that collection. What is absent is the step from the collection’s careful *distinctions* between kinds of derivative to existence, obstruction and support-control *theorems* for differential equations.

Three boundaries in the existing documentation delimit the gap precisely. The trigonometry report constructs finite phases and states explicitly that it chooses no derivation on **No**, and that its classification of global phase extensions is not a classification of differential-equation solutions [23]. The computer-algebra report separates an external-variable derivative, a formal monomial derivative and a derivation of the surreal scalar field, and its subsection “Differential equations must name their category” issues a uniqueness warning rather than developing a solution theory [25]. The foundations report insists that a fine derivative, a formal-series germ, a coherent Hahn datum and a scalar-field derivation stay four different interfaces [26]. The analysis report goes further still: its derivation-obstruction theorem already shows that one chosen canonical global surcomplex exponential is incompatible with an internal scalar derivation normalized by $\partial\omega = 1$ [24]. That existing theorem is prior coverage, acknowledged as such; Corollary 11.4 below is strictly stronger in its phase-independence and is compatible with it.

Every one of the seven audits was targeted: the catalogue, the specialist READMEs, and selected source passages, at one pinned revision. None was a line-by-line certification of the fifteen reports or of their archived sources, and none used the absence of a code-search hit as evidence of absence. Section 51 records those audit limits in full.

1.3 What is imported and what is proved

The deep input is the existence of the derivation, together with its strong additivity, its constant field, its compatibility with the surreal exponential, its surjectivity and its H -field positivity. These are imported from Berarducci and Mantova [1] and are not reproved; their construction through log-atomic numbers, path derivatives and nested truncation ranks is not reproduced. The positive-support summability lemma is imported in its standard form from Neumann [10], and its ordered-word core is additionally proved in full in appendix A because the original paper was not retrieved. Restricted composition and the global composition-incompatibility theorem for ω -series are separately imported from [2] and are flagged wherever used. The universal H -field results of [3] are background for Parts I, II, IV to VI and VIII, not premises; Part VII uses a different part of [3], the ordinal log-atomic formulas of its Section 2, as a premise, and not its elementary embedding; in Part III alone they become a premise, because Theorem 15.6 transfers along the elementary embedding of [3] two further published facts — splitting of monic scalar operators into first-order factors and surjectivity of nonzero scalar operators [4, Lemma 2.4.19]. That import is a genuine addition to the hypothesis base of this report and is flagged at every use (Warning 15.7, item N53). Ordinary complex initial-value theory [12] supplies the analytic ingredient of the coordinate half, and Picard–Vessiot terminology is used in the standard sense of [13, 14].

Parts V and VI add no new import of that kind. Part V uses the Berarducci–Mantova clauses *without* surjectivity and without Theorem 15.6 (Remark 23.9); classical regular-singular theory [15, 16, 17] is its acknowledged antecedent. Part VI is stated for the whole class of normalized derivations (Definition 29.1), whose one member needed for existence is the Berarducci–Mantova derivation. Surjectivity is assumed only for the global primitive-obstruction construction and its finite-primitive criterion; the explicit speed threshold needs no such assumption. The part imports the classical local normal forms and Rosenlicht’s algebraic-independence theorem as presented by Noordman, van der Put and Top [18], with the standard setting of abelian varieties [19] and the valuative reading of properness [20]. Part VII imports one input about the derivation that Theorem 3.1 does not list: the ordinal log-atomic tower and its derivatives, limit stages included [3, Section 2] (Proposition 35.3). It does not use surjectivity in its main construction, and it takes the classical Euler and Kneser–Weber–Hartman–Hille thresholds as its acknowledged antecedents [33].

Everything else is proved here. None of it is claimed to be new.

Remark 1.1 (Reading order). Part I fixes conventions, imports the derivation, derives smallness, and installs the vocabulary discipline. Readers who want only the headline should then read Section 8. Part II is the scalar theory over $\mathbf{No}[i]$; Part III raises it to finite matrix systems and contains the spectral–integral theorem Theorem 16.5; Part IV is its set-sized and Hahn workspaces; Part V classifies regular-singular matrix systems over those workspaces and over $\mathbf{No}[i]$; Part VI classifies first-order autonomous equations with constant coefficients for every normalized derivation; Part VII classifies one family of second-order equations along the transfinite log-atomic tower and continues the missing oscillator there; Part VIII is the separate coordinate theory, whose only dependence on Part II is negative — it explains what the scalar obstruction does *not* forbid. Part IX collects the computational contracts and the register of non-claims.

2 Conventions, classes, and normal forms

2.1 Class-theoretic ground rules

We reason classically: ZFC with definable classes, or NBG with choice, or any conservative class-theoretic presentation with enough replacement for the constructions displayed below. The symbols $\mathbf{No}$ and $\mathbf{No}[i]$ denote proper classes; a “class field” assertion means that the field operations are class functions satisfying the usual identities, and a class-level statement is a statement about elements.

Every individual number, every normal-form support, every family that is summed, every polynomial, and every matrix below is a *set*. No proper-class family of nonzero terms is summed and no sum is indexed by $\mathbf{No}$.

Warning 2.1 (No class of proper-class cosets). The suggestive notation $\mathbf{No}/\mathcal{O}$ must never be implemented as a collection whose members are cosets of $\mathcal{O}$: a coset of a proper class is not an object that can belong to another class. Throughout this report it denotes the *represented* quotient, namely the class $\mathcal{P}$ of purely infinite normal forms together with the projection pp of equation (2.3). Every obstruction value is an actual surreal number, hence a set. Likewise, automorphism groups over a proper-class base are encoded by constant multipliers of finitely many generators, not by forming a class whose members are proper-class maps; and rank-one equations are classified by their scalar coefficients together with an explicitly stated gauge relation, not by a class-sized collection of modules. Inside a fixed set-sized field, ordinary set quotients are of course legitimate; Section 20 supplies such fields.

2.2 Normal forms and the three-part splitting

$\mathbf{No}$ is real closed and $\mathbf{No}[i] = \mathbf{No}[i]$ is algebraically closed. We use the Conway normal form, the real surreal (Gonshor) exponential and its inverse log on positive elements as background [1, 7]. A real surreal has a unique expansion

$$x = \sum_{\alpha \in S} r_\alpha \omega^\alpha, \quad r_\alpha \in \mathbb{R} \setminus \{0\}, \quad (2.1)$$

with S a reverse well-ordered set of surreal exponents. For surcomplex numbers the coefficients lie in $\mathbb{C}$. It is often convenient to reverse the orientation and write

$$t^\gamma := \omega^{-\gamma}, \quad (2.2)$$

so that a support is well ordered in increasing γ and positive exponents describe infinitesimal monomials. We abbreviate $t = t^1 = \omega^{-1}$.

Warning 2.2 (Two different exponent constructions). In equation (2.1) and equation (2.2), $\gamma \mapsto \omega^{-\gamma}$ is the Conway *monomial map*. For an arbitrary surreal γ it must not be silently replaced by $\exp(-\gamma \log \omega)$: the omega-map and general surreal exponentiation are different constructions. The identity $\partial(\omega^r) = r\omega^{r-1}$, equivalently $\partial(t^r) = -r t^{r+1}$, is asserted for *ordinary real* r only, where the two interpretations agree. The safe universal identity is $\partial(\omega^\gamma) = \omega^\gamma \partial \log(\omega^\gamma)$ with the actual Gonshor logarithm of the monomial. A symbolic implementation needs different constructors for the two operations. For the record, the expression $\exp(\gamma \log \omega)$ has derivative $\exp(\gamma \log \omega)((\partial \gamma) \log \omega + \gamma/\omega)$, which is not a general rule for the monomial map.

Write

$$\mathcal{O} = \{x \in \mathbf{No} : |x| < n \text{ for some } n \in \mathbb{N}\}, \quad \mathfrak{m} = \{x \in \mathbf{No} : |x| < 1/n \text{ for every } n \geq 1\}.$$

Thus $\mathcal{O}$ is the convex valuation ring of finite elements and $\mathfrak{m}$ is its maximal ideal, with $\mathcal{O} = \mathbb{R} \oplus \mathfrak{m}$. Let $\mathcal{P}$ be the additive class of *purely infinite* normal forms: those all of whose exponents in equation (2.1) are strictly positive, together with 0. It is an additive real vector class. It is *not* the class of all infinite surreals. Splitting the support of equation (2.1) at exponent 0 gives the unique additive decomposition

$$x = \text{pp}(x) + \text{ct}(x) + \varepsilon, \quad \text{pp}(x) \in \mathcal{P}, \quad \text{ct}(x) \in \mathbb{R}, \quad \varepsilon \in \mathfrak{m}, \quad (2.3)$$

and we write $\text{fin}(x) = \text{ct}(x) + \varepsilon = x - \text{pp}(x)$. All of pp, ct and fin are additive and $\mathbb{R}$ -linear, and extend $\mathbb{C}$ -linearly to $\mathbf{No}[i]$ componentwise. So $\mathbf{No} = \mathcal{P} \oplus \mathbb{R} \oplus \mathfrak{m}$ and $\mathcal{O} = \mathbb{R} \oplus \mathfrak{m}$, and x is finite exactly when $\text{pp}(x) = 0$.

Warning 2.3 (ct is not a standard part and is not multiplicative). $\text{ct}(x)$ is the coefficient of the monomial $\omega^0 = 1$, defined for *every* surreal, including infinite ones. The standard part st is defined only on $\mathcal{O}$, where it is a ring homomorphism with kernel $\mathfrak{m}$; the two agree only there. ct is additive and real-linear but is not multiplicative: $\text{ct}(\omega) = \text{ct}(\omega^{-1}) = 0$ while $\text{ct}(\omega \cdot \omega^{-1}) = 1$. None of pp, ct, fin is a field homomorphism.

For complex values, conjugation fixes $\mathbf{No}$ and sends i to $-i$, and $|u + iv| = (u^2 + v^2)^{1/2}$. Put $\mathcal{O}_{\mathbb{C}} = \mathcal{O} + i\mathcal{O}$ and $\mathfrak{m}_{\mathbb{C}} = \mathfrak{m} + i\mathfrak{m}$; standard part is taken coordinatewise and lands in $\mathbb{C}$. An element of modulus one is necessarily finite, whatever the birthdays of its coordinates.

2.3 Summability is a support condition

Definition 2.4 (Strong summability). A *set*-indexed family $(x_j)_{j \in J}$ of Hahn normal forms is *summable* (equivalently *strongly summable*) when $\bigcup_j \text{supp } x_j$ is well ordered in the increasing-exponent convention of equation (2.2) and, at every exponent, only finitely many x_j have a nonzero coefficient. Its sum is formed coefficientwise, by those finite coefficient sums.

Warning 2.5 (What summability is not). Strong summability is a statement about supports. Increasing valuations alone do not establish it. It licenses coefficientwise rearrangement of a summable family; exchanging two infinite sums requires strong summability of the joint family, not merely of each row. And *writing* a formal geometric or exponential series is not by itself a summability proof. None of the Hahn identities below is asserted to be a limit of partial sums in the fine surreal topology, in which set-indexed convergence is degenerate. Conversely, a fixed-rank set-sized Hahn field does have nontrivial valuation-topological convergence, and full-class topological statements must not simply be transferred to it. The three notions — fine topology, fixed-rank valuation topology, strong summability — are never identified here.

A concrete separator: with $\Gamma = \mathbb{Z}^2$ lexicographically ordered, $v(t) = (0, 1)$ and $v(u) = (1, 0)$, one has $v(t^n) < v(u)$ for every ordinary n . The identity $\sum_{n \geq 0} t^n = (1 - t)^{-1}$ is strongly summable, yet its partial sums do not converge in that valuation topology. At higher rank $n\sigma$ need not become cofinal, and appendix A establishes no such cofinality; no argument in this report uses one.

The combinatorial engine is the positive-support lemma.

Lemma 2.6 (Positive-support calculus). *Let Γ be an ordered abelian group and let $S \subseteq \Gamma_{>0}$ be well ordered. Then the additive monoid*

$$S^* = \{0\} \cup S \cup (S + S) \cup (S + S + S) \cup \dots$$

is well ordered, and for each $\gamma \in \Gamma$ only finitely many finite ordered words $(s_1, \dots, s_n)$ in S satisfy $s_1 + \dots + s_n = \gamma$. Consequently, if h has positive support, then for ordinary real or complex coefficients a_n the family $(a_n h^n)_{n \geq 0}$ is summable.

This is the lemma underlying the classical Hahn–Neumann construction [10], used in the summability preliminaries of [1, 2]. Appendix A proves it in full, by way of Higman’s finite-word lemma [11], and stresses that the count is over *ordered* words: that stronger form is what noncommuting matrix products require in Sections 44 and 47, and an unordered or multiset count will not carry those theorems.

Warning 2.7 (Splitting a differentiated sum incurs two obligations). If a family $(a_k x^k)_k$ is summable, strong additivity gives $\partial \sum_k a_k x^k = \sum_k \partial(a_k x^k)$. To split the right-hand side into a coefficient-derivative sum and a variable-derivative sum, both separated families must *themselves* be summable. Summability of a family of sums does not imply summability of its summand families, because cancellation can hide a violation. Every Taylor evaluation used below carries the required positive-support certificate; Proposition 4.5 is the one place where the summability of the two split families is *derived* rather than assumed, and it says exactly which extra structure makes that possible.

3 The imported derivation, and smallness as a theorem

3.1 The imported package

Theorem 3.1 (Berarducci–Mantova: imported input). *There is a distinguished derivation $\partial : \mathbf{No} \rightarrow \mathbf{No}$ — the simplest Berarducci–Mantova derivation — with the following properties.*

- (BM1) *Leibniz*: $\partial(xy) = x\partial y + y\partial x$.
- (BM2) *Constant field*: $\ker \partial = \mathbb{R}$, *exactly*.
- (BM3) *Strong additivity*: $\partial(\sum_{j \in J} x_j) = \sum_{j \in J} \partial x_j$ for every set-indexed summable family, and the family of derivatives is itself summable.
- (BM4) *Exponential compatibility*: $\partial(\exp x) = (\exp x)\partial x$, for the real surreal exponential.
- (BM5) *Normalization*: $\partial \omega = 1$.
- (BM6) *Surjectivity*: $\partial \mathbf{No} = \mathbf{No}$.
- (BM7) *H-field positivity*: $x > \mathbb{R}$ implies $\partial x > 0$.

This is the package supplied by [1, Theorems A/6.30 and B/7.7]; the theorem numbering is that of the linked preprint. Its proof is deep and is not reproduced here. Assigning a derivative to a few familiar monomials is not a construction of a strongly additive derivation on every surreal: log-atomic numbers require extra initial data, the construction specifies a pre-derivation there and extends it through summable path derivatives, and both the summability proof and the surjectivity proof are substantial inputs rather than consequences of the product rule. This report uses their conclusions, not a pretended implementation.

Warning 3.2 (Two clauses that are easy to drop, and one that is not in the list). Clause (BM3) is load-bearing for every Hahn-sum computation below, including the infinitesimal exponential and logarithm (Lemma 6.1), the Laurent resolvents (Theorem 18.6), and the coefficient derivation on common-domain families (Theorem 48.4); it is *not* supplied by the Leibniz rule. Clause (BM7) is load-bearing for Lemma 3.8 and hence for the entire obstruction theory, and for the ordering obstruction of Theorem 21.10. Clause (BM5) is indispensable for discussing a *named* equation such as $\partial y = iy$: every named-equation statement below is a statement about this derivation with this normalization, and none of them is derivation-agnostic. (Scope amendment, added with member 11: the sentence just written holds for every part except Part VI, whose statements are proved for every derivation satisfying all of the clauses above except (BM6), and which says so at each statement; see Convention 29.2. No statement outside that part is thereby made derivation-agnostic.)

What is deliberately *not* in the list is *smallness*, the assertion $\partial \mathfrak{m} \subseteq \mathfrak{m}$. Three of the seven source manuscripts import it as a clause of the Berarducci–Mantova package, and one of those flags it as specific to the chosen derivation rather than a general fact. The other four derive it from (BM7). Section 3.3 derives it, for two reasons: advertising smallness as part of an imported theorem overstates the import when it follows from an axiom already assumed; and the derived version additionally yields strict order reversal on the infinitesimals (Lemma 3.8), which the imported version does not give and which Theorem 7.5 uses.

Warning 3.3 (No uniqueness, and no other derivation is classified). Other surreal derivations exist. Theorem 3.1 is a usable interface, not a uniqueness characterization, and the elementary properties listed do not characterize every possible surreal derivation. Nothing below classifies the alternatives. The word “canonical”, where it survives from a source, names this distinguished construction and nothing more. Every intrinsic statement in this report is relative to this one fixed ∂ .

Scope amendment, added with member 11. The last sentence is kept, and it now holds for every part except Part VI. That part works with the class of *normalized* derivations of Definition 29.1 — the clauses of Theorem 3.1 without surjectivity — and proves that one sector is the same for all of them: the solution set of every first-order autonomous equation $F(y, y') = 0$ with $F \in \mathbb{C}[Y, Z]$, and the solutions of every constant meromorphic vector field on a smooth projective curve (Theorem 31.4).

That is a statement about a class of derivations, not a classification of its members: it does not assert that the class has one member, that two members agree on $\mathbf{No}[i]$, or that any theorem of another part holds for another derivation. The only statements of other parts that it reproves for every normalized derivation are the first-order ones listed in Corollary 31.7.

From now on we write $x' = \partial x$ interchangeably. The elementary rules follow at once from Theorem 3.1: for $x \neq 0$, $u > 0$ and $r \in \mathbb{R}$,

$$\partial(x^{-1}) = -x^{-2}\partial x, \quad \partial\left(\frac{x}{y}\right) = \frac{y\partial x - x\partial y}{y^2}, \quad \partial \log u = \frac{\partial u}{u}, \quad \partial(u^r) = r u^{r-1}\partial u, \quad (3.1)$$

where $u^r = \exp(r \log u)$; in particular $\partial \omega^r = r \omega^{r-1}$ and $\partial t^r = -r t^{r-1}$ for ordinary real r (see Warning 2.2), $\partial t = -t^2$, and $\partial \log \omega = t$. With $\ell_0 = \omega$ and $\ell_{n+1} = \log \ell_n$, each ℓ_n is positive infinite and induction on equation (3.1) gives the iterated-logarithm derivatives

$$\partial \ell_n = \frac{1}{\ell_0 \ell_1 \cdots \ell_{n-1}} \quad (n \geq 1), \quad (3.2)$$

with the empty product understood as 1. A rational power rule $\partial(x^q) = qx^{q-1}\partial x$ for $q \in \mathbb{Q}$ and $x > 0$ is available purely algebraically, without the exponential.

For an algebraic element there is a corresponding implicit rule.

Proposition 3.4 (Algebraic implicit differentiation). *Let $F \subseteq \mathbf{No}[i]$ be a differential subfield and let α be algebraic over F with minimal polynomial $P = \sum_k a_k X^k \in F[X]$. Then*

$$\partial \alpha = -\frac{P^\partial(\alpha)}{P_X(\alpha)}, \quad P^\partial(X) = \sum_k (\partial a_k) X^k.$$

Proof. In characteristic zero an irreducible polynomial is separable, so $P_X(\alpha) \neq 0$. Differentiating the finite identity $P(\alpha) = 0$ gives $P^\partial(\alpha) + P_X(\alpha)\partial \alpha = 0$. $\square$

This is the derivative rule for an exact algebraic root object; it is not a choice of a numerical root branch.

3.2 The unique complexification

Proposition 3.5 (Unique extension to $\mathbf{No}[i]$). *There is exactly one derivation on $\mathbf{No}[i] = \mathbf{No}[i]$ extending ∂ , namely*

$$\partial(a + ib) = \partial a + i\partial b.$$

It commutes with conjugation, is strongly additive on set-indexed summable families in $\mathbf{No}[i]$, is surjective, and its constant field is exactly $\mathbb{C}$. Moreover, for $z \neq 0$ and $r = |z|$,

$$\frac{\partial r}{r} = \operatorname{Re}\left(\frac{\partial z}{z}\right). \quad (3.3)$$

Proof. Any extension satisfies $0 = \partial(i^2 + 1) = 2i\partial i$, hence $\partial i = 0$, which forces the displayed formula; direct expansion verifies the Leibniz rule. Conjugation commutes with it by inspection. For strong additivity, take real and imaginary parts: their derivative families are summable by (BM3), a finite union of well-ordered supports is well ordered, and each exponent retains finite multiplicity. The kernel consists of the elements both of whose components lie in $\ker \partial = \mathbb{R}$, that is exactly $\mathbb{C}$. Given $a + ib$, choose real primitives A, B by (BM6); then $\partial(A + iB) = a + ib$. Finally $r^2 = z\bar{z}$ gives $2r\partial r = (\partial z)\bar{z} + z\overline{\partial z}$; divide by $2r^2$ to obtain equation (3.3). $\square$

All seven manuscripts prove this proposition, by this four-line argument. Two consequences deserve emphasis. First, the constant field *changes* from $\mathbb{R}$ to $\mathbb{C}$ on complexification, and several later statements — the kernel of the logarithmic derivative, the fact that solution spaces are $\mathbb{C}$ -spans — depend on that. Second, ∂ is surjective on $\mathbf{No}[i]$ and $\mathbf{No}[i]$ is algebraically closed, and neither fact solves $\partial y = ay$ with $y \neq 0$: surjectivity solves $\partial y = f$, and algebraic closure supplies roots of polynomials in existing elements. Algebraic closure is not differential solution closure, and Section 8 measures the gap exactly.

Convention 3.6 (What “a differential equation over $\mathbf{No}[i]$ ” means). In Parts II and IV, a differential equation over $\mathbf{No}[i]$ is an algebraic equation in elements of $\mathbf{No}[i]$ and their iterated ∂ -images. It is *not* an equation for a class function of an independent variable. Part VIII treats the latter, under a different operator and with a different constant field, and Section 4 keeps them apart.

Every real coefficient is a logarithmic derivative in the real field.

Proposition 3.7 (Real exponential integration). *For every $p \in \mathbf{No}$, the element $s = \exp(\mathcal{I}_0 p)$ of Proposition 7.1 satisfies $\partial s = ps$ and $s > 0$. More generally, for $a, b \in \mathbf{No}$ the equation $\partial y = ay + b$ has surreal solutions, and with $h = \mathcal{I}_0 a$ all real solutions are $y = e^h(c + \mathcal{I}_0(e^{-h}b))$, $c \in \mathbb{R}$.*

Proof. Apply (BM4) to a primitive of p , respectively of a ; the product rule verifies the displayed formula, and the quotient of the difference of two solutions by e^h has derivative zero. $\square$

Notice the two distinct uses of structure: (BM6) supplies the primitive, and (BM4) supplies the exponential. Knowing only that ∂ is onto would not justify the second step. The difficulty in the complex case is therefore not additive integration; it is the absence of a suitable nonzero integrating factor, and Section 8 says exactly when one exists.

3.3 Smallness, derived, with strict inclusion

Lemma 3.8 (Order reversal on infinitesimals, and smallness). *If $0 < \varepsilon \in \mathfrak{m}$ then $\partial\varepsilon < 0$. Consequently ∂ is strictly order-reversing on $\mathfrak{m}$, and*

$$\partial\mathcal{O} = \partial\mathfrak{m} \subseteq \mathfrak{m}.$$

Proof. Since ε is a positive infinitesimal, $1/\varepsilon > \mathbb{R}$, so (BM7) and equation (3.1) give

$$0 < \partial(1/\varepsilon) = -\frac{\partial\varepsilon}{\varepsilon^2},$$

whence $\partial\varepsilon < 0$. For every positive integer n , the element $\omega + n\varepsilon$ is positive infinite, so (BM7) and (BM5) give $1 + n\partial\varepsilon > 0$. Therefore $-1/n < \partial\varepsilon < 0$ for every n , that is $\partial\varepsilon \in \mathfrak{m}$. Negative infinitesimals follow by additivity and 0 is immediate. Strict order reversal follows by applying the first assertion to the positive infinitesimal difference of two infinitesimals. Finally every finite element is $r + \varepsilon$ with $r \in \mathbb{R}$ and $\varepsilon \in \mathfrak{m}$, and $\partial r = 0$, so $\partial\mathcal{O} = \partial\mathfrak{m}$. $\square$

Remark 3.9 (An independent route to smallness, and a second route to the containment). One source manuscript instead *imports* the finite form $\partial\mathcal{O} \subseteq \mathcal{O}$ — which is what the argument needs — and strengthens it to $\partial\mathcal{O} = \partial\mathfrak{m} \subseteq \mathfrak{m}$ by a square-root trick: a positive $\varepsilon \in \mathfrak{m}$ is q^2 with $q = \sqrt{\varepsilon} \in \mathfrak{m}$, so $\partial q \in \mathcal{O}$ by the imported smallness and $\partial\varepsilon = 2q\partial q \in \mathfrak{m}$. That argument is correct and is recorded here as an alternative, but it starts from the containment rather than proving it, and it does not

give the order reversal. A third formulation of the same one-line proof, used by three of the seven, applies (BM7) to $\omega \pm nf$ for $f \in \mathcal{O}$, obtaining $1 \pm n\partial f > 0$ directly; a fourth applies it to $\omega/n \pm f$. All are the same argument, and none uses a mean-value theorem, a topological limit, or an estimate about numerical rounding. Smallness is a structural consequence of the displayed positivity axiom for this derivation.

Warning 3.10 (The containment is strict, and that gap is the whole subject). $\partial\mathfrak{m} \subseteq \mathfrak{m}$ is a *strict* inclusion. Indeed $\partial \log \omega = 1/\omega$, and every primitive of $1/\omega$ differs from $\log \omega$ by a real constant, so every primitive of $1/\omega$ is infinite and $1/\omega \notin \partial\mathfrak{m}$. On the other hand $\omega^{-2} = \partial(-\omega^{-1}) \in \partial\mathfrak{m}$. Testing only whether an imaginary coefficient is infinitesimal therefore loses the decisive information. The entire obstruction theory of Part II is the measurement of this gap, and Warning 5.4 states what goes wrong if it is collapsed.

A first, coarse consequence is already enough to kill a constant nonreal rate.

Proposition 3.11 (A preliminary obstruction). *For every $z \in \mathbf{No}[i]^\times$, the imaginary part of $\partial z/z$ is infinitesimal. Hence for $\lambda \in \mathbb{C} \setminus \mathbb{R}$,*

$$\partial z = \lambda z \implies z = 0.$$

Proof. Multiplying z by 1 or by i does not change $\partial z/z$, so we may write $z = u(1 + ih)$ with $u \in \mathbf{No}^\times$ and $h \in \mathcal{O}$, taking as u a nonzero coordinate of maximal absolute value. Then

$$\frac{\partial z}{z} = \frac{\partial u}{u} + \frac{i\partial h}{1 + ih} = \frac{\partial u}{u} + \frac{h\partial h}{1 + h^2} + i\frac{\partial h}{1 + h^2}.$$

By Lemma 3.8, ∂h is infinitesimal, and division by $1 + h^2 \geq 1$ preserves that. A nonzero ordinary imaginary part is not infinitesimal. $\square$

This already rules out a surcomplex value behaving as $e^{i\omega}$ under ∂ . It does *not* yet classify infinitesimal rates: the exact image is strictly smaller than $\mathbf{No} + i\mathfrak{m}$, and identifying it needs the finite-angle structure of Section 6.

4 One derivation, and seven other operators

Almost every apparent paradox in this subject is a type error. The seven manuscripts between them separate eight operators; the merged report keeps all eight distinct, with the symbols of Table 1. A formula can be correct for one and false for another with no contradiction, and several theorems below are true for one and false for another *on the same printed equation*.

Warning 4.1 (Seven separators, each of which has been mistaken for a paradox). (1) *t is a number, not a symbol.* With $t = \omega^{-1}$ one has $D_z(t^3) = 0$, $D_t(t^3) = 3t^2$ and $\partial(t^3) = -3t^4$, all correct. Treating t as a formal constant is a coefficient operation; substituting $t = \omega^{-1}$ and applying the number derivation gives $\partial t = -t^2$, not 0. Replacing D_t by ∂ in a displayed ordinary differential equation without changing the coefficient functions usually changes the equation. The notation D_t becomes related to ∂ only after an embedding is specified and the formal operations are verified to commute with its interpretation.

(2) *The constant function is not the number.* $F(Z) = \omega$ has $\partial_Z F = 0$, while the number ω has $\partial\omega = 1$. For $f(Z) = e^Z$ at a real point r , $f'(r) = e^r$ but $\partial(f(r)) = \partial(e^r) = 0$.

Table 1: The eight operators. Only the first is “the derivation”.

Symbol	What it does	What it holds fixed	Sample value
∂	The intrinsic Berarducci–Mantova derivation on a <i>number</i> of $\mathbf{No}$ or $\mathbf{No}[i]$	Exactly $\mathbb{R}$, or $\mathbb{C}$ after complexification	$\partial\omega = 1$, $\partial t = -t^2$, $\partial(t^3) = -3t^4$
∂_s	The rescaled derivation $\partial_s f = \partial f / \partial s$, for $\partial s \neq 0$	Same kernel; but not the same normalization	$\partial_s s = 1$; for $s = \log \omega$, $\partial s = t$
∂_Z	Formal differentiation of an indeterminate Z on $\mathbf{No}[i][[Z]]$	Every coefficient in $\mathbf{No}[i]$, however infinite	$\partial_Z(\omega Z) = \omega$, $\partial_Z \omega = 0$
$\tilde{\partial}$	Coefficientwise <i>intrinsic</i> differentiation on $\mathbf{No}[i][[Z]]$, $\tilde{\partial} \sum a_n Z^n = \sum (\partial a_n) Z^n$	The formal indeterminate Z	$\tilde{\partial}(\omega Z) = Z$
D_z	Coordinate differentiation of the holomorphic coefficient functions of a common-domain Hahn family, in an ordinary complex z	Every Hahn monomial and every scalar parameter	$D_z(t^3) = 0$, $D_z \omega = 0$, $D_z z = 1$
∂_H	Intrinsic differentiation of the Hahn <i>monomials</i> of a common-domain family	The ordinary variable z	$\partial_H(\omega z) = z$, $\partial_H(t^2 e^z) = -2t^3 e^z$
D_t	Formal differentiation in a Hahn parameter treated as an indeterminate	The coefficients	$D_t(t^3) = 3t^2$
F'_{fine}	The pointwise fine order-field derivative of a class <i>function</i> , an ϵ – δ difference quotient in the actual field variable	Determined by the function and its topology; not a field derivation	$\text{fin}'_{\text{fine}} \equiv 1$, $\text{pp}'_{\text{fine}} \equiv 0$

- (3) *A fine derivative does not determine an intrinsic one.* For every $x \in \mathbf{No}$ and every finite h , exactly,

$$\text{fin}(x + h) = \text{fin}(x) + h, \quad \text{pp}(x + h) = \text{pp}(x),$$

so the fine derivatives of $x \mapsto \text{fin}(x)$ and $x \mapsto \text{pp}(x)$ are 1 and 0 everywhere, with exact difference quotients for all $0 < |h| < 1$ and no limit subtlety. Yet $\partial(\text{fin}(\omega)) = \partial 0 = 0$ while $\partial(\text{pp}(\omega)) = \partial \omega = 1$. Hence a blanket identity $\partial(F(x)) = F'_{\text{fine}}(x) \partial x$ is *false* for arbitrary fine-differentiable F ; the missing hypothesis is a compatibility structure for F , and fine differentiability is not one. The same example explains why a zero fine derivative need not force a class function to be constant, whereas $\partial f = 0$ for a *scalar* means exactly $f \in \mathbb{C}$.

- (4) *A second fine-derivative separator, with a phase.* Let $H(x) = \text{cis}(\text{fin } x)$, using the finite-angle map of Definition 6.4 on finite angles only. Then $H(\omega) = 1$ and, for infinitesimal h , $H(\omega + h) = E(ih) = 1 + ih + O(h^2)$ as an ordered-field estimate, so $H'_{\text{fine}}(\omega) = i$. Nevertheless

$$\partial(H(\omega)) = \partial 1 = 0 \neq H'_{\text{fine}}(\omega) \partial \omega = i.$$

A locally valid Taylor rule grants no universal scalar chain rule for class functions.

- (5) *Omitting a variable, not using the wrong rule.* For $F(z, t) = tz$ at $z = \omega$, $t = \omega^{-1}$, the value is 1 and

$$\partial F = t \partial \omega + \omega \partial t = t - \omega t^2 = 0,$$

whereas the coordinate derivative $D_z F = t$ alone would return t . Here the ordinary multivariable chain rule is valid; what is wrong is dropping a variable. The same point in one variable: for $F(Z) = tZ$ and $x = \omega$, $\partial(F(\omega)) = (-t^2)\omega + t = 0$, as it must be.

- (6) *A coherent family is not a number.* For $F(z) = tz$ with z a fixed ordinary complex number, $D_z F = t$ while $\partial(F(z)) = -t^2 z$, because that fixed ordinary z is a constant of ∂ .
- (7) *The same printed equation, twice.* $D_z F = iF$, $F(0) = 1$ has the ordinary entire solution e^{iz} , a single-coefficient member of a common-domain Hahn ring. $\partial y = iy$ has no nonzero solution in $\mathbf{No}[i]$ (Corollary 8.8). There is no contradiction: one statement is about a holomorphic function of an ordinary coordinate, the other about a scalar element of a fixed differential field. The nonexistent bridge would be to evaluate the entire function at $z = \omega$, call the result a surcomplex scalar, and assume its scalar derivative comes from the coordinate chain rule. Neither the fixed-domain Hahn representation nor the ordinary initial-value theorem supplies that bridge.

Proposition 4.2 (Change of independent scale changes the solvability data). *Let $s \in \mathbf{No}$ with $\partial s \neq 0$. Then $\partial_s f = \partial f / \partial s$ is a derivation with $\partial_s s = 1$, and*

$$\partial_s y = qy \iff \partial y = (\partial s)qy.$$

Hence the phase test of Theorem 8.4 must be applied to $\text{Im}((\partial s)q)$, not to $\text{Im } q$. With $s = \log \omega$, so $\partial s = t$: the equation $\partial_s y = iy$ is the forbidden $\partial y = ity$, whereas $\partial_s y = ity$ becomes $\partial y = it^2 y$ and is solvable.

The rescaled operator preserves the field but not the normalization at ω , nor necessarily the same order conventions. A backend cannot change the independent scale while leaving differential solvability metadata untouched. At the scale $s = \tau = \log t = -\log \omega$ the rescaled derivation is the Euler derivation $\partial_\tau = -\omega \partial$ of Part V, which acts on real-exponent Hahn series by $t^a \mapsto a t^a$ and reverses the sign rule (BM7) (Lemma 23.1 and Warning 23.4).

Proposition 4.3 (The two formal derivations commute; the valid polynomial chain rule). *On $\mathbf{No}[i][[Z]]$ the derivations ∂_Z and $\tilde{\partial}$ commute, and $\tilde{\partial} + u\partial_Z$ is a derivation for every coefficient scalar u ; commutation with that sum need not hold when $\partial u \neq 0$. For a polynomial $F(Z) = \sum_{k=0}^n a_k Z^k \in \mathbf{No}[i][Z]$ and $x \in \mathbf{No}[i]$,*

$$\partial(F(x)) = (\tilde{\partial}F)(x) + (\partial_Z F)(x) \partial x. \quad (4.1)$$

The same identity holds for rational functions at arguments where the denominator is nonzero.

Proof. Commutation and the derivation property are coefficientwise identities using only finite convolution sums at each degree. For equation (4.1), apply the finite product rule to each $a_k x^k$, obtaining $(\partial a_k)x^k + k a_k x^{k-1} \partial x$, and add. Both the coefficient and the variable derivative obey the quotient rule, which gives the rational case. $\square$

Example 4.4 (Two worked checks of equation (4.1)). For $F(Z) = \omega Z^2 + \omega^{-1}$ at $x = \omega$: $F(x) = \omega^3 + \omega^{-1}$ and $\partial(F(x)) = 3\omega^2 - \omega^{-2}$, whereas the variable derivative alone gives only $2\omega^2$; the missing term is the derivative of the coefficients after evaluation. For $F(Z) = \omega Z + tZ^2$ and $h = \omega^2$: $F(h) = 2\omega^3$ so $\partial(F(h)) = 6\omega^2$, while on the right $(\tilde{\partial}F)(h) = h - t^2 h^2 = 0$ and $(\partial_Z F)(h) \partial h = (\omega + 2th)(2\omega) = 6\omega^2$. The cancellation in the coefficient term is a feature of that example, not a licence to omit coefficient differentiation.

An infinite series needs more, and the extra hypothesis can be made minimal.

Proposition 4.5 (Two-derivative chain rule for series, with derived summability). *Let $F(Z) = \sum_{n \geq 0} a_n Z^n$ with $a_n \in \mathbf{No}[i]$, and let $h \in \mathbf{No}[i]$. If the evaluated family $(a_n h^n)_{n \geq 0}$ is summable, then the two families defining $(\tilde{\partial}F)(h)$ and $(\partial_Z F)(h)$ are summable as well, and*

$$\partial(F(h)) = (\tilde{\partial}F)(h) + (\partial_Z F)(h) \partial h.$$

Proof. Assume first $h \neq 0$. Multiplying each member of a summable family by its ordinary integer index preserves summability, since it enlarges no support; multiplying the whole family by the fixed scalar h^{-1} preserves Hahn summability. Hence $(na_n h^{n-1})_{n \geq 1} = (nh^{-1}(a_n h^n))_{n \geq 1}$ is summable, and so is its multiple by the fixed ∂h . By (BM3) the family $(\partial(a_n h^n))_n$ is summable. Subtracting the already controlled variable-derivative family shows $((\partial a_n)h^n)_n$ is summable. The termwise product rule may now be summed. If $h = 0$, the evaluated series and both formal-derivative evaluations have at most one nonzero term each, and the identity reduces to $\partial a_0 = \partial a_0 + a_1 \partial 0$. $\square$

Remark 4.6 (Why this is stronger than three summability certificates). An arbitrary sum of two families can be summable through cancellation even when neither summand family is (Warning 2.7). The special power-series form is what lets the variable-derivative family be controlled *first*, after which the split is justified. So Proposition 4.5 assumes one certificate and derives two, which is strictly stronger than assuming three independent ones. It still does not assert that an arbitrary formal series is evaluable at an arbitrary scalar; see Section 48 for the relevant separations and for the counterexample of Warning 48.11.

Remark 4.7 (No general composition is assumed). No global operation “replace ω by x in every surreal” is ever used. Berarducci and Mantova construct a composition $f \circ x$ for f in the ω -series class and positive infinite x , identify $(\partial f) \circ x$ with the fine derivative of $x \mapsto f \circ x$, give a compatible chain rule inside that setting, and prove a Taylor theorem there [2, Cor. 7.6–7.7, Thms 7.14, 8.2]. They also prove that the simplest derivation on *all* of **No** admits no composition satisfying their composition and compatibility axioms [2, Thm 8.4]. Section 22 records both results with their hypotheses intact and explains the exact sense in which the two derivatives then agree. All differentiation in this report is either the fixed ∂ , finite algebra, or a summable infinitesimal formal substitution whose validity is proved.

5 The central hazard: three objects called “phase”

5.1 Three names, three symbols

Three different objects occur in the source manuscripts under the single word “phase”. Two of them are angles; the third is not an angle of anything, and does not even live in $\mathcal{O}$. Merging them — in a sentence, in a word, or in a symbol — produces false statements. This report therefore fixes the following vocabulary once and uses it without exception.

Convention 5.1 (The three phases). (a) The *finite angle* θ . The map cis of Definition 6.4 sends the *finite* surreals $\mathcal{O}$ onto the unit circle $\mathbb{T}(\mathbf{No})$ with kernel $2\pi\mathbb{Z}$ (Theorem 6.5). Its argument θ is the whole angle, and it is always finite: $\theta \in \mathcal{O}$. No value of a circular function at an infinite angle is ever used.

(b) The *infinitesimal phase residue* η . Every unit $u \in \mathbb{T}(\mathbf{No})$ factors *uniquely* as $u = cE(i\eta)$ with c an ordinary unit complex number and $\eta \in \mathfrak{m}$ (Theorem 6.3). The constant c carries all of the ordinary phase, $\partial c = 0$, and η carries all of the variable intrinsic phase: $u^\dagger = i\partial\eta$. The two are related to (a) by $\theta = \text{st}(\theta) + \eta$ and $c = e^{i\text{st}(\theta)}$, so (b) is the *infinitesimal part* of (a), not a synonym for it.

(c) The *accumulated phase* B . For a real coefficient b , an accumulated phase is any $B \in \mathbf{No}$ with $\partial B = b$; the normalized choice is $\mathcal{I}_0(b)$. Any two differ by a real constant. B is an arbitrary surreal. It need not be finite, it is not the angle of any surcomplex number, and when it is infinite it lies outside $\mathcal{O}$ entirely, with its purely infinite part $\text{pp}(B) \in \mathcal{P}$ — the

obstruction — nonzero. The whole theory of Part II is the statement that $\partial y = (a + ib)y$ is solvable exactly when the accumulated phase of b is finite, that is $B \in \mathcal{O}$. Only then does B acquire an angular reading at all: a finite B is the argument of $\text{cis } B$ in sense (a), with $\eta = B - \text{st}(B)$ its infinitesimal residue in sense (b). An infinite B is the argument of nothing. So (c) is not a third kind of angle; it is the object whose *finiteness* is what sense (a) requires, and the criterion is exactly the assertion that it can fail.

- (d) **A missing oscillatory generator.** In Section 21.4 and in the Picard–Vessiot sections the phrase *independent phases* counts something else again: the number of transcendental generators $\mathbb{T}$ that must be adjoined *outside* $\mathbf{No}[i]$ before a diagonal system has a full solution basis. That count is $d = m - \text{rank } \Lambda$, a rank of a lattice of obstructions, and it is not a count of angles, residues or accumulated phases. Wherever this report writes “independent phases” or “phase generators”, that fourth sense is meant, and it is never any of (a)–(c).
- (e) The *differential phase spectrum* $\text{Ph}(A)$. For a finite matrix system $\partial y = Ay$ over $\mathbf{No}[i]$, Theorem 15.9 attaches a *multiset* $\text{Ph}(A) = \{P_1, \dots, P_n\}$ of purely infinite surreals: the accumulated obstructions of the diagonal entries of its normal form. Its members are objects of sense (c) — more precisely, values of $\mathfrak{o}$ — so Ph adds no new kind of phase; what is new is that there are n of them at once and that the multiset, not its sum, is the complete invariant (Corollary 15.14). The eighth source manuscript writes Ψ for the map $b \mapsto \text{pp}(B)$ with $\partial B = b$. That map is exactly $\mathfrak{o}$ of Definition 7.4, so this report keeps the single symbol $\mathfrak{o}$ for it and does not adopt Ψ , in accordance with Convention 5.6. The adoption would in any case have collided: the letter Ψ already occurs once in this report, in Corollary 11.4, as a local placeholder for a putative derivation-compatible exponential $\mathbf{No}[i] \rightarrow \mathbf{No}[i]^\times$ — a map that is there proved not to exist, and that is not a phase map of any of the senses (a)–(e).

Warning 5.2 (“Spectrum” names three inequivalent things across the collection). The differential phase spectrum $\text{Ph}(A)$ of sense (e) is a multiset of *purely infinite surreal numbers*. It is not a list of eigenvalues: the companion report *Surcomplex Spectral Theory* [27] uses “spectrum” for the algebraic eigenvalue and singular-value lists of a finite matrix and their valuations, which are ordinary surcomplex numbers of the field, and Theorem 16.5 is precisely the theorem that relates the two — by integrating and taking purely infinite parts, not by identifying them. Example 16.10 shows that for a nonnormal coefficient the two are unrelated. Nor is $\text{Ph}(A)$ a set of angles: no member of a phase spectrum is ever routed through the finite-angle map cis of sense (a) or through the finite-angle constructions of [23], since a nonzero member of $\mathcal{P}$ is infinite and cis is defined on $\mathcal{O}$ only. And it is not a prime spectrum of a ring: where the collection’s analytic-geometry report discusses Spec of a coefficient ring, that is the ring-theoretic object and is written out as *prime spectrum* there.

Warning 5.3 (Where the senses collide in the sources). Two of the first seven manuscripts use “phase” for sense (a), four use it for sense (b), and all seven use “accumulated phase” for sense (c); the eighth introduces sense (e) and, with it, a fourth letter for the primitive obstruction. One of them prints (a) and (b) twenty lines apart, both called the phase. Sense (c) is the one that carries the obstruction, and it is the one that is *not* an angle: $\Phi(i) = \omega$ and $\Phi(i/\omega) = \log \omega$ are purely infinite surreals, not arguments of anything. A merged sentence such as “the phase of the solution is $\log \omega$ ” is meaningless; the correct sentence is “the accumulated phase of the coefficient i/ω is $\log \omega$, which is infinite, so there is no solution”. Similarly, “every surcomplex number has a finite phase” is a theorem about (a); “the variable phase of a unit is infinitesimal” is a theorem about (b); and neither implies anything about (c) without Theorem 8.4. Two later members add two further uses. Member 12 calls the ordinary real numbers $b = \text{Im } \lambda$ “phase labels”; they are none of (a)–(e),

Table 2: The phase objects, and where each source’s notation lands.

Name here	Symbol	Lives in	What it is	Source notations
Finite angle	θ	$\mathcal{O}$	The argument of cis; the <i>whole</i> angle of a unit, finite by theorem	θ, y
Infinitesimal phase residue	η	$\mathfrak{m}$	What is left of a unit after factoring out the constant ordinary unit complex number c	$\eta, \theta, \varepsilon, \delta$
Accumulated phase	B	No	A primitive of the imaginary coefficient; <i>not</i> an angle; obstructed exactly when $\text{pp}(B) \neq 0$	$B; \mathcal{I}_0(b)$; and four different letters for the primitive operator
Differential phase spectrum	$\text{Ph}(A)$	multisets in $\mathcal{P}$	The n accumulated obstructions of a finite matrix system, as a multiset; the complete invariant of differential equivalence	$\text{Ph}; \Psi(\lambda_j)$, where the map Ψ is this report’s σ ; member 12’s “phase labels” b , which are ordinary real residual frequencies with $\text{Ph} = \{b \log t\}$ (Warning 23.7)

this report calls them residual frequencies, and the frequency b corresponds to the phase $b \log t$ of sense (e) (Warning 23.7). Member 11 uses the word only to cite this report, and its one “finite angle” is sense (a); its abelian logarithmic derivative is a tangent vector, not an angle (Section 33.3). Member 13 uses “phase obstruction” for this report’s criterion and writes θ for a real infinitesimal primitive of the angular rate of a unit; that is sense (b), written η here (Convention 35.1).

5.2 Two load-bearing classes that differ by one operator

The second half of the hazard is not a word but a symbol. Four of the seven manuscripts print, within a few pages of each other, the two classes

$$\mathbf{No} + i\mathcal{A} \quad \text{and} \quad \mathcal{S} = \mathbf{No} + i\mathcal{O},$$

where $\mathcal{A} = \partial\mathfrak{m}$. The first is the exact image of the logarithmic derivative (Theorem 8.1); the second is the maximal domain of a derivation-compatible exponential (Theorem 11.2). They are different classes, and the difference is one application of ∂ .

Warning 5.4 (Do not collapse $\partial\mathcal{O}$ to $\mathcal{O}$). $\mathcal{A} = \partial\mathfrak{m}$ is a *strict* subclass of $\mathfrak{m}$ (Warning 3.10), and $\mathfrak{m}$ is a strict subclass of $\mathcal{O}$. If a merge collapses $\partial\mathcal{O}$ to $\mathcal{O}$, the rank-one criterion becomes “ b finite”. Even the stronger replacement “ b infinitesimal”, obtained by replacing $\mathcal{A}$ with $\mathfrak{m}$, would make $\partial y = i\omega^{-1}y$ solvable. Every one of the seven manuscripts refutes that conclusion, and the counterexample pair of Example 10.4 — $i/(\omega \log \omega)$ insoluble against $i/(\omega(\log \omega)^2)$ soluble — exists precisely to block it. The strip $\mathcal{S} = \mathbf{No} + i\mathcal{O}$ is ∂ -stable *because* $\partial\mathcal{O} \subseteq \mathfrak{m}$; that is a consequence of the difference between the two classes, not evidence that they are the same.

Warning 5.5 ($\mathbf{No} + i\partial\mathcal{O}$ and $\mathbf{No} + i\partial\mathfrak{m}$ are equal only after a lemma). The sources print the logarithmic-derivative image in both forms. They agree, but only because $\partial\mathcal{O} = \partial\mathfrak{m}$, which is a *theorem* (Lemma 3.8, with the independent square-root route of Remark 3.9). The two forms must never be presented as notational variants: one is the other after a proved step. This report uses the single symbol $\mathcal{A}$ for the common value and states the lemma before using it.

5.3 The letter J , and the word “obstruction”

Across the source batch the letter J denotes three unrelated objects: in one manuscript it is the normalized intrinsic primitive on $\mathbf{No}$; in another it is not an operator at all but the representative *class* of purely infinite surreals; and in a companion report that remains standalone — the rank-one Berkovich manuscript, on analytic functions on surcomplex disks and annuli — it is the zero-constant *coordinate* primitive of a convergent power series over a fixed coefficient field, whose “complete primitive obstruction” is an annular Laurent residue and has nothing to do with the phase obstruction here. Two further manuscripts write J for the inverse of $\partial|_{\mathfrak{m}}$.

Convention 5.6 (No letter J ; one symbol per object). This report uses no symbol J for any operator or class. The letter occurs only as a neutral index set for set-indexed families (Definition 2.4 and Theorem 3.1), where no confusion with the objects below is possible. The normalized intrinsic primitive is $\mathcal{I}_0$ (Proposition 7.1), with values in $\mathcal{P} \oplus \mathfrak{m}$, the zero-constant-term surreals. The purely infinite class $\mathcal{P}$ is the target of the obstruction map, not of $\mathcal{I}_0$. The inverse of $\partial|_{\mathfrak{m}}$ is written $\mathcal{I}_0|_{\mathcal{A}}$ and named the *unique infinitesimal primitive*. The obstruction on $\mathbf{No}$ is $\mathfrak{o}$ and its surcomplex companion is Φ (Definition 7.4). Cross-references to companion reports name the operator rather than the letter.

Remark 5.7 (The obstruction does not depend on the normalization). Adding a real constant to a primitive does not change its purely infinite part, so $\mathfrak{o}$ is insensitive to the normalization of $\mathcal{I}_0$. That insensitivity is what makes $\mathfrak{o}$ well defined, and it is why the normalization is a convenience rather than a hypothesis. It is also why a “primitive” defined only up to $\mathbb{R}$, or selected by a choice function, is a *different* object: the first would not give a well-defined primitive operator, and the second would not give a class function without a choice principle. $\mathcal{I}_0$ avoids both by singling out a unique element; see Proposition 7.1 and Warning 7.2.

Part II

The scalar theory over $\mathbf{No}[i]$

6 Infinitesimal exponential, finite angle, polar form

6.1 The infinitesimal exponential and logarithm

For $h \in \mathfrak{m}_{\mathbb{C}}$ define the Hahn sums

$$\mathbf{E}(h) = \sum_{n \geq 0} \frac{h^n}{n!}, \quad \mathbf{L}(1+h) = \sum_{n \geq 1} \frac{(-1)^{n+1}}{n} h^n. \quad (6.1)$$

Lemma 2.6 makes both families summable, because the support of a nonzero infinitesimal is positive and well ordered. These are defined by strong Hahn summation, without a topological limit. Their domain is *infinitesimal arguments only*: no attempt is made to sum the ordinary exponential Taylor series at a nonzero ordinary constant, still less at an infinite argument.

Lemma 6.1 (Formal identities survive infinitesimal evaluation). *The maps in equation (6.1) are inverse group isomorphisms*

$$(\mathfrak{m}_{\mathbb{C}}, +) \xrightleftharpoons[\mathbf{L}]{\mathbf{E}} (1 + \mathfrak{m}_{\mathbb{C}}, \cdot),$$

they commute with conjugation, and

$$\partial E(h) = E(h) \partial h, \quad \partial L(w) = \frac{\partial w}{w} \quad (w \in 1 + \mathfrak{m}_{\mathbb{C}}). \quad (6.2)$$

Moreover E is injective on $\mathfrak{m}_{\mathbb{C}}$: for $h \in \mathfrak{m}_{\mathbb{C}}$,

$$E(h) = 1 \iff h = 0. \quad (6.3)$$

More generally, for any $f(X) = \sum_{n \geq 0} c_n X^n$ with ordinary constant coefficients $c_n \in \mathbb{C}$ and any $h \in \mathfrak{m}_{\mathbb{C}}$, the family $(c_n h^n)_n$ is summable, its sum is denoted $f(h)$, and

$$\partial(f(h)) = f'_{\text{formal}}(h) \partial h. \quad (6.4)$$

Proof. The formal identities $\exp(U + V) = \exp(U) \exp(V)$, $\log(\exp U) = U$ and $\exp(\log(1 + U)) = 1 + U$ hold over the ordinary characteristic-zero coefficient field. After substitution at infinitesimals, Lemma 2.6 makes every coefficient calculation a finite one at each fixed exponent, which justifies the products and regroupings; this is a finite check per exponent, not an appeal to a limit. Conjugation acts on coefficients and preserves supports.

For equation (6.2) and equation (6.4), the finite product rule gives $\partial(h^n) = nh^{n-1} \partial h$, and (BM3) permits termwise differentiation, including the assertion that the differentiated family is summable. The family $(nc_n h^{n-1})_{n \geq 1}$ is separately summable by Lemma 2.6, and multiplication by the single fixed scalar ∂h preserves summability; so the factorisation

$$\partial E(h) = \partial h \sum_{m \geq 0} \frac{h^m}{m!}$$

and, for the logarithm, $\partial h \sum_{n \geq 1} (-1)^{n+1} h^{n-1} = \partial h / (1 + h)$, are both legitimate. For equation (6.3): if $h \neq 0$, every power h^n with $n \geq 2$ has valuation strictly larger than that of h , so the leading term of $E(h) - 1$ is the leading term of h . $\square$

Warning 6.2 (Three limits on Lemma 6.1). (i) The injectivity equation (6.3) is *local* to $\mathfrak{m}_{\mathbb{C}}$ and asserts nothing about an injective exponential on the whole class $\mathbf{No}[i]$. (ii) The argument does *not* extend to a Taylor series at an arbitrary infinite argument: the original family need not be summable there, and Remark 11.9 exhibits the failure. (iii) The evaluation rule $f(h)$ requires *constant* coefficients. A formal series with arbitrary coefficients in $\mathbf{No}[i]$ need not be summable at any given nonzero infinitesimal: $\sum_{n \geq 0} \omega^{n^2} X^n$ is not summable at $X = \omega^{-1}$, because its monomials ω^{n^2-n} are not reverse well ordered. The infinitesimal chain rule must not be exported to arbitrary $\mathbf{No}[i]$ -coefficient streams without a joint support certificate (Warning 48.11). No lower bound on an ordinary radius of convergence is needed, however: factorially growing complex coefficients remain coefficients at the same Hahn monomials.

6.2 The finite angle

Let

$$\mathbb{T}(\mathbf{No}) = \{u \in \mathbf{No}[i] : u\bar{u} = 1\}, \quad \mathbb{T}(\mathbb{C}) = \{c \in \mathbb{C} : |c| = 1\}.$$

Theorem 6.3 (Unique infinitesimal phase residue). *Every $u \in \mathbb{T}(\mathbf{No})$ has a unique representation*

$$u = c E(i\eta), \quad c \in \mathbb{T}(\mathbb{C}), \quad \eta \in \mathfrak{m}, \quad (6.5)$$

namely $c = \text{st}(u)$ and $i\eta = L(u/c)$; and

$$u^\dagger := \frac{\partial u}{u} = i \partial \eta. \quad (6.6)$$

Consequently $\mathbb{T}(\mathbf{No}) \cong \mathbb{T}(\mathbb{C}) \times (\mathfrak{m}, +)$ as abelian groups.

Proof. Write $u = a + ib$ with $a^2 + b^2 = 1$; both coordinates are finite, so $c = \text{st}(u)$ exists and $c\bar{c} = 1$, whence $c \neq 0$. Then $w = u/c$ lies in $1 + \mathfrak{m}_{\mathbb{C}}$ and satisfies $w\bar{w} = 1$, that is $\bar{w} = w^{-1}$. Put $q = L(w)$. By Lemma 6.1,

$$\bar{q} = L(\bar{w}) = L(w^{-1}) = -q,$$

so $q = i\eta$ with $\eta \in \mathfrak{m}$ real, and $w = E(i\eta)$. Conversely every expression equation (6.5) has modulus one, its standard part recovers c , and the injective local logarithm recovers η . Finally $\partial c = 0$, so equation (6.2) gives equation (6.6). Multiplication multiplies the c 's and adds the η 's. $\square$

The proof chooses no global value of $\sin \omega$, no branch of a global logarithm and no global circular character. The ordinary phase is stored as the *discrete datum* c , not as a selected real argument, so there is no ambiguity by multiples of 2π : the branch ambiguity of an angle is an ordinary constant $2\pi k$, which ∂ kills, and the infinitesimal residue in equation (6.5) is branch-free outright.

Definition 6.4 (The finite-angle map). For a *finite* real surreal $\theta = r + \eta$, with $r = \text{st}(\theta) \in \mathbb{R}$ and $\eta \in \mathfrak{m}$, set

$$\text{cis}(\theta) = e^{ir} E(i\eta),$$

where $e^{ir} \in \mathbb{C}$ is the ordinary complex exponential, a differential constant. The real and imaginary coordinates of cis are the canonical finite-angle cosine and sine, and they agree with their Taylor construction. The domain is $\mathcal{O}$ only; no value of a circular function at an infinite angle is used.

Theorem 6.5 (Finite-angle phase group, with its differential identity). *The map $\text{cis} : (\mathcal{O}, +) \rightarrow (\mathbb{T}(\mathbf{No}), \cdot)$ is a surjective group homomorphism with kernel exactly $2\pi\mathbb{Z}$, and*

$$\partial \text{cis}(\theta) = i(\partial\theta) \text{cis}(\theta), \quad \text{equivalently} \quad \frac{\partial \text{cis}(\theta)}{\text{cis}(\theta)} = i \partial\theta. \quad (6.7)$$

Consequently every nonzero $y \in \mathbf{No}[i]$ is $y = \rho \text{cis}(\theta)$ with $\rho > 0$ in $\mathbf{No}$ and θ finite.

Proof. The group law follows from Lemma 6.1 and the ordinary phase group law; conjugation inverts $E(i\eta)$, so the image lies in $\mathbb{T}(\mathbf{No})$. For surjectivity, let $u \in \mathbb{T}(\mathbf{No})$ and take c, η from Theorem 6.3; choosing an ordinary real r with $e^{ir} = c$ gives $u = \text{cis}(r + \eta)$. If $\text{cis}(r + \eta) = 1$, standard parts give $e^{ir} = 1$, so $r \in 2\pi\mathbb{Z}$, and then equation (6.3) gives $\eta = 0$; conversely these elements lie in the kernel. Since r and e^{ir} are differential constants, equation (6.2) gives equation (6.7). Applying surjectivity to $y/|y|$ gives the last assertion. $\square$

The surjectivity and kernel content of Theorem 6.5 is the finite-angle phenomenon already central to the repository's trigonometry report [23]; it is reproved here because the merged development needs it with the derivative identity attached. The differential clause equation (6.7) is the part the present theory adds, and it is where the obstruction originates: the factor $\partial\theta$ is indispensable, and removing it would replace an infinitesimal angular rate by the forbidden constant rate 1. In particular, for an infinitesimal ε the finite-angle sine and cosine satisfy $\partial \sin \varepsilon = (\cos \varepsilon) \partial \varepsilon$ and $\partial \cos \varepsilon = -(\sin \varepsilon) \partial \varepsilon$, so there is no conflict between the existence of finite-angle trigonometry and the nonexistence of an intrinsic oscillator (Corollary 8.8).

6.3 Polar decomposition and its differential content

Definition 6.6 (Logarithmic derivative). For $z \in \mathbf{No}[i]^\times$ write $z^\dagger = \partial z / z$. The notation does not presuppose the existence of a complex logarithm. It is a group homomorphism $(\mathbf{No}[i]^\times, \cdot) \rightarrow (\mathbf{No}[i], +)$: $(zw)^\dagger = z^\dagger + w^\dagger$, $(z^{-1})^\dagger = -z^\dagger$, and $\ker(\dagger) = \mathbb{C}^\times$ by Proposition 3.5.

Corollary 6.7 (Branch-free polar form). *Every $z \in \mathbf{No}[i]^\times$ has a unique decomposition*

$$z = \rho c E(i\eta), \quad \rho = |z| > 0, \quad c = \text{st}(z/|z|) \in \mathbb{T}(\mathbb{C}), \quad \eta \in \mathfrak{m}, \quad (6.8)$$

and

$$z^\dagger = \partial \log \rho + i \partial \eta = \frac{\partial \rho}{\rho} + i \partial \eta. \quad (6.9)$$

In coordinates $z = u + iv$,

$$\text{Re}(z^\dagger) = \frac{uu' + vv'}{u^2 + v^2}, \quad \text{Im}(z^\dagger) = \frac{uv' - vu'}{u^2 + v^2}. \quad (6.10)$$

Equivalently $z = \rho \text{cis}(\theta)$ with $\theta \in \mathcal{O}$, and then $\text{Re}(z^\dagger) = \partial|z|/|z|$ and $\text{Im}(z^\dagger) = \partial\theta$.

Proof. Apply Theorem 6.3 to $z/|z|$ and use $\partial \log \rho = \partial \rho / \rho$ together with equation (6.6); the coordinate formulas are direct computation. The cis form is Theorem 6.5, and $\partial\theta = \partial\eta$ because $\theta - \eta = \text{st}(\theta) \in \mathbb{R}$. $\square$

The first summand of equation (6.9) is a radial logarithmic rate and is unrestricted; the second is an angular rate, and its allowed primitives are constrained to finite angles. An enormous modulus does not create an enormous circular phase: the direction of $\omega + i$, for example, is a small perturbation of 1. Identity equation (6.9) is precisely the reason that an arbitrary radial logarithmic derivative is possible while an arbitrary imaginary one is not.

Warning 6.8 (A unit cannot hide an infinite accumulated phase). A unit u may be denoted by an expression involving a large parameter. As an element of $\mathbf{No}[i]$ it nevertheless has a finite angle θ , and hence $u^\dagger = i\partial\theta \in i\mathcal{A}$. A notation such as $e^{i \log \omega}$ cannot, by itself, turn an infinitesimal coefficient into a permissible logarithmic derivative. This is not a restriction on ordinary complex exponentiation, where the variable is a coordinate, and it is not a claim that global phase characters are impossible; it is a restriction on compatibility with the specified scalar derivation.

7 Primitives, accumulated phase, and the bounded-primitive ideal

7.1 The normalized primitive

Proposition 7.1 (Normalized primitive). *For each $b \in \mathbf{No}$ there is a unique $\mathcal{I}_0(b) \in \mathbf{No}$ with*

$$\partial \mathcal{I}_0(b) = b, \quad \text{ct}(\mathcal{I}_0(b)) = 0.$$

The map $\mathcal{I}_0$ is $\mathbb{R}$ -linear and satisfies $\partial \circ \mathcal{I}_0 = \text{id}$ and $\mathcal{I}_0 \circ \partial = \text{id} - \text{ct}$. Its componentwise extension $\mathcal{I}_0(a + ib) = \mathcal{I}_0(a) + i\mathcal{I}_0(b)$ is the unique $\mathbb{C}$ -linear map $\mathbf{No}[i] \rightarrow \mathbf{No}[i]$ with $\partial \mathcal{I}_0 = \text{id}$ and $\text{ct} \circ \mathcal{I}_0 = 0$, and it satisfies $\mathcal{I}_0(\partial f) = f - \text{ct}(f)$.

Proof. By (BM6) choose any primitive F of b and put $\mathcal{I}_0(b) = F - \text{ct}(F)$. Any two primitives differ by an element of $\ker \partial = \mathbb{R}$, and if both have vanishing ω^0 coefficient that difference is zero; this gives existence, uniqueness and independence of the choice of F . Linearity follows from uniqueness, since a sum or real multiple of normalized primitives has the required derivative and normalization. The identity $\mathcal{I}_0(\partial f) = f - \text{ct}(f)$ holds because $f - \text{ct}(f)$ is a primitive of ∂f with zero constant coefficient. The complex statements follow componentwise from Proposition 3.5. $\square$

Warning 7.2 ($\mathcal{I}_0$ is a normalization, not an algorithm). The definition is by a *uniqueness* property and makes no assertion about an effective primitive-finding procedure. The normalization refers to the coefficient of ω^0 , *not* to evaluation at an initial point; $\mathcal{I}_0$ is not asserted to be multiplicative, not asserted to preserve arbitrary Hahn sums, and not asserted to be computable from a supplied symbolic representation. It does not require a choice function on primitive families — the normalization singles out a unique element, so no choice principle is involved. Its value is semantic precision. Sample values: $\mathcal{I}_0(1) = \omega$, and $\mathcal{I}_0(\omega^{-1}) = \log \omega - \text{ct}(\log \omega)$, which is infinite regardless of its real constant term.

The integration constant is not evaluation at an endpoint, and one familiar algebraic identity accordingly acquires a correction term.

Proposition 7.3 (Integration by parts, and a Rota–Baxter correction). *For $f, g \in \mathbf{No}[i]$,*

$$\mathcal{I}_0(f\partial g + g\partial f) = fg - \text{ct}(fg), \quad (7.1)$$

whose final term is not in general $\text{ct}(f)\text{ct}(g)$; and

$$\mathcal{I}_0(f)\mathcal{I}_0(g) = \mathcal{I}_0(f\mathcal{I}_0(g) + \mathcal{I}_0(f)g) + \text{ct}(\mathcal{I}_0(f)\mathcal{I}_0(g)). \quad (7.2)$$

The correction term in equation (7.2) cannot in general be omitted.

Proof. Both identities follow by differentiating the left-hand sides and applying $\mathcal{I}_0 \circ \partial = \text{id} - \text{ct}$. For the last assertion take $f = 1$ and $g = t^2$: then $\mathcal{I}_0(f) = \omega$, $\mathcal{I}_0(g) = -t$, so $\mathcal{I}_0(f)\mathcal{I}_0(g) = -1$, while $f\mathcal{I}_0(g) + \mathcal{I}_0(f)g = -t + \omega t^2 = 0$. Without the correction the identity would assert $-1 = 0$. $\square$

In the terminology sometimes used for integral operators, $\mathcal{I}_0$ is therefore *not* a weight-zero Rota–Baxter operator on all of $\mathbf{No}[i]$, and equation (7.2) is the precise replacement. No such terminology is needed to use the formula; the point is that the selected constant coefficient is linear but not multiplicative (Warning 2.3), so a “zero constant of integration” convention is not automatically an integral based at a common evaluation point.

7.2 Accumulated phase and its obstruction

Definition 7.4 (Accumulated phase and obstruction). For $b \in \mathbf{No}$, an *accumulated phase* of b is any $B \in \mathbf{No}$ with $\partial B = b$; the normalized choice is $B = \mathcal{I}_0(b)$. Put

$$\mathcal{A} := \partial \mathbf{m} = \partial \mathcal{O}, \quad \mathfrak{o}(b) := \text{pp}(\mathcal{I}_0(b)) \in \mathcal{P},$$

and for $q \in \mathbf{No}[i]$ put

$$\Phi(q) := \mathfrak{o}(\text{Im } q) \in \mathcal{P}.$$

We call $\mathcal{A}$ the *bounded-primitive ideal* (equivalently, the class of *admissible angular rates*), and $\mathfrak{o}(b)$, $\Phi(q)$ the *phase obstruction*.

By Remark 5.7 the value $\mathfrak{o}(b) = \text{pp}(B)$ is the same for every accumulated phase B of b , so $\mathfrak{o}$ is well defined without the normalization. It has an actual surreal representative, not an ambiguous antiderivative choice. Sample values:

$$\mathfrak{o}(1) = \omega, \quad \mathfrak{o}(\omega^{-1}) = \log \omega, \quad \mathfrak{o}(\omega^{-2}) = 0. \quad (7.3)$$

The obstruction measures accumulated phase, not the instantaneous size of a coefficient: $\Phi(a) = 0$ for every real coefficient $a \in \mathbf{No}$, however large, whereas $\Phi(ib) = \mathfrak{o}(b)$ may be nonzero even for infinitesimal b .

Theorem 7.5 (Structure of the bounded-primitive ideal). (i) $\mathcal{A} = \partial\mathfrak{m} = \partial\mathcal{O}$, and $\partial : \mathfrak{m} \rightarrow \mathcal{A}$ is an $\mathbb{R}$ -linear bijection that reverses strict order.

(ii) For $b \in \mathbf{No}$ the following are equivalent: $b \in \mathcal{A}$; $\mathcal{I}_0(b)$ is infinitesimal; $\mathcal{I}_0(b)$ is finite; some primitive of b is finite; every primitive of b is finite; $\mathfrak{o}(b) = 0$.

(iii) $\mathcal{A}$ is convex in $\mathbf{No}$, is an additive subgroup, and is an $\mathcal{O}$ -submodule of $(\mathbf{No}, +)$, that is, an ideal of the valuation ring $\mathcal{O}$.

(iv) $0 \neq \mathcal{A} \subsetneq \mathfrak{m} \subsetneq \mathcal{O}$, with witnesses $\omega^{-2} = \partial(-\omega^{-1}) \in \mathcal{A}$ and $\omega^{-1} \in \mathfrak{m} \setminus \mathcal{A}$.

(v) There is an exact cut description

$$\mathcal{A} = \{b \in \mathbf{No} : |b| < \partial f \text{ for every } f > \mathbb{R}\}. \quad (7.4)$$

(vi) $\mathfrak{o} : \mathbf{No} \rightarrow \mathcal{P}$ is $\mathbb{R}$ -linear and surjective with kernel exactly $\mathcal{A}$; indeed $\mathcal{I}_0(\partial P) = P$ and $\mathfrak{o}(\partial P) = P$ for $P \in \mathcal{P}$.

(vii) As additive real vector classes,

$$\mathbf{No} = \partial\mathcal{P} \oplus \mathcal{A} = \partial\mathcal{P} \oplus \partial\mathfrak{m}, \quad (7.5)$$

with projection $b \mapsto \partial \text{pp} \mathcal{I}_0(b)$ onto the first summand; ∂ is injective on each of $\mathcal{P}$ and $\mathfrak{m}$ separately, and $\partial : \mathcal{P} \rightarrow \partial\mathcal{P}$ is an order-preserving $\mathbb{R}$ -linear bijection.

(viii) ∂ induces an order-preserving $\mathbb{R}$ -linear isomorphism of ordered additive quotients

$$\mathbf{No}/\mathcal{O} \longrightarrow \mathbf{No}/\mathcal{A}, \quad x + \mathcal{O} \longmapsto \partial x + \mathcal{A}, \quad (7.6)$$

and in particular $\partial^{-1}(\mathcal{A}) = \mathcal{O}$.

Proof. (i) The equality $\partial\mathcal{O} = \partial\mathfrak{m}$ and the order reversal are Lemma 3.8; the kernel of $\partial|_{\mathfrak{m}}$ is $\mathfrak{m} \cap \mathbb{R} = \{0\}$, giving injectivity, and surjectivity onto $\mathcal{A}$ is the definition.

(ii) A primitive in $\mathfrak{m}$ is finite; conversely a finite primitive becomes infinitesimal after subtracting its standard part. All primitives differ by a real constant, so either all are finite or none is. Since $\text{ct}(\mathcal{I}_0(b)) = 0$, the splitting equation (2.3) makes $\mathcal{I}_0(b)$ finite exactly when it is infinitesimal, exactly when $\text{pp} \mathcal{I}_0(b) = 0$.

(iii) Suppose $0 < b \leq d$ with $d \in \mathcal{A}$. Write $d = -\partial\varepsilon$ with $0 \leq \varepsilon \in \mathfrak{m}$, using (i), and let B be a primitive of b . If B were negative infinite then $-B$ would be positive infinite and (BM7) would give $b = \partial B < 0$, a contradiction. If B were positive infinite then $B + \varepsilon$ would be positive infinite, whence $\partial(B + \varepsilon) = b - d > 0$, contradicting $b \leq d$. So B is finite and $b \in \mathcal{A}$ by (ii). Sign symmetry and additivity give convexity in general. For $a \in \mathcal{O}$ choose a positive integer $n > |a|$; then $|ab| \leq n|b|$ for $b \in \mathcal{A}$, and convexity with real linearity gives $ab \in \mathcal{A}$.

- (iv) $\mathcal{A} \subseteq \mathfrak{m}$ is Lemma 3.8. Nonzero and strict: the displayed witnesses, using Warning 3.10.
- (v) If $b = \partial\varepsilon$ with $\varepsilon \in \mathfrak{m}$, then for every $f > \mathbb{R}$ both $f + \varepsilon$ and $f - \varepsilon$ are positive infinite, so their derivatives are positive and $\partial f > |b|$. Conversely let $b \notin \mathcal{A}$ and let B be a primitive; then B is infinite. Take $f = |B|/2 > \mathbb{R}$; then $\partial f = |b|/2$ by (BM7), so the required inequality $|b| < \partial f$ fails.
- (vi) Linearity is inherited from $\mathcal{I}_0$ and pp. If $P \in \mathcal{P}$ then $\text{ct}(P) = 0$, so P is the normalized primitive of ∂P and $\mathfrak{o}(\partial P) = \text{pp}(P) = P$; this gives surjectivity. The kernel is (ii).
- (vii) Decompose $\mathcal{I}_0(b) = P + \varepsilon$ with $P = \text{pp } \mathcal{I}_0(b) \in \mathcal{P}$ and $\varepsilon \in \mathfrak{m}$, and differentiate: $b = \partial P + \partial\varepsilon$. If $\partial P = \partial\varepsilon$ with $P \in \mathcal{P}$, $\varepsilon \in \mathfrak{m}$, then $P - \varepsilon \in \mathbb{R}$ and uniqueness of equation (2.3) forces $P = \varepsilon = 0$; the same argument with one term zero gives the injectivity assertions. A nonzero positive member of $\mathcal{P}$ is positive infinite, so its derivative is positive by (BM7); applying this to differences gives the order assertion.
- (viii) Surjectivity is inherited from (BM6), and $\partial^{-1}(\mathcal{A}) = \mathcal{O}$ is (ii) applied to ∂x . If $x + \mathcal{O}$ is positive then x is positive infinite, and for every $d = \partial\varepsilon \in \mathcal{A}$ the element $x - \varepsilon$ is positive infinite, so $\partial x > d$; hence the induced map preserves positivity, and the converse follows by bijectivity and sign symmetry. $\square$

Definition 7.6 (The unique infinitesimal primitive). By Theorem 7.5(i) the restriction $\partial|_{\mathfrak{m}}$ is a bijection onto $\mathcal{A}$; its inverse is $\mathcal{I}_0|_{\mathcal{A}} : \mathcal{A} \rightarrow \mathfrak{m}$, the *unique infinitesimal primitive*. Unlike a general primitive, it carries no residual additive-constant ambiguity. (This is the object that two source manuscripts call J ; see Convention 5.6.)

Remark 7.7 (The represented quotient, concretely). Equation (7.6) is the represented form of the familiar isomorphism “ $\mathbf{No}/\mathcal{O} \cong \mathbf{No}/\mathcal{A}$ ”. Its actual representatives are the classes $\mathcal{P}$ and $\partial\mathcal{P}$, in accordance with Warning 2.1; no proper-class coset is made into an object. Inside a set-sized field (Section 20) the corresponding ordinary set quotients are available.

8 The finite-primitive criterion

8.1 The exact image of the logarithmic derivative

Theorem 8.1 (Exact logarithmic-derivative image). *For the fixed derivation on $\mathbf{No}[i]$,*

$$\boxed{\{z^\dagger : z \in \mathbf{No}[i]^\times\} = \mathbf{No} + i\mathcal{A} = \mathbf{No} + i\partial\mathfrak{m} = \mathbf{No} + i\partial\mathcal{O}.} \quad (8.1)$$

The kernel of $\dagger$ is $\mathbb{C}^\times$, and $\partial : \mathfrak{m} \rightarrow \mathcal{A}$ is injective. The image is a proper subclass of $\mathbf{No}[i]$: $1 \in \mathbf{No}$ is in the image, being $(e^\omega)^\dagger$, while i is not.

Proof. If $z \neq 0$, Corollary 6.7 gives $z^\dagger = \partial \log|z| + i\partial\eta$ with $\eta \in \mathfrak{m}$, so the imaginary part lies in $\partial\mathfrak{m} = \mathcal{A}$; this is the inclusion from left to right.

For the reverse inclusion, let $a \in \mathbf{No}$ and $b \in \mathcal{A}$. Put $A = \mathcal{I}_0(a)$ and let $\eta \in \mathfrak{m}$ be the unique infinitesimal primitive of b (Definition 7.6). Then

$$y_0 = \exp(A) E(i\eta) \neq 0 \quad (8.2)$$

and, by (BM4) and equation (6.2), $y_0^\dagger = \partial A + i\partial\eta = a + ib$. The kernel assertion is Definition 6.6; injectivity of ∂ on $\mathfrak{m}$ is Theorem 7.5(i). Properness: $1 \notin \mathcal{A}$ by Lemma 3.8, since 1 is not infinitesimal. $\square$

Warning 8.2 (Two products of two different exponentials). The witness equation (8.2) is a product of *two* exponential constructions with explicitly different declared domains: $\exp$ is the real surreal exponential on $\mathbf{No}$, and E is the infinitesimal Hahn exponential of equation (6.1) on $\mathbf{m}_{\mathbb{C}}$. The product is not an abbreviation for an unspecified global surcomplex exponential, and the two are never glued. Section 11 determines exactly how far a single compatible exponential can reach, and Corollary 11.4 shows it cannot reach all of $\mathbf{No}[i]$.

Warning 8.3 (The image is real-linear, not complex-linear). $\mathbf{No} + i\mathcal{A}$ is closed under addition and under multiplication by real constants, but not under multiplication by i : every real coefficient has zero Φ -obstruction, while multiplying one by i may not. An implementation must not model this image as a complex vector subspace.

8.2 The criterion, and all solutions

Theorem 8.4 (Finite-primitive criterion for rank-one equations). *Let $q = a + ib \in \mathbf{No}[i]$ with $a, b \in \mathbf{No}$, and let $A, B \in \mathbf{No}$ satisfy $\partial A = a$ and $\partial B = b$. The following are equivalent.*

- (i) $\partial y = qy$ has a nonzero solution in $\mathbf{No}[i]$.
- (ii) $b \in \mathcal{A}$, that is, $b = \partial\eta$ for some $\eta \in \mathbf{m}$.
- (iii) b has a finite accumulated phase: $B \in \mathcal{O}$.
- (iv) Every accumulated phase of b is finite.
- (v) $\mathcal{I}_0(b)$ is finite, equivalently infinitesimal.
- (vi) $\text{pp}(B) = 0$, equivalently $\mathfrak{o}(b) = 0$, equivalently $\Phi(q) = 0$.

When these hold, the infinitesimal η with $\partial\eta = b$ is unique, namely $\eta = \mathcal{I}_0(b) = B - \text{st}(B)$, and all solutions are

$$y = c \exp(\mathcal{I}_0(a)) E(i\mathcal{I}_0(b)), \quad c \in \mathbb{C}, \quad (8.3)$$

the nonzero ones being those with $c \in \mathbb{C}^\times$. When they fail, the only solution is $y = 0$.

Proof. (i) $\Leftrightarrow$ (ii) is Theorem 8.1, since for $y \neq 0$ the equation says $y^\dagger = q$. Alternatively and directly: if $y \neq 0$ solves it, write $y = \rho \text{cis}(\theta)$ by Theorem 6.5; then equation (6.9) gives $\partial\rho/\rho = a$ and $\partial\theta = b$ with θ finite, which is (iii). The equivalences among (ii)–(vi) are Theorem 7.5(ii). The displayed family consists of solutions by equation (8.2); if y is any solution and y_0 is the nonzero one, the quotient rule gives $\partial(y/y_0) = 0$, so $y/y_0 \in \mathbb{C}$ by Proposition 3.5. The case $c = 0$ is the zero solution. $\square$

This is the theorem that all seven source manuscripts prove, in seven notations. Three points about it deserve separate statements.

Warning 8.5 (Integrated, not pointwise). The criterion is *not* “ b must be infinitesimal”. That condition is necessary and strictly weaker. The three equations

$$\partial y = iy, \quad \partial y = i\omega^{-1}y, \quad \partial y = i\omega^{-2}y$$

have, respectively: only the zero solution (the accumulated phase of 1 is ω); only the zero solution, although ω^{-1} is infinitesimal (its accumulated phase is $\log \omega$, infinite); and the solutions $c E(-i\omega^{-1})$, $c \in \mathbb{C}$. A sibling statement that “ $b \in \mathbf{m}$ is necessary” is a strictly weaker and different result.

Remark 8.6 (The real coefficient never obstructs). Adding any real coefficient a introduces no further obstruction in the full class: multiply the solution by $\exp(\mathcal{I}_0(a))$. Thus

$$\partial y = \left(\omega^3 + \frac{i}{\omega^2} \right) y \quad \text{has solutions} \quad y = c \exp(\omega^4/4) E(-i/\omega),$$

while replacing i/ω^2 by i/ω leaves only $y = 0$; and $\partial y = (\omega + i\omega^{-2})y$ has solutions $c \exp(\omega^2/2) E(-i/\omega)$. The real growth rate may be arbitrarily large. That assertion can fail in a *restricted workspace*, because A or e^A may leave the workspace; Section 19 computes that failure exactly.

Remark 8.7 (A one-dimensional equation with a zero-dimensional solution space). A one-dimensional linear equation need not have a one-dimensional nonzero solution space *inside this field*. The familiar dimension count is an upper bound until a sufficiently large differential extension is supplied; Section 21 supplies one.

8.3 No intrinsic oscillator: three independent proofs

Corollary 8.8 (The missing oscillator). *For $\lambda \in \mathbb{C}$, the equation $\partial y = \lambda y$ has a nonzero solution in $\mathbf{No}[i]$ if and only if $\lambda \in \mathbb{R}$, in which case the solutions are exactly $c e^{\lambda\omega}$ with $c \in \mathbb{C}$. In particular $\partial y = iy$ and $\partial y = -iy$ have only the zero solution, and for every ordinary real $\lambda \neq 0$ so do $\partial y = i\lambda y$ and $\partial^2 y + \lambda^2 y = 0$. In particular $\partial^2 y + y = 0$ has only $y = 0$ in $\mathbf{No}[i]$, while $\partial^2 y - y = 0$ has all solutions $Ae^\omega + Be^{-\omega}$ with $A, B \in \mathbb{C}$.*

First proof, from the criterion. Write $\lambda = \alpha + i\beta$. An accumulated phase of β is $\beta\omega$, which is finite exactly when $\beta = 0$; apply Theorem 8.4. When $\beta = 0$, dividing a solution by $e^{\alpha\omega}$ gives derivative zero. For the second-order equation, factor $\partial^2 + \lambda^2 = (\partial - i\lambda)(\partial + i\lambda)$ over $\mathbf{No}[i]$; both factors are injective by the first part, so their composite is. $\square$

Second proof, from smallness alone. Suppose $\partial y = i\beta y$ with $\beta \in \mathbb{R} \setminus \{0\}$ and $y \neq 0$. Conjugation gives $\partial \bar{y} = -i\beta \bar{y}$, so $\partial(y\bar{y}) = 0$ and $y\bar{y}$ is a positive real constant. Dividing by its positive square root gives $u + iv$ with

$$u^2 + v^2 = 1, \quad \partial u = -\beta v, \quad \partial v = \beta u.$$

Both u and v are finite, so both derivatives are infinitesimal by Lemma 3.8; dividing by the nonzero real constant β then makes both u and v infinitesimal, so $u^2 + v^2$ is infinitesimal and cannot equal 1. $\square$

Third proof, an energy argument for the second-order equation. First let $f \in \mathbf{No}$ be real with $\partial^2 f + f = 0$. The energy $E = f^2 + (\partial f)^2$ satisfies $\partial E = 2\partial f (f + \partial^2 f) = 0$, so E is a nonnegative real constant; hence f and ∂f are finite. By Lemma 3.8 both ∂f and $\partial^2 f$ are infinitesimal, and since $f = -\partial^2 f$ the element f is infinitesimal too. Then E is both real and infinitesimal, so $E = 0$, and positivity gives $f = 0$. For complex f , apply this to its real and imaginary parts. $\square$

The three proofs are genuinely different and all three survive. The first uses the whole criterion. The second uses only the constant field, the H -field positivity clause and the normalization $\partial\omega = 1$ — not the finite-angle parametrization, not surjectivity, and no logarithm — so the obstruction is provably not an artefact of a branch choice or of a chosen phase convention. The third uses only conservation of energy plus smallness, and isolates the role of smallness in the second-order equation; the factorization proof, however, scales more efficiently to general operators (Theorem 13.2).

Warning 8.9 (What the nonexistence statements are about). All of these are statements about surcomplex *number* solutions under the fixed ∂ . They do not deny ordinary oscillatory functions, do not say that the coordinate functions $\sin z$, $\cos z$ fail their ordinary differential equations, and do not deny algebraically defined global surcomplex phases. They do say that $\sin \omega$ and $\cos \omega$ are not Berarducci–Mantova differential elements of **No** in the sense of Convention 3.6: the system $c^2 + s^2 = 1$, $c' = -s$, $s' = c$ has no solution in **No**. See Corollary 8.10 and Section 11.3.

Corollary 8.10 (The unit-circle version). *For $x \in \mathbf{No}$, the system*

$$c^2 + s^2 = 1, \quad \partial c = -(\partial x)s, \quad \partial s = (\partial x)c$$

has a solution $c, s \in \mathbf{No}$ if and only if x is finite.

Proof. If c, s solve it then $z = c + is \neq 0$ satisfies $\partial z = i(\partial x)z$, so by Theorem 8.4 ∂x has a finite accumulated phase; all those are $x + r$ with $r \in \mathbb{R}$, so x is finite. Conversely, for finite x take the coordinates of $\text{cis}(x)$ and apply equation (6.7). $\square$

This concerns the derivatives of the *values* c, s . An externally differentiable sine function is a different object, and its derivative with respect to its argument must not be substituted for ∂ without a proved chain rule (Warning 4.1).

Corollary 8.11 (A Riccati transfer). *The only solutions in $\mathbf{No}[i]$ of $\partial u = 1 + u^2$ are $u = i$ and $u = -i$; in particular there is none in **No**.*

Proof. Both constants are solutions. For any other solution, $v = (u - i)/(u + i)$ is defined and nonzero, and

$$v^\dagger = \frac{\partial u}{u - i} - \frac{\partial u}{u + i} = \frac{2i \partial u}{u^2 + 1} = 2i,$$

contradicting Corollary 8.8. $\square$

The same Riccati equation for the variable derivative D_z , on strongly entire Hahn power series, has the same two solutions; see `hol:nl:prop:riccati` in the holonomic-rigidity report. The derivations differ, and neither result implies the other.

This is not a general nonlinear solvability theorem. It is a direct way to transfer the first-order phase obstruction to a familiar nonlinear equation. (Added with member 11: the sentence stands for this proof, but the class containing the equation is now classified. Theorem 30.1 describes all solutions of every first-order autonomous equation with constant coefficients, for every normalized derivation, and Example 32.4 recovers this corollary from it by a projective route. Beyond that class — variable coefficients, higher order, systems — no nonlinear solvability theorem is claimed; see items N21 and N25.)

Remark 8.12 (Algebraic closure is not differential closure). $\mathbf{No}[i]$ is algebraically closed and ∂ is additively surjective, yet $\partial y = iy$ has no nonzero solution. These are three different closure properties: $X^2 + 1 = 0$ has a solution, $\partial y = f$ has a solution, and $\partial y = iy$ with $y \neq 0$ does not. Adjoining an algebraic element cannot repair the last, since algebraic closedness leaves no proper algebraic extension; and taking an algebraic root cannot cancel a nonzero obstruction, because every nonzero integer multiple of a nonzero element of $\mathcal{P}$ is nonzero (Corollary 9.4). A genuine differential extension must introduce a transcendental element, and Section 21 constructs several. The nonzero requirement can itself be written differentially, by adding an unknown v and the equation $yv = 1$; so this is an explicit failure of differential closedness, not a complaint about an undefined expression such as $e^{i\omega}$.

9 Gauge normal form and the obstruction sequence

9.1 Gauge equivalence and the unique normal form

A change of unknown $y = gz$ with $g \in \mathbf{No}[i]^\times$ sends

$$\partial y = qy \quad \text{to} \quad \partial z = (q - g^\dagger)z. \quad (9.1)$$

Call $q, r \in \mathbf{No}[i]$ *gauge equivalent* when $q - r = g^\dagger$ for some $g \in \mathbf{No}[i]^\times$. A gauge transformation changes the coefficient by a logarithmic derivative, not by an arbitrary primitive, and it does not change whether a nonzero solution exists in the base field. No differential-module terminology is needed to use it.

Theorem 9.1 (Purely infinite phase normal form, with uniqueness). *The map $\Phi : \mathbf{No}[i] \rightarrow \mathcal{P}$, $\Phi(q) = \mathfrak{o}(\text{Im } q)$, is additive and $\mathbb{R}$ -linear, is surjective, and satisfies*

$$\Phi(q) = 0 \iff q \text{ is a logarithmic derivative.}$$

Two coefficients are gauge equivalent if and only if their obstructions agree. Explicitly, writing $q = a + ib$ and $\mathcal{I}_0(b) = P + \eta$ with $P = \Phi(q) \in \mathcal{P}$ and $\eta \in \mathfrak{m}$, the gauge

$$g = \exp(\mathcal{I}_0(a)) \mathbf{E}(i\eta)$$

transforms $\partial y = qy$ into

$$\partial z = i(\partial P)z. \quad (9.2)$$

Every equation $\partial y = qy$ is therefore gauge equivalent to exactly one equation of the form equation (9.2) with $P \in \mathcal{P}$, and the canonical parameter is $P = \Phi(q)$.

Proof. Linearity and surjectivity are Theorem 7.5(vi) composed with Im ; the kernel description is Theorem 8.4. The logarithmic derivative of the displayed g is $a + i\partial\eta$, and $b = \partial P + \partial\eta$, so equation (9.1) gives equation (9.2). If $q_1 - q_2 = g^\dagger$ then $\text{Im}(q_1 - q_2) \in \mathcal{A}$ and linearity of $\mathfrak{o}$ makes the obstructions equal; conversely equal obstructions put $\text{Im}(q_1 - q_2)$ in $\mathcal{A}$, and Theorem 8.1 supplies a g with $g^\dagger = q_1 - q_2$. Uniqueness: $\Phi(i\partial P) = \mathfrak{o}(\partial P) = P$ by Theorem 7.5(vi), so distinct P represent distinct classes. $\square$

The normal form says more than “no solution”. It isolates the precise accumulated phase that cannot be represented by a surcomplex direction: the real growth factor and the infinitesimal phase residue are exactly the removable parts, and the purely infinite part is exactly the irremovable one. The parameter P is *not* an angular value assigned to an infinite input; it records the part of an accumulated phase that no existing surcomplex phase can absorb, the absorbed factor carrying only the infinitesimal residue η (Convention 5.1).

Corollary 9.2 (Obstruction exact sequence). *The following is an exact sequence of class groups, multiplicative before $\dagger$ and additive after it:*

$$1 \longrightarrow \mathbb{C}^\times \longrightarrow \mathbf{No}[i]^\times \xrightarrow{\dagger} (\mathbf{No}[i], +) \xrightarrow{\Phi} (\mathcal{P}, +) \longrightarrow 0. \quad (9.3)$$

The corresponding real obstruction sequence is $0 \rightarrow \mathcal{A} \rightarrow \mathbf{No} \xrightarrow{\mathfrak{o}} \mathcal{P} \rightarrow 0$; it uses $\Phi(ib) = \mathfrak{o}(b)$, not the restriction $\Phi|_{\mathbf{No}} = 0$.

Proof. $\ker(\dagger) = \mathbb{C}^\times$ and $\text{im}(\dagger) = \mathbf{No} + i\mathcal{A}$ by Theorem 8.1; that class is $\ker \Phi$ by Theorem 8.4; and Φ is onto by Theorem 9.1. $\square$

Warning 9.3 (The sequence is elementwise). The word “class” in equation (9.3) is intentional, and the sequence is shorthand for the displayed kernel and image identities *about elements*. No set or class of all equivalence classes is formed, and $\mathcal{P}$ appears precisely so that no coset of $\mathcal{O}$ has to be an object (Warning 2.1). Every map in the sequence acts on set-sized elements, and Φ together with the representative $i\partial P$ carries all the content.

Corollary 9.4 (Tensor, dual, and no finite-order obstruction). *Under multiplication of scalar solution equations — equivalently, tensor product of the corresponding rank-one objects, represented by addition of coefficients — the obstructions add; under inversion, that is dualization, they negate. For an integer $n \neq 0$,*

$$\Phi(nq) = n\Phi(q), \quad \Phi(nq) = 0 \iff \Phi(q) = 0.$$

Every nonzero gauge class has infinite order. Hence raising a putative solution of an obstructed equation to a finite power does not remove the obstruction.

Proof. The coefficient of a product of nonzero solutions is the sum of their logarithmic derivatives, and of an inverse is the negative. The rest is $\mathbb{R}$ -linearity of Φ together with the absence of additive torsion in the real vector class $\mathcal{P}$. $\square$

The integer linearity just recorded — rather than real linearity — is what drives the differential Galois arguments of Sections 21 and 21.4: rational functions in solutions involve integral monomial exponents, so the relations that matter are integer relations. This rank-one classification is a differential gauge classification; it is not the holomorphic line-bundle or divisor classification of the repository’s global analytic reports, and no theory of higher-rank differential Galois groups is needed to state it.

10 Sharp thresholds and worked scales

10.1 The support threshold over a real-exponent Hahn field

Fix $t = \omega^{-1}$ and, for an additive subgroup $\Gamma \subseteq \mathbb{R}$, embed the Hahn fields

$$H_\Gamma = \mathbb{R}((t^\Gamma)), \quad K_\Gamma = \mathbb{C}((t^\Gamma)) = H_\Gamma[i]$$

into $\mathbf{No}$ and $\mathbf{No}[i]$ by Conway monomials. On these fields the real-exponent rule of equation (3.1) applies, so $\partial(t^\alpha) = -\alpha t^{\alpha+1}$ for ordinary real α and $\partial = -t^2 d/dt$ (Warning 2.2 applies: this formula is for real exponents only). The case $\Gamma = \mathbb{R}$ gives the largest real-exponent field; the cases $\Gamma = \mathbb{Z}$ and $\Gamma = \mathbb{Q}$ recur in Part IV.

Theorem 10.1 (Sharp support threshold). *Let $\Gamma \subseteq \mathbb{R}$ be an additive subgroup and let $b = \sum_{\alpha \in S} b_\alpha t^\alpha \in H_\Gamma$ with S well ordered. A primitive of b in $\mathbf{No}$ is*

$$B = \sum_{\substack{\alpha \in S \\ \alpha \neq 1}} \frac{b_\alpha}{1 - \alpha} t^{\alpha-1} + b_1 \log \omega, \quad (10.1)$$

and for every $p \in \mathbf{No}$ the equation $\partial y = (p + ib)y$ has a nonzero solution in $\mathbf{No}[i]$ if and only if

$$b = 0 \quad \text{or} \quad \min \text{supp}(b) > 1. \quad (10.2)$$

Equivalently, for real $f \in H_\Gamma$ one has $f \in \mathcal{A}$ if and only if $f = 0$ or $v(f) > 1$, where v is the least-exponent valuation. In the soluble case the sum in equation (10.1) is the unique infinitesimal accumulated phase of b .

Proof. Shifting a well-ordered support by -1 preserves well ordering, and omitting one exponent while adding one logarithmic term preserves summability, so equation (10.1) is a legitimate element of $\mathbf{No}$; the real-exponent rule and (BM3) give $\partial B = b$.

If every exponent exceeds 1, all powers in B have positive exponent, there is no logarithmic term, and B is infinitesimal. If the least exponent α_0 is less than 1, the corresponding primitive term has negative exponent and nonzero coefficient; it dominates the remaining power terms and also $\log \omega$, since $\log \omega \prec \omega^\delta$ for every real $\delta > 0$. So B is infinite. If the least exponent is exactly 1, the logarithmic term has nonzero coefficient while all remaining power terms are infinitesimal, so B is again infinite. Adding a real constant changes none of these conclusions. Apply Theorem 8.4. $\square$

The logarithmic comparison used here is a standard consequence of the surreal exponential; equivalently it follows from the normal-form description of $\log \omega$ [1, 7]. The threshold is a *strict* inequality at exponent 1, and the strictness is not an artefact of the statement: it is the $\log \omega$ term.

Corollary 10.2 (Power threshold). *For $p \in \mathbb{R}$, the equation $\partial y = i\omega^{-p}y$ has a nonzero surcomplex solution exactly when $p > 1$, and then all solutions are*

$$y = c E\left(\frac{i\omega^{1-p}}{1-p}\right), \quad c \in \mathbb{C}.$$

Equivalently $\omega^{-p} \in \mathcal{A} \iff p > 1$.

Proof. For $p \neq 1$ a primitive of ω^{-p} is $\omega^{1-p}/(1-p)$, which is infinitesimal for $p > 1$ and infinite for $p < 1$; for $p = 1$ a primitive is $\log \omega$, infinite. Apply Theorems 8.4 and 10.1. $\square$

Corollary 10.3 (Iterated-logarithm thresholds at every finite depth). *With $\ell_0 = \omega$, $\ell_{n+1} = \log \ell_n$ and, for an ordinary integer $n \geq 0$ and real p ,*

$$b_{n,p} = \frac{1}{\ell_0 \ell_1 \cdots \ell_{n-1} \ell_n^p}$$

(the prefix product empty for $n = 0$), the equation $\partial y = ib_{n,p}y$ has a nonzero solution in $\mathbf{No}[i]$ exactly when $p > 1$. An accumulated phase is $\ell_n^{1-p}/(1-p)$ for $p \neq 1$ and ℓ_{n+1} for $p = 1$.

Proof. By equation (3.2), $\partial \ell_n = 1/(\ell_0 \cdots \ell_{n-1})$; differentiate the displayed expressions using the real power and logarithm rules of equation (3.1). For $p > 1$ the accumulated phase is infinitesimal; for $p \leq 1$ it is infinite, each ℓ_n being positive infinite. Apply Theorem 8.4. $\square$

These are finite iterated constructions: no infinite product and no transfinite logarithm tower is used. (Added with member 13: Part VII imports that tower, and Corollary 37.1 carries the obstructed boundary case $p = 1$ to every ordinal depth.) The tests resemble familiar convergence thresholds for improper integrals, but their proof here is algebraic differentiation and comparison of surreal sizes. No improper real integral, no limiting process and no integration to a numerical endpoint occurs anywhere.

10.2 Why the criterion cannot be globalized by valuation

Example 10.4 (The separating pair). Put $L = \log \omega$. Then

$$\partial(\log L) = \frac{1}{\omega L}, \quad \partial(-L^{-1}) = \frac{1}{\omega L^2}.$$

The first accumulated phase, $\log \log \omega$, is infinite; the second, $-1/\log \omega$, is infinitesimal. Therefore

$$\partial y = \frac{i}{\omega \log \omega} y \text{ has no nonzero solution,} \quad \partial y = \frac{i}{\omega (\log \omega)^2} y \text{ does.} \quad (10.3)$$

Both coefficients have $v(\operatorname{Im} a) > 1$ in the full Hahn valuation normalized by $v(t) = 1$: indeed $v(L) < 0$, so $v(t/L) = 1 - v(L) > 1$.

Warning 10.5 (The threshold theorem is restricted, and must stay restricted). Theorem 10.1 is stated over a real-exponent Hahn field and the restriction is substantive. By Example 10.4, even $v(\operatorname{Im} a) > 1$ is *not* sufficient over all of **No**: the missing information lies in the slower logarithmic scale, which a real-exponent workspace cannot see. A computer-algebra backend must not apply Theorem 10.1 outside its declared coefficient domain, and the support threshold must not be merged into the general- Γ resolvent of Theorem 18.6, whose hypotheses are different. A primitive-based representation handles all of these scales uniformly; a single power cutoff does not.

Table 3: Distinguishable situations at two and three scales. $L = \log \omega$.

Imaginary coefficient b	An accumulated phase B	Nonzero solution of $\partial y = iby$?
1	ω	No
$t = \omega^{-1}$	L	No, although b is infinitesimal
$t^2 = \omega^{-2}$	$-t$	Yes: $E(-it)$
$t^{3/2}$	$-2t^{1/2}$	Yes: $E(-2it^{1/2})$
$1/(\omega L)$	$\log L$	No: a slower divergence still obstructs
$1/(\omega L^2)$	$-1/L$	Yes: $E(-i/L)$
$\partial \eta$ with $\eta \in \mathfrak{m}$	η	Yes: $E(i\eta)$
$\omega^3 + i\omega^{-2}$ (whole coefficient)	—	Yes: $c \exp(\omega^4/4)E(-i/\omega)$

10.3 An exact decision procedure for rational coefficients

The obstruction map is a semantic classification, not a universal effective procedure. For rational functions over the ordinary constants the condition is nevertheless decidable by finite algebra.

Theorem 10.6 (Rational phase criterion). *Let $b(X) \in \mathbb{R}(X)$ be nonzero, in reduced form $b = P/Q$. Then*

$$b(\omega) \in \mathcal{A} \iff \deg Q - \deg P \geq 2. \quad (10.4)$$

Hence for $q(\omega) = a(\omega) + ib(\omega)$ with $a, b \in \mathbb{R}(X)$ this is exactly the criterion for a nonzero solution of $\partial y = q(\omega)y$. The case $b = 0$ is always admissible.

Proof. Let $p = \deg Q - \deg P$. Expanding at infinity gives $b(\omega) = c\omega^p + (\text{higher integer powers of } \omega)$ with $c \in \mathbb{R}^\times$. If $p \geq 2$, write $b(\omega) = \sum_{n \geq 2} b_n \omega^n$; then $\eta = -\sum_{n \geq 2} b_n \omega^{n-1}/(n-1)$ is infinitesimal and

differentiates strongly to $b(\omega)$, so $b(\omega) \in \mathcal{A}$. If $p \leq 1$, the leading term gives $|b(\omega)| \geq |c|t^p/2$; were $b(\omega) \in \mathcal{A}$, real linearity and the convexity of Theorem 7.5(iii) would force $t^p \in \mathcal{A}$, contradicting Corollary 10.2. The assertion for q is Theorem 8.4. $\square$

Table 4: The rational criterion on examples.

$b(\omega)$	Finite accumulated phase?	Reason, or the phase
0	yes	0
1	no	ω is infinite
$1/\omega$	no	$\log \omega$ is infinite
$1/\omega^2$	yes	$-1/\omega$
$\omega/(1 + \omega^2)$	no	degree gap 1
$1/(1 + \omega^2)$	yes	degree gap 2
$(1 - \omega^2)/(1 + \omega^2)^2$	yes	$\omega/(1 + \omega^2)$

Warning 10.7 (The theorem is over $\mathbb{R}(X)$; the implementation is over $\mathbb{Q}(X)$). Theorem 10.6 is an algebraic statement over $\mathbb{R}(X)$. The delivered executable restricts its input to exact rational coefficients so that polynomial arithmetic and degree comparison are effective, and it *raises* on inputs outside $\mathbb{Q}(\text{variable})$ — other functions, nonrational coefficients, unbound parameters, non-numeric tokens — rather than reporting them as mathematical nonexistence. No algorithm for an arbitrary real coefficient is asserted. See Section 49.1 and appendix D.

Example 10.8 (An algebraic solution with a small angular rate). For $b = 1/(1 + \omega^2)$ the unit-modulus *algebraic* number

$$y = \frac{1 + i\omega}{\sqrt{1 + \omega^2}}$$

satisfies $\partial y = iby$, since

$$y^\dagger = \frac{i}{1 + i\omega} - \frac{\omega}{1 + \omega^2} = \frac{i}{1 + \omega^2}.$$

This example needs neither a named inverse trigonometric function nor an infinite-phase exponential.

Example 10.9 (A rational unit-circle (Cayley) coordinate). With $t = \omega^{-1}$ put $u = (1 + it)/(1 - it)$. Then $u\bar{u} = 1$, $\text{st}(u) = 1$, and

$$u^\dagger = \frac{-2it^2}{1 + t^2}.$$

Its infinitesimal phase residue is $2 \arctan(t)$, understood as the infinitesimal Taylor evaluation of Lemma 6.1; indeed $\partial(2 \arctan t) = -2t^2/(1 + t^2)$. So an admissible solution need not be stored as an infinite sum: an exact rational expression can already carry the required phase, and its full expansion is optional. The coefficient domain here is Gaussian rationals in t .

Example 10.10 (A solvable coefficient with an infinite Hahn expansion). For $q = i/\omega^2 = it^2$ the normalized solution is

$$y = E(-it) = 1 - it - \frac{t^2}{2} + \frac{it^3}{6} + \frac{t^4}{24} - \cdots,$$

whose support lies in $\mathbb{N}$, so the family is summable. The identity $\partial y = it^2 y$ is exact, not asymptotic. A finite truncation $E_N(t) = \sum_{n \leq N} (-i)^n t^n / n!$ has the exact residual

$$\partial E_N - it^2 E_N = -i \frac{(-i)^N}{N!} t^{N+2}, \quad (10.5)$$

which is what a finite verifier can check; it is not the whole solution.

11 The maximal derivation-compatible exponential strip

11.1 A surjective partial exponential

Set

$$\mathcal{S} := \mathbf{No} + i\mathcal{O} \subset \mathbf{No}[i].$$

This is an additive subgroup and a real vector subclass, and it is ∂ -stable precisely because $\partial\mathcal{O} \subseteq \mathfrak{m}$ (Lemma 3.8). It is *not* a subring: $\omega \in \mathcal{S}$ and $i \in \mathcal{S}$, while $i\omega \notin \mathcal{S}$. Note carefully that $\mathcal{S}$ involves $\mathcal{O}$, whereas the logarithmic-derivative image of Theorem 8.1 involves $\partial\mathcal{O} = \mathcal{A}$; see Warning 5.4. Define

$$E_{\mathcal{S}}(x + i\theta) = \exp(x) \operatorname{cis}(\theta) = \exp(x) e^{i \operatorname{st}(\theta)} E(i(\theta - \operatorname{st}(\theta))), \quad x \in \mathbf{No}, \theta \in \mathcal{O}. \quad (11.1)$$

Theorem 11.1 (The strip exponential). *$E_{\mathcal{S}} : (\mathcal{S}, +) \rightarrow (\mathbf{No}[i]^{\times}, \cdot)$ is a surjective group homomorphism with kernel exactly $2\pi i\mathbb{Z}$, it extends the ordinary complex exponential, and it satisfies*

$$\partial E_{\mathcal{S}}(w) = E_{\mathcal{S}}(w) \partial w \quad (w \in \mathcal{S}). \quad (11.2)$$

Hence $\mathcal{S}/(2\pi i\mathbb{Z}) \cong \mathbf{No}[i]^{\times}$ as abelian groups. Moreover $\partial\mathcal{S} = \mathbf{No} + i\mathcal{A}$.

Proof. Multiplicativity follows from the real exponential law, the additivity of standard part on finite elements and the group law of Theorem 6.5; the derivative identity follows from (BM4) and equation (6.7), noting that $\operatorname{st}(\theta)$ and $e^{i \operatorname{st}(\theta)}$ are differential constants. If $E_{\mathcal{S}}(x + i\theta) = 1$, taking moduli gives $\exp x = 1$, so $x = 0$, and then $\operatorname{cis} \theta = 1$ gives $\theta \in 2\pi\mathbb{Z}$; conversely these lie in the kernel. For surjectivity, write $z = \rho c E(i\eta)$ by Corollary 6.7 and choose an ordinary real r with $e^{ir} = c$; then $z = E_{\mathcal{S}}(\log \rho + i(r + \eta))$. This uses the ordinary circle only, not a branch on the whole surcomplex plane. Finally $\partial\mathbf{No} = \mathbf{No}$ and $\partial\mathcal{O} = \mathcal{A}$ give the image assertion. $\square$

So $E_{\mathcal{S}}$ reaches every nonzero surcomplex number while restricting the allowable *exponents*: surjectivity does not require an exponent for every element of $\mathbf{No}[i]$. Its kernel consists of ordinary integer periods, not of a surreal integer part. Every nonzero z has strip logarithms, all differing by an ordinary period $2\pi in$, and all with the *same* intrinsic derivative $z^{\dagger}$ — a conclusion that is canonical even when no choice of logarithm branch is.

11.2 Maximality is pointwise

Theorem 11.2 (Exact domain of the exponential differential equation). *For $w = x + iB \in \mathbf{No}[i]$ with $x, B \in \mathbf{No}$, the following are equivalent.*

- (1) *There exists $y \in \mathbf{No}[i]^{\times}$ with $y^{\dagger} = \partial w$.*
- (2) *B is finite.*
- (3) *$w \in \mathcal{S}$.*

Consequently no map taking nonzero surcomplex values and satisfying equation (11.2) at every point of its domain can extend $E_{\mathcal{S}}$ to even one exponent outside $\mathcal{S}$.

Theorem 11.3 (Maximality restated). *Equivalently, $\partial^{-1}(\mathcal{A}) = \mathcal{O}$: the condition in Theorem 11.2(1) is $\partial B \in \mathcal{A}$, and Theorem 7.5(viii) identifies its solution class as $\mathcal{O}$.*

Proof of Theorems 11.2 and 11.3. By Theorem 8.1, condition (1) is equivalent to $\text{Im}(\partial w) = \partial B \in \mathcal{A}$. Every accumulated phase of ∂B is $B + r$ with $r \in \mathbb{R}$, so by Theorem 8.4 such a phase is finite exactly when B is finite; equivalently $\partial^{-1}(\mathcal{A}) = \mathcal{O}$ by Theorem 7.5(viii). The remaining equivalence is the definition of $\mathcal{S}$, and Theorem 11.1 supplies a value at every point of $\mathcal{S}$. The reasoning is pointwise: it excludes even an isolated proposed value outside the strip, irrespective of any group law, homomorphism axiom, or regularity imposed on the map. $\square$

The claimed strength is exactly this pointwise form. It is stronger than the failure of one candidate formula: at an excluded point there is *no* admissible nonzero value at all. On the other hand the differential law alone is not claimed to pin down every normalization on $\mathcal{S}$: ratios of admissible values are differential constants, and equation (11.1) is the specified canonical construction with the ordinary complex normalization. A symbolic system should therefore attach the domain predicate $\text{Im } w \in \mathcal{O}$ to $E_{\mathcal{S}}$ rather than advertise an exponential *operation* on a differential subfield.

11.3 No globally compatible exponential, in three strengths

Corollary 11.4 (Phase-independent nonexistence). *There is no map $\Psi : \mathbf{No}[i] \rightarrow \mathbf{No}[i]^{\times}$ with*

$$\partial \Psi(z) = \Psi(z) \partial z \quad \text{for every } z \in \mathbf{No}[i]. \quad (11.3)$$

No homomorphism, continuity, analyticity, conjugation or normalization hypothesis is needed. Likewise there is no map $E : \mathbf{No} \rightarrow \mathbf{No}[i]^{\times}$ with $\partial(E(x)) = iE(x)\partial x$ for every $x \in \mathbf{No}$, and no extension of the finite-angle cis to all real surreals can satisfy $\partial \text{cis}(x) = i(\partial x) \text{cis}(x)$ everywhere.

Proof. Evaluate equation (11.3) at $z = i\omega$: the value $y = \Psi(i\omega)$ would be nonzero with $\partial y = iy$, contradicting Corollary 8.8 (or Proposition 3.11, or Theorem 11.2). For the second statement evaluate at $x = \omega$. $\square$

Because no structural hypothesis is used, this cannot be repaired by a cleverer choice of infinite circular phase. It is stronger in its phase-independence than an obstruction proved for one chosen canonical exponential, and it is compatible with the repository's existing derivation-obstruction theorem [24], which it strengthens rather than contradicts.

The defect can also be quantified, uniformly over all conventions of one natural shape.

Proposition 11.5 (Character-independent invariant defect). *Let $\chi : (\mathcal{P}, +) \rightarrow \mathbb{T}(\mathbf{No})$ be any group homomorphism, and for $y = P + r + \eta$ with $P \in \mathcal{P}$, $r \in \mathbb{R}$, $\eta \in \mathfrak{m}$ define the phase extension*

$$E_{\chi}(x + iy) = \exp(x) e^{ir} E(i\eta) \chi(P).$$

Put $\delta_{\chi}(z) = E_{\chi}(z)^{\dagger} - \partial z$ for $z = x + iy$. Then

$$\mathfrak{o}(\text{Im } \delta_{\chi}(z)) = -\text{pp}(y). \quad (11.4)$$

Proof. By Theorem 6.3 the individual value $\chi(P)$ is $c_P E(i\eta_P)$ with $c_P \in \mathbb{T}(\mathbb{C})$ and $\eta_P \in \mathfrak{m}$; we differentiate that value only, and never differentiate χ as a function. Its logarithmic derivative is $i\partial\eta_P$, so $E_{\chi}(z)^{\dagger} = \partial x + i\partial\eta + i\partial\eta_P$ and $\delta_{\chi}(z) = i(\partial\eta_P - \partial P)$. Apply $\mathfrak{o}$, using $\partial\eta_P \in \mathcal{A}$ and $\mathfrak{o}(\partial P) = P$ from Theorem 7.5(vi). $\square$

So no choice of character repairs the obstruction at a purely infinite imaginary exponent: the defect may change with χ , but its purely infinite class cannot. The trivial character is one admissible convention; nothing here assumes that every conceivable global exponential has this form.

Warning 11.6 (What is not being denied). Ehrlich and Kaplan construct canonical global surcomplex exponential and circular functions using an initial integer part, with kernel described by $2\pi i$ times that integer part [8, §11]; the repository carries that construction alongside its phase-extension discussion [23]. These are algebraic and structural choices, not assertions of compatibility with equation (11.2). The no-go theorems above do not invalidate those functions, do not say that a symbol such as $\text{Exp}(i\omega)$ cannot be assigned a surcomplex value, and do not settle their separate robustness questions. They say only that *whatever* nonzero value is assigned, its Berarducci–Mantova derivative cannot equal i times itself. The trigonometry report’s global phase constructions likewise do not assert the scalar chain rule and are not refuted.

Example 11.7 (A phase convention is not a differential solution). A convention that discards the purely infinite part of an angle may assign phase 1 to ω . That value is algebraically meaningful under its own convention, but its intrinsic derivative is $\partial 1 = 0$, whereas differential exponential compatibility at $i\omega$ would require $\partial 1 = i$. The constant-value convention makes the distinction especially transparent: a nonzero ordinary constant has derivative zero, not i times itself. So a phase convention and a differential solution are different objects even when both are written $e^{i\omega}$.

Remark 11.8 (There is no finite clock). Ordinary sine and cosine lifts on finite halos obey the chain rule with the factor ∂z , and for finite z that factor is infinitesimal: there is no finite surreal z with $\partial z = 1$. The forbidden oscillator would require a genuinely infinite intrinsic clock.

Remark 11.9 (The formal exponential series is not summable at ω). The formal series $F(Z) = \sum_{n \geq 0} (iZ)^n / n! \in \mathbb{C}[[Z]]$ satisfies $\partial_Z F = iF$, an ordinary formal differential equation with no bearing on $\partial y = iy$ until one evaluates. At $Z = \omega$ the terms carry monomials $\omega^n = t^{-n}$, whose additive valuation support $\{0, -1, -2, \dots\}$ is *not* well ordered; the family therefore has no Hahn sum and does not define an element of $\mathbf{No}[i]$. The same holds for $\sum_{n \geq 0} (i\omega)^n / n! = \sum_n (i^n / n!) t^{-n}$, and choosing a larger exponent group does not repair that support defect. At an infinitesimal argument the series *is* summable, but then the chain rule involves the derivative of that infinitesimal argument rather than the value 1. This is how formal analytic solutions are reconciled with the obstruction.

12 Inhomogeneous equations: existence is a separate question

Theorem 12.1 (Variation of constants). *Suppose $y_0 \neq 0$ satisfies $\partial y_0 = qy_0$ in $\mathbf{No}[i]$ — equivalently, by Theorem 8.4, suppose $\Phi(q) = 0$ and take $y_0 = \exp(\mathcal{I}_0 a)E(i\mathcal{I}_0 b)$ for $q = a + ib$. Then for every $f \in \mathbf{No}[i]$ all solutions of $\partial y - qy = f$ are*

$$y = y_0(\mathcal{I}_0(f/y_0) + c), \quad c \in \mathbb{C}. \quad (12.1)$$

In particular $\partial - q$ is surjective on $\mathbf{No}[i]$ in this case. If the homogeneous equation has no nonzero solution, then for a given f the inhomogeneous equation has at most one solution in $\mathbf{No}[i]$.

Proof. Put $y = y_0 u$; since $\partial y_0 = qy_0$ the equation becomes $y_0 \partial u = f$, and (BM6) with Proposition 7.1 gives $u = \mathcal{I}_0(f/y_0) + c$ with $c \in \mathbb{C}$, these being all primitives by Proposition 3.5. Substitution verifies equation (12.1). For the last clause, the difference of two solutions would solve the obstructed homogeneous equation. $\square$

Warning 12.2 (“At most one” must not be replaced by “none”). The absence of a nonzero homogeneous solution does *not* imply inhomogeneous nonexistence. Particular solutions may exist and then be unique:

$$\begin{aligned} (\partial - i)i &= 1, & (\partial - i)(i\omega + 1) &= \omega, \\ \partial y - iy &= 1 \text{ has the unique } y = i, & \partial y &= iy - i \text{ has the solution } y = 1, \\ \partial^2 y + y &= \omega^2 \text{ has the unique } y = \omega^2 - 2, & \partial^2 y + y &= \omega^3 \text{ has the unique } y = \omega^3 - 6\omega, \end{aligned}$$

existence verified by differentiation and uniqueness by Corollary 8.8. An integrating-factor expression using an undefined value $e^{i\omega}$ is unnecessary and would obscure the correct reasoning; the integrating factor simply does not exist in $\mathbf{No}[i]$, so variation of constants yields no existence conclusion there. This argument alone gives no general inhomogeneous existence theorem outside $\Phi(q) = 0$. The additional import in Theorem 15.6 will supply existence for every forcing in Part III. Note also that having a trivial kernel is not the same as failing to be surjective: Theorem 18.6 exhibits an operator that is bijective on a workspace although its homogeneous equation has no nonzero solution anywhere in $\mathbf{No}[i]$.

Proposition 12.3 (Polynomial forcing, and a terminating inverse). *Let $f \in \mathbb{C}[\omega]$ of degree N and $\lambda \in \mathbb{C}^\times$. Then*

$$y = - \sum_{k=0}^N \frac{\partial^k f}{\lambda^{k+1}} \quad (12.2)$$

solves $(\partial - \lambda)y = f$; for $f = 0$ read the sum as 0. If λ is nonreal this is the unique solution in $\mathbf{No}[i]$; if λ is real, all solutions are obtained by adding $c \exp(\lambda\omega)$, $c \in \mathbb{C}$. In particular, with $\lambda = i$,

$$(\partial - i)^{-1}f = i \sum_{k=0}^N (-i)^k \partial^k f. \quad (12.3)$$

More generally, for real μ and nonzero $P \in \mathbb{C}[X]$, $P(\partial)(e^{\mu\omega}q(\omega)) = e^{\mu\omega}P(\mu + D)q(\omega)$ where D is ordinary polynomial differentiation; writing $P(\mu + X) = X^r Q(X)$ with $Q(0) \neq 0$, the operator $Q(D)^{-1}$ is a finite nilpotent expansion on a finite-dimensional polynomial space and repeated polynomial integration solves the remaining D^r equation. Hence every forcing term that is a finite sum of real-exponent exponential polynomials has a particular solution of the same kind.

Proof. Applying $\partial - \lambda$ to the finite sum equation (12.2) makes all intermediate terms cancel, and $\partial^{N+1}f = 0$ leaves f . For equation (12.3) use $\partial - i = -i(1 + i\partial)$ together with $\partial^{N+1}f = 0$, so the finite geometric identity applies: this is a true terminating formula, with no infinite summation and no asymptotic truncation. On ordinary polynomials in ω the operator ∂ is ordinary polynomial differentiation, since $\partial\omega = 1$; the displayed conjugation identity then follows from (BM4), and Corollary 8.8 and Theorem 13.2 give the uniqueness statements. $\square$

13 All homogeneous constant-coefficient equations

Every ordinary complex number is a differential constant, so for $P(T) \in \mathbb{C}[T]$ the substitution $T \mapsto \partial$ gives a well-defined $\mathbb{C}$ -linear operator on $\mathbf{No}[i]$.

Lemma 13.1 (Kernels of powers and real shifts). *For every ordinary integer $m \geq 1$,*

$$\ker(\partial^m) = \text{span}_{\mathbb{C}}\{1, \omega, \dots, \omega^{m-1}\} = \{P(\omega) : P \in \mathbb{C}[X], \deg P < m\},$$

with unique coefficients. For $\lambda \in \mathbb{R}$, $\ker(\partial - \lambda)^m = e^{\lambda\omega} \ker(\partial^m)$. For $\lambda \in \mathbb{C} \setminus \mathbb{R}$, every power of $\partial - \lambda$ is injective.

Proof. The case $m = 1$ is the constant-field statement. Inductively, if $\partial^m y = 0$ then $\partial^{m-1} y = c \in \mathbb{C}$, and $y - c\omega^{m-1}/(m-1)!$ is killed by ∂^{m-1} , hence is a polynomial of degree at most $m-2$; the converse is repeated differentiation, using $\partial\omega = 1$. Coefficients are unique because a nonzero ordinary-coefficient polynomial cannot vanish at the infinite ω , its highest-degree term dominating all lower ones; equivalently, ω is transcendental over the ordinary constants. For real λ , (BM4) gives

$$(\partial - \lambda)(e^{\lambda\omega} f) = e^{\lambda\omega} \partial f, \quad (13.1)$$

and iteration proves the second assertion. For nonreal λ , Proposition 3.11 or Corollary 8.8 makes $\partial - \lambda$ injective — alternatively multiply by $e^{-\alpha\omega}$ to reduce to $\partial v = i\beta v$ — and a finite power of an injective operator is injective. $\square$

Theorem 13.2 (Constant-coefficient solution classification). *Let $P \in \mathbb{C}[T]$ be nonzero and let m_λ be the multiplicity of a root λ . The complete solution space in $\mathbf{No}[i]$ of $P(\partial)y = 0$ is*

$$\ker P(\partial) = \bigoplus_{\substack{\lambda \in \mathbb{R} \\ P(\lambda)=0}} e^{\lambda\omega} \operatorname{span}_{\mathbb{C}}\{1, \omega, \dots, \omega^{m_\lambda-1}\}, \quad (13.2)$$

of complex dimension $\sum_{\lambda \in \mathbb{R}, P(\lambda)=0} m_\lambda$. Nonreal characteristic roots contribute nothing; a nonzero constant polynomial gives the zero space. In particular an operator of positive order may have zero kernel, and the solution dimension may be strictly less than $\deg P$.

Proof. Factor P over $\mathbb{C}$ into pairwise coprime primary factors $p_\lambda(T) = (T - \lambda)^{m_\lambda}$; a nonzero scalar factor does not change the kernel. For pairwise coprime $f, g \in \mathbb{C}[T]$ choose A, B with $Af + Bg = 1$; then for $v \in \ker(f(\partial)g(\partial))$,

$$v = A(\partial)f(\partial)v + B(\partial)g(\partial)v,$$

whose first summand is killed by $g(\partial)$ and second by $f(\partial)$, and whose two kernels intersect in 0 by the same identity. This coprime-kernel decomposition is valid for any linear operator on any vector space; no finite-dimensionality is needed. Induction over the primary factors, or equivalently the Chinese-remainder idempotents modulo P , decomposes $\ker P(\partial)$ as the direct sum of the primary kernels. Apply Lemma 13.1: nonreal-root summands vanish and real-root summands are those displayed. (Equivalently, factor $P = cQR$ with Q the product of the real-root factors and R that of the nonreal ones; $R(\partial)$ is injective and commutes with $Q(\partial)$, so $\ker P(\partial) = \ker Q(\partial)$.) Each displayed summand has dimension m_λ , and directness gives the count. $\square$

Example 13.3 (Order four, dimension two; order five, dimension three). In $\mathbf{No}[i]$:

$$\begin{aligned} \partial^2 y + y = 0 &\implies y = 0, \\ \partial^2 y - y = 0 &\iff y = Ae^\omega + Be^{-\omega}, \\ (\partial - 1)^2(\partial^2 + 1)y = 0 &\iff y = (A + B\omega)e^\omega, \\ (\partial - 2)^2(\partial^2 + 1)y = 0 &\iff y = (A + B\omega)e^{2\omega}, \\ (\partial - 2)^3(\partial + 1)(\partial^2 + 1)y = 0 &\iff y = e^{2\omega}(A + B\omega + C\omega^2) + De^{-\omega}, \\ (\partial - 2)^2(\partial + 1)(\partial^2 + 1)y = 0 &\iff y = (A + B\omega)e^{2\omega} + Ce^{-\omega}, \\ \partial(\partial - 1)^2(\partial^2 + 1)y = 0 &\iff y = A + e^\omega(B + C\omega), \end{aligned}$$

with all constants in $\mathbb{C}$. The missing modes are in every case exactly the oscillatory ones. There is no contradiction with the ordinary theorem that an order- n linear equation has an n -dimensional local solution space: that theorem concerns functions of a coordinate, or solutions in a sufficiently large differential extension, not solutions in an arbitrary fixed differential field.

Remark 13.4 (Real solutions, and the coefficient hypothesis). If $P \in \mathbb{R}[T]$ and real surreal solutions are wanted, the constants in equation (13.2) must be taken in $\mathbb{R}$: the imaginary part is a linear combination of the same independent real basis elements and vanishes only when all its coefficients do. The hypothesis that the coefficients of P are *ordinary complex constants* is essential: for $a \in \mathbf{No}[i] \setminus \mathbb{C}$ the operators ∂ and multiplication by a do not commute, so polynomial factorization in a commuting symbol cannot be transported to differential operators. Theorem 13.2 says nothing about variable-coefficient operators.

Corollary 13.5 (Constant matrix systems). *Let $A \in M_n(\mathbb{C})$ and consider $\partial Y = AY$ with $Y \in \mathbf{No}[i]^n$. The solution space has complex dimension equal to the total algebraic multiplicity of the real eigenvalues of A . On a Jordan block $\lambda I + N$ of size m with $\lambda \in \mathbb{R}$, all solutions are*

$$Y = e^{\lambda\omega} \left(\sum_{k=0}^{m-1} \frac{\omega^k N^k}{k!} \right) c, \quad c \in \mathbb{C}^m, \quad (13.3)$$

a finite polynomial in the nilpotent N , not an infinite matrix exponential requiring convergence; its inverse replaces ω by $-\omega$. On a block with nonreal eigenvalue only $Y = 0$ occurs. An invertible fundamental matrix in $\mathrm{GL}_n(\mathbf{No}[i])$ exists if and only if every eigenvalue of A is real.

Proof. An ordinary complex similarity is constant for ∂ , so pass to Jordan form. On a block with nonreal λ the last coordinate satisfies $(\partial - \lambda)y_m = 0$ and is zero by Lemma 13.1; upward substitution through the block makes every coordinate zero. On a real block, the displayed finite matrix polynomial B satisfies $\partial B = NB$ and is invertible, so multiplication by $e^{\lambda\omega}B$ carries constant vectors to solutions and its inverse carries solutions to constant vectors. Add the block dimensions. $\square$

For the real rotation matrix $\begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$ the solution space is zero, while a unipotent Jordan block has a fundamental matrix that is a polynomial in ω . The contrast isolates oscillation, not general complexity, as the obstruction, and it already rules out a universal matrix-exponential constructor in the intrinsic field.

14 Triangular systems, and the limits of a determinant test

Let $A = (a_{jk})$ be an upper-triangular $n \times n$ matrix over $\mathbf{No}[i]$, with n an ordinary positive integer. A *fundamental matrix* is $Y \in \mathrm{GL}_n(\mathbf{No}[i])$ with $\partial Y = AY$, entrywise.

Theorem 14.1 (Triangular fundamental-matrix criterion). *A fundamental matrix in $\mathrm{GL}_n(\mathbf{No}[i])$ exists if and only if*

$$\Phi(a_{jj}) = 0 \quad (1 \leq j \leq n). \quad (14.1)$$

When it does, an upper-triangular one can be constructed by finitely many first-order homogeneous integrations and additive primitives.

Proof. Sufficiency. Choose $s_j \neq 0$ with $\partial s_j = a_{jj}s_j$ by Theorem 8.4, put $Y_{jj} = s_j$ and $Y_{ij} = 0$ for $i > j$. For each column j , construct the entries above the diagonal in *descending* order of i by

$$Y_{ij} = s_i \mathcal{I}_0 \left(s_i^{-1} \sum_{k=i+1}^j a_{ik} Y_{kj} \right), \quad i < j. \quad (14.2)$$

The entries on the right are already constructed. The product rule gives $\partial Y_{ij} = a_{ii}Y_{ij} + \sum_{k=i+1}^j a_{ik}Y_{kj}$, so $\partial Y = AY$, and $\det Y = \prod_j s_j \neq 0$.

Necessity, by induction on n . The last row of any fundamental matrix satisfies $\partial Y_{nj} = a_{nn}Y_{nj}$, and at least one entry is nonzero, so $\Phi(a_{nn}) = 0$. All last-row entries are complex constant multiples of one nonzero solution, so a change of columns by an invertible constant matrix makes the last row $(0, \dots, 0, s_n)$; the result is still fundamental and has block form $\begin{pmatrix} Z & v \\ 0 & s_n \end{pmatrix}$ with $\det Z \neq 0$, whose upper-left block satisfies the system for the leading $(n-1) \times (n-1)$ block of A . Induction gives the remaining conditions. $\square$

In particular every strictly upper-triangular system has a unipotent fundamental matrix over $\mathbf{No}[i]$: this is a finite iterated-integration fact, and no infinite exponential series for an arbitrary matrix is needed.

Proposition 14.2 (The trace test is necessary and not sufficient). *For any finite matrix system, triangular or not, Jacobi's formula gives $\partial \det Y = (\operatorname{tr} A) \det Y$ for a fundamental Y , so*

$$\Phi(\operatorname{tr} A) = 0 \quad (14.3)$$

is necessary. It is not sufficient: for $A = \operatorname{diag}(i, -i)$ the trace vanishes, yet neither diagonal equation has a nonzero solution and the vector system has no nonzero solution at all.

Opposite phase obstructions can cancel in a determinant without disappearing from the individual modes, so a determinant-only test discards essential differential information. For a general matrix system the trace condition is only the first invariant. The complete one is the *multiset* of phases, and Part III supplies it: Theorem 15.9 shows that every finite system over $\mathbf{No}[i]$ has a well-defined differential phase spectrum $\operatorname{Ph}(A) \subset \mathcal{P}$ which is a complete invariant of gauge equivalence, and equation (15.5) recovers equation (14.3) from it as the sum

$$\Phi(\operatorname{tr} A) = \sum_P m_P(A) P,$$

whose failure to determine the multiset is exactly why the trace test is insufficient. Two of the phase spectrum's consequences belong beside the trace test. First, by Corollary 15.11 a fundamental matrix exists in $\operatorname{GL}_n(\mathbf{No}[i])$ if and only if every member of $\operatorname{Ph}(A)$ is zero, which extends Theorem 14.1 beyond the triangular case. Second, by Corollary 16.7, perturbing every entry of a Hermitian H by a coefficient with a finite primitive leaves the entire differential unitary class of $\partial y = iHy$ unchanged, with *no* spectral-gap hypothesis — a statement the trace invariant cannot see at all, and one that Proposition 16.9 shows is sharp. For non-Hermitian coefficients no such spectral formula holds, and Example 16.10 is the counterexample that marks the line; Section 51.5 separates this limitation from the complete finite-system classification.

Part III

Finite matrix systems over $\mathbf{No}[i]$

15 The differential phase spectrum of a finite system

15.1 From one necessary invariant to a complete one

Proposition 14.2 leaves a finite matrix system with exactly one invariant, $\Phi(\mathrm{tr} A)$, and with a proof that it is not enough: $\mathrm{diag}(i, -i)$ has vanishing trace obstruction and no nonzero solution. The missing information is a *multiset*, not a sum. This part supplies it for every finite system, and computes it from ordinary algebraic eigenvalues when the coefficient is iH with H Hermitian.

The eighth source manuscript is the origin of this part (appendix C). Its own audit names the gap in this report's terms — general nontriangular systems and higher-rank differential modules left outside the classification — and states that the rank-one phase criterion, the bounded-primitive ideal and the diagonal Galois tori of Part II and Section 21 are prior coverage rather than discoveries of its own. That is how they are used below: nothing already proved in Part II is reproved here, and every statement of this part that coincides with one already in the report is cited, not restated.

Two agreements are worth printing as agreements before anything new is proved.

Remark 15.1 (The eighth manuscript refutes the same collapse). Warning 5.4 is load-bearing: $\mathcal{A} = \partial\mathfrak{m} = \partial\mathcal{O}$ is a *strict* subclass of $\mathfrak{m}$, so the criterion is never “the coefficient is infinitesimal”. The eighth manuscript opens with exactly that point and exactly that witness: i/ω is obstructed because its imaginary part has the infinite primitive $\log \omega$. Its structure lemma for the ideal — order-convexity, the $\mathcal{O}$ -module property, containment in $\mathfrak{m}$, and the separating pair $\omega^{-2} \in \mathcal{A}$, $\omega^{-1} \notin \mathcal{A}$ — is Theorem 7.5(iii),(iv), proved there by the same order-theoretic route from (BM7). Nothing in this part weakens that warning, and Proposition 16.9 below shows that the strictness is sharp for matrices as well as for scalars.

Remark 15.2 (The rank-one criterion is imported from Part II). The eighth manuscript reproves the exact logarithmic-derivative image equation (8.1) by polar decomposition and a formal Hahn logarithm, and credits it to this collection. It is Theorem 8.1 and Theorem 8.4, with the same proof idea as Corollary 6.7; it is not reproved here. Every uniqueness argument below rests on the following special case of Theorem 8.4, which is used often enough to deserve a number.

Lemma 15.3 (The rank-one separation test). *Let $P, Q \in \mathcal{P}$ and let $x \in \mathbf{No}[i]$ satisfy $\partial x = i\partial(P - Q)x$. If $x \neq 0$ then $P = Q$ and $x \in \mathbb{C}^\times$.*

Proof. By Theorem 8.4 a nonzero solution forces $\Phi(i\partial(P - Q)) = \mathfrak{o}(\partial(P - Q)) = 0$, which is $P - Q = 0$ by Theorem 7.5(vi). Then $\partial x = 0$, so $x \in \mathbb{C}^\times$ by the constant-field statement of Theorem 3.1. $\square$

We also need the order behaviour of $\mathfrak{o}$ itself, which Theorem 7.5 contains but does not display.

Corollary 15.4 (The obstruction preserves order). *The map $\mathfrak{o} : \mathbf{No} \rightarrow \mathcal{P}$ is order preserving: $b \leq c$ implies $\mathfrak{o}(b) \leq \mathfrak{o}(c)$. Consequently, for $P, Q \in \mathcal{P}$,*

$$P < Q \iff \partial P < \partial Q,$$

and an ordered tuple of real coefficients has an ordered tuple of obstructions.

Proof. The displayed equivalence is Theorem 7.5(vii), which states that $\partial : \mathcal{P} \rightarrow \partial\mathcal{P}$ is an order-preserving bijection. For the first assertion, suppose $b \leq c$ and put $P = \mathfrak{o}(b)$, $Q = \mathfrak{o}(c)$. If $Q < P$ then $\partial Q < \partial P$, and $b - \partial P \in \mathcal{A}$, $c - \partial Q \in \mathcal{A}$ by Theorem 7.5(vi),(vii); so $b - c = (\partial P - \partial Q) + d$ with $d \in \mathcal{A}$ and $\partial P - \partial Q > 0$. Since $\partial P - \partial Q = \partial(P - Q)$ with $P - Q \in \mathcal{P} \setminus \{0\}$, it does not lie in $\mathcal{A}$, and convexity of $\mathcal{A}$ (Theorem 7.5(iii)) then gives $b - c > 0$, contradicting $b \leq c$. $\square$

Convention 15.5 (The alphabet of this part). Throughout Part III, n is an ordinary positive integer and all matrices are $n \times n$ over $\mathbf{No}[i]$ unless said otherwise; ∂ acts entrywise, A^* is the conjugate transpose for the conjugation of Proposition 3.5, and I is the identity matrix.

- A is a coefficient matrix, as in Section 14; Y is a fundamental matrix.
- H is a *Hermitian* matrix, $H^* = H$. In this report the letter H otherwise occurs only in the hyphenated adjective “ H -field” and in the subscripted coefficient fields $H_\Gamma = \mathbb{R}((t^\Gamma))$; a bare H in this part is always a matrix and never either of those.
- $G \in \mathrm{GL}_n(\mathbf{No}[i])$ is a gauge matrix — the matrix upgrade of the scalar gauge g of equation (9.1) — and W is a gauge that is *unitary*, $W^*W = I$. The letter U is reserved for an ordinary complex domain (Warning B.1) and is not used for a unitary matrix; $\mathrm{U}_n(\mathbf{No}[i])$ denotes the unitary group.
- $\mathcal{M}$ is a horizontal Hermitian metric (Definition 15.15), and $\mathcal{R} = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}$ is the rotation generator. The sources write J for the latter; this report uses no letter J (Convention 5.6).
- $\mathcal{T}$ is the differential field of transseries, which occurs only in Theorem 15.6; it is never the unit circle $\mathbb{T}(\mathbf{No})$ and never a transcendental generator $\mathbb{T}$.
- Π is the additive *phase lattice* generated inside $\mathcal{P}$ by the phases of a system, and $\Lambda \subseteq \mathbb{Z}^n$ remains the integer *relation* lattice of equation (21.6). The two are different objects over different groups and are never written with the same letter.

15.2 Matrix gauge equivalence and the complete phase normal form

A change of unknown $y = Gz$ with $G \in \mathrm{GL}_n(\mathbf{No}[i])$ carries $\partial y = Ay$ to $\partial z = A^G z$, where

$$A^G = G^{-1}AG - G^{-1}\partial G. \quad (15.1)$$

Call A, B *differentially equivalent* when $B = A^G$ for some $G \in \mathrm{GL}_n(\mathbf{No}[i])$, and *differentially unitarily equivalent* when G may be taken unitary. The connection term $G^{-1}\partial G$ is not optional: similarity of matrices alone is not differential equivalence, and Example 16.10 below turns on exactly that.

Two inputs are needed that Part II does not contain. Both are imported, and the import is flagged here and in Section 51.9.

Theorem 15.6 (Imported: splitting and first-order surjectivity, transferred). *The following two properties hold over $\mathcal{T}[i]$, where $\mathcal{T}$ is the field of transseries, and are recorded in the asymptotic-differential-algebra literature [4, Lemma 2.4.19]: every monic scalar differential operator of positive order splits into first-order factors, and every nonzero scalar linear differential operator is surjective. Both transfer to $\mathbf{No}[i]$. In particular, for all $a, f \in \mathbf{No}[i]$ there is $u \in \mathbf{No}[i]$ with*

$$\partial u - au = f. \quad (15.2)$$

Proof. For each fixed order the two assertions are first-order sentences of the ordered differential-field language: expanding a product of first-order operators uses only the finite rule $\partial a = a\partial + \partial(a)$, so the coefficient identities are polynomial identities in finitely many coefficients and their derivatives, and surjectivity of a given order is a first-order sentence with a nonzero leading coefficient as hypothesis. A complex coefficient is an ordered pair of real ones, and complex addition, multiplication and differentiation are definable in that interpretation, so the sentences about $\mathcal{T}[i]$ are sentences about $\mathcal{T}$. The elementary embedding $\mathcal{T} \preceq \mathbf{No}$ of [3] transfers them to $\mathbf{No}$, hence to $\mathbf{No}[i]$, for arbitrary surreal coefficients and not merely for coefficients in the embedded copy of $\mathcal{T}$. $\square$

Warning 15.7 (What this import does and does not add). Theorem 15.6 is an *additional imported input*, on the same footing as Theorem 3.1 and not a consequence of anything proved in Part II. It must not be read as making $\mathbf{No}[i]$ differentially closed, and it must not be read as a homogeneous existence theorem: “linearly closed” there means the splitting property, not that every homogeneous equation has a full solution set in the base field. Reading it the second way would make $\partial y = iy$ appear solvable and contradict Corollary 8.8. It also yields no algorithm: no effective factorization of an arbitrary surreal-coefficient operator follows from a first-order transfer (Section 51.4). What it does supply is *inhomogeneous existence* for first-order scalar equations, which Theorem 12.1 deliberately did not assert outside the admissible locus $\Phi(q) = 0$; see Corollary 15.11 and the amended item N22.

Lemma 15.8 (Cyclic vector, and a rank-one filtration). *Let M be a differential module over $\mathbf{No}[i]$ of ordinary finite dimension $n \geq 1$, that is, a free $\mathbf{No}[i]$ -module of rank n with an additive ∇ satisfying $\nabla(fv) = (\partial f)v + f\nabla v$. Then M has a vector v for which $v, \nabla v, \dots, \nabla^{n-1}v$ is a basis, and M has a filtration by differential submodules whose successive quotients have dimension one.*

Proof. Fix a basis $e_0, \dots, e_{n-1}$ and extend ∇ to $M[T]$ by giving the indeterminate derivative 1. Define $v_0 = e_0$ and $v_k = e_k - \sum_{j < k} \binom{k}{j} \nabla^{k-j} v_j$, and put $v(T) = \sum_{j < n} T^j v_j / j!$. The Leibniz rule gives $\nabla^k v(T)|_{T=0} = e_k$ for $k < n$, so the determinant of the columns $v(T), \dots, \nabla^{n-1}v(T)$ is a polynomial in T with constant term 1, hence nonzero. Substituting $T = \omega - c$ for a real c respects differentiation because $\partial(\omega - c) = 1$, and a nonzero polynomial has only finitely many roots while the elements $\omega - c$ are pairwise distinct; choose c outside that finite exceptional set to get a cyclic vector.

For the filtration, a cyclic vector presents M as $\mathbf{No}[i][\partial]/\mathbf{No}[i][\partial]L$ with L monic of order n . By Theorem 15.6 write $L = (\partial - b)R$ with R a product of $n - 1$ first-order factors. The class of R is nonzero, spans a one-dimensional differential submodule and satisfies $\partial R \equiv bR$ modulo L ; the quotient is the module with annihilator R , and induction on n finishes. The dimension counts follow from division by a monic operator, whose remainder has strictly smaller order. $\square$

Theorem 15.9 (Complete phase normal form for finite systems). *For every $A \in \text{Mat}_n(\mathbf{No}[i])$ there are $G \in \text{GL}_n(\mathbf{No}[i])$ and $P_1, \dots, P_n \in \mathcal{P}$ with*

$$A^G = i \text{diag}(\partial P_1, \dots, \partial P_n). \quad (15.3)$$

The multiset

$$\text{Ph}(A) := \{P_1, \dots, P_n\} \subset \mathcal{P}$$

does not depend on G , and it is a complete invariant of differential equivalence: two finite systems of the same size are differentially equivalent if and only if their multisets agree. We call $\text{Ph}(A)$ the differential phase spectrum of A , in the sense of Convention 5.1(e).

Proof. Apply Lemma 15.8 to the connection $\nabla = \partial - A$ on $\mathbf{No}[i]^n$; an adapted basis makes the system upper triangular. Suppose inductively that the first $n-1$ coordinates have been diagonalized, leaving

$$A = \begin{pmatrix} D & b \\ 0 & a \end{pmatrix}, \quad D = \text{diag}(d_1, \dots, d_{n-1}).$$

For $S = \begin{pmatrix} I & u \\ 0 & 1 \end{pmatrix}$ the off-diagonal column of A^S is $b + Du - ua - \partial u$, and each scalar equation $\partial u_j - (d_j - a)u_j = b_j$ has a solution by equation (15.2). This kills the column, with no hypothesis on repetitions among the diagonal entries: when $d_j = a$ the equation is ordinary intrinsic integration, available by (BM6). Induction gives a diagonal form, and Theorem 9.1 applied to each diagonal entry supplies a further diagonal gauge bringing it to equation (15.3) with $P_j = \Phi(a_j)$.

For independence and completeness, suppose $D_P = i \text{diag}(\partial P_j)$ and $D_Q = i \text{diag}(\partial Q_k)$ satisfy $D_Q = (D_P)^\Xi$ for an invertible Ξ . Then $\partial \Xi = D_P \Xi - \Xi D_Q$, that is

$$\partial \Xi_{jk} = i \partial(P_j - Q_k) \Xi_{jk} \quad (15.4)$$

entrywise. By Lemma 15.3 an entry can be nonzero only when $P_j = Q_k$, and is then a complex constant. Invertibility of Ξ forces the two multisets to have the same multiplicities. Conversely equal multisets give the same normal form after a constant permutation. $\square$

Remark 15.10 (This is the matrix upgrade of Theorem 9.1, and of nothing else). For $n = 1$, Theorem 15.9 is Theorem 9.1. For $n > 1$ it is a genuinely stronger statement, and it is the first classification in this report of a system that is neither triangular nor constant. It is still *not* a module-theoretic filtration theorem with computable data: the normal form is obtained from an existence theorem, the phases are semantic values of $\mathfrak{o}$, and Section 51.9 records what remains open.

15.3 Solution spaces, projectors, and the refined trace identity

For $P \in \mathcal{P}$ write $m_P(A)$ for the multiplicity of P in $\text{Ph}(A)$, zero if P does not occur.

Corollary 15.11 (Existence for every forcing, and the exact kernel dimension). *For every $A \in \text{Mat}_n(\mathbf{No}[i])$ and every $f \in \mathbf{No}[i]^n$ the equation $\partial y - Ay = f$ has a solution in $\mathbf{No}[i]^n$, and its solution set is an affine space over $\mathbb{C}$ of dimension $m_0(A)$. In particular*

$$\dim_{\mathbb{C}} \ker(\partial - A)|_{\mathbf{No}[i]^n} = m_0(A),$$

and $\partial - A$ is bijective on $\mathbf{No}[i]^n$ exactly when $0 \notin \text{Ph}(A)$.

Proof. Transport by the gauge of Theorem 15.9 and solve each scalar inhomogeneous equation by equation (15.2). In the homogeneous equation a coordinate can be nonzero only at phase 0, where it is a complex constant, by Lemma 15.3. The difference of two particular solutions is homogeneous. $\square$

Warning 15.12 (What Corollary 15.11 changes, and what it does not). Theorem 12.1 proves surjectivity of $\partial - q$ only on the admissible locus $\Phi(q) = 0$, and Warning 12.2 states the correct weaker conclusion elsewhere: *at most one*, not none. Corollary 15.11 upgrades “at most one” to “exactly one” in the obstructed case, and it does so *only* by invoking the separately imported first-order surjectivity of Theorem 15.6. It does not follow from the rank-one criterion, it does not follow from Theorem 12.1, and it does not make the homogeneous equation solvable. The examples of Warning 12.2 are unaffected and remain correct; what changes is that their uniqueness is now

known to be accompanied by existence. Item N22 of Section 51 is amended accordingly and not deleted.

Proposition 15.13 (Canonical phase projectors). *Let E_P be the coordinate projector onto the phase- P block of a normal form and put $\Pi_P = GE_PG^{-1}$ for a normalizing gauge G . Then Π_P is independent of the choice of G , and*

$$\Pi_P \Pi_Q = \delta_{PQ} \Pi_P, \quad \sum_P \Pi_P = I, \quad \partial \Pi_P = A \Pi_P - \Pi_P A.$$

The algebra of horizontal endomorphisms of the system — matrices X with $\partial X = AX - XA$ — is isomorphic to $\prod_P \text{Mat}_{m_P(A)}(\mathbb{C})$, and for two systems

$$\dim_{\mathbb{C}} \text{Hom}_{\partial}(A, B) = \sum_P m_P(A) m_P(B).$$

Direct sums unite phase multisets, tensor products add phases pairwise, and duals negate them.

Proof. After aligning the ordered phase blocks, equation (15.4) makes any change between normalizing gauges a constant block-diagonal matrix, so it commutes with every E_P ; the displayed identities follow by multiplying and differentiating $\Pi_P = GE_PG^{-1}$ and using $\partial G = AG - GD_P$. The same entrywise calculation, applied to $\partial X = AX - XB$ in normal-form coordinates, identifies the horizontal homomorphisms with constant matrices supported on matching phase blocks. The last three statements are the corresponding computations for $A \oplus B$, $A \otimes I + I \otimes B$ and $-A^T$, in each case reduced to the scalar addition rule of Corollary 9.4. $\square$

Corollary 15.14 (The determinant sees only the phase sum). *For every finite A ,*

$$\Phi(\text{tr } A) = \sum_P m_P(A) P. \quad (15.5)$$

Proof. Take traces in $A = (\partial G)G^{-1} + GD_PG^{-1}$ and use Jacobi's identity $\text{tr}((\partial G)G^{-1}) = (\det G)^\dagger$, whose obstruction is zero by Theorem 8.4. $\square$

Equation (15.5) contains Proposition 14.2 and explains it: a fundamental matrix exists only if every phase is zero, so $\Phi(\text{tr } A) = 0$ is necessary, and it fails to be sufficient precisely because a multiset is not recoverable from its sum. The witness $\text{diag}(i, -i)$ of Proposition 14.2 has $\text{Ph} = \{\omega, -\omega\}$, which sums to zero while $\{0, 0\}$ does too; the two systems are not differentially equivalent, and Theorem 15.9 says so. The trace condition is the first invariant; the phase multiset is the complete one.

15.4 Horizontal metrics and the unitary normal form

Definition 15.15 (Horizontal metric). A Hermitian $\mathcal{M} \in \text{Mat}_n(\mathbf{No}[i])$ is *horizontal* for A when

$$\partial \mathcal{M} + A^* \mathcal{M} + \mathcal{M} A = 0, \quad (15.6)$$

and *positive definite* when $v^* \mathcal{M} v > 0$ for every nonzero $v \in \mathbf{No}[i]^n$.

Equation (15.6) says formally that $y^* \mathcal{M} y$ is constant along a solution, in any differential extension carrying the compatible conjugation. It does not assert that the system has nonzero solutions in $\mathbf{No}[i]^n$; by Corollary 15.11 it usually does not.

Theorem 15.16 (The complete metric cone). *Let G normalize A as in Theorem 15.9. Every horizontal Hermitian matrix is uniquely*

$$\mathcal{M} = G^{-*} \left(\bigoplus_P C_P \right) G^{-1}, \quad C_P \in \text{Herm}_{m_P(A)}(\mathbb{C}), \quad (15.7)$$

and is positive definite exactly when every C_P is. In particular every finite system has a positive horizontal metric, and the real dimension of its space of horizontal Hermitian forms is $\sum_P m_P(A)^2$.

Proof. Put $B = G^* \mathcal{M} G$. Since $\partial G = AG - GD_P$, differentiating gives $\partial B = D_P B - B D_P$, that is $\partial B_{jk} = i \partial(P_j - P_k) B_{jk}$, so Lemma 15.3 makes every off-phase entry zero and every within-phase entry a complex constant. Hermitian symmetry leaves exactly the blocks C_P . Congruence preserves definiteness, and for a *constant* Hermitian matrix positivity over $\mathbf{No}[i]$ is ordinary complex positive definiteness: diagonalize by an ordinary constant unitary matrix and read off its real eigenvalues. Taking every $C_P = I$ gives existence, and $\dim_{\mathbb{R}} \text{Herm}_m(\mathbb{C}) = m^2$. $\square$

Warning 15.17 (A positive horizontal metric is not a stability statement). For

$$A = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}, \quad G = e^\omega \begin{pmatrix} 1 & \omega \\ 0 & 1 \end{pmatrix},$$

one has $\partial G = AG$, so both phases are zero, and

$$\mathcal{M} = e^{-2\omega} \begin{pmatrix} 1 & -\omega \\ -\omega & \omega^2 + 1 \end{pmatrix}$$

is horizontal with $\det \mathcal{M} = e^{-4\omega} > 0$. The ordinary differential equation with the same printed coefficients has growing solutions. There is no contradiction, and no Lyapunov or uniform-equivalence claim is made anywhere in this part: the invariant metric itself has a strongly varying scale, here $e^{-2\omega}$, and Theorem 15.16 has no hypothesis relating it to a fixed norm. This is the matrix form of the separation already made in Warning 4.1(7) between a printed equation read in $\mathbf{No}[i]$ and the same equation read in an ordinary coordinate.

Theorem 15.18 (Unitary phase normal form). *If $A^* = -A$, the normalizing gauge of Theorem 15.9 can be taken unitary. For a fixed ordering of the phases, any two unitary normalizing gauges differ by a constant block unitary matrix, one block per repeated phase, and the projectors of Proposition 15.13 are then Hermitian orthogonal projectors.*

Proof. The identity matrix is horizontal for a skew-Hermitian A , so Theorem 15.16 applied to any normalizing G makes $G^* G = C$ a constant positive definite block Hermitian matrix. Let $R = C^{-1/2}$, computed in the ordinary finite-dimensional complex block matrices; then $\partial R = 0$ and R commutes with D_P , so $W = GR$ is still normalizing and $W^* W = I$. Uniqueness is equation (15.4) again, with unitarity making the constant diagonal blocks unitary, and $W E_P W^*$ is Hermitian and idempotent. $\square$

The square root in that proof is of a *constant complex* matrix. No analytic functional calculus on a class-sized space is used, and the theorem is about differential diagonalization: the term $W^* \partial W$ has not been removed and reappears, essentially, in equation (16.5).

16 The spectral–integral theorem for Hermitian coefficients

16.1 Finite Hermitian algebra over a real closed field

The two lemmas of this subsection hold over any real closed field and its complexification, and use no property of ∂ at all. Related finite spectral material for $\mathbf{No}[i]$ is developed in the companion report *Surcomplex Spectral Theory* [27], whose “spectrum” is the algebraic eigenvalue list; see Convention 5.1(e) for why that is not the object named in Theorem 16.5.

Lemma 16.1 (Finite spectral theorem with attained min–max). *Every Hermitian $H \in \text{Mat}_n(\mathbf{No}[i])$ has a unitary eigenbasis and eigenvalues in $\mathbf{No}$. Writing $\lambda_1(H) \leq \dots \leq \lambda_n(H)$,*

$$\lambda_k(H) = \min_{\dim S=k} \max_{v \in S, v^*v=1} v^*Hv, \quad (16.1)$$

and both extrema are attained.

Proof. $\mathbf{No}[i]$ is algebraically closed, so there is an eigenvector $v \neq 0$ with eigenvalue λ ; since $v^*v > 0$ and v^*Hv is fixed by conjugation, $\lambda = v^*Hv/v^*v \in \mathbf{No}$. Normalize v by the positive square root of v^*v , available because $\mathbf{No}$ is real closed; the orthogonal complement is H -invariant, and finite induction gives a unitary eigenbasis. For a fixed k -dimensional S the compression of H to an orthonormal basis of S is again finite Hermitian, so its largest eigenvalue attains the inner maximum; the span of the first k eigenvectors attains the value $\lambda_k(H)$, and every k -dimensional S meets the span of the last $n - k + 1$ eigenvectors nontrivially, where a unit vector has Rayleigh quotient at least $\lambda_k(H)$. No compactness and no completeness is used; the argument is finite algebra over an ordered field. $\square$

Lemma 16.2 (An entrywise Weyl bound, and its ideal form). *For Hermitian $H, E \in \text{Mat}_n(\mathbf{No}[i])$ put $m = \max_{j,k} |E_{jk}|$. Then*

$$|\lambda_j(H + E) - \lambda_j(H)| \leq nm \quad (1 \leq j \leq n). \quad (16.2)$$

If every entry of E lies in $\mathcal{A}_{\mathbb{C}} := \mathcal{A} + i\mathcal{A}$, then every eigenvalue difference in equation (16.2) lies in $\mathcal{A}$.

Proof. For $v^*v = 1$ the triangle and Cauchy–Schwarz inequalities over the real closed field $\mathbf{No}$ give $|v^*Ev| \leq m(\sum_j |v_j|)^2 \leq nm$. Using the span of the first j eigenvectors of H in equation (16.1) gives the upper bound, and interchanging H and $H + E$ the lower one. If $E_{jk} = a + ib$ with $a, b \in \mathcal{A}$ then $|E_{jk}| \leq |a| + |b| \in \mathcal{A}$, so convexity of $\mathcal{A}$ (Theorem 7.5(iii)) puts $|E_{jk}|$ in $\mathcal{A}$; a finite maximum and the integer multiple nm stay in $\mathcal{A}$, and one more application of convexity finishes the proof. $\square$

16.2 Bounded gauges have integrable connection terms

Lemma 16.3 (Bounded-gauge estimate). *Let $W \in \text{U}_n(\mathbf{No}[i])$. Then every entry of W and of W^* is finite, every entry of ∂W lies in $\mathcal{A}_{\mathbb{C}}$, every entry of $W^*\partial W$ lies in $\mathcal{A}_{\mathbb{C}}$, and $W^*\partial W$ is skew-Hermitian.*

Proof. A unitary column has squared moduli summing to 1, so each entry has modulus at most 1 and is finite. Differentiating a finite real or imaginary coordinate lands in $\partial\mathcal{O} = \mathcal{A}$ by Theorem 7.5(i). Since $\mathcal{A}$ is an $\mathcal{O}$ -module (Theorem 7.5(iii)), finite sums of products of entries of W^* with entries of ∂W stay in $\mathcal{A}_{\mathbb{C}}$. Differentiating $W^*W = I$ gives $(\partial W)^*W + W^*\partial W = 0$. $\square$

Warning 16.4 (The conclusion is about primitives, not about size). It is the conclusion $W^*\partial W \in \text{Mat}_n(\mathcal{A}_{\mathbb{C}})$, and not the weaker statement that its entries are infinitesimal, that makes Theorem 16.5 true. The two are different by Warning 5.4: an infinitesimal entry may accumulate an infinite phase, as ω^{-1} does. This is the precise point at which the strictness $\mathcal{A} \subsetneq \mathfrak{m}$ is load-bearing for the matrix theory.

16.3 The theorem

Theorem 16.5 (Spectral–integral classification of Hermitian systems). *Let $H = H^* \in \text{Mat}_n(\mathbf{No}[i])$ with algebraic eigenvalues ordered $\lambda_1 \leq \dots \leq \lambda_n$ in $\mathbf{No}$, and put*

$$P_j := \mathfrak{o}(\lambda_j) \in \mathcal{P}.$$

Then $P_1 \leq \dots \leq P_n$ and

$$\text{Ph}(iH) = \{P_1, \dots, P_n\}. \quad (16.3)$$

There is a unitary $W \in \text{U}_n(\mathbf{No}[i])$ with

$$W^*(iH)W - W^*\partial W = i \text{diag}(\partial P_1, \dots, \partial P_n), \quad (16.4)$$

and two Hermitian coefficients give differentially unitarily equivalent systems $\partial y = iH_1 y$ and $\partial y = iH_2 y$ if and only if their ordered algebraic eigenvalues have identical images under $\mathfrak{o}$, counted with multiplicity. Neither commuting coefficients nor a spectral gap is assumed.

Proof. The matrix iH is skew-Hermitian, so Theorem 15.18 supplies a unitary normalizing gauge W with phases $Q_1 \leq \dots \leq Q_n$, ordered by Corollary 15.4. Rearranging its defining identity gives

$$W^*HW = \text{diag}(\partial Q_1, \dots, \partial Q_n) + E, \quad E := -iW^*\partial W. \quad (16.5)$$

By Lemma 16.3 the matrix E is Hermitian with every entry in $\mathcal{A}_{\mathbb{C}}$. Conjugation by a unitary matrix does not change eigenvalues, and $\partial Q_1 \leq \dots \leq \partial Q_n$ by Corollary 15.4; so Lemma 16.2, applied to the diagonal matrix and the perturbation E , gives $\lambda_j(H) - \partial Q_j \in \mathcal{A}$ for every j . Apply $\mathfrak{o}$: its kernel is exactly $\mathcal{A}$ by Theorem 7.5(vi), and $\mathfrak{o}(\partial Q_j) = Q_j$ by the same item, so $\mathfrak{o}(\lambda_j(H)) = Q_j$. This is equations (16.3) and (16.4), and the ordering of the P_j is Corollary 15.4. The equivalence statement is Theorem 15.9 together with Theorem 15.18, which makes the normalizing gauges of two skew-Hermitian systems unitary, so that the equivalence between them is unitary as well. $\square$

Warning 16.6 (The connection term is not removed). Theorem 16.5 does *not* say that pointwise diagonalization commutes with differentiation; it does not. The term $W^*\partial W$ in equation (16.5) is present, and the theorem says only that it is invisible in the ordered additive quotient by $\mathcal{A}$ described in Theorem 7.5(viii). Nor is the classification the assertion that every phase grade has a horizontal vector in $\mathbf{No}[i]^n$: by Corollary 15.11 only the grade $P = 0$ does. The formula is exact for arbitrary surreal scales and arbitrary ordinary finite multiplicities; there is no hypothesis that the λ_j be distinct, that gaps dominate a derivative, or that H commute with its derivative or with any of its primitives.

16.4 Gap-free perturbation, the exact kernel count, and sharpness

Corollary 16.7 (Finite-primitive perturbation invariance, with no gap hypothesis). *If H_1, H_2 are Hermitian and every entry of $H_1 - H_2$ lies in $\mathcal{A}_{\mathbb{C}}$, then $\partial y = iH_1 y$ and $\partial y = iH_2 y$ are differentially unitarily equivalent.*

Proof. Lemma 16.2 puts corresponding ordered eigenvalue differences in $\mathcal{A} = \ker \mathfrak{o}$; apply Theorem 16.5. $\square$

Corollary 16.8 (Exact kernel count). *For Hermitian H ,*

$$\dim_{\mathbb{C}}\{y \in \mathbf{No}[i]^n : \partial y = iHy\} = \#\{j : \lambda_j(H) \in \mathcal{A}\}, \quad (16.6)$$

counted with algebraic multiplicity. The forced equation $\partial y - iHy = f$ is solvable for every $f \in \mathbf{No}[i]^n$, and uniquely so exactly when no eigenvalue of H lies in $\mathcal{A}$.

Proof. Combine Theorem 16.5 and Corollary 15.11 with $\ker \mathfrak{o} = \mathcal{A}$. $\square$

For $H = \text{diag}(\omega^{-2}, \omega^{-1}, 1)$ the kernel has complex dimension one, because $\omega^{-2} \in \mathcal{A}$ while $\omega^{-1} \notin \mathcal{A}$ and $1 \notin \mathcal{A}$ (equation (7.3)). Replacing H by any Hermitian matrix differing from it entrywise by $\mathcal{A}_{\mathbb{C}}$ preserves that answer, even when the perturbation is far larger than a gap at some more remote infinitesimal scale — which is exactly what a gap hypothesis would forbid.

Proposition 16.9 (Sharpness of the perturbation class). *For real $h, k \in \mathbf{No}$ the equations $\partial y = ihy$ and $\partial y = iky$ are differentially unitarily equivalent exactly when $h - k \in \mathcal{A}$. Hence no strictly larger additive class of scalar perturbations preserves the phase of every Hermitian system.*

Proof. A scalar unitary gauge equivalence is $i(h - k) = u^\dagger$ with $u\bar{u} = 1$; Theorem 8.1 together with Theorem 6.3 gives necessity and constructs a unit solution for sufficiency. If $b \notin \mathcal{A}$ the one-dimensional systems with coefficients 0 and b have obstructions 0 and $\mathfrak{o}(b) \neq 0$. $\square$

“Infinitesimal perturbation” therefore cannot replace “finite-primitive perturbation”: the scalar change $0 \mapsto \omega^{-1}$ already moves the phase from 0 to $\log \omega$. This is Warning 5.4 again, now in its sharpest form: $\mathcal{A}$ is not merely a convenient subclass of $\mathfrak{m}$, it is the exact class for which Corollary 16.7 is true.

16.5 The two boundaries, exhibited

Example 16.10 (A nonnormal system whose eigenvalues predict the wrong phases). Let $x = \omega$ and

$$G = \begin{pmatrix} 1 - x^2 & x \\ -x & 1 \end{pmatrix}, \quad A = \begin{pmatrix} -x & 1 + x^2 \\ -1 & x \end{pmatrix}.$$

Then $\det G = 1$ and $\partial G = AG$, so A has a fundamental matrix in $\text{GL}_2(\mathbf{No}[i])$ and, by Corollary 15.11, $\text{Ph}(A) = \{0, 0\}$. Yet

$$\det(\zeta I - A) = \zeta^2 + 1,$$

so the instantaneous algebraic eigenvalues of A are i and $-i$, and taking obstructions of their imaginary parts would predict $\{\omega, -\omega\}$. The prediction is wrong by Theorem 15.9.

Warning 16.11 (What the counterexample rules out, and what it marks). Example 16.10 is not a curiosity; it is the boundary of Theorem 16.5. It shows that the Hermitian hypothesis cannot be dropped and that the main formula does not extend to arbitrary matrices by extracting imaginary parts of eigenvalues. The mechanism is exactly the one Lemma 16.3 controls in the Hermitian case: the similarity gauges needed for a general nonnormal matrix need not be entrywise finite, so their connection term need not lie in $\text{Mat}_n(\mathcal{A}_{\mathbb{C}})$ and can carry a nonzero phase of its own. Here G has the infinite entry $1 - \omega^2$.

The complete classification of Theorem 15.9 still applies to this non-Hermitian system: its phase multiset is $\{0, 0\}$. What fails is computing that multiset from the instantaneous eigenvalues by the Hermitian formula. Theorem 16.5 applies $\mathfrak{o}$ to the algebraic eigenvalues of H when $A = iH$; the general normal form instead uses differential factorization and gauges. Effective construction of those data on specified input languages remains a separate question (Section 51.13).

For the power-Hahn regular-singular systems of Part V, $\text{Ph}(-tA) = \{(\text{Im } \lambda_j) \log t\}$, with λ_j the eigenvalues of the ordinary residual matrix (Proposition 25.12). This provides another explicitly computable non-Hermitian class; Example 16.10 lies outside it.

Example 16.12 (A genuinely coupled unitary system with two independent phases). Let $x = \omega$ and

$$a = \frac{1 - x^2}{1 + x^2}, \quad b = \frac{2x}{1 + x^2}, \quad W = \begin{pmatrix} a & -b \\ b & a \end{pmatrix}, \quad D = i \text{diag}(x, x^{-1}),$$

so that W is a real orthogonal matrix over $\mathbf{No}$ and $A := (\partial W)W^\top + WDW^\top$ satisfies $A^* = -A$. Explicitly

$$A = \begin{pmatrix} i(a^2x + b^2x^{-1}) & -\frac{2}{1+x^2} + iab(x - x^{-1}) \\ \frac{2}{1+x^2} + iab(x - x^{-1}) & i(b^2x + a^2x^{-1}) \end{pmatrix},$$

which is neither diagonal nor a constant-matrix system. Its exact normal form is D , so

$$\text{Ph}(A) = \left\{ \frac{1}{2}\omega^2, \log \omega \right\},$$

using $\mathfrak{o}(\omega) = \frac{1}{2}\omega^2$ and $\mathfrak{o}(\omega^{-1}) = \log \omega$ (equation (7.3)). Writing $H = -iA$, one computes

$$\text{tr } H = x + x^{-1}, \quad \det H = 1 - \frac{4}{(1+x^2)^2},$$

so the instantaneous eigenvalues of H are *not* x and x^{-1} . Their discrepancies lie in $\mathcal{A}$, so Theorem 16.5 recovers the same two phases from them. Both phases are nonzero, so by Corollary 15.11 every forced system $\partial y - Ay = f$ has a unique solution in $\mathbf{No}[i]^2$; and the two phases are $\mathbb{Q}$ -linearly independent, which Section 17.6 uses.

17 Consequences of the matrix normal form

17.1 Normalized non-Abelian integration

Write

$$\mathfrak{u}_n(\mathcal{A}_{\mathbb{C}}) = \{A \in \text{Mat}_n(\mathcal{A}_{\mathbb{C}}) : A^* = -A\},$$

and let $U_n(1 + \mathfrak{m}_{\mathbb{C}})$ be the unitary matrices W with $W - I \in \text{Mat}_n(\mathfrak{m}_{\mathbb{C}})$. The latter is a congruence subgroup, not the requirement that off-diagonal entries be near 1.

Theorem 17.1 (Normalized compact integration). *The logarithmic-derivative map*

$$U_n(1 + \mathfrak{m}_{\mathbb{C}}) \longrightarrow \mathfrak{u}_n(\mathcal{A}_{\mathbb{C}}), \quad W \longmapsto (\partial W)W^*, \quad (17.1)$$

is a bijection of classes. Equivalently, for skew-Hermitian A the following are equivalent:

- (i) *every entry of A has finite real and imaginary primitives, that is $A \in \text{Mat}_n(\mathcal{A}_{\mathbb{C}})$;*

- (ii) $\partial y = Ay$ has a fundamental matrix in $\mathrm{GL}_n(\mathbf{No}[i])$;
- (iii) $\partial W = AW$ has a unitary solution with $\mathrm{st}(W) = I$ entrywise.

The solution in (iii) is unique.

Proof. Lemma 16.3 makes equation (17.1) well defined and gives (iii) $\Rightarrow$ (i). Assume (i) and put $H = -iA$, Hermitian with entries in $\mathcal{A}_{\mathbb{C}}$. Comparing H with the zero matrix, Lemma 16.2 places every eigenvalue of H in $\mathcal{A}$, so Theorem 16.5 gives $\mathrm{Ph}(A) = \{0, \dots, 0\}$ and a unitary V with $A^V = 0$, that is $\partial V = AV$. Taking standard parts entrywise in $V^*V = I$ — legitimate because every entry is finite by Lemma 16.3 — shows that $C = \mathrm{st}(V)$ is an ordinary complex unitary matrix, and $W = VC^{-1}$ has $\partial W = AW$ and $\mathrm{st}(W) = I$.

If W_1, W_2 are two normalized solutions then $\partial(W_1^{-1}W_2) = 0$, so $W_1^{-1}W_2$ is a constant complex matrix with standard part I , hence I . That is uniqueness and injectivity of equation (17.1). Clearly (iii) $\Rightarrow$ (ii); conversely a fundamental matrix makes every phase zero by Corollary 15.11, and for a skew-Hermitian system Theorem 15.18 then supplies a unitary fundamental matrix, which after multiplication by the inverse of its standard part is normalized. $\square$

Warning 17.2 (This is an existence theorem, not an exponential formula). Theorem 17.1 is not the assertion that $W = \exp(\int A)$, and there is no ordering prescription, no chosen contour, and no claimed convergence of Picard iteration. Its normalization is residue data, $\mathrm{st}(W) = I$, not an initial value at a real or surreal time — the same distinction Warning 7.2 makes for $\mathcal{I}_0$. For the unique solution W_A the ordinary product rule gives

$$(\partial(W_A W_B))(W_A W_B)^* = A + W_A B W_A^*, \quad (17.2)$$

whose right-hand side again lies in $\mathbf{u}_n(\mathcal{A}_{\mathbb{C}})$. So the bijection transports the group law to $A \star B = A + W_A B W_A^*$, which is equal to $A + B$ exactly when $W_A B = B W_A$. Commutation of A and B alone is not sufficient, as the following example also shows.

Example 17.3 (An exact noncommuting Hahn solution). Let $t = \omega^{-1}$, so $\partial t = -t^2$, and take the constant skew-Hermitian matrices

$$X = \begin{pmatrix} 0 & i \\ i & 0 \end{pmatrix}, \quad Z = \begin{pmatrix} i & 0 \\ 0 & -i \end{pmatrix}.$$

For $A = -t^2 X - 2t^3 Z$ — every entry of which lies in $\mathcal{A}_{\mathbb{C}}$ — the normalized solution of Theorem 17.1 is the Hahn series $W = \sum_{n \geq 0} W_n t^n$ determined by

$$W_0 = I, \quad (n+1)W_{n+1} = XW_n + 2ZW_{n-1}, \quad W_{-1} = 0, \quad (17.3)$$

whose first terms are

$$W = I + tX + t^2\left(\frac{1}{2}X^2 + Z\right) + t^3\left(\frac{1}{6}X^3 + \frac{1}{3}XZ + \frac{2}{3}ZX\right) + \dots$$

This differs from $\exp(tX + t^2Z)$ at order t^3 by $(ZX - XZ)/6 \neq 0$. The support is contained in $\mathbb{N}$, so the series is strongly summable by Lemma 2.6 and the recurrence proves the equation exactly; $(\partial(W^*W)) = 0$ with standard part I makes it unitary, so it is the unique solution. This is a computation in a restricted Hahn workspace and is not a claim that such a recurrence is effectively available for a general coefficient matrix.

For the product law take $B = A$, so $[A, B] = 0$ automatically. The displayed coefficients of $W = W_A$ give

$$[t^4](W_A A W_A^* - A) = -[X, Z] \neq 0.$$

Indeed the two contributions at this order are $[tX, -2t^3Z] = -2t^4[X, Z]$ and $[t^2Z, -t^2X] = t^4[X, Z]$; the terms involving only X cancel. Thus $A \star A \neq 2A$, even though the coefficient matrices A and B commute.

Example 17.3 is the same phenomenon as the two-scale computation of Section 47, which refutes the naive exponential for the *coordinate* derivative D_z ; the two live in different categories (Table 10) and neither is evidence for the other. What they share is the mechanism: a noncommuting pair of constant matrices produces an exact third-order defect that no exponential of an integral can absorb.

17.2 Real systems: rotation blocks and metric signatures

For a real system, conjugation sends phase P to phase $-P$, so the two multiplicities are equal.

Theorem 17.4 (Real rotation-block classification). *For every $A \in \text{Mat}_n(\mathbf{No})$ there is $G \in \text{GL}_n(\mathbf{No})$ with*

$$A^G = 0_{m_0} \oplus \bigoplus_{P>0} ((\partial P)\mathcal{R})^{\oplus m_P}, \quad n = m_0 + 2 \sum_{P>0} m_P, \quad (17.4)$$

where $\mathcal{R} = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}$. The positive phases $P \in \mathcal{P}$ and their multiplicities are unique. Every finite real differential module over $\mathbf{No}$ is a direct sum of trivial one-dimensional modules and nontrivial two-dimensional rotation modules, and if $A^\top = -A$ then G can be chosen orthogonal.

Proof. Uniqueness of the complex phase subspaces (Proposition 15.13) makes conjugation interchange the P and $-P$ subspaces. On the zero-phase subspace the horizontal complex space is conjugation stable, so a real basis of its fixed part is a complex basis after complexification. For $P > 0$ take normalizing columns $v_1, \dots, v_{m_P}$ of phase P ; their conjugates span phase $-P$. Write $v_j = a_j + ib_j$. From $\partial v_j = Av_j - i(\partial P)v_j$,

$$\partial a_j = Aa_j + (\partial P)b_j, \quad \partial b_j = Ab_j - (\partial P)a_j,$$

which is the block $(\partial P)\mathcal{R}$. The real columns are independent because the change between $(v_j, \bar{v}_j)$ and (a_j, b_j) is invertible over $\mathbb{C}$. The only invariant real subspaces inside one nonzero phase pair are 0 and the whole pair, since a real subspace complexifies to a conjugation-stable sum of two distinct phase lines. For skew-symmetric A the identity metric is horizontal, so $G^\top G$ is a positive constant matrix commuting with the rotation blocks, by Theorem 17.5 below; its positive constant square root commutes with them too, and multiplying G by that inverse square root makes it orthogonal without changing the normal form. $\square$

Theorem 17.5 (The real metric cone and its parity). *Every real finite system has a positive symmetric horizontal metric. With the multiplicities of equation (17.4), the cone of positive symmetric horizontal metrics is isomorphic to*

$$\text{Sym}_{m_0}^+(\mathbb{R}) \times \prod_{P>0} \text{Herm}_{m_P}^+(\mathbb{C}), \quad (17.5)$$

a product over a finite set, empty products being a point. The real vector space of all horizontal symmetric forms has dimension $\frac{1}{2}m_0(m_0 + 1) + \sum_{P>0} m_P^2$, and on each nonzero real phase block both indices of a nondegenerate horizontal symmetric form are even.

Proof. Pull a real horizontal form back by the gauge of Theorem 17.4. Complex phase separation (Theorem 15.16) eliminates all terms between different positive phase pairs and between such a pair and phase zero, and leaves an arbitrary constant real symmetric matrix on the zero block. Group a repeated positive phase as $(\partial P)\mathcal{R}_m$ with $\mathcal{R}_m = \begin{pmatrix} 0 & I_m \\ -I_m & 0 \end{pmatrix}$; the real symmetric horizontal matrices are then the constant matrices commuting with $\mathcal{R}_m$, that is

$$C = \begin{pmatrix} S & T \\ -T & S \end{pmatrix}, \quad S^\top = S, \quad T^\top = -T,$$

which is the realification of the complex Hermitian matrix $S - iT$. A constant complex unitary diagonalization of $S - iT$ becomes a real orthogonal diagonalization of C with every eigenvalue repeated twice; this gives positivity, the cone formula and the parity statement. Congruence by G^{-1} returns to the original coordinates. $\square$

There is no integration over phases in equation (17.5): the products are over a finite multiset of constant-matrix cones. In particular a real irreducible two-dimensional system has a one-dimensional space of horizontal symmetric forms, hence a single positive ray. That fact is the source of the uniqueness in Theorem 17.6.

17.3 A positive Ermakov–Pinney dichotomy

For $q \in \mathbf{No}$ consider the intrinsic equations

$$\partial^2 y + qy = 0 \tag{17.6}$$

and

$$\partial^2 \rho + q\rho = \rho^{-3}, \quad \rho > 0. \tag{17.7}$$

Both derivatives are the fixed ∂ . The positive sign on the right of equation (17.7) is part of the statement; the sign convention in the classical literature varies [6].

Theorem 17.6 (Positive Pinney existence and uniqueness). *For every $q \in \mathbf{No}$, equation equation (17.7) has a positive solution in $\mathbf{No}$, and exactly one of the following holds.*

- (i) *Equation (17.6) has no nonzero solution in $\mathbf{No}$. Then equation (17.7) has exactly one positive solution in $\mathbf{No}$, and $\partial^2 y + qy = f$ has a unique solution in $\mathbf{No}$ for every $f \in \mathbf{No}$.*
- (ii) *Equation (17.6) has a real fundamental pair in $\mathbf{No}$. Then the positive solutions of equation (17.7) form the classical positive quadratic-form family equation (17.10), with two real parameters after determinant normalization.*

Proof. Let $A_q = \begin{pmatrix} 0 & 1 \\ -q & 0 \end{pmatrix}$. By Theorem 17.5 it has a positive symmetric horizontal metric $\mathcal{M}$, and equation (15.6) gives $\partial \det \mathcal{M} = -2(\text{tr } A_q) \det \mathcal{M} = 0$, so $\det \mathcal{M}$ is a positive real constant; scale it to 1. Writing $\mathcal{M} = \begin{pmatrix} a & b \\ b & c \end{pmatrix}$, the horizontality equations are

$$\partial a = 2qb, \quad \partial b = qc - a, \quad \partial c = -2b. \tag{17.8}$$

Since $c > 0$ and $\mathbf{No}$ is real closed, put $\rho = \sqrt{c}$. The last equation gives $b = -\rho \partial \rho$ and $\det \mathcal{M} = 1$ gives $a = (\partial \rho)^2 + \rho^{-2}$; the middle equation becomes exactly equation (17.7). Conversely a positive solution of equation (17.7) produces the positive determinant-one horizontal metric

$$\mathcal{M}_\rho = \begin{pmatrix} (\partial \rho)^2 + \rho^{-2} & -\rho \partial \rho \\ -\rho \partial \rho & \rho^2 \end{pmatrix}, \tag{17.9}$$

positive because its lower diagonal entry is positive and its determinant is 1. So positive Pinney solutions correspond bijectively to positive determinant-one horizontal metrics.

In dimension two the real normal form equation (17.4) has either two zero phases or one nonzero conjugate pair. If equation (17.6) has no nonzero solution in $\mathbf{No}$ it must be the second case, the positive metric cone is a single ray by equation (17.5), and the determinant-one condition selects one point. In the zero-phase case a real fundamental matrix exists and the cone is the three-dimensional positive symmetric constant-matrix cone, whose determinant-one slice is the asserted two-parameter family. In the nonzero-phase case Corollary 15.11 gives a unique solution in $\mathbf{No}[i]^2$ of every forced system, and conjugating shows it is real when f is.

There is no intermediate one-dimensional case: if $u \in \mathbf{No}^\times$ solves equation (17.6), choose B with $\partial B = u^{-2}$ by (BM6) and put $v = uB$; then $u\partial v - v\partial u = 1$, so u, v are an independent real fundamental pair. $\square$

With u, v a real fundamental pair of constant Wronskian $\text{Wr} = u\partial v - v\partial u \neq 0$, alternative (ii) reads

$$\rho = \sqrt{\alpha u^2 + 2\beta uv + \gamma v^2}, \quad \alpha, \gamma > 0, \quad \alpha\gamma - \beta^2 = \text{Wr}^{-2}, \quad \alpha, \beta, \gamma \in \mathbb{R}. \quad (17.10)$$

This is the classical Pinney parametrization [6], here read in an ordered differential field; no priority is claimed for it, nor for the abstract Schwarz-closed-field existence principle that underlies it [4]. What is new to this field is that alternative (i) can occur, and that it collapses the positive family to a single point. For recent existence and sharp uniqueness results for amplitude–phase representations over Hardy fields see [5]; no solution of the named Boshernitzan conjecture discussed there is claimed here.

For $q = 1$ alternative (i) holds by Corollary 8.8, and the unique positive Pinney solution is $\rho = 1$. For $q = -1$ the linear equation has the fundamental pair $e^\omega, e^{-\omega}$ and alternative (ii) gives many positive amplitudes. A unique positive solution of a second-order equation is not a contradiction to ordinary initial-value theory: intrinsic differentiation has no two-real-parameter initial-data interpretation (Warning 4.1). (Added with member 13: for the supercritical critical potentials of Part VII the unique positive solution and the canonical phase are explicit; see Remark 38.5.)

Theorem 17.7 (Canonical amplitude–phase reduction). *Call $q \in \mathbf{No}$ intrinsically oscillatory when alternative (i) of Theorem 17.6 holds; this means only that the nonzero homogeneous oscillations are missing from $\mathbf{No}$, and says nothing about oscillation of a function of a real variable. Let ρ be the unique positive Pinney solution, $h = \rho^{-2}$, and $B = \mathcal{I}_0(h) = P + \eta$ with $P \in \mathcal{P}$ and $\eta \in \mathfrak{m}$ (equation (2.3)). Then $P > 0$, and with*

$$G_\rho = \begin{pmatrix} \rho & 0 \\ \partial\rho & \rho^{-1} \end{pmatrix}, \quad R(\eta) = \text{Re}(\mathcal{E}(i\eta)) I + \text{Im}(\mathcal{E}(i\eta)) \mathcal{R},$$

the matrix $S_q = G_\rho R(\eta) \in \text{SL}_2(\mathbf{No})$ satisfies

$$A_q^{S_q} = (\partial P) \mathcal{R}. \quad (17.11)$$

The amplitude ρ , the normalized primitive B , the phase P and this amplitude-times-infinitesimal-rotation gauge are canonical; no uniqueness among all symplectic gauges is asserted.

Proof. A direct multiplication using equation (17.7) gives $A_q^{G_\rho} = \rho^{-2} \mathcal{R} = h \mathcal{R}$. The rotation $R(\eta)$ is a well-defined real matrix with entries the infinitesimal cosine and sine of Theorem 6.3; it has determinant one, commutes with $\mathcal{R}$, and satisfies $R(\eta)^{-1} \partial R(\eta) = (\partial \eta) \mathcal{R}$. Subtracting gives

equation (17.11). If $P = 0$ then $B = \eta$ is finite and the same calculation reduces the system to zero, producing a real fundamental matrix and contradicting alternative (i). Since $\partial B = h > 0$, B cannot be negative infinite by (BM7), so its nonzero purely infinite part is positive. The uniqueness assertions are Theorem 17.6 and the normalization $\text{ct}(B) = 0$ of Proposition 7.1. $\square$

The conserved quadratic expression attached to ρ is

$$\mathcal{E}_\rho(y, \partial y) = (\rho \partial y - y \partial \rho)^2 + (y/\rho)^2, \quad (17.12)$$

whose derivative vanishes formally along equation (17.6) by equation (17.9). It is meaningful in an oscillatory extension of the kind constructed in Section 21 even when the only solution in $\mathbf{No}$ is zero. It is an exact invariant, not an averaged energy.

17.4 The oscillatory Airy equation as an exact Hahn example

Let $x = \omega$ and consider

$$\partial^2 y + xy = 0. \quad (17.13)$$

Put $P = \frac{2}{3}x^{3/2}$, so $\partial P = x^{1/2}$ by the real-power rule (Warning 2.2, applied at the ordinary real exponent $\frac{3}{2}$). Let $\mathbb{T}_P$ denote a formal oscillatory symbol with $\partial \mathbb{T}_P = i(\partial P)\mathbb{T}_P$ in the sense of Theorem 21.1; it is a new differential element and never a surcomplex number. Seek a solution $a \mathbb{T}_P$ with

$$a = x^{-1/4} \sum_{n \geq 0} c_n x^{-3n/2}, \quad c_0 = 1. \quad (17.14)$$

Substituting into equation (17.13) gives

$$\partial^2 a + 2ix^{1/2}\partial a + \frac{i}{2x^{1/2}} a = 0, \quad (17.15)$$

and with $\alpha_n = -\frac{1}{4} - \frac{3}{2}n$ coefficient comparison yields

$$c_{n+1} = \frac{\alpha_n(\alpha_n - 1)}{3i(n+1)} c_n = -i \frac{(6n+1)(6n+5)}{48(n+1)} c_n, \quad c_n = (-i)^n \left(\frac{3}{4}\right)^n \frac{(1/6)_n (5/6)_n}{n!}. \quad (17.16)$$

The exponents in equation (17.14) decrease strictly to $-\infty$, so the displayed expression is a legitimate surcomplex Hahn sum however fast the ordinary coefficients grow, and its termwise differentiation is justified by (BM3). This is the same phenomenon as the factorially divergent solutions of Theorem 18.13, and it carries the same disclaimer (Warning 18.14): a strictly descending support is a summability certificate, not convergence in the fine topology on $\mathbf{No}[i]$.

Put $b = \partial a + ix^{1/2}a$ and $G = \begin{pmatrix} a & \bar{a} \\ b & \bar{b} \end{pmatrix}$. The two columns satisfy $\partial v = A_x v \mp i(\partial P)v$ with $A_x = \begin{pmatrix} 0 & 1 \\ -x & 0 \end{pmatrix}$, so $\partial G = A_x G - G \text{diag}(i\partial P, -i\partial P)$. The determinant $\text{Wr} = a\bar{b} - \bar{a}b$ has zero derivative, hence lies in $\mathbb{C}$, and its leading term is $-2i$ because $a \sim x^{-1/4}$ and $b \sim ix^{1/4}$; so $\text{Wr} = -2i$ exactly, G is invertible, and

$$\text{Ph}(A_x) = \left\{ \frac{2}{3}\omega^{3/2}, -\frac{2}{3}\omega^{3/2} \right\}. \quad (17.17)$$

By Corollary 15.11 equation (17.13) has no nonzero solution in $\mathbf{No}[i]$, and $\partial^2 y + \omega y = f$ has a unique solution in $\mathbf{No}[i]$ for every f . As always in this report that is a statement about surcomplex *number* solutions under ∂ ; it does not deny the ordinary Airy functions of a complex coordinate (Warning 8.9).

The unique positive solution of $\partial^2 \rho + \omega \rho = \rho^{-3}$ is

$$\rho = \sqrt{a\bar{a}}. \quad (17.18)$$

Indeed, with $v = \begin{pmatrix} a \\ b \end{pmatrix}$ the real matrix $C = \operatorname{Re}(vv^*)$ satisfies $\partial C = A_x C + C A_x^\top$ — the scalar imaginary phase cancels — and $\det C = (\operatorname{Im}(a\bar{b}))^2 = 1$ since $\operatorname{Wr} = -2i$; so $\mathcal{M} = C^{-1}$ is a positive determinant-one horizontal metric and the bijection of Theorem 17.6 gives equation (17.18), with uniqueness because the phases equation (17.17) are nonzero. Its expansion begins

$$\rho = \omega^{-1/4} \left(1 - \frac{5}{64}\omega^{-3} + \frac{2285}{8192}\omega^{-6} - \frac{1690275}{524288}\omega^{-9} + \frac{10371973875}{134217728}\omega^{-12} + \dots \right), \quad (17.19)$$

these being coefficients of an exact Hahn series and not estimates for a convergent ordinary power series. The normalized primitive of ρ^{-2} is $B = \frac{2}{3}\omega^{3/2} - \frac{5}{48}\omega^{-3/2} + \dots$, whose purely infinite part is exactly the positive phase in equation (17.17); the remaining correction is infinitesimal and is absorbed by the rotation $R(\eta)$ of Theorem 17.7.

For a truncation $a_N = \sum_{n \leq N} c_n x^{\alpha_n}$ the recurrence cancels everything but the last second derivative, giving the exact residual

$$\partial^2 a_N + 2ix^{1/2}\partial a_N + \frac{i}{2x^{1/2}}a_N = \alpha_N(\alpha_N - 1)c_N x^{\alpha_N - 2}, \quad (17.20)$$

which the delivered suite checks at $N = 16$ (appendix D). As Warning 49.3 requires, the residual records the *type* of the omitted information: it is a formal remainder at a stated order, and the infinite-support statement is proved by the recurrence together with strict descent of exponents, not by the finite check.

17.5 First integrals from phase resonance

Let $y_1, \dots, y_n$ be algebraically independent formal variables and extend ∂ to $\mathbf{No}[i][y_1, \dots, y_n]$ by

$$\mathcal{D}_A f = \partial_{\text{coeff}} f + \sum_j (A y)_j \frac{\partial f}{\partial y_j},$$

where ∂_{coeff} differentiates coefficients only. A polynomial or rational *first integral* is an element killed by $\mathcal{D}_A$. This definition presupposes no nonzero solution in $\mathbf{No}[i]^n$. In phase coordinates $z = G^{-1}y$ one has $\mathcal{D}_A z_j = i(\partial P_j)z_j$.

Put

$$\Pi = \sum_{j=1}^n \mathbb{Z}P_j \subseteq \mathcal{P}, \quad \Lambda = \left\{ m \in \mathbb{Z}^n : \sum_j m_j P_j = 0 \right\}, \quad \Lambda^+ = \Lambda \cap \mathbb{N}^n,$$

and $r = \dim_{\mathbb{Q}} \operatorname{span}_{\mathbb{Q}} \{P_1, \dots, P_n\}$. This Λ is the relation lattice of equation (21.6) for the diagonal system $\operatorname{Ph}(A)$; it is saturated by Lemma 21.11, so $\operatorname{rank} \Lambda = n - r$ and Π is free abelian of rank r .

Theorem 17.8 (The resonance algebra). *In phase coordinates,*

$$\ker \mathcal{D}_A|_{\mathbf{No}[i][z]} = \mathbb{C}[z^m : m \in \Lambda^+], \quad \ker \mathcal{D}_A|_{\mathbf{No}[i](z)} = \mathbb{C}(z^m : m \in \Lambda). \quad (17.21)$$

The polynomial algebra is finitely generated, and the rational field is purely transcendental over $\mathbb{C}$ of transcendence degree $n - r$.

Proof. For $f = \sum_{\alpha} c_{\alpha} z^{\alpha}$ the derivative vanishes exactly when $\partial c_{\alpha} + i(\partial \sum_j \alpha_j P_j) c_{\alpha} = 0$ for each α , and Lemma 15.3 makes a nonzero coefficient require $\alpha \in \Lambda^+$ and $c_{\alpha} \in \mathbb{C}$.

For the rational statement work in the Laurent ring and write f/g with coprime nonzero Laurent polynomials. A zero derivative gives $f \mid \mathcal{D}_A f$ and $g \mid \mathcal{D}_A g$; since $\mathcal{D}_A$ enlarges no Laurent support and the highest and lowest degrees in each variable are additive under multiplication, one gets $\mathcal{D}_A f = hf$ and $\mathcal{D}_A g = hg$ for a common $h \in \mathbf{No}[i]$, the case $h = 0$ included. Comparing two coefficients of f shows their ratio has logarithmic derivative $i\partial(\beta \cdot P - \alpha \cdot P)$, so by Lemma 15.3 their phases agree and the ratio is constant; thus all monomials of f share one phase, and after factoring out one coefficient and one monomial the rest has constant coefficients and exponents in Λ . The same holds for g , and equality of the two values of h forces the factored phases to agree and the factored coefficient ratio to be constant. Hence f/g is a rational function of monomials with exponents in Λ ; the reverse inclusion is immediate.

For finite generation, call $0 \neq u \in \Lambda^+$ indecomposable when it is not a sum of two nonzero members of Λ^+ . Two distinct indecomposables are incomparable in the coordinatewise order on $\mathbb{N}^n$, since $u < v$ would give $v - u \in \Lambda^+ \setminus \{0\}$. To see that such an antichain is finite, take any infinite sequence of distinct members of $\mathbb{N}^n$. Its first coordinates admit an infinite nondecreasing subsequence: either one value occurs infinitely often, or every tail is unbounded and an increasing subsequence can be chosen recursively. Repeat this selection for each of the remaining finitely many coordinates. The result contains two coordinatewise comparable members, a contradiction. This is Dickson's finite-antichain lemma. Hence there are finitely many indecomposables, and induction on total degree writes every member of Λ^+ as a sum of them. Finally a $\mathbb{Z}$ -basis of Λ gives $n - r$ Laurent monomials generating the rational field, algebraically independent because distinct integer combinations of basis exponents give distinct monomials. No analytic finiteness principle is used. $\square$

Example 17.9 (Complete reducibility does not make the invariant algebra a polynomial ring). Let $x = \omega$ and

$$G = \begin{pmatrix} 1 & x^{-1} & 0 \\ 0 & 1 & x^{-2} \\ 0 & 0 & 1 \end{pmatrix}, \quad D = \text{diag}(i, i, -2i), \quad A = (\partial G)G^{-1} + GDG^{-1},$$

so that

$$A = \begin{pmatrix} i & -x^{-2} & x^{-4} \\ 0 & i & (-3ix - 2)x^{-3} \\ 0 & 0 & -2i \end{pmatrix} \quad (17.22)$$

and $\text{Ph}(A) = \{\omega, \omega, -2\omega\}$, so by Corollary 15.11 the system has no nonzero solution in $\mathbf{No}[i]^3$. Its formal first integrals are nevertheless nontrivial: with $z = G^{-1}y$ the resonance condition is $\alpha_1 + \alpha_2 = 2\alpha_3$ and

$$\ker \mathcal{D}_A|_{\mathbf{No}[i][z]} \cong \mathbb{C}[\mathcal{U}, \mathcal{V}, \mathcal{W}]/(\mathcal{U}\mathcal{W} - \mathcal{V}^2), \quad \mathcal{U} = z_1^2 z_3, \quad \mathcal{V} = z_1 z_2 z_3, \quad \mathcal{W} = z_2^2 z_3, \quad (17.23)$$

the rational invariant field being $\mathbb{C}(z_1/z_2, z_2^2 z_3)$ of transcendence degree two. Here $r = 1$, matching Example 21.13(i) in kind: ω and -2ω are $\mathbb{Q}$ -dependent. The three first-integral identities are checked in the *original* coupled coordinates by the delivered suite.

17.6 The phase lattice as a differential Galois torus

Theorem 21.12 classifies the Picard–Vessiot extension of a *diagonal* system by its integer relation lattice. Theorem 15.9 reduces every finite system to a diagonal one, so that theorem now applies to

all of them. The point of this subsection is only to record what the reduction gives, and to add the two ingredients the diagonal treatment did not need: an explicit group-algebra presentation in rank r , and the real form. Nothing here is new Galois theory, and the eighth manuscript says so of itself.

Corollary 17.10 (Every finite system has a phase torus). *Let $A \in \text{Mat}_n(\mathbf{No}[i])$ with $\text{Ph}(A) = \{P_1, \dots, P_n\}$, phase lattice $\Pi = \sum_j \mathbb{Z}P_j$ of rank r , and relation lattice Λ as in equation (21.6). Then the number $d = n - \text{rank } \Lambda$ of independent phase generators of Convention 5.1(d) equals r , and Theorem 21.12 applies to the normal form of A : a Picard–Vessiot extension for $\partial y = Ay$ is obtained by adjoining r algebraically independent generators, and its differential Galois group is $\mathbf{G}_m(\mathbb{C})^r \cong \text{Hom}(\Pi, \mathbb{C}^\times)$.*

Proof. Λ is the kernel of the surjection $\mathbb{Z}^n \rightarrow \Pi$, $m \mapsto \sum m_j P_j$, so $\mathbb{Z}^n / \Lambda \cong \Pi$ is free of rank r by Lemma 21.11, giving $d = r$; and $\text{Hom}(\Pi, \mathbb{C}^\times) \cong (\mathbb{C}^\times)^r$ is the description of equation (21.7) in coordinates dual to a $\mathbb{Z}$ -basis of Π . The Picard–Vessiot statement is Theorem 21.12 applied to $i \text{diag}(\partial P_1, \dots, \partial P_n)$, transported back by the gauge of Theorem 15.9, which lies in $\text{GL}_n(\mathbf{No}[i])$ and therefore generates no new constants. $\square$

Proposition 17.11 (The group algebra of the phase lattice, and its real form). *Choose a $\mathbb{Z}$ -basis $Q_1, \dots, Q_r$ of Π and form the Laurent differential algebra*

$$\mathcal{L}_\Pi = \mathbf{No}[i][\mathbb{T}_1^{\pm 1}, \dots, \mathbb{T}_r^{\pm 1}], \quad \partial \mathbb{T}_k = i(\partial Q_k) \mathbb{T}_k, \quad (17.24)$$

writing $\mathbb{T}_Q = \prod_k \mathbb{T}_k^{m_k}$ for $Q = \sum_k m_k Q_k \in \Pi$, so that $\mathbb{T}_Q \mathbb{T}_{Q'} = \mathbb{T}_{Q+Q'}$ and $\partial \mathbb{T}_Q = i(\partial Q) \mathbb{T}_Q$. Then:

- (i) $\mathcal{L}_\Pi$ has no nonzero proper differential ideal;
- (ii) if G normalizes A then $Y = G \text{diag}(\mathbb{T}_{P_1}, \dots, \mathbb{T}_{P_n})$ is a fundamental matrix for $\partial y = Ay$, and its entries together with $(\det Y)^{-1}$ generate $\mathcal{L}_\Pi$ over $\mathbf{No}[i]$;
- (iii) for a real system Π is stable under negation, the involution $\overline{c \mathbb{T}_Q} = \bar{c} \mathbb{T}_{-Q}$ commutes with ∂ , the fixed combinations $(\mathbb{T}_Q + \mathbb{T}_{-Q})/2$ and $(\mathbb{T}_Q - \mathbb{T}_{-Q})/2i$ supply the entries of a real fundamental matrix in the normal form equation (17.4), and the real form of the torus is $(\text{Res}_{\mathbb{C}/\mathbb{R}}^1 \mathbf{G}_m)^r$, whose real points are $(S^1)^r$.

Every element of $\mathcal{L}_\Pi$ is a finite sum; no Hahn summability question arises, and no exponential e^{iP} is defined in $\mathbf{No}[i]$.

Proof. (i) Let $\mathfrak{a} \neq 0$ be a differential ideal and choose $f \in \mathfrak{a}$ with the fewest nonzero monomials; multiplying by a Laurent monomial and a nonzero scalar, arrange $f = 1 + \sum_{Q \neq 0} c_Q \mathbb{T}_Q$. Differentiation kills the constant term and introduces no new monomials, so ∂f has fewer terms and must vanish; then $c_Q^\dagger = -i\partial Q$ for each remaining Q , impossible for $Q \neq 0$ by Theorem 8.4. Hence $f = 1$. This is the rank- r form of the rank-one statement proved in Theorem 21.2.

(ii) $\partial G = AG - iG \text{diag}(\partial P_j)$ gives $\partial Y = AY$ at once. Conversely the entries of $G^{-1}Y$ supply every $\mathbb{T}_{P_j}$, and $(\det Y)^{-1}$ supplies the inverse of their product, so multiplying by all but one supplies each $\mathbb{T}_{P_j}^{-1}$; these characters generate the group algebra of Π .

(iii) Stability under negation is the conjugation symmetry used in Theorem 17.4, and $\overline{\partial(c \mathbb{T}_Q)} = (\partial c + ic \partial Q) \mathbb{T}_Q = (\partial \bar{c} - i \bar{c} \partial Q) \mathbb{T}_{-Q} = \partial(\bar{c} \mathbb{T}_{-Q})$. The fixed combinations satisfy ∂ -relations of the rotation block $(\partial Q)\mathcal{R}$, by the computation of Proposition 21.9 applied with $a = i\partial Q$. An automorphism commuting with the involution has $\chi(Q)\overline{\chi(Q)} = 1$, which is the asserted real form. $\square$

Warning 17.12 (What the torus statement is about). Corollary 17.10 and Proposition 17.11 concern the very large base field $\mathbf{No}[i]$, with the factorization and surjectivity properties of Theorem 15.6. They say nothing comparable for the same printed equation over a small rational-function base: in particular equation (17.17) is *not* a claim that the ordinary Airy differential Galois group over $\mathbb{C}(x)$ is a torus. Enlarging the base to $\mathbf{No}[i]$ has already supplied the nonoscillatory gauge entries. Nor is the real form of Proposition 17.11(iii) a construction of trigonometric functions on surreals; it is an algebraic-group statement, and Theorem 21.10 records what such an extension costs. Finally, when $\Pi \neq 0$ no differential embedding $\mathcal{L}_\Pi \rightarrow \mathbf{No}[i]$ over $\mathbf{No}[i]$ exists, by Corollary 21.8: the extension represents the obstruction, it does not remove it.

17.7 What this part does not settle

Warning 17.13 (The boundaries of the matrix theory). Part III classifies systems $\partial y = Ay$ for A an $n \times n$ matrix over $\mathbf{No}[i]$ with n an ordinary finite integer, and computes the invariant from algebraic eigenvalues only when $A = iH$ with H Hermitian. It does not classify non-Hermitian coefficients by their spectra — Example 16.10 refutes that — does not treat infinite matrices, does not supply an algorithm for the phase map or the normalizing gauge, and does not settle which nontriangular differential modules admit a filtration whose rank-one factors carry computable obstructions, which needs genuine module-theoretic information rather than the eigenvalues of an instantaneous coefficient matrix. Every ordinary finite rank over $\mathbf{No}[i]$ is covered; classifications over smaller prescribed bases and in infinite dimension are outside this theorem. All statements are for the one fixed ∂ of Theorem 3.1 and for no other derivation, no class-function derivative and none of the coordinate operators of Table 1. The consolidated list is Section 51.9. (Added with member 12: for one class of coefficients the filtration question and the algorithm question are answered in part. Part V splits every power-Hahn regular-singular system over $K_{\mathbb{R}}[\log t]$ into rank-one factors whose obstructions are read off from the residual matrix, and for finite rational supports with algebraic coefficients computes its Hahn invariants by a finite procedure (Remark 28.6). Everything in this warning stands outside that class.)

Remark 17.14 (Proper classes, and a set-sized reading). $\mathbf{No}$ is a proper class, and Section 2.1 fixes the framing this report uses. Nothing in Part III makes a proper class an element of a set or needs a class-sized basis of $\mathcal{P}$: every phase lattice Π is finitely generated and every element of $\mathcal{L}_\Pi$ is a finite sum. There is also a set-sized reading of each finite-data argument, in the spirit of Theorem 20.8: starting from a set of given coefficients together with $\mathbb{R} \cup \{\omega\}$, form a countable chain of set-sized subfields of $\mathbf{No}$, adjoining at each stage the real algebraic roots, derivatives, real exponentials, purely infinite parts, primitives and finite angles needed by the previous stage, together with witnesses for all finite factorizations and first-order inhomogeneous equations of Theorem 15.6 with coefficients at that stage. These are a *set* of requests, each specified by a finite tuple; there need not be only finitely many requests. Witnesses can be chosen set-theoretically: for each request take the least ordinal birthday bound at which a witness tuple exists, then choose from the nonempty set of witness tuples at that bound. Replacement and set choice therefore produce a set of witnesses. The union is a real closed differential subfield $\mathbf{No}_0$ inheriting the H -field rule and the designated witnesses, whose complexification has constant field exactly $\mathbb{C}$; every finite tuple appears at some stage and its witnesses at a later one, so the elementary lemmas of Section 7 and all the finite matrix arguments hold internally. Proposition 17.11 may therefore be read as an ordinary set-sized Picard–Vessiot statement over such an $\mathbf{No}_0[i]$.

This is a size justification for the proofs and nothing more. It does *not* say that any fixed Hahn workspace already contains the required primitives or solutions — Example 18.2 shows one that does not — and it gives no effective presentation, no birthday bound and no minimal such subfield.

Part IV

Workspaces: Laurent, rational, Hahn, and set-sized

18 The Laurent parent and its resolvents

18.1 A differential subfield with a simple valuation

Throughout this section $t = \omega^{-1}$ and, unless stated otherwise, $\Gamma \subseteq \mathbb{R}$ is an additive subgroup with $1 \in \Gamma$, so that $\Gamma + 1 = \Gamma$. Put

$$K_\Gamma = \mathbb{C}((t^\Gamma)) \subseteq \mathbf{No}[i], \quad H_\Gamma = \mathbb{R}((t^\Gamma)) \subseteq \mathbf{No},$$

embedded by Hahn summation of Conway monomials, and write $K_0 = \mathbb{C}((t))$ for the case $\Gamma = \mathbb{Z}$, the ordinary formal Laurent series field with ordinary complex coefficients. For nonzero f let $v(f)$ be the least exponent with nonzero coefficient, with $v(0) = +\infty$.

Proposition 18.1 (The derivation on a real-exponent workspace). *K_Γ is ∂ -stable, its constants are exactly $\mathbb{C}$, and*

$$\partial \left(\sum_{r \in S} a_r t^r \right) = - \sum_{r \in S} r a_r t^{r+1}, \quad \text{so} \quad v(\partial f) \geq v(f) + 1, \quad (18.1)$$

and hence $v(\partial^n f) \geq v(f) + n$ for every $n \geq 0$. On K_0 this is exactly the operator

$$D := \partial|_{K_0} = -t^2 \frac{d}{dt}. \quad (18.2)$$

Furthermore

$$\partial K_\Gamma = \{f \in K_\Gamma : [t^1]f = 0\}, \quad (18.3)$$

and for $f = \sum a_r t^r$ with $a_1 = 0$ a primitive inside K_Γ is $\sum_{r \neq 1} \frac{a_r}{1-r} t^{r-1}$; for general $f \in K_\Gamma$ the normalized primitive in the ambient $\mathbf{No}[i]$ is

$$\mathcal{I}_0(f) = a_1 \log \omega + \sum_{r \neq 1} \frac{a_r}{1-r} t^{r-1}, \quad (18.4)$$

which is normalized because $\log \omega$ is purely infinite and so has zero constant coefficient.

Proof. The real-exponent rule $\partial t^r = -r t^{r+1}$ of equation (3.1) and strong additivity (BM3) give equation (18.1); translation of a well-ordered support by 1 preserves well ordering and distinct exponents stay distinct, and $\Gamma + 1 = \Gamma$ keeps the result in the field. A nonconstant term has a nonzero derivative at a unique shifted exponent, so the constants are the exponent-0 terms, that is $\mathbb{C}$. The only term that could contribute at exponent 1 has exponent 0, whose multiplier is zero, which proves necessity in equation (18.3); conversely the displayed primitive has a support that is a translate of a subset of $\text{supp } f$, hence well ordered and inside Γ , and termwise differentiation returns f . For equation (18.4) use $\partial \log \omega = t$. $\square$

Example 18.2 (A missing logarithmic primitive). $K_0 = \mathbb{C}((t))$ is a differential field, yet t has *no* primitive in it: by equation (18.1) the coefficient of t^1 in ∂f is always zero, since it could only arise from exponent 0, whose derivative coefficient vanishes. In the full surreal field $\partial \log \omega = t$. The same holds over $H_{\mathbb{Q}} = \mathbb{R}((t^{\mathbb{Q}}))$, which is ∂ -stable while t has no primitive in it, its ambient primitives being $\log \omega + c$ with $c \in \mathbb{R}$.

Warning 18.3 (Differentiation closure is not integration closure). So ∂ is *not* surjective on K_{Γ} , even though ∂ is surjective on $\mathbf{No}[i]$. This is a failure of closure of a chosen representation parent, not a failure of ambient surjectivity, and the whole-field surjectivity clause (BM6) must not be silently transported to a localized workspace. Algebraic localization, algebraic closure, derivative stability, and closure under primitives are four separate properties; see Section 20 for the counterexamples separating them. There is a formal differential-form reading of the obstruction: since $d\omega = -t^{-2} dt$, the coefficient of $t^{-1} dt$ in $f d\omega$ is $-a_1$. That is a formal Laurent/Hahn residue statement only; it is not an identification with a cluster residue or an actual-point contour residue from the repository's analytic reports.

Over the real form, the ideal $\mathcal{A}$ intersects the Laurent field exactly.

Corollary 18.4 (A valuation test, on real exponents only). *For real $f \in H_{\Gamma}$ with $\Gamma \subseteq \mathbb{R}$,*

$$f \in \mathcal{A} \iff f = 0 \text{ or } v(f) > 1,$$

where $\mathcal{A}$ is computed in the full surreal field. In particular $t^2 \mathbb{R}[[t]] = \mathcal{A} \cap \mathbb{R}((t))$ on the real Laurent field.

Proof. This is Theorem 10.1 restated through equation (18.4). If $v(f) > 1$ every exponent of the primitive is positive, so it is infinitesimal. If the least exponent r is below 1, the primitive has the nonzero leading infinite term $a_r t^{r-1}/(1-r)$, which dominates any logarithmic term and cannot be removed by a real constant. If the least exponent is exactly 1 the logarithmic coefficient is nonzero and all other primitive terms are infinitesimal, so the primitive is again infinite. A logarithmic term cannot cancel a Laurent principal part, and primitives differ only by real constants. $\square$

Warning 18.5 (Not a valuation test for arbitrary coefficients). Corollary 18.4 uses *real exponents relative to $t = \omega^{-1}$* . It is not a valuation-only classification of arbitrary surreal coefficients: Example 10.4 is exactly the information that this real-exponent workspace cannot see.

18.2 Resolvents as exact Hahn sums

Theorem 18.6 (First-order resolvent on a real-exponent workspace). *Let $\Gamma \subseteq \mathbb{R}$ be an additive subgroup with $1 \in \Gamma$, and let $\lambda \in \mathbb{C}^{\times}$. Then $\partial - \lambda$ is a bijection of K_{Γ} , with inverse the strongly summable operator series*

$$(\partial - \lambda)^{-1} f = - \sum_{n \geq 0} \lambda^{-n-1} \partial^n f, \tag{18.5}$$

and for the truncation $y_N = - \sum_{n=0}^N \lambda^{-n-1} \partial^n f$ the residual is exactly

$$(\partial - \lambda)y_N - f = -\lambda^{-(N+1)} \partial^{N+1} f. \tag{18.6}$$

Proof. For $f \neq 0$ put $S = \text{supp } f$ and $s_0 = \min S$. By equation (18.1), $\text{supp } \partial^n f \subseteq S + n$, and $S + \mathbb{N}$ is well ordered: a decreasing sequence in it would have integer parts bounded above by its first term minus s_0 , and an infinite subsequence with the same integer part would be a decreasing sequence in

S . At a fixed exponent γ , a contribution from $\partial^n f$ requires $\gamma - n \in S$, hence $n \leq \gamma - s_0$, and $\gamma - s_0$ is an *ordinary real number*, so only finitely many n occur. This proves strong summability. Strong additivity (BM3) allows ∂ to be applied termwise, and the terms cancel pairwise after reindexing, leaving f ; the finite identity equation (18.6) is the same cancellation truncated. For injectivity, if $y \neq 0$ then either $\partial y = 0$ or $v(\partial y) \geq v(y) + 1 > v(y)$ by equation (18.1), so the leading term of λy cannot be cancelled; hence $(\partial - \lambda)y \neq 0$. $\square$

Warning 18.7 (The real-exponent hypothesis is substantive). The step “ $\gamma - s_0$ is an ordinary real number, so only finitely many n occur” is where $\Gamma \subseteq \mathbb{R}$ is used. In a higher-rank value group a fixed increment 1 need not dominate all relevant scales, and no claim is made here that successive valuation improvements always converge. A support argument has not been replaced by a convergence claim, and Theorem 18.6 is not asserted for an arbitrary set-sized ordered Γ .

Proposition 18.8 (Polynomial Laurent inverse). *Let $P \in \mathbb{C}[T]$ with $P(0) \neq 0$ and formal reciprocal $1/P(T) = \sum_{n \geq 0} q_n T^n \in \mathbb{C}[[T]]$. For every $f \in K_0$ the family $(q_n \partial^n f)_{n \geq 0}$ is strongly summable, $P(\partial) : K_0 \rightarrow K_0$ is bijective, and*

$$P(\partial)^{-1} f = \sum_{n \geq 0} q_n \partial^n f. \quad (18.7)$$

If $Q_N(T) = \sum_{n=0}^N q_n T^n$ and $y_N = Q_N(\partial)f$, then $P(T)Q_N(T) - 1$ has order at least $N + 1$, so

$$v(P(\partial)y_N - f) \geq v(f) + N + 1. \quad (18.8)$$

Proof. Summability is the valuation bound of equation (18.1) exactly as in Theorem 18.6. Strong additivity permits applying each of the finitely many derivatives in $P(\partial)$ termwise; at every exponent the calculation is finite, so the formal identity $P(T) \sum q_n T^n = 1$ gives $P(\partial)y = f$. For injectivity, the leading term of $P(\partial)y$ is $P(0)$ times the leading term of y , every positive-order derivative having strictly larger valuation. Alternatively one may factor P over $\mathbb{C}$ and compose the first-order resolvents of Theorem 18.6 with multiplicities, which gives the same conclusion; an effective implementation then needs exact representations of the roots, or can derive a coefficient recurrence from P directly. $\square$

The hypothesis $P(0) \neq 0$ is essential, and its failure is exactly Example 18.2: when $P(0) = 0$ the factor ∂ carries the obstruction equation (18.3), which must not be handled by a division by zero inside a resolvent routine. Proposition 18.8 covers both $P(T) = T - i$ and $P(T) = T^2 + 1$. So the absence of a nonzero homogeneous oscillator coexists with unique solvability of *every* inhomogeneous oscillator equation whose forcing lies in K_0 — an explicit instance of Warning 12.2.

Remark 18.9 (Parent selection changes homogeneous solution spaces). For nonreal λ , $\partial - \lambda$ has trivial kernel in all of $\mathbf{No}[i]$ and Theorem 18.6’s uniqueness is uniqueness in $\mathbf{No}[i]$. For real nonzero λ the Laurent solution is unique *inside* K_Γ , while the full field also contains $ce^{\lambda\omega}$, which lies outside K_Γ . The same inverse formula holds in both cases. A theorem whose uniqueness statement changes when the ambient field is enlarged must carry its parent as part of its statement, and an algorithm must return that parent.

18.3 Factorially divergent exact solutions

Since $\partial^n t = (-1)^n n! t^{n+1}$, the resolvents produce exact Hahn series whose coefficients grow factorially. Three instances occur in the source manuscripts and all three are recorded here.

Example 18.10 $((\partial - i)y = t)$. Applying Theorem 18.6 with $\lambda = i$ and $f = t$, or equivalently $1/(s - i) = i \sum_{n \geq 0} (-i)^n s^n$ in equation (18.7), gives the unique solution in K_0

$$y = \sum_{n \geq 0} i^{n+1} n! t^{n+1} = it - t^2 - 2it^3 + 6t^4 + 24it^5 - 120t^6 - \dots, \quad (18.9)$$

which by Corollary 8.8 is the unique solution in all of $\mathbf{No}[i]$. Its truncation residual is equation (18.6); explicitly, writing y_N for the sum through $n = N$, $(\partial - i)y_N - t = -i^{-(N+1)}(-1)^{N+1}(N+1)!t^{N+2}$.

Example 18.11 $((\partial + 1)Y = t$, and a beyond-all-orders ambiguity). With $\lambda = -1$ and $f = t$,

$$Y_H = \sum_{n \geq 0} n! t^{n+1}, \quad (\partial + 1)Y_H = t, \quad (18.10)$$

with support $\{1, 2, 3, \dots\}$, each exponent occurring once; and for $Y_N = \sum_{n=0}^{N-1} n! t^{n+1}$ the exact residual is

$$(\partial + 1)Y_N - t = -N! t^{N+1}. \quad (18.11)$$

Proposition 18.12 (Uniqueness depends on the ambient field). *Equation (18.10) has exactly one solution in K_Γ , but all its solutions in $\mathbf{No}[i]$ are*

$$Y_H + C e^{-\omega}, \quad C \in \mathbb{C}. \quad (18.12)$$

Every nonzero correction $C e^{-\omega}$ is smaller in modulus than t^r for every ordinary real r , and lies outside K_Γ .

Proof. Workspace uniqueness is Theorem 18.6. In the full field the difference of two solutions satisfies $\partial Z = -Z$, so Corollary 8.8 gives $Z = C e^{-\omega}$. From $\log \omega \prec \omega$ one gets $-\omega < -r \log \omega$ for every real r after allowing any fixed real multiplicative coefficient; exponentiating gives $|C| e^{-\omega} \prec t^r$. A nonzero real-exponent Hahn series has a least real exponent with a nonzero ordinary leading coefficient, so it cannot be smaller than every real power of t . $\square$

The whole sequence of algebraic-order asymptotic coefficients therefore fails to distinguish the full-field solutions in equation (18.12). The exact Hahn workspace selects one solution by its allowed support, not by any limiting process. No Borel summation, no Stokes multiplier and no analytic continuation rule is asserted by this example; those would require additional data and separate theorems.

Theorem 18.13 (An exact factorially divergent forced oscillator). *The equation*

$$\partial^2 y + y = \omega^{-1} \quad (18.13)$$

has the unique solution in $\mathbf{No}[i]$

$$y = \sum_{n \geq 0} (-1)^n (2n)! \omega^{-(2n+1)}, \quad (18.14)$$

and for $N \geq 1$ its N -term truncation y_N satisfies exactly

$$y_N'' + y_N - \omega^{-1} = (-1)^{N-1} (2N)! \omega^{-(2N+1)}. \quad (18.15)$$

Proof. The formal inverse of $1 + T^2$ is $\sum_{n \geq 0} (-1)^n T^{2n}$; apply Proposition 18.8 to $f = t$, using $\partial^k t = (-1)^k k! t^{k+1}$. Existence can also be checked directly: $\partial^2 t^{2n+1} = (2n+1)(2n+2)t^{2n+3}$, so every positive odd exponent beyond the first cancels between $\partial^2 y$ and y ; the same cancellation for a finite truncation leaves exactly equation (18.15). The difference of two surcomplex solutions satisfies $h'' + h = 0$, hence is zero by Theorem 13.2, so uniqueness holds in the whole class field and not merely in K_0 . $\square$

Warning 18.14 (An exact value that is not a limit of its truncations). The ordinary scalar series $\sum (-1)^n (2n)! t^{2n+1}$ has radius of convergence zero: for nonzero ordinary complex t the ratio of successive term magnitudes is $(2n+2)(2n+1)|t|^2$, which grows without bound. That causes no difficulty for equation (18.14): its coefficients are ordinary complex numbers and its support is the well-ordered set $\{1, 3, 5, \dots\}$, which is all Hahn summation requires. Conversely the existence of the surreal value does not make the scalar series an ordinary holomorphic germ, nor is the value a fine-topology limit of its truncations: the tail $y - y_N$ has leading term $(-1)^N (2N)! \omega^{-(2N+1)}$, so

$$|y - y_N| > \omega^{-\omega} \quad \text{for every } N \geq 1,$$

and the errors never enter that fixed infinitesimal neighbourhood of zero. The residual identities are exact algebraic identities and scale certificates; they certify a formal remainder at a stated order, not convergence of the sequence (y_N) in the fine surreal topology. A numerical substitution such as $t = 10^{-6}$ would not verify a Hahn identity and is not used anywhere in this report. A closed formula for a coefficient stream is not a finite list of every coefficient.

Remark 18.15 (The coefficient recursion, and its exact guarantee). Writing $(\partial - \lambda)y = f$ coefficientwise in K_0 gives

$$-\lambda y_n - (n-1)y_{n-1} = f_n, \quad \text{that is} \quad y_n = -\lambda^{-1}(f_n + (n-1)y_{n-1}), \quad (18.16)$$

and below a known lower exponent bound all coefficients vanish. Each requested finite prefix costs one inversion of λ and a linear number of coefficient operations, and the recursion works for negative lower bounds. It does *not* make equality of arbitrary program-generated coefficient streams decidable.

19 The rational-exponent workspace

A particularly informative set-sized workspace is

$$\mathcal{H} := \mathbb{C}((t^{\mathbb{Q}})), \quad t = \omega^{-1}, \quad \mathcal{H}_{\mathbb{R}} := \mathbb{R}((t^{\mathbb{Q}})), \quad (19.1)$$

embedded in $\mathbf{No}[i]$ through normal forms. Every element has a well-ordered *rational* support: this is a *full* Hahn field, not merely the union of the finite-ramification Laurent-series fields, and in particular well-ordered supports need not have a common denominator. By Proposition 18.1 it is ∂ -stable with $\partial(\sum c_q t^q) = \sum (-q) c_q t^{q+1}$, so the restriction of the abstract derivation is completely explicit here. That explicitness does not mean the full Berarducci–Mantova construction has been implemented: the special rational monomial subgroup is doing essential work (Warning 2.2).

Remark 19.1 (A companion report that uses this workspace). The companion report *Tail-Spans and Differential Transcendence in Surreal and Surcomplex Hahn Fields* ([docs/surreal/tail-spans-and-differential-transcendence/](#), [29]) takes equation (19.1) and Proposition 18.1 as its starting point rather than reproving them, and works inside the real-algebraic subfield $\mathbb{A}_{\mathbb{R}}((t^{\mathbb{Q}})) \subseteq \mathcal{H}_{\mathbb{R}}$,

where the displayed formula still applies and keeps that subfield ∂ -stable, since it multiplies each coefficient by a rational number and shifts its exponent by one. Over the base $\mathbb{Q}((t^{\mathbb{Q}}))$ it produces a family of actual positive infinitesimal surreal numbers, of cardinality $2^{\aleph_0}$, whose ∂ -jets are all algebraically independent; the numbers are $\sum_{n \in A} \sqrt{p_n} \omega^{-n}$ for an explicit almost disjoint family of infinite sets $A \subseteq \mathbb{N}_{>0}$. That is a statement about differential transcendence over a *set-sized* Hahn base and does not bear on the image and primitive questions settled here, nor does it supply any derivative formula for a monomial outside the rational exponents (Warning 2.2 still applies).

Theorem 19.2 (Exact derivative image: one resonant coefficient). *For $f = \sum_q c_q t^q \in \mathcal{H}$,*

$$f \in \partial(\mathcal{H}) \iff [t^1]f = 0. \quad (19.2)$$

If $c_1 = 0$ the normalized primitive inside $\mathcal{H}$ is $-\sum_{q \neq 1} \frac{c_q}{q-1} t^{q-1}$; for arbitrary $f \in \mathcal{H}$ the normalized primitive in the ambient $\mathbf{No}[i]$ is $c_1 \log \omega - \sum_{q \neq 1} \frac{c_q}{q-1} t^{q-1}$. Hence $\mathcal{H}$ is closed under differentiation but not under antidifferentiation, and $\log \omega \notin \mathcal{H}$.

Proof. This is Proposition 18.1 with $\Gamma = \mathbb{Q}$. The formula remains valid for infinite rational supports: translation of an already well-ordered support is the precise support certificate, and no denominator bound or discrete increasing sequence of exponents is needed. No appeal to topology, cardinality or computational limitation is involved; the obstruction is the single resonant coefficient $[t]f$. $\square$

Theorem 19.3 (Exact logarithmic-derivative image in $\mathcal{H}$).

$$\{y^\dagger : y \in \mathcal{H}^\times\} = \{-rt + g : r \in \mathbb{Q}, g = 0 \text{ or } v(g) > 1\}. \quad (19.3)$$

For $q = -rt + g$ in this class, put

$$k = -\sum_{\alpha > 1} \frac{[t^\alpha]g}{\alpha - 1} t^{\alpha-1}, \quad (19.4)$$

which has positive valuation (or is 0) and satisfies $\partial k = g$. Then all solutions in $\mathcal{H}$ of $\partial y = qy$ are

$$y = c t^r E(k), \quad c \in \mathbb{C}. \quad (19.5)$$

Proof. A nonzero $y \in \mathcal{H}$ has a unique leading-term factorization $y = c t^r (1 + h)$ with $c \in \mathbb{C}^\times$, $r \in \mathbb{Q}$ and $v(h) > 0$ (possibly $h = 0$), and then

$$y^\dagger = -rt + \frac{\partial h}{1+h} = -rt + \partial L(1+h)$$

by Lemma 6.1. Since $L(1+h)$ has positive valuation or is zero, every nonzero term of its derivative has exponent strictly greater than 1 by equation (18.1). This gives containment in the right-hand class. Conversely, for $v(g) > 1$ the series equation (19.4) has positive valuation and $\partial k = g$, so $E(k) \in \mathcal{H}$ is defined by a positive-support Hahn sum and $(t^r E(k))^\dagger = -rt + g$. The ratio of two nonzero solutions lies in the constant field $\mathbb{C}$. $\square$

Proposition 19.4 (Workspace versus ambient). *Let $q = a + ib \in \mathcal{H}$ with $a, b \in \mathcal{H}_{\mathbb{R}}$. A nonzero solution of $\partial y = qy$ exists in the ambient $\mathbf{No}[i]$ if and only if*

$$b = 0 \quad \text{or} \quad v(b) > 1. \quad (19.6)$$

Proof. This is Corollary 18.4 with $\Gamma = \mathbb{Q}$, equivalently Theorem 10.1; the growth comparison $\log \omega \ll \omega^\delta$ for positive rational δ is what makes a nonzero $\log \omega$ term infinite. $\square$

Table 5: Where a solution lives. $\mathcal{H} = \mathbb{C}((t^{\mathbb{Q}}))$, $t = \omega^{-1}$.

Coefficient q in $\partial y = qy$	Location	Explanation, or one solution
1	$\mathbf{No}[i]$, not $\mathcal{H}$	e^{ω} ; the exponent-0 coefficient violates equation (19.3)
$\sqrt{2}t$	$\mathbf{No}[i]$, not $\mathcal{H}$	$\exp(\sqrt{2}\log \omega)$; the exponent-1 coefficient is irrational
it	Neither	The accumulated phase is $\log \omega$
it^2	$\mathcal{H}$	$E(-it)$
$-\frac{3}{2}t + (1 + 2i)t^2$	$\mathcal{H}$	$t^{3/2}E(-(1 + 2i)t)$

So there are *three* distinct outcomes, not a binary solvable/unsolvable flag: a solution in the declared workspace, a solution only in the ambient field, and no nonzero solution at all.

The first two rows mean that no *nonzero* solution belongs to $\mathcal{H}$; multiplication by an ordinary complex constant cannot remove either obstruction, since it does not change a logarithmic derivative. Note that the second row is the workspace failure announced in Remark 8.6: the real coefficient never obstructs *ambiently*, but A or e^A may leave a declared workspace.

Remark 19.5 (A resonant normal form). Theorem 19.3 has a normal-form reading. A scalar equation over $\mathcal{H}$ has a fundamental solution in the same field only when its logarithmic coefficient splits into a rational monomial contribution $-rt$ and a higher-valuation perturbation: the monomial supplies the exponent-1 residue, and the infinitesimal exponential absorbs the perturbation. Lower exponents require larger-scale real exponentials, and lower *imaginary* exponents face the stronger ambient phase obstruction. The condition at exponent 1 is much more restrictive for logarithmic than for additive integration: a logarithmic derivative in $\mathcal{H}$ has no exponents below 1, its exponent-1 coefficient is a real *rational* number, and its higher exponents may carry arbitrary ordinary complex coefficients. This resembles a one-dimensional normal-form calculation, but every assertion is algebraic and exact: there is no asymptotic error term in equation (19.5), and the rational exponent r is not an approximation to a more general exponent.

20 Differentially stable workspaces for set-sized data

The repository's localization principle is algebraic: a set of surcomplex values fits inside a set-sized Hahn workspace [26]. That is a *size* statement. It does not say that the workspace contains the derivatives, primitives, exponentials or finite phases of its members. Two genuinely different refinements are available, and both survive the merge: a *full Hahn field* indexed by a closed exponent group, and a set-sized *subfield* closed under a specified list of finitary operations. They prove different things and neither implies the other.

20.1 Full Hahn fields: the monomial criterion and the hull

Definition 20.1. A set-sized additive subgroup $\Gamma \subseteq \mathbf{No}$ is *differentially closed on monomials* if $\text{supp } \partial(t^\gamma) \subseteq \Gamma$ for every $\gamma \in \Gamma$, the support being taken in the t -exponent convention of equation (2.2).

Lemma 20.2 (Monomial criterion). *For a set-sized additive group $\Gamma \subseteq \mathbf{No}$,*

$$\partial K_\Gamma \subseteq K_\Gamma \iff \Gamma \text{ is differentially closed on monomials,}$$

and the same equivalence holds for $H_\Gamma = \mathbb{R}((t^\Gamma))$.

Proof. Necessity: apply the closure condition to each t^γ . Sufficiency: for $x = \sum_{\gamma \in S} c_\gamma t^\gamma$ with $S \subseteq \Gamma$, strong additivity (BM3) gives the summable family $(c_\gamma \partial t^\gamma)_{\gamma \in S}$ and $\partial x = \sum_{\gamma \in S} c_\gamma \partial t^\gamma$. Every term has support in Γ , so the sum does too; ordinary real and complex coefficients are constants of ∂ . $\square$

Theorem 20.3 (Differential Hahn hull). *Let $\Gamma_0 \subseteq \mathbf{No}$ be a set-sized additive subgroup. Define*

$$\Gamma_{n+1} = \text{span}_{\mathbb{Q}} \left(\Gamma_n \cup \bigcup_{\gamma \in \Gamma_n} \text{supp } \partial(t^\gamma) \right), \quad \Gamma_* = \bigcup_{n < \omega} \Gamma_n. \quad (20.1)$$

Then Γ_ is a set-sized divisible additive group, K_{Γ_*} is algebraically closed, and $\partial K_{\Gamma_*} \subseteq K_{\Gamma_*}$. It is the least such divisible monomial-closed exponent group containing Γ_0 . Consequently every set $A \subseteq \mathbf{No}[i]$ lies in a set-sized algebraically closed Hahn field stable under ∂ : take Γ_0 to be the group generated by the union of the normal-form supports of the members of A .*

Proof. Each $\partial(t^\gamma)$ is an individual surreal and has a set support, so by replacement and union the displayed union is a set, as is its rational span. The stages increase, so their countable union is a set-sized additive group; every member lies in a stage whose successor contains all its rational multiples, so the union is divisible. If $\gamma \in \Gamma_*$ it lies in some Γ_n , and then $\text{supp } \partial(t^\gamma) \subseteq \Gamma_{n+1}$. Now take any $f = \sum_{\gamma \in S} c_\gamma t^\gamma \in K_{\Gamma_*}$: strong additivity gives the summable family $(c_\gamma \partial t^\gamma)_{\gamma \in S}$, each member has support in Γ_* , so their sum has support in Γ_* and equals ∂f . Divisibility makes H_{Γ_*} real closed and K_{Γ_*} algebraically closed; this is the standard Hahn field fact, an *algebraic* statement, not a differential one. Minimality: any divisible monomial-closed group containing Γ_0 contains each stage by induction, hence their union. $\square$

Warning 20.4 (The proof must not use a union of fields). The last step of the proof appeals to strong additivity applied to every series over Γ_* *simultaneously*. It does not use

$$K_{\bigcup_n \Gamma_n} = \bigcup_n K_{\Gamma_n},$$

which is generally *false*: one Hahn series can use exponents appearing at arbitrarily late stages. The correct route is the one displayed, through the monomial supports.

Warning 20.5 (Three scope limits on Theorem 20.3). (i) Minimality is only among *full Hahn fields indexed by divisible exponent groups*, not among all differential subfields. (ii) The hull group Γ_* need not be countable: it is the number of closure *stages* that is countable, and the countably many stages do not constitute a terminating algorithm for every input representation. (iii) The theorem is a semantic existence construction, not an effective algorithm for computing $\partial(t^\gamma)$ at arbitrary surreal exponents, and it supplies no optimal birthday or cardinal bound. It is not primitive closure, and it asserts no closure under real exponentiation, arbitrary logarithms, or solutions of all first-order linear equations; Example 18.2 is the witness.

Proposition 20.6 (The real-exponent case, exactly). *Let $0 \neq \Gamma \subseteq \mathbb{R}$ be an additive subgroup. Then K_Γ is ∂ -stable if and only if $1 \in \Gamma$; and if Γ is divisible its differential Hahn hull is $\Gamma + \mathbb{Q}$. The group $\{0\}$ gives the constant field and is already stable.*

Proof. For nonzero $\gamma \in \Gamma$, equation (3.1) gives $\partial t^\gamma = -\gamma t^{\gamma+1}$, so closure requires $\gamma + 1 \in \Gamma$, hence $1 \in \Gamma$; conversely containing 1 provides every exponent $\gamma + 1$. The hull assertion is Theorem 20.3. $\square$

Example 20.7 (Divisible, algebraically closed, and not derivative-stable). $\Gamma = \mathbb{Q}\sqrt{2}$ is divisible, so K_Γ is algebraically closed and contains $t^{\sqrt{2}}$; but

$$\partial t^{\sqrt{2}} = -\sqrt{2} t^{\sqrt{2}+1} \notin K_{\mathbb{Q}\sqrt{2}}, \quad \text{since } \sqrt{2} + 1 \notin \mathbb{Q}\sqrt{2}.$$

Its hull is $\mathbb{Q} + \mathbb{Q}\sqrt{2}$. So a divisible Hahn value group need not define a derivative-stable workspace, and an arbitrary algebraically closed Hahn workspace does not automatically support the ambient derivation. Together with Example 18.2 this separates *four* requirements: input containment, algebraic closure, derivative stability, and closure under primitives. This example does not dispute the repository's theorem that any set of surcomplex inputs fits inside *some* set-sized Hahn workspace [26]; it shows that the workspace must be chosen with the derivation in view. Closure properties of surreal subfields under exponential, logarithmic, derivative and antiderivative operations are studied much more finely by Bournez and Guilmant [9]; that literature addresses constraints not imposed by the elementary constructions here, and no theorem of it is used to justify Theorem 20.3.

20.2 Set-sized subfields closed under a named operation list

Theorem 20.8 (Closure localization). *Let $S \subseteq \mathbf{No}[i]$ be a set. There is a set-sized real closed subfield $F \subseteq \mathbf{No}$ with $\mathbb{R} \cup \{\omega\} \subseteq F$ and $S \subseteq F[i]$ such that F is closed under*

$$\partial, \quad \mathcal{I}_0, \quad \text{pp}, \text{ct}, \text{fin}, \quad \exp, \quad \log \text{ on positive inputs},$$

and under the real and imaginary components of E and L on their stated domains — equivalently, under finite-angle sine and cosine on $F \cap \mathcal{O}$. One may arrange

$$|F| \leq \kappa := \max\{|S|, |\mathbb{R}|, \aleph_0\}.$$

Then $L = F[i]$ is algebraically closed, is a differential subfield of $\mathbf{No}[i]$ with $\partial L = L$ and constants exactly $\mathbb{C}$, and is closed under componentwise normalized primitives; $\ker(\partial|_F) = \mathbb{R}$ and $\partial F = F$. Moreover the criterion of Theorem 8.4 holds internally: for $a, b \in F$ a nonzero solution of $\partial y = (a + ib)y$ exists in L if and only if b has a finite accumulated phase in $\mathbf{No}$, and then $\mathfrak{o}$ takes values in $F \cap \mathcal{P}$ and is onto it, so the gauge classification and the exact sequence of Section 9 read as ordinary set-level results for this workspace.

One may in addition require that $f(h) \in L$ for every $f \in \mathbb{C}[[X]]$ with ordinary complex coefficients and every $h \in L \cap \mathfrak{m}_{\mathbb{C}}$, at the same cardinality bound, using $|\mathbb{C}|^{\aleph_0} = |\mathbb{R}|$ to count the formal series.

Proof. Let F_0 be the real closure inside $\mathbf{No}$ of the field generated by $\mathbb{R}$, ω and the real and imaginary parts of the members of S . Given F_n , adjoin the images of all eligible finite tuples under the listed class functions — with their declared domains: positive arguments for $\log$, finite arguments for the phase components, infinitesimal arguments for E, L and for the optional formal-series values — take the field generated by those images, and let F_{n+1} be its real closure inside $\mathbf{No}$, that is, adjoin all real roots in $\mathbf{No}$ of nonzero polynomials over the current field.

Each stage is a set: images under fixed definable class functions are sets by replacement, field generation uses finite expressions, and a set of nonzero polynomials of finite degree has a set of roots, finitely many each. Each stage has size at most κ , since the operations have finite arity; for the optional clause there are $|\mathbb{C}|^{\aleph_0} = |\mathbb{R}|$ formal series, so at most κ elements are adjoined per stage and a real algebraic closure of a field of that cardinality has the same cardinality.

Put $F = \bigcup_{n < \omega} F_n$, a set of size at most κ . Any input to one of the operations lies in a stage and its image in the next; any finite tuple of polynomial coefficients lies in a single stage and all its

real algebraic roots are adjoined at the next. So F has the advertised closures and is real closed, and $L = F[i]$ is algebraically closed. Differential closure holds componentwise, $\partial F = F$ because $\mathcal{I}_0 F \subseteq F$, and the constant fields are $F \cap \mathbb{R} = \mathbb{R}$ and $L \cap \mathbb{C} = \mathbb{C}$ because $\mathbb{R} \subseteq F$.

For the internal criterion, necessity is inherited from the ambient statement. For sufficiency, if $b \in F$ has a finite accumulated phase then $\mathcal{I}_0(b) \in F \cap \mathfrak{m}$ by Theorem 7.5(ii); for $a \in F$ we have $\exp(\mathcal{I}_0 a) \in F$, and $E(i\mathcal{I}_0 b) \in L$ because its real and imaginary coordinates are finite-angle cosine and sine of an element of $F \cap \mathcal{O}$. So the witness equation (8.2) lies in L . Whenever a nonzero solution exists, *every* solution lies in L as well, since the multipliers are in $\mathbb{C} \subseteq L$; the same applies to the solution spaces of Theorem 13.2 and Corollary 13.5. Finally pp and $\mathcal{I}_0$ preserve F , so $\mathfrak{o}$ maps F into $F \cap \mathcal{P}$, and $\mathfrak{o}(\partial P) = P$ for $P \in F \cap \mathcal{P}$ gives surjectivity. $\square$

Warning 20.9 (What Theorem 20.8 does not give). It adjoins values of *specified finitary operations*, some of which are defined in the ambient field by Hahn sums. It does *not* assert closure under all set-indexed summable families: an arbitrary-support closure is a different and strictly stronger demand, and F need not contain every Hahn series over its own monomials. The optional clause adjoins formal-series values with *constant* coefficients at a *single* infinitesimal argument; that is a specified set of operations, not Hahn completeness. A particular coherent family and its requested evaluations can be included as set-sized data, but arbitrary future families are not automatically covered. The theorem is not proved to produce a full Hahn field of the form $\mathbb{C}((t^\Gamma))$ — Theorem 20.3 is the construction that does that, and the two are not interchangeable — and it provides no finite data representation, no algorithm for the adjoined maps, no birthday bound, no computable enumeration, and no termination claim for symbolic integration. Normalization of primitives need not be computable from a supplied symbolic representation. Structured fields with simultaneous exponential, logarithmic, differential and integration closure are studied much more finely in [9].

Remark 20.10 (A weaker but sometimes sufficient variant). If a full Hahn field is not needed, a lighter construction suffices: start from the field generated by the data and the ordinary constants; at each finite stage adjoin the values of a specified finitary operation set — for instance ∂ , $\mathcal{I}_0$, conjugation, and real $\exp$ and $\log$ where defined — and take the generated field; the union over $\mathbb{N}$ is a set-sized field stable under each listed operation, because any finite tuple from the union occurs at a common stage and its operation values at the next. This variant is *not* proved to be a full Hahn field and need not contain every summable family of its own members. The distinction is useful for formalization: a small operational workspace can support a particular proof even when it is not closed under all possible Hahn sums.

Remark 20.11 (Localization cannot supply an oscillator). Adjoining a solution of $T' = iT$ cannot be accomplished by passing to a larger workspace of either kind *inside* $\mathbf{No}[i]$: the proof of Corollary 8.8 already applies to the full class. The next section changes the differential field itself rather than its set-sized container.

21 Where an oscillatory solution can live

We use Picard–Vessiot terminology in the standard sense of [13, 14]: a solution-generated differential extension with *no new constants*. Five explicit constructions occur across the source manuscripts, over four base fields (with two constructions over $\mathbf{No}[i]$). They are genuinely different theorems and all are recorded; none of them appeals to an unqualified existence theorem for differential extensions of a proper-class field. Other parts compute further differential Galois groups: the phase-lattice torus of an arbitrary finite system (Corollary 17.10), the additive group of one logarithm (Theorem 26.3),

and, over a set-sized Hahn base at every limit stage of the log-atomic tower, the Borel subgroup of $\mathrm{SL}_2(\mathbb{C})$ (Theorem 41.2), the only non-commutative one in this report.

21.1 The missing-solution extension over $\mathbf{No}[i]$

Theorem 21.1 (Adjoining one missing rank-one solution). *Let $a \in \mathbf{No}[i]$ with $\Phi(a) \neq 0$. Introduce a transcendental T and extend ∂ to the rational function field $\mathbf{No}[i](\mathsf{T})$ by*

$$\partial \mathsf{T} = a\mathsf{T}, \quad (21.1)$$

using the product rule on polynomials and then the quotient rule. Then the constant field of $\mathbf{No}[i](\mathsf{T})$ is exactly $\mathbb{C}$; $\mathbf{No}[i](\mathsf{T})$ is a Picard–Vessiot extension for $\partial y = ay$; its differential automorphisms over $\mathbf{No}[i]$ are exactly $\mathsf{T} \mapsto c\mathsf{T}$ with $c \in \mathbb{C}^\times$; and its differential Galois group is $\mathbf{G}_m(\mathbb{C})$. If $\Phi(a) = 0$ the corresponding extension is trivial.

Proof. The product and quotient rules extend the derivation uniquely from $\mathbf{No}[i][\mathsf{T}]$ to $\mathbf{No}[i](\mathsf{T})$. Let $R = P/Q \neq 0$ have derivative zero, with nonzero coprime $P, Q \in \mathbf{No}[i][\mathsf{T}]$. Then $Q\partial P = P\partial Q$, and coprimality gives $P \mid \partial P$ and $Q \mid \partial Q$; since $\partial \mathsf{T} = a\mathsf{T}$, differentiation does not increase the degree in T , so $\partial P = hP$ and $\partial Q = hQ$ for the same $h \in \mathbf{No}[i]$. If $p_j \mathsf{T}^j$ and $p_k \mathsf{T}^k$ are two nonzero terms of P with $j \neq k$, comparing coefficients gives $\partial p_j/p_j + ja = h = \partial p_k/p_k + ka$, that is

$$(p_j/p_k)^\dagger = (k - j)a.$$

Applying Φ and using Corollary 9.4 yields $(k - j)\Phi(a) = 0$, impossible since $\mathcal{P}$ is torsion-free. So P is a single term, and likewise Q , whence $R = c\mathsf{T}^m$ with $c \in \mathbf{No}[i]^\times$ and $m \in \mathbb{Z}$. The constancy equation is $c^\dagger + ma = 0$; applying Φ again forces $m = 0$, and then $\partial c = 0$ gives $c \in \mathbb{C}^\times$.

An automorphism over $\mathbf{No}[i]$ sends T to another nonzero solution of equation (21.1); the quotient by T has derivative zero, hence lies in $\mathbb{C}^\times$. Conversely every such scaling is a differential automorphism. When $\Phi(a) = 0$ a nonzero solution already lies in $\mathbf{No}[i]$ by Theorem 8.4. $\square$

Transcendence is necessary: algebraic closedness of $\mathbf{No}[i]$ leaves no nontrivial algebraic extension in which a missing solution could appear, and an algebraic root cannot cancel a nonzero obstruction (Corollary 9.4). T is a genuinely new differential element, not an alternative notation for a pre-existing surcomplex number.

Theorem 21.2 (The same extension, with differential simplicity). *For $a = i$ the conclusions of Theorem 21.1 hold, and in addition the Laurent polynomial ring $\mathbf{No}[i][\mathsf{T}, \mathsf{T}^{-1}]$ has no nonzero proper differential ideal; so $\mathbf{No}[i](\mathsf{T})$ has the explicit Picard–Vessiot properties for $\partial y = iy$ with differential Galois group $\mathbb{C}^\times$. A second, independent proof of the constant-field statement runs as follows.*

Proof. Let $R \in \mathbf{No}[i](\mathsf{T})^\times$ be constant. Since $\mathbf{No}[i]$ is algebraically closed, factor numerator and denominator:

$$R = c \mathsf{T}^m \prod_{j=1}^k (\mathsf{T} - a_j)^{n_j},$$

with $c \in \mathbf{No}[i]^\times$, $m \in \mathbb{Z}$, the $a_j \in \mathbf{No}[i]^\times$ distinct and the n_j nonzero integers. Logarithmic differentiation gives

$$0 = c^\dagger + i \left(m + \sum_j n_j \right) + \sum_j \frac{n_j (ia_j - \partial a_j)}{\mathsf{T} - a_j}. \quad (21.2)$$

Each $ia_j - \partial a_j$ is nonzero by Corollary 8.8, and distinct simple poles cannot cancel, so $k = 0$. What remains is $c^\dagger = -im$, which Theorem 8.4 excludes unless $m = 0$; then $c \in \mathbb{C}^\times$.

For differential simplicity, suppose $\mathfrak{q}$ is a nonzero proper differential ideal of $\mathbf{No}[i][\mathbb{T}, \mathbb{T}^{-1}]$. Choose a nonzero element of $\mathfrak{q}$ with the fewest nonzero monomials and multiply by a unit to make its constant coefficient 1. If it has another term $a_j \mathbb{T}^j$ then $j \neq 0$ and $\partial(a_j \mathbb{T}^j) = (\partial a_j + i j a_j) \mathbb{T}^j \neq 0$, again by Theorem 8.4. Its derivative is therefore a nonzero element of $\mathfrak{q}$ with one fewer nonzero monomial, the constant term differentiating to zero, contradicting minimality. So the chosen element is 1 and $\mathfrak{q}$ is the whole ring. $\square$

21.2 Set-sized bases, and a non-embedding

The proper-class base can be avoided entirely. Each of the following is a complete construction over a set-sized base, with its own no-new-constants proof.

Theorem 21.3 (Over a localized workspace). *Let F be as in Theorem 20.8, put $L = F[i]$, and let $\mathbb{T}$ be transcendental over L with $\partial \mathbb{T} = i\mathbb{T}$, the derivation extended by the product and quotient rules. Then the constant field of $L(\mathbb{T})$ is exactly $\mathbb{C}$, $L(\mathbb{T})$ is a scalar Picard–Vessiot extension for $\partial y = iy$, and its differential automorphisms over L are exactly $\mathbb{T} \mapsto c\mathbb{T}$ with $c \in \mathbb{C}^\times$, so the group is $\mathbf{G}_m(\mathbb{C})$.*

Proof. Every rational $R(\mathbb{T}) \in L(\mathbb{T})$ has an injective formal Laurent expansion at $\mathbb{T} = 0$, $R = \sum_{n \geq n_0} a_n \mathbb{T}^n \in L((\mathbb{T}))$, with supports bounded below in $\mathbb{Z}$; the induced derivation is

$$\partial R = \sum_{n \geq n_0} (\partial a_n + ni a_n) \mathbb{T}^n,$$

which agrees with the rational-function derivation by the product and inverse rules. If $\partial R = 0$ then $\partial a_n = -ni a_n$ for every n . For $n \neq 0$ the constant $-ni$ is nonreal, so $a_n = 0$ by Proposition 3.11 applied in the ambient field. For $n = 0$, $\partial a_0 = 0$ gives $a_0 \in \mathbb{C}$. Hence $R = a_0$ and there are no new constants. The automorphism statement is as in Theorem 21.1. No black-box existence theorem is used: the field and its constants have been constructed and checked. $\square$

Theorem 21.4 (Over the rational Hahn field). *Let $K_0^\mathbb{Q} = \mathbb{C}((t^\mathbb{Q}))$ with its explicit coefficientwise derivation $\partial(\sum_q c_q t^q) = \sum_q (-q) c_q t^{q+1}$. This field is set-sized and algebraically closed, its differential constants are $\mathbb{C}$, it contains $\omega = t^{-1}$ with $\partial \omega = 1$, and ω being a differential subfield of $\mathbf{No}[i]$ — it contains no nonzero solution of $\partial y = niy$ for $0 \neq n \in \mathbb{Z}$. Adjoin a transcendental $\mathbb{T}$ with $\partial \mathbb{T} = i\mathbb{T}$. Then the differential constants of $K_0^\mathbb{Q}(\mathbb{T})$ are exactly $\mathbb{C}$, it is a Picard–Vessiot field for $\partial y = iy$, and its differential Galois group is $\mathbf{G}_m$ over $\mathbb{C}$.*

Proof. Let $f \in K_0^\mathbb{Q}(\mathbb{T})^\times$ have $\partial f = 0$. Write $f = cP/Q$ with $c \in (K_0^\mathbb{Q})^\times$ and monic coprime $P, Q \in K_0^\mathbb{Q}[\mathbb{T}]$ of degrees m, n . Clearing denominators gives

$$\frac{\partial c}{c} PQ + (\partial P)Q - P(\partial Q) = 0,$$

and reduction modulo P and modulo Q gives $P \mid \partial P$ and $Q \mid \partial Q$. The derivation does not raise the degree in $\mathbb{T}$, and on a monic polynomial of degree m its leading coefficient is im . Therefore

$$\partial P = imP, \quad \partial Q = inQ, \quad \frac{\partial c}{c} = i(n - m),$$

and the last equation is impossible in $K_0^\mathbb{Q}$ unless $m = n$, whence $c \in \mathbb{C}^\times$ by the constant-field statement. Comparing the coefficient of $\mathbb{T}^j$ in P for $j < m$ gives $\partial a_j = i(m - j)a_j$, again impossible

for a nonzero a_j ; so $P = T^m$, similarly $Q = T^n$, and coprimality with $m = n$ forces $m = n = 0$. Hence $f \in \mathbb{C}^\times$. The automorphism computation is as before. $\square$

Remark 21.5 (Field-transcendental is not differentially transcendental). Since $K_0^{\mathbb{Q}}$ is already algebraically closed, T is necessarily transcendental as a *field* element. It is nevertheless *differentially algebraic*, satisfying the nonzero differential polynomial $\partial T - iT = 0$. The two notions must not be confused. One might try to identify T with $\sum_{n \geq 0} (i\omega)^n/n!$; that fails for the support reason of Remark 11.9, and choosing a larger exponent group does not repair the defect. The differential extension is therefore not obtained by appending a missing value to a list of permitted exponents.

Theorem 21.6 (Over the rational function field $\mathbb{C}(X)$). *Let $F = \mathbb{C}(X)$ with $\partial X = 1$, let T be transcendental over F , and define a derivation on $E = F(T)$ by $\partial X = 1$, $\partial T = iT$, extended by the polynomial and quotient rules; this is well defined because X and T are algebraically independent. Then the differential constant field of E is exactly $\mathbb{C}$, E is a rank-one Picard–Vessiot field for $\partial Y = iY$ over F , its differential automorphisms over F are exactly $T \mapsto cT$ with $c \in \mathbb{C}^\times$, and its differential Galois group is $\mathbf{G}_m(\mathbb{C})$.*

Lemma 21.7. *For $r \in \mathbb{C}(X)^\times$, the identity $r'/r = c$ with $c \in \mathbb{C}^\times$ is impossible.*

Proof. Write $r = p/q$ with nonzero complex polynomials of degrees d, e . Expansion at infinity gives

$$\frac{r'}{r} = \frac{p'}{p} - \frac{q'}{q} = \frac{d-e}{X} + O(X^{-2}),$$

an identity of formal Laurent expansions in $1/X$ whose constant term is zero, so it cannot equal a nonzero constant. Equivalently, compare degrees after clearing denominators in $p'q - pq' = cpq$. $\square$

Proof of Theorem 21.6. Expand $R \in E$ at $T = 0$ as an injective formal Laurent series $R = \sum_{n \geq n_0} r_n(X)T^n \in \mathbb{C}(X)((T))$ — a formal expansion, with no analytic convergence asserted — on which the derivation acts by

$$\partial \left(\sum_n r_n T^n \right) = \sum_n (r'_n + in r_n) T^n,$$

compatible with the rational subfield. If $\partial R = 0$ then $r'_n + in r_n = 0$ for all n ; for $n \neq 0$ Lemma 21.7 forces $r_n = 0$, and for $n = 0$ the equation $r'_0 = 0$ gives $r_0 \in \mathbb{C}$. Injectivity of the expansion gives $R \in \mathbb{C}$. The automorphism statement is as before. $\square$

Corollary 21.8 (No differential embedding). *$F = \mathbb{C}(X)$ embeds differentially into $\mathbf{No}[i]$ by $X \mapsto \omega$, since $\partial \omega = 1$. That embedding does not extend to $E = \mathbb{C}(X)(T)$: the image of T would be a nonzero solution of $\partial Y = iY$, contradicting Corollary 8.8. The same argument applies to the extension $\mathbb{C}(\omega)(\mathcal{E})$ with $\partial \omega = 1$ and $\partial \mathcal{E} = i\mathcal{E}$, defined by the polynomial product rule and then the quotient rule: there is no differential embedding of it into $\mathbf{No}[i]$ fixing $\mathbb{C}(\omega)$, and this conclusion requires no assertion whatever about that extension's full constant field. Likewise no differential embedding of $\mathbf{No}[i](T)$ into $\mathbf{No}[i]$ over $\mathbf{No}[i]$ exists, and T cannot be algebraic over $\mathbf{No}[i]$.*

So a field can be algebraically closed, contain the required base coefficients, and possess additive primitives, and still fail to contain this simplest oscillatory differential extension. The generator T is transcendental as a field element, but *differentially algebraic*, since it satisfies $\partial T = iT$. It is not an algebraic root missing from $\mathbf{No}[i]$, and no change of normal-form encoding inside the same field can make it exist. A symbolic constructor for such an extension is mathematically reasonable, but it must announce a change of ambient differential field: calling its new element a surcomplex number would be a mathematical error, not a harmless choice of representation.

Table 6: Five Picard–Vessiot constructions for $\partial y = iy$ (or $\partial y = ay$).

Base field	Size	Route to “no new constants”	Reference
$\mathbf{No}[i] = \mathbf{No}[i]$	proper class	Coprimalty, then Φ on coefficient ratios, using torsion-freeness of $\mathcal{P}$	Theorem 21.1
$\mathbf{No}[i] = \mathbf{No}[i]$	proper class	Partial fractions: distinct simple poles cannot cancel; plus differential simplicity of $\mathbf{No}[i][T, T^{-1}]$	Theorem 21.2
$L = F[i]$, F localized	set	Formal Laurent expansion at $T = 0$, coefficientwise, plus Proposition 3.11	Theorem 21.3
$\mathbb{C}((t^{\mathbb{Q}}))$	set	Degree and divisibility in $K_0^{\mathbb{Q}}[T]$, leading coefficient <i>im</i>	Theorem 21.4
$\mathbb{C}(X)$, $\partial X = 1$	set	Formal Laurent expansion at $T = 0$ plus Lemma 21.7	Theorem 21.6

21.3 The recovered oscillator, its real form, and what it costs

Inside any of the extensions above, with $\partial T = iT$, put

$$C = \frac{T + T^{-1}}{2}, \quad S = \frac{T - T^{-1}}{2i}. \quad (21.3)$$

Proposition 21.9 (A genuine oscillator). *The elements equation (21.3) satisfy*

$$\partial C = -S, \quad \partial S = C, \quad C^2 + S^2 = 1.$$

They are linearly independent over $\mathbb{C}$, and every solution of $\partial^2 y + y = 0$ in the extension is a $\mathbb{C}$ -linear combination of them.

Proof. $\partial(T^{-1}) = -iT^{-1}$ gives the differential identities, and direct expansion the quadratic one. T and T^{-1} are $\mathbb{C}$ -independent, since a nontrivial relation would make the transcendental T algebraic, and the linear change equation (21.3) is invertible. For the spanning statement, coprimeness of $\partial - i$ and $\partial + i$ decomposes the oscillator kernel as in Theorem 13.2; any solution of $\partial y = iy$ is a constant multiple of T and any solution of $\partial y = -iy$ is a constant multiple of T^{-1} , because the quotient has derivative zero and the constants are exactly $\mathbb{C}$. $\square$

These are genuine oscillator elements in a field that actually contains them. They are *not* values of surcomplex sine and cosine at ω : identifying T with an element of $\mathbf{No}[i]$ would contradict Proposition 3.11. Calling T an exponential is harmless only when its status as a new differential element is retained.

For the real form, extend conjugation by $\bar{T} = T^{-1}$. This is an involution commuting with the derivation, since $\partial \bar{T} = i\bar{T} = -iT^{-1} = \partial(T^{-1})$, and it fixes C and S . Two equivalent real presentations occur in the sources. First, over a real base F one may take a transcendental $T_{\mathbb{R}}$ with

$$\partial T_{\mathbb{R}} = \frac{1 + T_{\mathbb{R}}^2}{2}, \quad C = \frac{1 - T_{\mathbb{R}}^2}{1 + T_{\mathbb{R}}^2}, \quad S = \frac{2T_{\mathbb{R}}}{1 + T_{\mathbb{R}}^2}, \quad (21.4)$$

the three identities following by the quotient rule, with $T = (1 + iT_{\mathbb{R}})/(1 - iT_{\mathbb{R}})$; the real rational function field then has constant field $\mathbb{R}$. Second, and equivalently, the conjugation-fixed Cayley coordinate

$$v = \frac{T - 1}{i(T + 1)}, \quad T = \frac{1 + iv}{1 - iv}, \quad C = \frac{1 - v^2}{1 + v^2}, \quad S = \frac{2v}{1 + v^2}, \quad \partial v = \frac{1 + v^2}{2}, \quad (21.5)$$

is transcendental over F , so $L(\mathbb{T}) = F(\mathbf{v})[i]$ and the fixed field of the involution is $F(\mathbf{v})$: an ordinary rational function field over an ordered field, which admits orderings extending the order of F .

Theorem 21.10 (Oscillation cannot retain the H -field ordering rule). *Let $F \subseteq \mathbf{No}$ be an ordered differential subfield containing $\mathbb{R} \cup \{\omega\}$, as for the real bases above. No ordering of $F(\mathbf{v})$ extending the given order of F can satisfy*

$$h > \mathbb{R} \implies \partial h > 0$$

with the inherited derivation and $\partial\omega = 1$.

Proof. In any such ordering the identity $\mathbf{C}^2 + \mathbf{S}^2 = 1$ bounds $\mathbf{C}$ and $\mathbf{S}$ in absolute value by 1. The proof of Lemma 3.8 uses the displayed positivity rule, $\partial\omega = 1$, the inverse rule (3.1) and $\ker \partial \supseteq \mathbb{R}$, and nothing else; all four hold in this ordered extension, so the lemma applies there and forces $\partial\mathbf{C}$ and $\partial\mathbf{S}$ to be infinitesimal relative to $\mathbb{R}$. Since $\partial\mathbf{C} = -\mathbf{S}$ and $\partial\mathbf{S} = \mathbf{C}$, both $\mathbf{C}$ and $\mathbf{S}$ are then infinitesimal, so $\mathbf{C}^2 + \mathbf{S}^2$ is infinitesimal and cannot equal 1. $\square$

Equivalently, in the presentation equation (21.4): $F(\mathbb{T}_{\mathbb{R}})$ admits orders extending that of F , but no such realization can preserve $\partial\mathcal{O} \subseteq \mathfrak{m}$ with constant field $\mathbb{R}$, both oscillator coordinates being bounded by 1 and so having infinitesimal derivatives. So it is not the bare possibility of ordering a real field that blocks oscillation. The obstruction is the *compatibility* of that ordering with the Hardy-type differential positivity rule. One may enlarge the differential field to obtain the oscillator, but one cannot keep all the structure that made the base a surreal H -field. This says precisely what must be surrendered.

21.4 Several phases: the differential Galois torus

Let $a_1, \dots, a_m \in \mathbf{No}[i]$ and consider the diagonal system $\partial y_j = a_j y_j$. Write $\Phi_j = \Phi(a_j) \in \mathcal{P}$ and define the *phase-relation lattice*

$$\Lambda = \left\{ n = (n_1, \dots, n_m) \in \mathbb{Z}^m : \sum_j n_j \Phi_j = 0 \right\}. \quad (21.6)$$

If $n \in \Lambda$ then $\sum_j n_j a_j$ is a logarithmic derivative in $\mathbf{No}[i]$, so a product of integer powers of solutions can be normalized to an element of the base field. These are *integer* relations because rational functions in solutions involve integral monomial exponents; real linear dependence of the obstructions is not enough.

Lemma 21.11 (The lattice is saturated). *Λ is saturated in $\mathbb{Z}^m$: if $kn \in \Lambda$ for a nonzero integer k then $n \in \Lambda$. Hence $\mathbb{Z}^m/\Lambda$ is free abelian of rank $d = m - \text{rank } \Lambda$.*

Proof. If $k \sum_j n_j \Phi_j = 0$, torsion-freeness of the real vector class $\mathcal{P}$ gives $\sum_j n_j \Phi_j = 0$. So the quotient is finitely generated and torsion-free, hence free; equivalently the elementary integer normal-form theorem extends a basis of the saturated subgroup to a basis of $\mathbb{Z}^m$. $\square$

Theorem 21.12 (The diagonal phase torus). *The diagonal system has a Picard–Vessiot extension obtained by adjoining d algebraically independent solution symbols after base-field gauges and an invertible integer monomial change of coordinates. Its differential Galois group is isomorphic to $\mathbf{G}_m(\mathbb{C})^d$, and in the original coordinates its action on the solutions is described by*

$$G = \left\{ (c_1, \dots, c_m) \in (\mathbb{C}^\times)^m : \prod_j c_j^{n_j} = 1 \text{ for every } n \in \Lambda \right\}. \quad (21.7)$$

Proof. Choose an integer basis of $\mathbb{Z}^m$ whose first $\ell = \text{rank } \Lambda$ vectors form a basis of Λ (Lemma 21.11). The associated unimodular matrix changes the diagonal coefficients to integer linear combinations $a'_1, \dots, a'_m$. The first ℓ have zero obstruction, hence nonzero solutions $g_j \in \mathbf{No}[i]$ by Theorem 8.4. The remaining obstructions $\Phi'_{\ell+1}, \dots, \Phi'_m$ are $\mathbb{Z}$ -independent, since any relation among them would be an element of Λ with nonzero complementary coordinates.

For those d coefficients introduce algebraically independent symbols $T_1, \dots, T_d$ with $\partial T_k = a'_{\ell+k} T_k$. If a nonzero constant of $\mathbf{No}[i](T_1, \dots, T_d)$ is written P/Q with coprime multivariate polynomials, the argument of Theorem 21.1 gives $\partial P = hP$ and $\partial Q = hQ$ with $h \in \mathbf{No}[i]$, total degree not increasing under this derivation. For distinct nonzero terms $p_\alpha T^\alpha$ and $p_\beta T^\beta$ of P , coefficient comparison gives

$$(p_\alpha/p_\beta)^\dagger = \sum_{k=1}^d (\beta_k - \alpha_k) a'_{\ell+k},$$

and applying Φ contradicts integer independence unless $\alpha = \beta$. So P and Q are monomials, and the same argument on their quotient shows that a rational constant has zero exponent vector and a coefficient in $\mathbb{C}$.

Combining the g_j with the T_k and inverting the unimodular monomial change supplies nonzero solutions of all the original diagonal equations, generating the same differential field. Each independent symbol may be scaled by an arbitrary element of $\mathbb{C}^\times$, giving $\mathbf{G}_m(\mathbb{C})^d$; in the original coordinates the trivialized integer combinations must stay fixed, which is exactly equation (21.7), and those equations leave precisely d independent constant multipliers. $\square$

Example 21.13 (Three separations). (i) For $a_1 = i$, $a_2 = 2i$ the obstructions are ω and 2ω , the lattice is generated by $(2, -1)$, one can choose solutions with $y_2 = y_1^2$, and $G = \{(c, c^2) : c \in \mathbb{C}^\times\}$ is one-dimensional. (ii) For $a_1 = i$, $a_2 = \sqrt{2}i$ the obstructions ω and $\sqrt{2}\omega$ are dependent over $\mathbb{R}$ but have no nonzero integer relation, so there are two algebraically independent generators and the group is $\mathbf{G}_m(\mathbb{C})^2$; the field exponential on real arguments does not supply a canonical irrational power of a new oscillatory symbol inside a rational Picard–Vessiot extension. (iii) For $a_1 = i/\omega$, $a_2 = i/(\omega \log \omega)$ the obstructions are $\log \omega$ and $\log \log \omega$; the first dominates the second by an infinite factor, so no nonzero integer relation exists and the torus again has dimension two — smallness of both coefficients does not reduce the number of independent missing phases.

The theorem treats diagonal systems completely and supplies useful information for triangular extensions. Taken by itself it is *not* a classification of Picard–Vessiot groups for arbitrary nontriangular matrices over $\mathbf{No}[i]$. It becomes one in combination with the matrix normal form of Part III: Theorem 15.9 reduces an arbitrary finite system to a diagonal one over $\mathbf{No}[i]$, and Corollary 17.10 then reads off its torus, identifying the count d of Lemma 21.11 with the rational rank r of the phase lattice $\Pi \subseteq \mathcal{P}$ generated by $\text{Ph}(A)$. Proposition 17.11 adds the group-algebra presentation in rank r and the real form; the differential simplicity proved there in rank r is the general form of the rank-one statement of Theorem 21.2. For a computational backend the lattice is additional required data: treating $i\omega$ and $2i\omega$ as independent phases would produce the wrong Galois group, and treating $i\omega$ and $i\sqrt{2}\omega$ as algebraically dependent merely because their obstructions are real multiples of one another would also be wrong.

22 The one genuine bridge: ω -series and composition

There is an established setting in which intrinsic differentiation really is recoverable as differentiation of a represented function. It is not all of $\mathbf{No}$ with an arbitrary choice of functions.

Let $\mathscr{W} = \mathbb{R}\langle\omega\rangle$ denote the ω -series: the smallest subfield of **No** containing $\mathbb{R}$ and ω and closed under exponential, positive logarithm, and set-indexed summable families. It is a *proper class*, explicitly distinct from the set-sized workspaces of Sections 19 and 20. Berarducci and Mantova construct a composition $f \circ x$ for $f \in \mathscr{W}$ and positive infinite x ; their results identify $(\partial f) \circ x$ with the *fine* derivative of $x \mapsto f \circ x$ and give a compatible chain rule inside that setting, and their Taylor theorem states that for sufficiently small h

$$f \circ (x + h) = \sum_{n \geq 0} \frac{(\partial^n f) \circ x}{n!} h^n, \quad (22.1)$$

the sum being support-controlled and the permitted size of h depending on f and x [2, Cor. 7.6–7.7, Thms 7.14, 8.2].

Remark 22.1 (How the two derivatives agree, and where). For $f = \log \omega$ the represented function is $x \mapsto \log x$ and $\partial f = 1/\omega$; composition sends the latter to $1/x$, exactly its fine function derivative. For $f = \omega^3$ both descriptions give $3x^2$. Equation (22.1) explains this agreement in a much larger class than finite elementary expressions. This is a different construction from a formal polynomial: in $P(Z) = \omega Z$ the symbol ω is a coefficient held fixed by ∂_Z , whereas in the ω -series $f = \omega^2$ both occurrences belong to the distinguished compositional scale and $f \circ x = x^2$. A representation must record which interpretation is intended before any chain rule can be chosen. The local agreement of derivative operations does not say that every surreal scalar is an ω -series or that every primitive belongs to that class: the full-field surjectivity clause (BM6) and the restricted composition theorem have different domains and must be applied separately.

Warning 22.2 (Imported, with its hypotheses intact). The composition, Taylor and incompatibility results are *separate imported theorems*; they do not follow from Theorem 3.1, and they are not hypotheses hidden inside the phase criterion. In particular $f \circ x$ is an additional operation, not the result of substituting into an arbitrary normal form. Berarducci and Mantova also prove that the simplest derivation on *all* of **No** admits no composition compatible with the composition and compatibility axioms of that paper [2, Thm 8.4]. Their hypotheses matter — the compatibility includes a constant rule, an order condition for positive derivative, and a chain rule at positive infinite arguments — and the theorem is quoted here for this particular derivation. It is *not* strengthened into an assertion about all conceivable notions of composition, or about every other choice of derivation, and nothing here settles alternative global composition structures. The present finite-phase obstruction is a direct differential-algebraic argument, not a restatement or a strengthening of that composition theorem.

The practical lesson is to retain at least three input kinds: a formal expression with named coefficient constants, an ω -series with a distinguished compositional scale, and a general scalar with an intrinsic derivation. They can overlap as *values* while supporting different operations, and identifying their carriers in memory does not justify identifying their semantic interfaces.

Remark 22.3 (Four tempting inferences, each refuted by a witness). The elementary embedding of the ordered differential field of logarithmic-exponential transseries into $(\mathbf{No}, \partial)$ [3] places this derivation in a powerful nonoscillatory asymptotic setting; it is a theorem about a specified ordered differential language and must not be silently upgraded to a statement about every simultaneously added analytic, compositional or complex exponential operation, and it is not a theorem that every algebraic differential equation has a solution. Four specific inferences fail, each with an explicit witness:

- (1) algebraic closedness does not imply surjectivity of the logarithmic derivative — i is missing from the image (Theorem 8.1);

- (2) surjectivity of ∂ does not imply closure under exponential integrals — $i\omega$ is a primitive of i , and no compatible exponential of it exists in $\mathbf{No}[i]$ (Corollary 11.4);
- (3) differential stability of a Hahn workspace does not imply primitive closure — $\log \omega \notin H_{\mathbb{Q}}$ (Example 18.2);
- (4) an infinitesimal imaginary part is necessary but not sufficient for a nonzero rank-one solution — i/ω is excluded while i/ω^2 is included (Warning 8.5).

Part V

Regular-singular systems over real-exponent Hahn workspaces

This part is the contribution of member 12 (appendix C). It treats the finite matrix equations

$$\partial y = -t Ay, \quad A = A_0 + A_+, \quad A_0 \in \text{Mat}_n(\mathbb{C}), \quad A_+ \in \text{Mat}_n(K_{\mathbb{R}}^{>0}),$$

whose coefficient is a real-power Hahn series of order at least one in t with an ordinary complex leading matrix, and it classifies them completely, both over the Hahn field $K_{\mathbb{R}} = \mathbb{C}((t^{\mathbb{R}}))$ and over the whole of $\mathbf{No}[i]$. Three relations to the rest of the report fix its place.

- It is the *matrix version of the rank-one resonant normal form* of Remark 19.5 and Theorem 19.3: in rank one it returns that theorem (Remark 28.2), and in rank n it replaces the single exponent-1 coefficient by a residual matrix and its Levelt resonances.
- Over $\mathbf{No}[i]$ it computes one explicit *non-Hermitian* sector of the differential phase spectrum of Part III: $\text{Ph}(-tA) = \{(\text{Im } \lambda_j) \tau\}$ with $\tau = \log t$ and λ_j the eigenvalues of A_0 (Proposition 25.12). It does so by an elementary route that uses neither the import of Theorem 15.6 nor the surjectivity clause (BM6). It therefore answers item N24 and the corresponding open continuation *for this class only*: an explicit phase formula is available here without the matrix import. The abstract classification already covers all finite systems over $\mathbf{No}[i]$; computable formulas for broader non-Hermitian classes remain open (Section 51.13).
- It supplies, for this class, an explicit splitting whose rank-one factors carry obstructions read off from A_0 alone, together with a finite exact procedure for finite rational supports with algebraic coefficients (Section 28.3). That is a partial answer to item N57 and to the “computable filtration” question of Warning 17.13; it is *not* an answer to item N41, which concerns irregular singularities of *coordinate* equations.

23 The Euler derivation, the coefficient class, and three hazards

23.1 The Euler derivation is a rescaled derivation

Put

$$\tau := \log t = -\log \omega, \quad \partial \tau = -t \neq 0. \quad (23.1)$$

The rescaled derivation of Proposition 4.2 with $s = \tau$ is

$$\partial_{\tau} f = \frac{\partial f}{\partial \tau} = -\omega \partial f. \quad (23.2)$$

It is a derivation of $\mathbf{No}[i]$ with the same constant field $\mathbb{C}$, it satisfies $\partial_\tau \tau = 1$, and it is *not* normalized at ω : $\partial_\tau \omega = -\omega$.

Lemma 23.1 (The Euler derivation on real-exponent Hahn fields). *For every additive subgroup $\Gamma \subseteq \mathbb{R}$, the field $K_\Gamma = \mathbb{C}((t^\Gamma)) \subseteq \mathbf{No}[i]$ is ∂_τ -stable and*

$$\partial_\tau \left(\sum_{\gamma} c_\gamma t^\gamma \right) = \sum_{\gamma} \gamma c_\gamma t^\gamma, \quad \partial_\tau \tau = 1. \quad (23.3)$$

Its constants in K_Γ are exactly $\mathbb{C}$, and ∂_τ maps $K_\Gamma^{\geq 0} = \{f : f = 0 \text{ or } v(f) \geq 0\}$ into $K_\Gamma^{> 0} = \{f : f = 0 \text{ or } v(f) > 0\}$. In particular $\partial = -t \partial_\tau$ preserves K_Γ exactly when $\Gamma = \{0\}$ or $1 \in \Gamma$, whereas ∂_τ preserves every K_Γ . The nonzero-group case is Proposition 20.6.

Proof. For ordinary real a the real-power rule of equation (3.1), which Warning 2.2 allows at ordinary real exponents, gives $\partial t^a = -a t^{a+1}$, that is $\partial_\tau t^a = a t^a$. Strong additivity (BM3) extends this to Hahn sums; the support can only shrink, so every K_Γ is stable. Finally $\partial_\tau \tau = -\omega \partial(-\log \omega) = \omega t = 1$. The coefficient γc_γ vanishes only at $\gamma = 0$, which gives the constants and the valuation statement. If $\Gamma = \{0\}$, then $K_\Gamma = \mathbb{C}$ and ∂ vanishes on it. If $1 \in \Gamma$, the formula $\partial = -t \partial_\tau$ preserves K_Γ . Conversely, if $\Gamma \neq \{0\}$ and ∂ preserves K_Γ , choose $0 \neq \gamma \in \Gamma$. The nonzero derivative $\partial t^\gamma = -\gamma t^{\gamma+1}$ belongs to K_Γ , so $\gamma + 1 \in \Gamma$ and hence $1 \in \Gamma$. $\square$

Remark 23.2 (One operator, several names). ∂_τ is the row ∂_s of Table 1 at the scale $s = \tau$. Member 12 writes it $\delta = -\omega \partial$ and writes ℓ for τ ; the companion report on tail-spans calls the same operator the Euler derivation and records $\partial|_{H_0} = -t$ times it [29]. This report writes ∂_τ and τ only: the letter δ already occurs in Table 2 as a source notation for the infinitesimal phase residue, and $\ell_1 = \log \omega = -\tau$ is the first iterated logarithm of equation (3.2). Equation (23.3) lets one calculate as though $\partial_\tau = t d/dt$, and it is exactly the formula of Proposition 18.1 divided by $-t$.

Remark 23.3 (Not a coordinate derivative, and the cross-category pair). The symbol t names the actual surreal number ω^{-1} , not a moving ordinary complex variable (Warning 4.1(1)). Equation (23.3) is an identity for the intrinsic derivation; no differentiability in the fine order topology is assumed, and where the verification program of member 12 uses a formal variable it checks the algebraic identities that this embedding then realizes. The contrast with the coordinate theory is sharp. For $\varepsilon \in K_\mathbb{R}^{> 0}$ the *intrinsic* scalar equation $\partial_\tau y = (\lambda + \varepsilon)y$ has a nonzero solution in $\mathbf{No}[i]$ exactly when $\text{Im } \lambda = 0$ (Theorem 25.3), while the *coordinate* equation $D_z Y = (\lambda + \varepsilon)Y/z$ of Example 46.2 has the coherent solution $z^\lambda E(\varepsilon \log z)$ for every $\lambda \in \mathbb{C}$. This is the cross-category pair of Warning 46.3 once more. Likewise the coordinate Frobenius theory of the companion non-Abelian-support report concerns d/dz with positive-support residue, a different operator, and is not merged here.

Warning 23.4 (Orientation). ∂_τ reverses the sign rule (BM7): if $x > \mathbb{R}$ then $\partial x > 0$ and hence $\partial_\tau x < 0$. So $\partial_\tau y = ay$ with real $a > 0$ has the *infinitesimal* solution t^a , whereas $\partial y = \lambda y$ has the infinitesimal solutions $c e^{\lambda \omega}$ only for $\lambda < 0$, and the same holds for every normalized derivation in Part VI (Lemma 30.4). The positivity $\lambda - \mu > 0$ in the Levelt resonances below is oriented to ∂_τ . No sign statement is moved between the two frames without the translation $\partial_\tau = -\omega \partial$.

23.2 The coefficient class and the alphabet of this part

For $\Gamma \subseteq \mathbb{R}$ we keep the least-exponent valuation v of Section 18, the valuation ring $K_\Gamma^{\geq 0}$ and its maximal ideal $K_\Gamma^{> 0}$ from Lemma 23.1. In the embedding into $\mathbf{No}[i]$, their elements are finite, respectively infinitesimal.

Definition 23.5 (Power-Hahn regular-singular system). A *power-Hahn regular-singular system* is an equation

$$\partial_\tau y = Ay, \quad A = A_0 + A_+, \quad A_0 \in \text{Mat}_n(\mathbb{C}), \quad A_+ \in \text{Mat}_n(K_{\mathbb{R}}^{>0}), \quad (23.4)$$

equivalently $\partial y = -t Ay$. Its *residual matrix* is A_0 . The unknown y ranges over all of $\mathbf{No}[i]^n$, with no support restriction.

The adjective is a definition of the coefficient class, not an assertion that arbitrary finite surreal matrices have such an expansion; Proposition 28.4 makes the distinction indispensable.

Convention 23.6 (The alphabet of this part). Throughout Part V:

- $K_{\mathbb{R}} = \mathbb{C}((t^{\mathbb{R}}))$, and $K_{\Gamma} = \mathbb{C}((t^{\Gamma}))$ for an *arbitrary* additive subgroup $\Gamma \subseteq \mathbb{R}$. The standing hypothesis $1 \in \Gamma$ of Section 18 is dropped in this part, because ∂_τ preserves every K_{Γ} (Lemma 23.1); contrast Example 20.7.
- $\text{Eig}(A_0)$ is the set of ordinary complex eigenvalues of A_0 , written $\lambda = a_\lambda + ib_\lambda$. The word “spectrum” is not used for it (Warning 5.2).
- $G_0 \in \text{GL}_n(\mathbb{C})$ is a constant gauge bringing A_0 to a Jordan form $A_0^\circ = G_0^{-1} A_0 G_0$, with generalized eigenspaces $V = \mathbb{C}^n = \bigoplus_\lambda V_\lambda$, $d_\lambda = \dim V_\lambda$, and $A_0^\circ|_{V_\lambda} = \lambda I + N_\lambda^\circ$ with N_λ° nilpotent; a single N_λ° may contain several Jordan blocks. The source writes J for A_0° and J_λ for N_λ° ; this report uses no letter J (Convention 5.6).
- $S \subset \mathbb{R}_{>0}$ is the positive support of $G_0^{-1} A G_0 - A_0^\circ$ (a constant change of basis redistributes coefficients among entries but does not change the union of supports), S^* is its additive monoid as in Lemma 2.6, and $S_{>0}^* = S^* \setminus \{0\}$. The source writes $M = \langle S \rangle_{\mathbb{N}}$.
- $\text{Lev}(A_0) = \{\lambda - \mu \in \mathbb{R}_{>0} : \lambda, \mu \in \text{Eig}(A_0)\}$ is the finite set of *Levelt resonances*. The source writes $\mathcal{R}$, which is the rotation generator $\mathcal{R}$ here.
- $G_{\natural}$ is the normalizing Hahn gauge and $A^{\natural}$ the normalized coefficient of Theorem 24.5 (the source’s H and B : a bare H is a Hermitian matrix in Part III, and B is an accumulated phase throughout); pr_γ is a block projection (the source’s Π_γ ; Π is the phase lattice); $\mathcal{H}_\gamma$ and $\mathcal{N}_{\lambda\mu}$ are the homological operators (the source’s L_γ and $K_{\lambda\mu}$).
- $D = \text{diag}(b_\lambda I_{d_\lambda})$ and $D_{\text{re}} = \text{diag}(a_\lambda I_{d_\lambda})$ are real diagonal matrices, $t^{D_{\text{re}}} = \text{diag}(t^{a_\lambda} I_{d_\lambda})$ is the real-monomial shear, $A^{\flat} = iD + N$ is the constant normal form, $F = G_0 G_{\natural} t^{D_{\text{re}}}$ and $G_\tau = F \exp(\tau N)$ (the source’s T , R_0 , C , PHT and Q).
- hob_A is the *Hahn obstruction* of Theorem 26.5 (the source’s ob_A); it is not the phase obstruction $\mathfrak{o}$ or Φ . The operator $y \mapsto \partial_\tau y - Ay$ is written out; the source’s name $\mathcal{D}_A$ for it is the first-integral derivation of Section 17.5 here.

For a ∂_τ -stable subfield $\mathbb{C} \subseteq E \subseteq \mathbf{No}[i]$ with $A \in \text{Mat}_n(E)$ and $G \in \text{GL}_n(E)$, the substitution $y = Gz$ carries equation (23.4) to

$$\partial_\tau z = A^G z, \quad A^G = G^{-1} A G - G^{-1} \partial_\tau G. \quad (23.5)$$

Multiplying by $-t$ gives equation (15.1) for the ∂ -coefficient $-tA$ as an identity in $\mathbf{No}[i]$. If $t \in E$, then E is stable under both derivations, and E -gauge equivalence of the two systems is the same in either frame. Without this hypothesis the ∂_τ -system can be defined over E while its ∂ -coefficient lies outside E : for $E = K_{\mathbb{Z}\sqrt{2}}$, the equation $\partial_\tau y = y$ has coefficient $1 \in E$ but its translated coefficient

$-t$ is not in E . A morphism from a system B' to a system A , of possibly different sizes, is a rectangular X over the stated field with $\partial_\tau X = AX - XB'$. Because the field contains $\mathbb{C}$, these morphism spaces are $\mathbb{C}$ -vector spaces.

Warning 23.7 (Residual frequencies are not phases). The source calls the real numbers $b = \text{Im } \lambda$ “phase labels”, and speaks of “phase multiplicities” and “mixed-phase” systems. Such a b is an ordinary real number. It is none of the senses (a)–(e) of Convention 5.1, and this report calls it a *residual frequency*. The relation to sense (e) is a translation, not an identification: $\partial_\tau z = ibz$ is $\partial z = i(-bt)z$, whose accumulated phase is $-b \log \omega = b\tau$. So the frequency b corresponds to the purely infinite phase $b\tau \in \mathcal{P}$, and Proposition 25.12 shows that the multiset of frequencies determines $\text{Ph}(-tA) = \{b_j\tau\}$.

Warning 23.8 (Three senses of “resonance”). With this part the word acquires a third meaning, and every occurrence is qualified. (1) A *Levelt resonance* is a positive real difference $\lambda - \mu$ of two residual eigenvalues; the set $\text{Lev}(A_0)$ is used only in this part. (2) *Phase resonance* is an integer relation among the phases of a finite system, and the resonance algebra of Theorem 17.8 is its invariant ring. (3) The *resonant coefficient* is the single coefficient $[t^1]f$ of Theorem 19.2, and the resonant normal form of Remark 19.5 is its logarithmic version. In ∂ -form the residual matrix is the t^1 -coefficient $-A_0$ of $-tA$; so sense (3) is the rank-one residual coefficient, sense (1) concerns differences of residual eigenvalues and is empty in rank one, and sense (2) is unrelated to both.

Remark 23.9 (What this part imports). The hypotheses used in this part are the clauses (BM1), (BM2), (BM3), (BM4), (BM5) and (BM7) of Theorem 3.1, the complexification of Proposition 3.5 (whose extension formula and constant field do not use surjectivity), and the ordered-word count of Lemma 2.6, proved in appendix A. Neither (BM6) nor the transferred splitting and surjectivity of Theorem 15.6 is used, and no linear-operator factorization theorem is needed. Where results of earlier parts are cited below, the cited steps use only these clauses; the source proves each such step directly and the report records the agreement instead of printing it twice. Classical regular-singular theory — Levelt’s Jordan decomposition for singular differential operators [15], its exposition for formal connections by Kamgarpour and Weatherhog [16], and van der Hoeven’s operator methods on generalized power series [17] — is the direct antecedent of the homological reduction; the ordinary formal theory over $\mathbb{C}((t))$ does not by itself give the support statements below for an arbitrary well-ordered subset of $\mathbb{R}_{>0}$.

24 The Hahn–Levelt normal form and finite resonance data

24.1 Supports, words, and the resonance-reaching letters

Lemma 24.1 (Finite words at a fixed weight, with a length bound). *Let $S \subset \mathbb{R}_{>0}$ be well ordered. Then S^* is well ordered, and for each $r \in \mathbb{R}$ only finitely many ordered words $(s_1, \dots, s_k)$ in S satisfy $s_1 + \dots + s_k = r$. If $S \neq \emptyset$ and $\varepsilon = \min S$, all such words have length at most $\lfloor r/\varepsilon \rfloor$.*

Proof. The first two assertions are Lemma 2.6, proved in appendix A; the length bound follows by adding $s_j \geq \varepsilon$. $\square$

Here is member 12’s second, Higman-free proof for real exponents. If $S = \emptyset$, only the empty word occurs and $S^* = \{0\}$. Otherwise put $\varepsilon = \min S > 0$. Every sequence in S has an infinite nondecreasing subsequence, as proved in appendix A.1. Applying this extraction successively to the k coordinates gives a componentwise nondecreasing subsequence of any sequence in S^k . If infinitely

many distinct words of length k had one sum, this subsequence would be constant: increasing any coordinate strictly would increase the sum. This contradicts distinctness. The length bound leaves only finitely many possible k , proving finiteness at each weight.

For well-ordering, suppose $r_1 > r_2 > \dots$ were word sums. They are nonnegative, and all chosen representing words have length at most $\lfloor r_1/\varepsilon \rfloor$. Some length occurs infinitely often; coordinatewise extraction at that length makes the corresponding sums nondecreasing, a contradiction. The argument uses the Archimedean property of $\mathbb{R}$. At higher rank, $k\varepsilon \leq r$ need not bound k : in $\mathbb{Z} \oplus_{\text{lex}} \mathbb{Z}$, take $\varepsilon = (0, 1)$ and $r = (1, 0)$. The general finite-word lemma still gives finitely many representations of each weight; this numerical length estimate does not replace it (Warning 2.5).

Corollary 24.2 (Products, geometric inverses, and well-founded recursion). *If X is a finite matrix whose entries are supported in S , then the family $(X^k)_{k \geq 0}$ is strongly summable and $(I - X)^{-1} = \sum_{k \geq 0} X^k$ has support in S^* . For each $\gamma \in S^*$ only finitely many pairs $(\alpha, \beta) \in (S_{>0}^*)^2$ satisfy $\alpha + \beta = \gamma$, and each member of such a pair is strictly smaller than γ . Consequently a coefficient recursion using only those pairs is a well-defined transfinite recursion on S^* .*

Proof. A contribution to a fixed coefficient of a matrix power is labelled by an ordered word in S together with a finite sequence of matrix indices, so Lemma 24.1 makes the number of contributions finite; this is where *ordered* words are needed (Remark A.3). The geometric identity follows coefficientwise by finite cancellation. The pair assertion is Lemma A.1 applied to the well-ordered set $S_{>0}^*$, and positivity of the other member gives the strict inequality. $\square$

Definition 24.3 (Resonance-reaching support). For a finite set $R \subset \mathbb{R}_{>0}$ let $S_{\text{crit}}(R)$ be the set of letters $s \in S$ that occur in some finite ordered word in S whose sum lies in R .

Lemma 24.4 (Finite dependence at finitely many weights). *$S_{\text{crit}}(R)$ is finite. Any coefficient expression formed from input matrices A_s by addition, fixed linear operators and ordered multiplication that adds weights, and having total weight in R , depends only on the A_s with $s \in S_{\text{crit}}(R)$, provided every term has positive input weight and the expression is defined by the well-founded recursion of Corollary 24.2.*

Proof. If $S = \emptyset$, there are no positive-weight inputs and $S_{\text{crit}}(R) = \emptyset$. Otherwise the union of the words with sums in the finite set R is finite by Lemma 24.1, so their letters form a finite set.

To expand a coefficient of weight r , replace each recursively defined factor by its defining expression. Every product splits its weight into two positive members of S^* , each at least $\varepsilon = \min S$. Thus following either factor decreases the remaining weight by at least ε . A dependency branch has at most $\lfloor r/\varepsilon \rfloor$ such steps, and at each step there are only finitely many predecessor pairs. The entire dependency tree is therefore finite. Addition and fixed linear operators introduce no new input letters or weights. At its leaves, every term is labelled by an ordered word of input letters summing to r , so if $r \in R$ those letters all belong to $S_{\text{crit}}(R)$. $\square$

24.2 The homological operator and the normalized resonant form

Replace A by $G_0^{-1}AG_0$ and write $A = A_0^\circ + \sum_{s \in S} A_s t^s$, with $A_\gamma = 0$ for $\gamma \notin S$. For $\gamma > 0$ define

$$\mathcal{H}_\gamma(X) = \gamma X - A_0^\circ X + X A_0^\circ, \quad \mathcal{N}_{\lambda\mu}(X) = N_\lambda^\circ X - X N_\mu^\circ \quad \text{on } \text{Hom}(V_\mu, V_\lambda). \quad (24.1)$$

On $\text{Hom}(V_\mu, V_\lambda)$ one has $\mathcal{H}_\gamma = (\gamma - \lambda + \mu)I - \mathcal{N}_{\lambda\mu}$ and $\mathcal{N}_{\lambda\mu}^{d_\lambda+d_\mu-1} = 0$, by expanding a power of the two commuting operators $X \mapsto N_\lambda^\circ X$ and $X \mapsto -XN_\mu^\circ$. If $c = \gamma - \lambda + \mu \neq 0$ the inverse is the finite operator

$$\mathcal{H}_\gamma^{-1} = \sum_{k=0}^{d_\lambda+d_\mu-2} c^{-k-1} \mathcal{N}_{\lambda\mu}^k. \quad (24.2)$$

No bound on the ordinary complex numbers c^{-1} is imposed: a Hahn series admits arbitrary complex coefficients, and a numerical small-divisor estimate is not a condition in this category. Let pr_γ be the projection of $\text{Mat}_n(\mathbb{C})$ onto the direct sum of the *whole* blocks $\text{Hom}(V_\mu, V_\lambda)$ with $\gamma = \lambda - \mu \in \mathbb{R}_{>0}$. The whole block is retained, not only the kernel of $\mathcal{H}_\gamma$ on it; on the other blocks $\mathcal{H}_\gamma$ is invertible by equation (24.2).

The distinction matters for a nonsemisimple residual matrix. Take

$$A_0^\circ = \begin{pmatrix} 1 & 1 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{pmatrix}, \quad A = A_0^\circ + tE_{23},$$

where E_{23} has its only nonzero entry, equal to 1, in row 2, column 3. On the resonant block $\text{Hom}(V_0, V_1) \cong \mathbb{C}^2$ at weight 1, write a column as $u = (u_1, u_2)^\top$. Then $\mathcal{H}_1 u = (-u_2, 0)^\top$ and $\ker \mathcal{H}_1 = \mathbb{C}(1, 0)^\top$. If the retained coefficient b were required to lie only in this kernel, the equation $\mathcal{H}_1 u = (0, 1)^\top - b$ would be impossible in its second coordinate. Retaining the whole block and setting $u = 0$ gives $b = (0, 1)^\top$. This normalization does not claim the smallest possible resonant representative; it specifies a unique one by fixing the whole resonant gauge block to zero.

Theorem 24.5 (Normalized resonant form). *There is a unique pair*

$$G_\natural = I + \sum_{\gamma \in S_{>0}^*} G_\gamma t^\gamma, \quad A^\natural = A_0^\circ + \sum_{\gamma \in S_{>0}^*} A_\gamma^\natural t^\gamma \quad (24.3)$$

satisfying

$$\partial_\tau G_\natural = AG_\natural - G_\natural A^\natural, \quad \text{pr}_\gamma G_\gamma = 0, \quad A_\gamma^\natural = \text{pr}_\gamma A_\gamma^\natural. \quad (24.4)$$

The inverse of $G_\natural$ has support in S^* , and $A^\natural - A_0^\circ$ has finite support contained in $\text{Lev}(A_0) \cap S^*$.

Proof. Suppose the coefficients of weight below γ have been constructed. The coefficient of t^γ in equation (24.4) is

$$\mathcal{H}_\gamma(G_\gamma) = R_\gamma - A_\gamma^\natural, \quad R_\gamma = A_\gamma + \sum_{\substack{\alpha+\beta=\gamma \\ \alpha, \beta > 0}} (A_\alpha G_\beta - G_\alpha A_\beta^\natural), \quad (24.5)$$

where α, β range over $S_{>0}^*$. The sum is finite, and every recursively constructed coefficient on the right has smaller weight (Corollary 24.2); A_γ is given input. On resonant blocks impose $G_\gamma = 0$, which forces the corresponding part of $A_\gamma^\natural$ to equal R_γ ; on the other blocks impose $A_\gamma^\natural = 0$ and invert $\mathcal{H}_\gamma$:

$$\boxed{A_\gamma^\natural = \text{pr}_\gamma R_\gamma, \quad G_\gamma = \mathcal{H}_\gamma^{-1}(I - \text{pr}_\gamma)R_\gamma,} \quad (24.6)$$

the inverse acting on the complementary block space only. Transfinite recursion on S^* defines all coefficients; their supports lie in the well-ordered S^* , so they are Hahn matrices, and the same equations prove the identity coefficient by coefficient and uniqueness under the stated normalization. Since $G_\natural - I$ has strictly positive support in S^* , the geometric series for its inverse is strongly summable and stays in S^* . Finally $\text{pr}_\gamma = 0$ unless $\gamma \in \text{Lev}(A_0)$, a finite set. If $S = \emptyset$, the construction is simply $G_\natural = I$ and $A^\natural = A_0^\circ$. $\square$

Already at this stage, if $S \subseteq \Gamma$ for an additive subgroup $\Gamma \subseteq \mathbb{R}$, then $S^* \subseteq \Gamma$ and the gauge, its inverse and the normalized coefficient all lie over K_Γ . There is no need for $1 \in \Gamma$ in the Euler frame. The later real-monomial shear can require additional exponents, as recorded in Theorem 28.1 and Remark 28.3.

Remark 24.6 (Where the difficulty lies). Solving a formal Sylvester equation at each exponent would not prove Theorem 24.5. One must also show that the recursion has only finite coefficient contributions, produces a well-ordered support and gives an invertible Hahn matrix; those assertions are what permit a support that accumulates below a finite exponent. Conversely there is no analytic norm estimate to prove: arbitrarily large coefficients do not invalidate a Hahn series.

24.3 Finite resonance dependence, not merely high-tail invariance

Theorem 24.7 (Finite resonance theorem). *Fix A_0° , its generalized-eigenspace decomposition, and a well-ordered positive support set S . The matrices $A_\gamma^\natural$ of Theorem 24.5 depend only on the input coefficient matrices at the finite set $S_{\text{crit}} = S_{\text{crit}}(\text{Lev}(A_0))$. Every entry of every $A_\gamma^\natural$ is a polynomial in those input entries, with coefficients obtained from the nilpotent blocks N_λ° and the inverses of the nonzero numbers $\gamma' - \lambda + \mu$ occurring in the finite dependency calculation. If $S \neq \emptyset$ and $\varepsilon = \min S$, the polynomial for $A_r^\natural$, $r \in \text{Lev}(A_0)$, has degree at most $\lfloor r/\varepsilon \rfloor$. In particular, replacing A by*

$$A^{\text{crit}} = A_0^\circ + \sum_{s \in S_{\text{crit}}} A_s t^s \quad (24.7)$$

does not change $A^\natural$.

Proof. When $S = \emptyset$, both systems in the assertion are A_0° and there is no positive coefficient to compute. Otherwise put $\varepsilon = \min S$ and give A_s the input weight s . Equations (24.5) and (24.6) build G_γ and $A_\gamma^\natural$ by addition, weight-preserving fixed linear operators and weight-adding products, so each positive-weight coefficient is a polynomial in input entries all of whose terms have total input weight γ ; finiteness and termination are Lemmas 24.1 and 24.4, and each term has at most $\lfloor \gamma/\varepsilon \rfloor$ input factors. The only possibly nonzero positive coefficients of $A^\natural$ sit at $\gamma \in \text{Lev}(A_0)$, where all letters belong to S_{crit} by Lemma 24.4; setting every other input coefficient to zero leaves these polynomials unchanged, and uniqueness in Theorem 24.5 gives the same $A^\natural$ for equation (24.7). $\square$

Corollary 24.8 (Gauge-equivalent finite coefficient replacement). *The system A is $K_\mathbb{R}$ -gauge equivalent to the finite-support system A^{crit} . More generally, two systems with the same residual matrix, a common well-ordered positive support container S , and the same coefficients on $S_{\text{crit}}(\text{Lev}(A_0))$ are $K_\mathbb{R}$ -gauge equivalent. If $\text{Lev}(A_0) \cap S^* = \emptyset$ then $A^\natural = A_0^\circ$.*

Proof. If $G_\natural$ and $G_\natural^{\text{crit}}$ normalize the two systems to the same $A^\natural$, put $X = G_\natural(G_\natural^{\text{crit}})^{-1}$. Differentiating the inverse gives

$$\partial_\tau X = AX - XA^{\text{crit}},$$

since the two terms containing $A^\natural$ cancel. Thus $y = Xz$ carries the A^{crit} -system to the A -system. These matrices are in the common residual Jordan basis fixed above; conjugation by G_0 returns the statement to the original basis. The general assertion has the same proof. The gauge also lies in K_Γ whenever the common support container lies in Γ . If no Levelt resonance lies in S^* , every projection used in the recursion vanishes. $\square$

Corollary 24.9 (A useful but weaker cutoff statement). *If $\text{Lev}(A_0) \neq \emptyset$, changing or adding coefficients at exponents strictly greater than $\max \text{Lev}(A_0)$ cannot change $A^\natural$, provided the union support stays well ordered and positive. If $\text{Lev}(A_0) = \emptyset$, every positive perturbation normalizes to A_0° .*

Proof. A letter larger than $\max \text{Lev}(A_0)$ cannot occur in a positive word whose sum lies in $\text{Lev}(A_0)$; apply Theorem 24.7 to the union support. $\square$

Theorem 24.7 is stronger than Corollary 24.9. A support such as $\{1 - 1/j : j \geq 2\}$ has infinitely many members below 1, yet only finitely many of them can occur in exact word sums reaching a fixed finite list of resonances. The critical set is sufficient, not necessarily minimal: matrix products can vanish and word contributions can cancel, and the definition deliberately does not predict those coefficient-dependent cancellations from the support alone (Example 27.2).

25 Spectral selection and the two classifications

25.1 The real-monomial shear

Every real power belongs to $K_{\mathbb{R}}$, so $t^{D_{\text{re}}} \in \text{GL}_n(K_{\mathbb{R}})$ and $\partial_\tau t^{D_{\text{re}}} = D_{\text{re}} t^{D_{\text{re}}}$ by Lemma 23.1.

Theorem 25.1 (Constant frequency-and-nilpotent form). *For the original system, $F = G_0 G_\natural t^{D_{\text{re}}} \in \text{GL}_n(K_{\mathbb{R}})$ satisfies*

$$\partial_\tau F = AF - FA^\flat, \quad A^\flat = iD + N, \quad (25.1)$$

where $N \in \text{Mat}_n(\mathbb{C})$ is nilpotent, $N^n = 0$, and $[D, N] = 0$. The diagonal generalized-eigenspace blocks of N are the N_λ° ; its other possibly nonzero blocks are exactly the constant matrices obtained from the resonant $A_\gamma^\natural$ by the shear.

Proof. After G_0 and $G_\natural$ the coefficient is $A^\natural$, and the shear gives $A^\flat = t^{-D_{\text{re}}} A^\natural t^{D_{\text{re}}} - D_{\text{re}}$. The diagonal block $\lambda I + N_\lambda^\circ$ becomes $ib_\lambda I + N_\lambda^\circ$. A positive coefficient block from V_μ to V_λ is nonzero only if $\gamma = \lambda - \mu \in \mathbb{R}_{>0}$; then $b_\lambda = b_\mu$ and $\gamma = a_\lambda - a_\mu$, so conjugation by the shear multiplies its monomial by $t^{-a_\lambda + a_\mu}$ and makes it constant. Every remaining coefficient is therefore constant and connects only spaces with the same frequency, which proves $[D, N] = 0$. Within one frequency space, order the distinct values a_λ increasingly; every off-diagonal block of N maps a smaller value to a larger one. Thus N is block triangular with nilpotent diagonal blocks. Its characteristic polynomial is the product of their characteristic polynomials, hence is X^n ; Cayley–Hamilton gives $N^n = 0$. The gauge identity follows from the product rule. $\square$

For a fixed residual Jordan basis, N is determined polynomially by the finite resonance data of Theorem 24.7. Its block similarity invariants do not depend on the choices made, by Theorem 25.7 below; the displayed *entries* of N are not asserted to be canonical across choices of Jordan basis.

25.2 Euler modes in the ambient field

This is the point at which the ambient field, not an abstract Hahn field, becomes essential: it excludes solutions with completely arbitrary surreal supports.

Lemma 25.2 (Euler modes). (i) *If $x \in \mathbf{No}$ is finite, then $\partial_\tau x$ is infinitesimal.*

- (ii) For $b \in \mathbb{R} \setminus \{0\}$ the equation $\partial_\tau z = ibz$ has only $z = 0$ in $\mathbf{No}[i]$; for $b = 0$ its solutions are exactly $\mathbb{C}$.
- (iii) For $\lambda = a + ib \in \mathbb{C}$, the solutions in $\mathbf{No}[i]$ of $\partial_\tau z = \lambda z$ are $\mathbb{C}t^a$ if $b = 0$, and 0 if $b \neq 0$.

Proof. (i) For every positive real r both $\log \omega \pm rx$ are positive infinite, so (BM7) gives $t \pm r\partial x > 0$; hence $|\omega \partial x| < 1/r$ for every r . This strengthens $\partial \mathcal{O} \subseteq \mathfrak{m}$ of Lemma 3.8 to $\omega \partial \mathcal{O} \subseteq \mathfrak{m}$, and it is the forward half of the cut description equation (7.4) with $f = r^{-1} \log \omega$. (ii) This is the second proof of Corollary 8.8 with ∂_τ in place of ∂ and (i) in place of Lemma 3.8: since ∂_τ commutes with conjugation, $\partial_\tau(z\bar{z}) = 0$, so $z\bar{z}$ is a nonnegative ordinary real constant. If $z \neq 0$, this constant is positive, so the real and imaginary parts u, v are finite. By (i), $\partial_\tau u = -bv$ and $\partial_\tau v = bu$ are infinitesimal. Since b is a nonzero ordinary real, both u and v are infinitesimal, making $u^2 + v^2$ infinitesimal, contrary to its positive constant value. For $b = 0$, the assertion is the constant-field statement for ∂_τ . Equivalently $\partial z = i(-bt)z$, whose accumulated phase $b\tau$ is infinite for $b \neq 0$; this is the case $p = 1$ of Corollary 10.2 scaled by $-b$, but the route above needs no primitive and no (BM6). (iii) Write $z = t^a w$; then $\partial_\tau w = ibw$, and apply (ii). $\square$

Member 12 states that this scalar obstruction is not new: it is the Euler-scale specialization of the finite-primitive criterion of Part II, proved again only so that the ambient exclusion below rests on the published derivation clauses alone.

25.3 Spectral selection over the whole surcomplex field

Theorem 25.3 (Spectral selection and explicit solution envelope). *For a power-Hahn regular-singular system equation (23.4) put*

$$m_{\mathbb{R}}(A_0) = \sum_{\lambda \in \text{Eig}(A_0) \cap \mathbb{R}} \text{algmult}_{A_0}(\lambda).$$

The explicitly constructed $G_\tau = F \exp(\tau N) \in \text{GL}_n(K_{\mathbb{R}}[\tau])$ and the real diagonal matrix D , whose entries are the imaginary parts of the eigenvalues of A_0 with algebraic multiplicity, satisfy

$$\partial_\tau G_\tau = AG_\tau - G_\tau iD. \quad (25.2)$$

Every surcomplex solution, without any support restriction on the unknown, is uniquely of the form

$$y = G_\tau c, \quad c \in \ker D \subseteq \mathbb{C}^n. \quad (25.3)$$

Hence $\dim_{\mathbb{C}} \text{Sol}_{\mathbf{No}[i]}(A) = m_{\mathbb{R}}(A_0)$, a fundamental matrix exists in $\text{GL}_n(\mathbf{No}[i])$ if and only if every eigenvalue of A_0 is real, and every solution already lies in $K_{\mathbb{R}}[\tau]^n$.

Proof. Take F, D, N from Theorem 25.1. Since $N^n = 0$, $\exp(\tau N) = \sum_{k=0}^{n-1} \tau^k N^k / k!$ is a finite polynomial matrix in the actual surreal τ , with inverse $\exp(-\tau N)$, with $\partial_\tau \exp(\tau N) = N \exp(\tau N)$, and commuting with D ; no complex exponential at an infinite imaginary argument is used. The product rule gives $\partial_\tau G_\tau = AF \exp(\tau N) - F(iD + N) \exp(\tau N) + FN \exp(\tau N) = AG_\tau - G_\tau iD$. For $y \in \mathbf{No}[i]^n$ the substitution $y = G_\tau w$ turns the system into $\partial_\tau w = iDw$, and Lemma 25.2(ii) kills the coordinates with nonzero frequency and leaves the others arbitrary constants. The columns of G_τ indexed by zero frequencies are independent over $\mathbf{No}[i]$, hence over $\mathbb{C}$, and there are $m_{\mathbb{R}}(A_0)$ of them. If that number is n , G_τ is a fundamental matrix; otherwise all solution columns lie in a proper $\mathbf{No}[i]$ -subspace and no invertible solution matrix exists. $\square$

Corollary 25.4 (Perturbation invariance in the specified sector). *The dimension of the full surcomplex solution space is unchanged by arbitrary positive-order power-Hahn perturbations of the coefficient, even when the perturbation does not commute with A_0 or changes the resonant nilpotent matrix. Two systems with the same residual matrix are $\mathbf{No}[i]$ -gauge equivalent.*

Proof. D depends only on the residual eigenvalues, and the gauges G_τ of the two systems reduce both to the same iD . $\square$

Corollary 25.5 (Real systems). *If A has entries in $\mathbb{R}((t^{\mathbb{R}}))$, then $\dim_{\mathbb{R}} \text{Sol}_{\mathbf{No}}(A) = m_{\mathbb{R}}(A_0)$, a real fundamental matrix exists if and only if every residual eigenvalue is real, and all real solutions lie in $\mathbb{R}((t^{\mathbb{R}}))[\tau]^n$.*

Proof. If $y = u + iv$ solves the real system, so do u and v ; the complex solution space is the complexification of the real one. The envelope follows by taking real and imaginary parts of the $K_{\mathbb{R}}[\tau]$ representation. When every residual eigenvalue is real, choose G_0 real and run every recursion over the real coefficient field. $\square$

Remark 25.6 (The solution collection is a set). Before Theorem 25.3 one may regard $\text{Sol}_{\mathbf{No}[i]}(A)$ as a class of tuples. Equation (25.3) identifies it with the image of the ordinary set $\mathbb{C}^{m_{\mathbb{R}}(A_0)}$, so it is a set and its dimension is an ordinary finite vector-space dimension; no dimension theory for a proper-class vector space is invoked. The fields $K_{\mathbb{R}}$ and $K_{\mathbb{R}}(\tau)$ are sets.

25.4 Classification over the Hahn field and over the whole field

Decompose the constant form by residual frequency, $V = \bigoplus_b V_b$ with $A^b|_{V_b} = ibI + N_b$, N_b nilpotent, b ranging over the finite set $\text{Im Eig}(A_0)$, and $m_b = \dim V_b$.

Theorem 25.7 (Classification over the power-Hahn field). *Let $A^b = iD + N$ and $A^b = iD' + N'$ be two constant forms, possibly of different sizes. Every $K_{\mathbb{R}}$ -valued solution X of $\partial_\tau X = A^b X - X A^b$ is a constant matrix, vanishes between distinct residual frequencies, and satisfies*

$$N_b X_b = X_b N'_b \quad (25.4)$$

on equal-frequency blocks. Consequently two power-Hahn regular-singular systems are $K_{\mathbb{R}}$ -gauge equivalent exactly when, for every b , their nilpotent matrices N_b have the same Jordan block sizes: the residual frequencies with these partitions form a complete invariant.

Proof. On a block from V'_c to V_b expand $X = \sum_\gamma X_\gamma t^\gamma$. Coefficient comparison gives

$$(\gamma - i(b - c))X_\gamma = N_b X_\gamma - X_\gamma N'_c. \quad (25.5)$$

The operator $X \mapsto N_b X - X N'_c$ has $(m_b + m'_c - 1)$ -st power zero: its two summands commute, and every term in that power contains either $N_b^{m_b}$ or $(N'_c)^{m'_c}$. If the scalar $\gamma - i(b - c)$ is nonzero, a finite geometric inverse therefore forces $X_\gamma = 0$. This scalar vanishes exactly when $\gamma = 0$ and $b = c$, because γ, b, c are real. The surviving coefficient satisfies equation (25.4), and conversely every such constant matrix is a morphism.

An invertible morphism is invertible on each frequency space, where equation (25.4) makes the nilpotent matrices similar; similarity of nilpotent complex matrices is determined by Jordan block sizes. For the original systems, transport a constant intertwiner C to $FC(F')^{-1}$. Thus constancy is a statement in the reduced frames; the original morphism need not have constant entries. $\square$

Corollary 25.8 (Exact morphism dimensions over $K_{\mathbb{R}}$). *If the nilpotent Jordan block sizes at frequency b are $\nu_{b,1}, \dots, \nu_{b,r_b}$ for A and $\nu'_{b,1}, \dots, \nu'_{b,r'_b}$ for A' , then*

$$\dim_{\mathbb{C}} \operatorname{Hom}_{K_{\mathbb{R}}}(A', A) = \sum_b \sum_{j=1}^{r_b} \sum_{k=1}^{r'_b} \min(\nu_{b,j}, \nu'_{b,k}).$$

Proof. A map between nilpotent Jordan blocks of sizes p, q is a $\mathbb{C}[X]$ -linear map $\mathbb{C}[X]/(X^q) \rightarrow \mathbb{C}[X]/(X^p)$, determined by the image of 1. That image must be annihilated by X^q , and the permitted subspace has basis

$$X^{\max(p-q,0)}, X^{\max(p-q,0)+1}, \dots, X^{p-1}.$$

There are $\min(p, q)$ basis elements. Add over block pairs with the same frequency; distinct frequencies contribute zero. $\square$

Theorem 25.9 (Classification over the whole surcomplex field). *Two power-Hahn regular-singular systems are $\mathbf{No}[i]$ -gauge equivalent if and only if the multisets*

$$\{\operatorname{Im} \lambda : \lambda \in \operatorname{Eig}(A_0), \text{ counted with algebraic multiplicity}\}$$

agree. More generally $\dim_{\mathbb{C}} \operatorname{Hom}_{\mathbf{No}[i]}(A', A) = \sum_b m_b m'_b$, and every ambient morphism already has entries in $K_{\mathbb{R}}[\tau]$.

Proof. With G_{τ}, G'_{τ} from Theorem 25.3, a morphism is $X = G_{\tau} Z (G'_{\tau})^{-1}$ with $\partial_{\tau} Z = iDZ - ZiD'$, whose scalar entries solve $\partial_{\tau} z = i(b - c)z$. By Lemma 25.2(ii) unequal frequencies give zero and equal ones give arbitrary complex constants. Hence the morphism space is the direct sum of the matrix spaces $\operatorname{Hom}_{\mathbb{C}}(V'_b, V_b)$, of dimensions $m_b m'_b$. An invertible morphism exists exactly when $m_b = m'_b$ for every b , which is equality of the frequency multisets. Both $G_{\tau} = F \exp(\tau N)$ and its inverse $\exp(-\tau N) F^{-1}$ have entries in $K_{\mathbb{R}}[\tau]$, so the same is true of every transported morphism. $\square$

Remark 25.10 (What the two classifications distinguish). The real parts of the residual eigenvalues are already removable over $K_{\mathbb{R}}$, because every real monomial t^a is available. Nilpotent Levelt data are not in general removable over $K_{\mathbb{R}}$, but become removable after adjoining τ . The residual frequencies remain invariants even over $\mathbf{No}[i]$. Passing from $K_{\mathbb{R}}$ to $\mathbf{No}[i]$ therefore erases exactly the nilpotent data in this class, and not the frequency multiplicities.

For a two-dimensional example, take the constant Euler coefficients

$$N = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix} \quad \text{and} \quad 0_2.$$

Both frequency multisets are $\{0, 0\}$. Over $K_{\mathbb{R}}$, their Jordan partitions are (2) and $(1, 1)$, so they are not gauge equivalent. Over $\mathbf{No}[i]$, the matrix $I + \tau N$, with inverse $I - \tau N$, satisfies $\partial_{\tau}(I + \tau N) = N(I + \tau N)$ and gives an equivalence. The endomorphisms of the first system over $K_{\mathbb{R}}$ are precisely $\begin{pmatrix} a & b \\ 0 & a \end{pmatrix}$ with $a, b \in \mathbb{C}$, a two-dimensional space. Over $\mathbf{No}[i]$ they are $(I + \tau N)C(I - \tau N)$ for arbitrary $C \in \operatorname{Mat}_2(\mathbb{C})$, a four-dimensional space.

The normal forms also make tensor constructions explicit. For systems of sizes n, n' , the tensor coefficient is $A \otimes I_{n'} + I_n \otimes A'$, as the product rule on $y \otimes y'$ shows. The notation $A \otimes A'$ for a system in the next corollary means this differential tensor product, not the matrix $A \otimes A'$ alone. If A and A' have frequency blocks (b, N_b) and (c, N'_c) , their tensor product has blocks $(b + c, N_b \otimes I + I \otimes N'_c)$, the nilpotent sum being nilpotent because its two terms commute; a dual has frequencies $-b$ and nilpotents $-N_b^{\top}$. These follow by differentiating a tensor product and a pairing.

Corollary 25.11 (Solution dimensions of tensor products). *Over $\mathbf{No}[i]$,*

$$\dim_{\mathbb{C}} \text{Sol}_{\mathbf{No}[i]}(A \otimes A') = \sum_b m_b m'_{-b}.$$

In particular two rank-one systems with no nonzero ambient solution, with frequencies $b \neq 0$ and $-b$, have a tensor product with a fundamental solution.

Proof. Use the tensor gauge $G_\tau \otimes G'_\tau$. It reduces the coefficient to $i(D \otimes I + I \otimes D')$, whose scalar frequencies are the pairwise sums $b + c$. The Euler-mode lemma leaves exactly the constant coordinates with $b + c = 0$, of total dimension $\sum_b m_b m'_{-b}$. This proves the formula, including the stated rank-one case. $\square$

This does not contradict the obstruction: a nonzero horizontal vector of a tensor product need not be a tensor product of horizontal vectors of its factors, and it is the frequency sum, not the separate existence of solutions, that controls the answer. It agrees with the rank-one statement of Corollary 9.4 (obstructions add) and with the tensor rule of Proposition 15.13 (phases add pairwise), and it is proved here without the import of Theorem 15.6.

25.5 The phase spectrum of a regular-singular system

Proposition 25.12 (An explicit non-Hermitian sector of the phase spectrum). *For every power-Hahn regular-singular system, with $\lambda_1, \dots, \lambda_n$ the eigenvalues of A_0 counted with algebraic multiplicity,*

$$\text{Ph}(-tA) = \{(\text{Im } \lambda_1) \tau, \dots, (\text{Im } \lambda_n) \tau\} \subset \mathcal{P}, \quad (25.6)$$

and the gauge $G_\tau \in \text{GL}_n(K_{\mathbb{R}}[\tau])$ of Theorem 25.3 realizes the normal form of Theorem 15.9 explicitly: $(-tA)^{G_\tau} = i \text{diag}(\partial((\text{Im } \lambda_j) \tau))$. Consequently $m_0(-tA) = m_{\mathbb{R}}(A_0)$, and Theorem 25.9 is the completeness statement of Theorem 15.9 for this class.

Proof. Multiply equation (25.2) by $-t$: $\partial G_\tau = (-tA)G_\tau - G_\tau(-itD)$, so $(-tA)^{G_\tau} = -itD = i \text{diag}(-b_j t)$, and $-b_j t = \partial(b_j \tau)$ by equation (23.1). Each $b_j \tau = -b_j \log \omega$ lies in $\mathcal{P}$. The multiset is independent of the gauge by the uniqueness half of Theorem 15.9, which is Lemma 15.3 and uses no import. $\square$

Only the *existence* half of Theorem 15.9 rests on the imported Theorem 15.6; for this class it is replaced by the explicit G_τ . The coefficient $-tA$ is in general neither Hermitian nor normal, so Proposition 25.12 is a genuinely non-Hermitian sector in which the phase spectrum is computed from finite data: the residual eigenvalues. It answers item N24 and the open continuation “compute the phase spectrum for broader non-Hermitian classes” for this class and for no other. Member 12 states the comparison in the same terms: Part III gives an abstract phase-multiset normal form and a constructive Hermitian sector, and this part computes one concrete non-Hermitian sector by a different, elementary route.

Remark 25.13 (The eigenvalue recipe in this class — an observation of the merge). The following observation is not stated by either source; it is recorded, with its proof, because it locates Proposition 25.12 relative to Theorem 16.5 and Example 16.10. In this class the naive recipe — integrate the imaginary parts of the instantaneous eigenvalues of the ∂ -coefficient and keep purely infinite parts — returns the correct spectrum. The characteristic polynomial of A is monic with coefficients in the valuation ring $K_{\mathbb{R}}^{\geq 0}$ of the algebraically closed field $K_{\mathbb{R}}$, and it reduces under st to the characteristic

polynomial of A_0 . To see that every root μ is finite, suppose $v(\mu) < 0$. In a monic equation of degree n , the term μ^n then has strictly smaller valuation than every lower-degree term, whose coefficient has nonnegative valuation; a unique least term cannot cancel. Thus every eigenvalue μ_j belongs to $K_{\mathbb{R}}^{\geq 0}$. Reducing the factorization $\prod_j (X - \mu_j)$ coefficientwise gives $\prod_j (X - \text{st}(\mu_j))$, the characteristic polynomial of A_0 . Its root multiset is therefore the residual eigenvalue multiset, including multiplicities. After renumbering, $\mu_j - \lambda_j \in K_{\mathbb{R}}^{> 0}$. The eigenvalues of $-tA$ are $-t\mu_j$, and $\text{Im}(-t\mu_j) = -tb_j - t \text{Im}(\mu_j - \lambda_j)$, whose second term is a real Hahn series of valuation greater than 1, hence in $\mathcal{A}$ by Corollary 18.4. So $\mathfrak{o}(\text{Im}(-t\mu_j)) = \mathfrak{o}(-b_j t) = b_j \tau$. Example 16.10 lies outside this class: the ∂_τ -coefficient $-\omega A$ of its system has infinite entries. The observation does not extend the recipe beyond the class. Finding other classes with computable phase formulas remains open; existence and completeness of the abstract normal form are already supplied by Theorem 15.9.

26 Hahn solutions, the logarithm, and forced equations

Let V_0 be the zero-frequency space and N_0 the restriction of N to it; it may have dimension zero.

Theorem 26.1 (The Hahn solution space). $\text{Sol}_{K_{\mathbb{R}}}(A) = F \ker N_0$ and $\dim_{\mathbb{C}} \text{Sol}_{K_{\mathbb{R}}}(A) = \dim_{\mathbb{C}} \ker N_0$. A fundamental matrix exists in $\text{GL}_n(K_{\mathbb{R}})$ if and only if $D = 0$ and $N = 0$, and

$$\dim_{\mathbb{C}} \text{Sol}_{\mathbf{No}[i]}(A) - \dim_{\mathbb{C}} \text{Sol}_{K_{\mathbb{R}}}(A) = \dim_{\mathbb{C}} V_0 - \dim_{\mathbb{C}} \ker N_0$$

counts exactly the additional solutions supplied by the logarithm.

Proof. Apply equation (25.5) to a column whose source is the trivial rank-one system: only its exponent-zero, zero-frequency coefficient survives, and N_0 must annihilate it. A fundamental matrix needs n independent columns, which happens exactly when every frequency vanishes and $N = 0$; then F itself is one. $\square$

Lemma 26.2 (The logarithm is transcendental over $K_{\mathbb{R}}$). No element of $K_{\mathbb{R}}$ has ∂_τ -derivative 1, and τ is transcendental over $K_{\mathbb{R}}$.

Proof. $\partial_\tau f = 1$ means $\partial f = -t$, and $-t \notin \partial K_{\mathbb{R}}$ by equation (18.3) (Example 18.2); directly, the t^0 -coefficient of $\partial_\tau f$ vanishes by equation (23.3).

If τ were algebraic, let $P(X) = X^d + a_{d-1}X^{d-1} + \cdots + a_0$ be its monic minimal polynomial over $K_{\mathbb{R}}$, with $d \geq 1$. Differentiating $P(\tau) = 0$ and using $\partial_\tau \tau = 1$ gives

$$(P^{\partial_\tau} + P')(\tau) = 0,$$

where P^{∂_τ} differentiates the coefficients and P' is the formal polynomial derivative. The polynomial in parentheses has degree at most $d - 1$, so minimality forces it to vanish identically. Its X^{d-1} -coefficient is $\partial_\tau a_{d-1} + d$. Hence $\partial_\tau(-a_{d-1}/d) = 1$ with $-a_{d-1}/d \in K_{\mathbb{R}}$, a contradiction. This argument uses characteristic zero and the coefficient formula, without a trace or algebraic-closedness assertion. $\square$

Theorem 26.3 (Minimal logarithmic repair). Assume every residual eigenvalue is real, so $D = 0$. If $N = 0$ a fundamental matrix already lies in $\text{GL}_n(K_{\mathbb{R}})$. If $N \neq 0$, let ν be its nilpotence index. Then:

- (i) every ambient fundamental matrix is $F \exp(\tau N) C_0$ with $C_0 \in \text{GL}_n(\mathbb{C})$, of polynomial degree exactly $\nu - 1$ in τ over $K_{\mathbb{R}}$;

- (ii) the field over $K_{\mathbb{R}}$ generated by the entries of a fundamental matrix and its inverse determinant is exactly $K_{\mathbb{R}}(\tau)$;
- (iii) this extension has no new constants, and its group of differential $K_{\mathbb{R}}$ -automorphisms is $(\mathbb{C}, +)$, acting by $\tau \mapsto \tau + c$.

So one actual logarithm repairs every finite nilpotent Levelt resonance; no tower of logarithms is needed in this class.

Proof. (i) follows from equation (25.3) with $D = 0$. The coefficient of $\tau^{\nu-1}$ is $FN^{\nu-1}C_0/(\nu-1)! \neq 0$, and transcendence (Lemma 26.2) makes polynomial degree unambiguous. (ii) The field generated contains $\exp(\tau N) = F^{-1}(\cdot)C_0^{-1}$; the finite logarithm identity $\sum_{j=1}^{n-1}(-1)^{j+1}(E-I)^j/j = \tau N$ for $E = \exp(\tau N)$ recovers τ from a nonzero entry of N . The identity is the formal identity $\log(\exp X) = X$ modulo X^n , evaluated at the nilpotent matrix τN ; every displayed sum is finite. The reverse inclusion is the fundamental-matrix formula. (iii) As a differential subfield of $\mathbf{No}[i]$, $K_{\mathbb{R}}(\tau)$ has constants $\mathbb{C}$. A differential automorphism sends τ to another primitive of 1, that is to $\tau + c$ with $c \in \mathbb{C}$; conversely transcendence makes each translation a field automorphism, and $\partial_{\tau}(\tau + c) = 1$ makes it differential. $\square$

Remark 26.4 (An additive extension inside $\mathbf{No}[i]$). Theorem 26.3 is the usual additive Picard–Vessiot extension, proved directly. It contrasts with the oscillatory extensions of Section 21: their groups are tori $\mathbf{G}_m(\mathbb{C})^r$ and they admit no differential embedding into $\mathbf{No}[i]$ (Corollary 21.8), whereas the group here is $(\mathbb{C}, +)$ and the extension $K_{\mathbb{R}}(\tau)$ already lies inside $\mathbf{No}[i]$. The logarithm is missing from the small base $K_{\mathbb{R}}$, not from the ambient field. If only some frequencies vanish, the logarithmic statement applies to the zero-frequency summand; adjoining τ does *not* manufacture a surcomplex fundamental matrix on a nonzero-frequency summand, because the obstruction of Lemma 25.2(ii) is untouched.

We now solve $\partial_{\tau}y = Ay + f$ for arbitrary $f \in K_{\mathbb{R}}^n$, which separates an obstruction to a Hahn solution from an obstruction to any surcomplex solution. For $f \in K_{\mathbb{R}}^n$ put $g = F^{-1}f$, write $g_b = \sum_{\gamma} g_{\gamma,b} t^{\gamma}$ on the frequency space V_b , and define the *Hahn obstruction*

$$\text{hob}_A(f) = [g_{0,0}] \in V_0/N_0V_0. \quad (26.1)$$

The map depends on the chosen normalizing identification; its vanishing and the resulting cokernel do not.

Theorem 26.5 (Exact Hahn obstruction sequence). *There is an exact sequence of $\mathbb{C}$ -vector spaces*

$$0 \longrightarrow \ker N_0 \xrightarrow{v \mapsto Fv} K_{\mathbb{R}}^n \xrightarrow{\partial_{\tau} - A} K_{\mathbb{R}}^n \xrightarrow{\text{hob}_A} V_0/N_0V_0 \longrightarrow 0. \quad (26.2)$$

So $\partial_{\tau} - A$ has finite-dimensional kernel and cokernel of equal dimension, and a Hahn solution of $\partial_{\tau}y - Ay = f$ exists if and only if $g_{0,0} \in N_0V_0$. Every forcing $f \in K_{\mathbb{R}}^n$ nevertheless has a solution in $K_{\mathbb{R}}[\tau]^n$, of τ -degree at most the nilpotence index ν_0 of N_0 ($\nu_0 = 1$ when $N_0 = 0$). If y_p is the constructed particular solution, all ambient solutions are

$$y = y_p + G_{\tau}c, \quad c \in \ker D \subseteq \mathbb{C}^n.$$

They all belong to $K_{\mathbb{R}}[\tau]^n$ and have the same degree bound. If $V_0 = 0$ there is a unique solution in $K_{\mathbb{R}}^n$, and it is also the unique solution in $\mathbf{No}[i]^n$.

Proof. The substitution $y = Fz$ gives $\partial_\tau(Fz) - AFz = F(\partial_\tau z - A^b z)$. For $(\gamma, b) \neq (0, 0)$ the coefficient equation has the unique solution

$$z_{\gamma,b} = ((\gamma - ib)I - N_b)^{-1} g_{\gamma,b} = \sum_{k=0}^{m_b-1} \frac{N_b^k g_{\gamma,b}}{(\gamma - ib)^{k+1}}, \quad (26.3)$$

with support inside $\text{supp } g_b$: at each exponent it applies a finite ordinary complex linear map to the coefficient vector and introduces no new exponent. Taking the finite union over the frequency blocks therefore gives a Hahn vector, even if the inverse coefficients are unbounded as ordinary complex numbers. At $(0, 0)$ the equation is $-N_0 z_{0,0} = g_{0,0}$, solvable exactly when $g_{0,0} \in N_0 V_0$, with homogeneous freedom $\ker N_0$. This proves exactness at the two copies of $K_{\mathbb{R}}^n$ and on the left; surjectivity of hob_A follows from $f = Fv$ with constant $v \in V_0$, and rank-nullity for N_0 gives equal dimensions. For an arbitrary resonant forcing, replace the missing coefficient solution by

$$p(\tau) = \sum_{k=0}^{\nu_0-1} \frac{\tau^{k+1}}{(k+1)!} N_0^k g_{0,0}. \quad (26.4)$$

Indeed,

$$\partial_\tau p = \sum_{k=0}^{\nu_0-1} \frac{\tau^k}{k!} N_0^k g_{0,0}, \quad N_0 p = \sum_{k=1}^{\nu_0} \frac{\tau^k}{k!} N_0^k g_{0,0}.$$

The last term of the second sum vanishes and all positive-degree terms cancel, leaving $(\partial_\tau - N_0)p = g_{0,0}$. Add p to the nonresonant Hahn solution and multiply by F to obtain a particular solution in $K_{\mathbb{R}}[\tau]^n$. The difference of any two ambient solutions is homogeneous, so Theorem 25.3 gives exactly the stated affine family. Its homogeneous term uses only N_0 and has degree at most $\nu_0 - 1$, proving the common degree bound. If $V_0 = 0$, nothing is missing and the homogeneous solution space is zero, giving uniqueness even in $\mathbf{No}[i]^n$. $\square$

The degree bound for forcing is sharp. Let $A = N$ be the nilpotent Jordan block of size ν , with $Ne_j = e_{j-1}$ for $j > 1$ and $Ne_1 = 0$, and force by the constant vector e_ν . Formula (26.4) gives

$$p(\tau) = \tau e_\nu + \frac{\tau^2}{2!} e_{\nu-1} + \cdots + \frac{\tau^\nu}{\nu!} e_1.$$

Every ambient solution differs from p by $\exp(\tau N)c$, $c \in \mathbb{C}^\nu$, which has degree at most $\nu - 1$. The leading coefficient $e_1/\nu!$ cannot cancel, so every solution has degree exactly ν over $K_{\mathbb{R}}$. This includes $\nu = 1$, where the equation is simply $\partial_\tau y = 1$.

Remark 26.6 (Algebraic, not topological, Fredholm terminology; relation to item N22). The equal-dimension statement may be called an algebraic index-zero statement. No Banach space, closed-range argument, compact operator or topology on $\mathbf{No}[i]$ is involved, and the existence after adjoining τ is proved only for forcing in $K_{\mathbb{R}}^n$; nothing is inferred for arbitrary surcomplex forcing. For this class and this forcing, Theorem 26.5 gives inhomogeneous existence *without* the import of Theorem 15.6: the existence statement of Corollary 15.11 for $\partial y - (-tA)y = -tf$ is recovered import-free when $f \in K_{\mathbb{R}}^n$. Item N22 is amended accordingly and not deleted.

27 Worked examples and exact resonance certificates

Write E_{jk} for a matrix unit. All examples use $\partial_\tau t^a = at^a$ and $\partial_\tau \tau = 1$.

Example 27.1 (A logarithm created by two nonresonant coefficients). Let

$$A = \text{diag}(2, 1, 0) + t^{1/2}E_{12} + t^{3/2}E_{23}.$$

Here $\text{Lev}(A_0) = \{1, 2\}$ and $S = \{\frac{1}{2}, \frac{3}{2}\}$. The first coefficient connects eigenvalues with difference 1 at weight $\frac{1}{2}$, the second at weight $\frac{3}{2}$; neither lies in its resonant block. Their ordered product maps the 0-space to the 2-space at weight 2. The exact normalized form is

$$G_{\natural} = I - 2t^{1/2}E_{12} + 2t^{3/2}E_{23}, \quad A^{\natural} = \text{diag}(2, 1, 0) + 2t^2E_{13},$$

the resonant coefficient arising from $A_{1/2}G_{3/2} = 2E_{13}$; no resonant input coefficient was present. With $t^{D_{\text{re}}} = \text{diag}(t^2, t, 1)$ the constant nilpotent is $N = 2E_{13}$, and

$$Y = G_{\natural}t^{D_{\text{re}}}(I + 2\tau E_{13}) = \begin{pmatrix} t^2 & -2t^{3/2} & 2\tau t^2 \\ 0 & t & 2t^{3/2} \\ 0 & 0 & 1 \end{pmatrix}$$

is a fundamental matrix with $\det Y = t^3$; its third column differentiates to $(2t^2 + 4\tau t^2, 3t^{3/2}, 0)^{\text{T}}$, which is A times the column. The surcomplex solution space has dimension 3, the Hahn solution space dimension $\dim \ker(2E_{13}) = 2$, and since $N^2 = 0 \neq N$ one logarithm of degree one is unavoidable. The whole input is critical; adding an arbitrary coefficient at weight $\frac{7}{3}$, or any positive well-ordered family entirely above 2, leaves N unchanged, although $G_{\natural}$ changes.

Example 27.2 (Accumulating small divisors without a logarithm). Let $A = \text{diag}(1, 0) + fE_{12}$ with $f = \sum_{j \geq 2} t^{1-1/j}$, whose support increases to 1 and is well ordered. Put $h = -\sum_{j \geq 2} j t^{1-1/j}$. Coefficientwise $((1 - \frac{1}{j}) - 1)(-j) = 1$ gives $(\partial_{\tau} - 1)h = f$, and since $E_{12}^2 = 0$, $G_{\natural} = I + hE_{12}$ satisfies $\partial_{\tau}G_{\natural} = AG_{\natural} - G_{\natural}A_0$. So $A^{\natural} = A_0$, $N = 0$, and $G_{\natural} \text{diag}(t, 1)$ is already a Hahn fundamental matrix. The divisors $(1 - \frac{1}{j}) - 1$ tend to zero and the coefficients of h grow without bound; neither is an obstruction. Every letter is at least $\frac{1}{2}$, so a word summing to 1 has length at most two. There is no letter 1, and a length-two word must have both letters equal to $\frac{1}{2}$. Its matrix product E_{12}^2 vanishes, so $S_{\text{crit}} = \{\frac{1}{2}\}$ is a valid finite dependence set while the minimal one here is empty.

Example 27.3 (A nonsemisimple residual matrix). Take $A = E_{12} + tE_{21}$. Both residual eigenvalues are 0 and $\text{Lev}(A_0)$ is empty, so $A^{\natural} = A_0^{\circ} = E_{12}$ and $N = E_{12}$. In the first step $\mathcal{N}(X) = E_{12}X - XE_{12}$ has $\mathcal{N}^3 = 0$ and

$$G_1 = (I - \mathcal{N})^{-1}E_{21} = E_{21} + \text{diag}(1, -1) - 2E_{12};$$

the recursion supplies all later coefficients. The ambient dimension is 2, the Hahn dimension 1, and the fundamental matrix needs τ -degree one. Dividing entrywise by eigenvalue differences would miss the nilpotent terms of equation (24.2).

Example 27.4 (A noncommuting mixed-frequency system). Let $A = \text{diag}(0, i) + t(\frac{1}{3} \frac{2}{4})$. The two displayed matrices do not commute, the residual matrix is semisimple, and no difference of distinct eigenvalues is a positive real, so $A^{\natural} = \text{diag}(0, i)$ and $N = 0$, with first normalizing coefficient

$$G_1 = \begin{pmatrix} 1 & 1-i \\ \frac{3}{2}(1+i) & 4 \end{pmatrix};$$

the remaining G_j exist in $\mathbb{C}[[t]]$ by the same recursion. Exactly one column of $G_{\natural}$ gives a nonzero family of homogeneous surcomplex solutions; the other residual mode acquires none, even from an arbitrary surreal support. The dimension is 1, not 2, and $\text{Ph}(-tA) = \{0, \tau\}$ by Proposition 25.12.

Example 27.5 (A forced equation with no Hahn solution). For $N = E_{12}$ and $g = (0, 1)^\top$, the equation $\partial_\tau z = Nz + g$ has no solution in $K_{\mathbb{R}}^2$, since its second row would be $\partial_\tau z_2 = 1$ (Lemma 26.2); equivalently $g \notin NV$. It has the particular solution $z = (\tau^2/2, \tau)^\top$ in $K_{\mathbb{R}}[\tau]^2$, since $\partial_\tau z = (\tau, 1)^\top = Nz + g$. The τ -degree two for this forcing exceeds the degree one needed for a homogeneous fundamental matrix, as equation (26.4) allows.

Example 27.6 (A residue change can change the answer discontinuously). The scalar equation $\partial_\tau y = (a + ib)y$ has a one-dimensional solution space for $b = 0$ and a zero one for every nonzero real b , however small in the ordinary metric. The invariance of Corollary 25.4 is invariance under positive *Hahn-order* perturbations, not under numerically small changes of the residual matrix; confusing the two notions of smallness would give a false stability claim.

28 Smaller workspaces, the power-Hahn boundary, and an exact procedure

28.1 Finite monomial adjunctions

Theorem 28.1 (Finite-monomial and one-logarithm envelope). *Let $\Gamma \subseteq \mathbb{R}$ be any additive subgroup, let $A \in \text{Mat}_n(K_{\mathbb{R}}^{\geq 0})$, and let its residual eigenvalues be $\lambda = a_\lambda + ib_\lambda$. The normalizing gauge $G_{\mathfrak{h}}$ of Theorem 24.5 and its inverse lie in $\text{Mat}_n(K_\Gamma)$, and the full constant normal form is obtained over the finitely generated extension $K_\Gamma(t^{a_\lambda} : \lambda \in \text{Eig}(A_0))$. Every ambient homogeneous solution already lies in*

$$K_\Gamma(t^a : a \in \text{Eig}(A_0) \cap \mathbb{R})[\tau]^n. \quad (28.1)$$

So finitely many actual real monomials and one logarithm suffice; no general differential closure is needed.

Proof. The recursion only adds positive input exponents, so all coefficients of $G_{\mathfrak{h}}$ and $G_{\mathfrak{h}}^{-1}$ have exponents in Γ , and so does the finite $A^{\mathfrak{h}}$. The shear uses only the listed monomials. For an ambient solution only the zero-frequency columns of $G_0 G_{\mathfrak{h}} t^{D_{\text{re}}} \exp(\tau N)$ occur; N preserves frequency spaces, so these columns use only generalized eigenspaces with real residual eigenvalues, whose shear entries are the monomials in equation (28.1). $\square$

These adjunctions are ordinary field generation inside $\mathbf{No}[i]$: an element of $K_\Gamma(t^{a_1}, \dots, t^{a_k})$ is a rational expression in finitely many added monomials with coefficients in K_Γ . The assertion does not replace this field by the entire Hahn field on $\Gamma + \mathbb{Z}a_1 + \dots + \mathbb{Z}a_k$, or claim closure under arbitrary Hahn sums there. The generated field is ∂_τ -stable, since $\partial_\tau t^{a_j} = a_j t^{a_j}$ and the product and quotient rules preserve rational expressions; adjoining τ , with $\partial_\tau \tau = 1$, also preserves this stability.

The monomial adjunction can be necessary. For $\Gamma = \mathbb{Q}$ the equation $\partial_\tau y = \sqrt{2}y$, that is $\partial y = -\sqrt{2}ty$, has the nonzero ambient solutions $ct^{\sqrt{2}}$, none of which lies in $K_{\mathbb{Q}}$. This is the second row of Table 5 with $\sqrt{2}$ replaced by $-\sqrt{2}$ — the same obstruction, an irrational exponent-1 coefficient in ∂ -form. Together with Example 20.7, where ∂ fails to preserve $K_{\mathbb{Q}\sqrt{2}}$, it shows why this part lets Γ be arbitrary: ∂_τ preserves every K_Γ , and the obstruction moves from stability of the workspace to membership of the monomial t^a .

Remark 28.2 (Rank one is the resonant normal form). For $n = 1$ and $\Gamma = \mathbb{Q}$ there are no Levelt resonances, and Theorems 25.3 and 28.1 say: $\partial_\tau y = (\lambda + g)y$ with $\lambda \in \mathbb{C}$ and $g \in K_{\mathbb{Q}}^{\geq 0}$ has a nonzero solution in $K_{\mathbb{Q}}$ exactly when $\lambda \in \mathbb{Q}$, the solutions being $cG_{\mathfrak{h}}t^\lambda$ with $G_{\mathfrak{h}} \in 1 + K_{\mathbb{Q}}^{\geq 0}$. In ∂ -form the

coefficient is $-\lambda t - tg$ with $v(tg) > 1$, and this is exactly Theorem 19.3: the logarithmic-derivative image of $\mathcal{H} = \mathbb{C}((t^{\mathbb{Q}}))$ is $\{-rt + g' : r \in \mathbb{Q}, g' = 0 \text{ or } v(g') > 1\}$, every member of which has valuation at least 1. So Part V is the matrix version of Remark 19.5: the rank-one exponent-1 residue becomes the residual matrix $-A_0$, its rational real part becomes the real parts a_λ to be sheared away, its imaginary part becomes the residual frequencies that no ambient gauge removes, and the new phenomenon in rank n is the finite nilpotent Levelt data N .

Remark 28.3 (Do not transfer the simplified classification backwards). Over K_Γ the real parts need not all be removable; even rank one retains the real exponent modulo Γ . The complete invariants of Theorem 25.7 are for $K_{\mathbb{R}} = K_{\mathbb{R}}$ only. Applied unchanged to $K_{\mathbb{Q}}$ they would identify $\partial_\tau y = \sqrt{2}y$ with the trivial equation, which is false over $K_{\mathbb{Q}}$.

More precisely, two constant rank-one Euler systems with coefficients $\lambda, \mu \in \mathbb{C}$ are K_Γ -gauge equivalent exactly when $\lambda - \mu \in \Gamma$. A gauge $x \neq 0$ must solve $\partial_\tau x = (\lambda - \mu)x$. Writing $x = \sum_{\gamma \in \Gamma} c_\gamma t^\gamma$ gives $(\gamma - \lambda + \mu)c_\gamma = 0$ at every exponent. Thus a nonzero gauge exists only when $\lambda - \mu = \gamma \in \Gamma$, and all such gauges are ct^γ with $c \in \mathbb{C}^\times$. Conversely these monomials are gauges. The invariant is therefore the imaginary part together with the real part modulo Γ . Rank-one positive perturbations first normalize over K_Γ , so the same criterion applies to their residual coefficients.

28.2 The power-Hahn hypothesis cannot be dropped

The theorems above are not a principle that “the standard part of every finite surreal coefficient controls all solutions”. Logarithmically slow scales refute that.

Proposition 28.4 (Infinitesimal coefficient, zero residue, no solution). *The scalar equation*

$$\partial_\tau y = \frac{i}{\log \omega} y \quad (28.2)$$

has no nonzero solution in $\mathbf{No}[i]$, although its coefficient is infinitesimal with standard part zero. The coefficient does not belong to $K_{\mathbb{R}}$.

Proof. In ∂ -form equation (28.2) is $\partial y = -iy/(\omega \log \omega)$, the first equation of Example 10.4 with b replaced by $-b$; its accumulated phase $-\log \log \omega$ is infinite, and a nonzero solution would make it finite by the polar decomposition (Theorem 8.4, whose direction (i) $\Rightarrow$ (iii) uses no primitive). Member 12’s own route: the derivation $\partial_1 = (\log \omega)\partial_\tau = -\omega \log \omega \partial$ satisfies $\partial_1 x \in \mathfrak{m}$ for finite real x , by (BM7) applied to the positive infinite $\log \log \omega \pm rx$, and the norm-square argument of Lemma 25.2(ii) applied to $\partial_1 y = iy$ excludes every nonzero solution. Finally $\log \omega < \omega^a$ for every real $a > 0$, a property of the surreal exponential [1, 3], so $1/\log \omega$ exceeds every t^a while being infinitesimal. If $1/\log \omega$ were in $K_{\mathbb{R}}$, its least exponent would be some $a_0 > 0$. Dividing by t^{a_0} would give a positive finite element with nonzero ordinary residue, hence $1/\log \omega < Ct^{a_0}$ for some ordinary $C > 0$. But $Ct^{a_0} < t^{a_0/2} < 1/\log \omega$, a contradiction. The use of $a_0/2$ makes the comparison independent of the leading coefficient. $\square$

Remark 28.5 (What the counterexample shows, and what it does not). Proposition 28.4 refutes the extension of this part from positive power-Hahn order to arbitrary infinitesimal surreal coefficients. It does *not* refute a classification based on richer logarithmic or differential phase data, and the scalar criterion of Part II already predicts the obstruction: in ∂ -form the imaginary coefficient $-1/(\omega \log \omega)$ has the infinite primitive $-\log \log \omega$ (Corollary 10.3). A layered analogue for coefficient fields containing $\log \omega$, $\log \log \omega$ and their real powers is recorded as an open continuation in Section 51.13. (Added with member 13: the equation is the case $s = \ell_1$, $\nu = -i$ of Corollary 37.1, the first-order obstruction at stage one of the log-atomic tower.)

28.3 An exact finite procedure, and what it does not compute

A concrete effective input class consists of finite positive rational supports and complex algebraic matrix entries, represented exactly with their algebraic roots specified. In this class one can compute a residual Jordan basis and eigenvalues, decide equality and whether an eigenvalue difference is positive real, and perform exact field arithmetic and matrix rank calculations. These coefficient operations are part of the input contract: finite rational support alone does not supply them for arbitrary named complex constants. The same finite procedure applies to another coefficient presentation only when it supplies these operations.

Fix the resulting A_0° and finite rational S . Write $R = \text{Lev}(A_0)$. If $R \neq \emptyset$, let $r_{\max} = \max R$ and choose a common denominator d with $S \subseteq d^{-1}\mathbb{Z}_{>0}$. Every monoid exponent at most $r_{\max}$ belongs to the finite grid

$$\{j/d : 0 \leq j \leq \lfloor dr_{\max} \rfloor\}.$$

Starting at 0 and adding letters of S while staying in this grid enumerates the required exponents. Apply equations (24.5) and (24.6) in increasing order, record the nonzero $A_r^\natural$, perform the shear and compute the Jordan forms of the N_b . If $R = \emptyset$, the normalized coefficient is already A_0° ; the nilpotent blocks of that residual matrix must still be retained. Empty support requires no recursion either. This determines all $K_{\mathbb{R}}$ -gauge invariants; the ambient invariants need only the residual eigenvalues.

A smaller certificate keeps only the weights

$$W = \{\gamma \in S^* : \text{there is } r \in R \text{ with } r - \gamma \in S^*\}.$$

These are precisely the partial sums of all words reaching a resonance, including their empty and full prefixes. The finite-word lemma makes W finite, even for an accumulating support when those words have been determined. If $\gamma \in W$ and $\alpha + \beta = \gamma$ with $\alpha, \beta \in S_{>0}^*$, then both $\alpha, \beta \in W$: for a witnessing r , $r - \alpha = (r - \gamma) + \beta$ and $r - \beta = (r - \gamma) + \alpha$ lie in S^* . Thus the recursion on these weights is closed under every predecessor pair. Moreover $S \cap W = S_{\text{crit}}(R)$, since a letter is in W exactly when it can be completed to a word reaching R . The certificate consists of these critical input coefficients, the homological inverses at positive weights in W , the resulting $A_r^\natural$ and the nilpotent Jordan data, each verified by finite identities. At a resonant weight the inverse is restricted to the complementary block space, as in equation (24.6). For example, if $S = \{\frac{2}{3}, 1, \frac{3}{2}\}$ and $R = \{1, 2\}$, then $W = \{0, \frac{2}{3}, 1, \frac{4}{3}, 2\}$ and $S_{\text{crit}}(R) = \{\frac{2}{3}, 1\}$. The coefficient at $\frac{3}{2}$ is below the largest resonance but cannot affect the normalized resonant data.

Three limits travel with this procedure. The full gauge $G_{\natural}$ may still have infinite support; a finite certificate for the invariants is not a finite listing of that gauge, whose exact specification is the support theorem and the recursion. Already the scalar coefficient $A = t$ has no Levelt resonances and normalizes to 0, but its normalized gauge is $\sum_{k \geq 0} t^k/k!$, with infinite support. For an arbitrary well-ordered support supplied only indirectly the critical set is finite, but finding all exact word sums that hit a resonance need not be an effective task under an unspecified representation; finitely many black-box coefficient queries cannot certify that an unseen coefficient at a resonant weight is absent, so Theorem 24.7 is a finite-dependence theorem and not an oracle algorithm. And an accumulating support means that “iterate through all exponents below $\max \text{Lev}(A_0)$ ” need not terminate, which is exactly why the finite-word form of Theorem 24.7 is stronger than a bounded-exponent prescription; for a symbolic family such as Example 27.2 the support description must be used to find the words.

Remark 28.6 (Relation to the open effectivity questions). Item N57 poses the identification of restricted, support-certified input languages on which eigenvalue primitives, their purely infinite

parts and a normalizing *unitary* gauge become effectively computable, and Warning 17.13 leaves open which nontriangular differential modules admit a filtration whose rank-one factors carry computable obstructions. For power-Hahn regular-singular systems this part answers both *in part*: the gauge G_τ splits the module over $K_{\mathbb{R}}[\tau]$ into rank-one factors whose obstructions $(\operatorname{Im} \lambda_j)\tau$ are read off from the residual matrix, and for finite rational supports with the effective coefficient operations specified above (in particular, algebraic coefficients), the procedure computes the $K_{\mathbb{R}}$ -invariants by a finite calculation. The gauge G_τ is not unitary, $G_{\mathfrak{h}}$ is not finitely listed, and nothing is said about other input languages; the unitary-gauge question of item N57 and the general filtration question stay open.

Table 7: Proof dependencies of Part V, after member 12’s audit register.

Result	Essential inputs	What is not used
Hahn normal form (Theorem 24.5)	Ordered-word lemma; Sylvester block inverse; transfinite recursion	Norm convergence; semisimple residue
Finite resonance data (Theorem 24.7)	Finite words at a fixed total weight; polynomial recursion	Finite support below a cutoff
Constant reduction (Theorem 25.1)	Real powers; finite directed resonance blocks	Analytic continuation; global complex exponential
Ambient mode exclusion (Lemma 25.2)	(BM7); constants; conjugation	Prescribed support for the unknown
Spectral selection (Theorem 25.3)	Constant reduction; finite nilpotent exponential; mode exclusion	Operator factorization or transferred surjectivity
Hahn classification (Theorem 25.7)	Coefficient comparison; scalar-plus-nilpotent inverse	Completeness of the ambient field
Ambient classification (Theorem 25.9)	Explicit logarithmic gauge; scalar mode exclusion	The abstract normal form Theorem 15.9
Forced equation (Theorem 26.5)	Finite coefficient resolvents; polynomial resonant primitive	An ambient forcing theorem
Minimal logarithm (Theorem 26.3)	Minimal-polynomial differentiation; finite matrix logarithm; constants	Algebraic closedness of the Hahn field

Several details are easy to miss and are recorded as the source records them. The sign in $\partial_\tau = -\omega\partial$ makes $\partial_\tau t = t$ and $\partial_\tau \tau = 1$. In equation (24.5) the second product is $G_\alpha A_\beta^{\mathfrak{h}}$, in that order and with a minus sign. The retained resonant space is a whole generalized-eigenspace Hom block. A Levelt resonance requires a *positive real* eigenvalue difference; an imaginary or negative difference cannot meet a positive real exponent. The nilpotent exponential is a finite polynomial. The solution count uses algebraic multiplicity, not the number of distinct real eigenvalues. And $\operatorname{Sol}_{K_{\mathbb{R}}}(A)$ can be smaller than $\operatorname{Sol}_{\mathbf{No}[i]}(A)$ although both fields have the constant field $\mathbb{C}$.

Part VI

Autonomous first-order equations: curves, derivation independence, and abelian logarithmic derivatives

This part is the contribution of member 11 (appendix C). It answers, for one precisely delimited nonlinear class, a question this report had explicitly declined: which solutions does a fixed autonomous algebraic first-order equation with ordinary complex coefficients have in $\mathbf{No}[i]$? The answer is complete for equations $F(y, \partial y) = 0$ with $F \in \mathbb{C}[Y, Z]$, and more generally for constant meromorphic vector fields on smooth projective complex curves (Theorem 30.1). Three features set the part apart.

- It is proved for *every normalized derivation* ∂ (Definition 29.1), not only for the fixed ∂ of Theorem 3.1, and it shows that the solution sets in this sector do not depend on the choice (Theorem 31.4). That is a scoped amendment to the one-derivation rule of Warnings 3.2 and 3.3 and items N2–N3, which are amended and not deleted (Convention 29.2).
- It lifts, for this class, the non-claims N21 and N25 and the remark after Corollary 8.11 that the Riccati corollary “is not a general nonlinear solvability theorem”. Those statements are kept as provenance and re-scoped; beyond the class they stand (Section 51.5).
- It adds a projective obstruction for $(\partial y)^2 = Q(y)$ and the exact logarithmic-derivative image of a constant abelian variety, the proper-group analogue of Theorem 8.1.

The part depends only on Parts I and II and on classical curve and formal-group theory.

29 Normalized derivations

Definition 29.1 (Normalized surreal derivation). A *normalized surreal derivation* is an additive map $\partial : \mathbf{No} \rightarrow \mathbf{No}$ with

$$\partial(xy) = x\partial y + y\partial x, \quad \partial(e^x) = e^x \partial x, \quad \ker \partial = \mathbb{R}, \quad x > \mathbb{R} \Rightarrow \partial x > 0, \quad \partial \omega = 1,$$

and *strong additivity*: the derivative family of every Hahn-summable family is summable and $\partial(\sum_j x_j) = \sum_j \partial x_j$. It is extended to $\mathbf{No}[i]$ by $\partial(x + iy) = \partial x + i\partial y$; the extension has constant field $\mathbb{C}$ and commutes with conjugation. In the numbering of Theorem 3.1 these are the clauses (BM1), (BM2), (BM3), (BM4), (BM5) and (BM7), without the surjectivity clause (BM6).

The Berarducci–Mantova derivation ∂ is one normalized derivation, and it is also surjective; its existence is the input for this part [1].

Convention 29.2 (The symbol ∂ , and the scope of the one-derivation rule). Three rules fix the use of the new symbol.

- (1) Throughout Part VI, ∂ denotes an *arbitrary* normalized derivation, and every statement holds for every such ∂ unless it is flagged. The abelian finite-primitive criterion (Corollary 33.10) assumes that ∂ is surjective; the explicit speed threshold of Corollary 33.11 needs no such assumption and holds for every normalized derivation. The source writes D . In this report D

never denotes a derivation other than the fixed one (Warning B.1), and the new symbol keeps the ∂ -specific statements of the other parts visibly distinct.

- (2) Outside this part every intrinsic statement remains relative to the one fixed ∂ (Warning 3.3, items N2–N3). No theorem of another part is upgraded to all normalized derivations. Where a theorem of this part reproves a ∂ -statement for every ∂ it says so, and it does so only for the first-order autonomous sector (Corollary 31.7).
- (3) Nothing here classifies normalized derivations, asserts that **No** has only one, or asserts that two of them agree on **No**[i]: they are shown to agree on the solutions of every nonzero $F(y, \partial y) = 0$ and on the fields K_ξ of Theorem 31.1, and nowhere else.

Convention 29.3 (The alphabet of this part). $\mathcal{X}$ is a smooth connected projective curve over $\mathbb{C}$ and $\mathbb{C}(\mathcal{X})$ its function field (not the rational function field $\mathbb{C}(X)$ of Theorem 21.6); ξ is a nonzero meromorphic vector field, that is a derivation of $\mathbb{C}(\mathcal{X})$, and φ_ξ the meromorphic differential with $\varphi_\xi(\xi) = 1$; σ is a local parameter at a point of $\mathcal{X}(\mathbb{C})$; points of $\mathcal{X}(\mathbf{No}[i])$ or of an abelian variety are bold, $\mathbf{y}$, and $\text{st}(\mathbf{y})$ is their reduction (the source writes $\bar{P}$, but a bar is complex conjugation here); ι is a differential embedding; φ, φ_j are 1-forms (the source’s η, θ_j , letters reserved here for the phase residue and the finite angle); **A** is a constant abelian variety of dimension d (the source’s A), with formal group law $\mathcal{F}$, formal logarithm $\log_{\mathcal{F}}$ and formal exponential $\exp_{\mathcal{F}}$; Log is a strip logarithm (Lemma 29.7); $\mathbf{t}$ is a time primitive; dlog_{∂} is the abelian logarithmic derivative; $\mathfrak{o}_{\partial}^{\mathbb{C}}$ is a complex primitive obstruction (Section 33.4); $\mathcal{P}_{\mathbb{C}} = \mathcal{P} + i\mathcal{P}$. For $v \neq 0$ we write $u \prec v$ when $u/v \in \mathfrak{m}_{\mathbb{C}}$ and $u \sim v$ when $u/v \in 1 + \mathfrak{m}_{\mathbb{C}}$.

Lemma 29.4 (Finite coordinates have infinitesimal derivatives). *For every normalized derivation, $\partial\mathcal{O}_{\mathbb{C}} \subseteq \mathfrak{m}_{\mathbb{C}}$, $\partial\mathfrak{m}_{\mathbb{C}} \subseteq \mathfrak{m}_{\mathbb{C}}$ and $\partial\mathcal{O}_{\mathbb{C}} = \partial\mathfrak{m}_{\mathbb{C}}$.*

Proof. For finite real x and real r the number $\omega - rx$ is positive infinite, so $1 - r\partial x > 0$; as r is arbitrary, $\partial x \in \mathfrak{m}$. Apply this to both coordinates, and use $\partial(c + \varepsilon) = \partial\varepsilon$ for $c \in \mathbb{C}$. $\square$

This is the “third formulation” of Remark 3.9. As the proof of Theorem 21.10 already notes, the argument of Lemma 3.8 uses only the positivity rule, $\partial\omega = 1$ and the constant field, so it holds verbatim for every normalized derivation; no surjectivity is used.

Lemma 29.5 (Formal evaluation and the chain rule). *For a finite tuple $\varepsilon = (\varepsilon_1, \dots, \varepsilon_r) \in \mathfrak{m}_{\mathbb{C}}^r$ there is a ring homomorphism $\mathbb{C}[[T_1, \dots, T_r]] \rightarrow \mathcal{O}_{\mathbb{C}}$, $f \mapsto f(\varepsilon)$. It preserves the constant coefficient modulo $\mathfrak{m}_{\mathbb{C}}$, respects formal composition when the inner series have zero constant term, and satisfies*

$$\partial f(\varepsilon) = \sum_{j=1}^r (\partial_{T_j} f)(\varepsilon) \partial \varepsilon_j. \quad (29.1)$$

Formal maps with zero constant term and invertible linear part have formal inverses, and the two maps evaluate to mutually inverse maps on infinitesimal tuples.

Proof. Zero coordinates can first be set to zero in the formal series; if all coordinates vanish, evaluation is the constant coefficient and the claims are immediate. Otherwise the union S of their supports is a well-ordered set of positive exponents in the convention of equation (2.2). For a fixed output exponent, the ordered-word lemma Lemma 2.6 gives only finitely many contributing words in S . Each occurrence of a letter has at most r labels, one for each input coordinate. Thus the fully expanded family of monomials ε^ν has finitely many contributions at each exponent and support in

the well-ordered monoid S^* . Ordinary complex coefficients add no exponents. Every nonconstant monomial has positive exponent, proving the residue assertion.

This finiteness licenses the Cauchy product and proves multiplicativity. For composition, let each inner series have zero constant term. In a fully expanded term, each inner factor then contributes at least one input letter. A fixed output exponent permits only finitely many input words, hence only finitely many partitions into nonempty inner factors and finitely many formal monomials at each resulting degree. The composed evaluation is therefore strongly summable before regrouping, and agrees with the formal composition.

Strong additivity and Leibniz give

$$\partial(\varepsilon^\nu) = \sum_{\substack{1 \leq j \leq r \\ \nu_j > 0}} \nu_j \varepsilon^{\nu - e_j} \partial \varepsilon_j.$$

For fixed j , the resulting sum is the evaluation of $\partial_{T_j} f$ multiplied by the fixed element $\partial \varepsilon_j$. There are only finitely many j , giving equation (29.1). Finally, formal inversion is constructed degree by degree: at each positive degree the unknown vector of coefficients is multiplied by the same invertible ordinary linear matrix. The two inverse identities then evaluate by the composition clause. $\square$

For $r = 1$ and $\partial = \partial$ this is Lemma 6.1; the multivariable form and the composition clause are what the curve and abelian arguments need, and Warning 6.2(iii) applies unchanged: the coefficients are ordinary constants.

Lemma 29.6 (Formal implicit uniqueness in a monad). *Let $\Theta(X_1, \dots, X_r, W) \in \mathbb{C}[[X_1, \dots, X_r, W]]$ with $\Theta(0, 0) = 0$ and $\partial_W \Theta(0, 0) = a \neq 0$. There is a unique $W = w(X) \in \mathbb{C}[[X]]$ with zero constant term and $\Theta(X, w(X)) = 0$. For every $\varepsilon \in \mathfrak{m}_{\mathbb{C}}^r$, $w(\varepsilon)$ is the unique $u \in \mathfrak{m}_{\mathbb{C}}$ with $\Theta(\varepsilon, u) = 0$.*

Proof. Solve by homogeneous degree: the unknown degree- n term occurs with coefficient a , and all other terms of that degree are determined by lower degrees. Evaluation (Lemma 29.5) gives a solution. For $u, v \in \mathfrak{m}_{\mathbb{C}}$, use the formal divided difference

$$\Theta(X, W) - \Theta(X, W') = (W - W')Q(X, W, W'), \quad Q(0, 0, 0) = a.$$

Indeed, each difference $W^k - (W')^k$ is divisible by $W - W'$; the quotient of the linear term contributes a , and every other term has positive total degree. Evaluation therefore gives $\Theta(\varepsilon, u) - \Theta(\varepsilon, v) = (u - v)(a + \varepsilon')$ with $\varepsilon' \in \mathfrak{m}_{\mathbb{C}}$. The factor $a + \varepsilon'$ is a unit, so two roots must coincide. The argument also applies after translating the unknown to any ordinary constant value. $\square$

This is a formal coefficient argument plus a unit argument; it is not a claim that a countable Newton iteration converges in the fine topology.

Lemma 29.7 (Strip logarithms, for every normalized derivation). *For $z \in \mathbf{No}[i]^\times$ write its leading-monomial decomposition $z = a\mu(1 + \varepsilon)$ with $a \in \mathbb{C}^\times$, $\mu > 0$ a monomial and $\varepsilon \in \mathfrak{m}_{\mathbb{C}}$, choose an ordinary logarithm $\log a$ of a , and put*

$$\text{Log } z = \log a + \log \mu + L(1 + \varepsilon). \quad (29.2)$$

Then $\text{Log } z \in \mathcal{S}$ and $E_{\mathcal{S}}(\text{Log } z) = z$, so $\text{Log } z$ is a strip logarithm in the sense of Theorem 11.1, and every choice satisfies $\partial \text{Log } z = \partial z/z$. For $z \in \mathfrak{m}_{\mathbb{C}} \setminus \{0\}$ and real $q > 0$, $\text{Log } z \prec |z|^{-q}$. If $z = \beta\omega^{-1/m}(1 + \varepsilon)$ with $\beta \in \mathbb{C}^\times$ and $m \geq 1$, then $\text{Log } z = \log \beta - \frac{1}{m} \log \omega + L(1 + \varepsilon)$.

Proof. The real part of $\text{Log } z$ is $\log|a| + \log \mu + \text{Re } L(1 + \varepsilon) = \log|z|$ and its imaginary part $\arg a + \text{Im } L(1 + \varepsilon)$ is finite; since the real surreal exponential agrees with E on real infinitesimals [7], equation (11.1) gives $E_S(\text{Log } z) = a\mu E(L(1 + \varepsilon)) = z$. Differentiate the three summands: the first is a constant, the second uses exponential compatibility, and the third uses Lemma 29.5; the logarithmic derivatives add. For the estimate, the last summand is infinitesimal and $\log a$ ordinary, and $|\log \rho| \prec \rho^{-q}$ for positive infinitesimal ρ follows from the domination of every real power by the surreal exponential [1]. The last formula is the normalized monomial decomposition of $\beta\omega^{-1/m}(1 + \varepsilon)$. $\square$

For $\partial = \partial$ the derivative statement is the remark after Theorem 11.1: all strip logarithms of z have the same intrinsic derivative $z^\dagger$. Log is used pointwise or on a specified leading-term class; it is not a global additive logarithm and it does not define a global surcomplex exponential. With the leading decomposition fixed, changing the ordinary choice of $\log a$ adds $2\pi ik$ for an ordinary integer k ; it changes no derivative or growth estimate.

30 The realization classification on curves

30.1 Solutions, and the statement

A *nonconstant solution* of $(\mathcal{X}, \xi)$ is a differential-field embedding

$$\iota : (\mathbb{C}(\mathcal{X}), \xi) \hookrightarrow (\mathbf{No}[i], \partial), \quad \partial \iota(g) = \iota(\xi g) \quad (g \in \mathbb{C}(\mathcal{X})), \quad (30.1)$$

equivalently a nonconstant point $\mathbf{y} \in \mathcal{X}(\mathbf{No}[i])$ with $\partial \mathbf{y} = \xi(\mathbf{y})$. Here a nonconstant point means one not in $\mathcal{X}(\mathbb{C})$. Its Zariski closure over $\mathbb{C}$ is the whole curve: a proper closed subset of a smooth connected curve consists of finitely many closed points, all ordinary because $\mathbb{C}$ is algebraically closed. Consequently no nonzero rational-function denominator vanishes there, and evaluation gives an embedding of $\mathbb{C}(\mathcal{X})$. Constant points are treated separately. This curve-and-derivation formalism, with classical formal local normal forms, is that of Noordman, van der Put and Top [18], whose local discussion concerns a formal time parameter at zero; here the time is the fixed infinite surreal ω with $\partial \omega = 1$, and the realization and exclusion conditions are different.

Every point of $\mathcal{X}(\mathbf{No}[i])$ has an ordinary reduction $p \in \mathcal{X}(\mathbb{C})$ (Lemma 30.2). Choose a local parameter σ at p and write

$$\xi(\sigma) = f(\sigma) = c_k \sigma^k + c_{k+1} \sigma^{k+1} + \cdots, \quad c_k \neq 0, \quad k \in \mathbb{Z}. \quad (30.2)$$

At a simple zero ($k = 1$) the coefficient $\lambda = c_1$ does not depend on the local parameter. Indeed, another parameter has expansion $\tilde{\sigma} = a\sigma + O(\sigma^2)$ with $a \in \mathbb{C}^\times$, and $\xi(\tilde{\sigma}) = (a + O(\sigma))(\lambda\sigma + O(\sigma^2)) = \lambda\tilde{\sigma} + O(\tilde{\sigma}^2)$.

Theorem 30.1 (Autonomous realization classification). *Let ∂ be a normalized derivation. The nonconstant solutions of $(\mathcal{X}, \xi)$ reducing to p are as follows.*

- (i) *If $k \leq 0$ there are none.*
- (ii) *If $k = 1$ there are solutions exactly when $\lambda = c_1 \in \mathbb{R}_{<0}$. With the normalized formal linearizer h_p of Proposition 30.5 they are in bijection with $c \in \mathbb{C}^\times$, with local parameter $\sigma = h_p^{-1}(c e^{\lambda\omega})$.*
- (iii) *If $k = m + 1 \geq 2$, put $c_* = -1/(mc_{m+1})$. There are exactly m branches, indexed by $\beta^m = c_*$, each in bijection with $C \in \mathbb{C}$, and every local parameter has the form*

$$\sigma = \beta \omega^{-1/m} w_{p,\beta,C} \left(\omega^{-1/m}, \frac{\log \omega}{\omega} \right), \quad w_{p,\beta,C} \in 1 + (X, Y)\mathbb{C}[[X, Y]], \quad (30.3)$$

the series being specified uniquely by the formal implicit equation of Proposition 30.7.

Distinct listed parameters give distinct solutions, and all nonconstant solutions occur in the list. The constant solutions of $\partial \mathbf{y} = \xi(\mathbf{y})$ are exactly the ordinary zeros of ξ .

Three features make this more than local existence. Projective reduction puts *every* solution, including those with infinite affine coordinates, into the local classification. Nonreal simple eigenvalues are excluded, not merely those with nonnegative real part. And the formulas bound the scales of every solution independently of its parameters and of the normalized derivation (Section 31).

30.2 Projective reduction

Lemma 30.2 (Reduction of projective points). *Every point $\mathbf{y}$ of a projective variety $\mathcal{Y}/\mathbb{C}$ over $\mathbf{No}[i]$ has a well-defined reduction $\text{st}(\mathbf{y}) \in \mathcal{Y}(\mathbb{C})$, and regular morphisms over $\mathbb{C}$ commute with reduction. For a smooth curve $\mathcal{X}$ and $p \in \mathcal{X}(\mathbb{C})$, any local parameter σ at p gives a bijection $\{\mathbf{y} \in \mathcal{X}(\mathbf{No}[i]) : \text{st}(\mathbf{y}) = p\} \leftrightarrow \mathfrak{m}_{\mathbb{C}}$, whose inverse is given by the formal local coordinate expansions at p .*

Proof. Embed $\mathcal{Y}$ in $\mathbb{P}_{\mathbb{C}}^N$ and write $\mathbf{y} = [z_0 : \cdots : z_N]$. Choose a coordinate of maximal modulus, say z_j ; the ratios z_i/z_j are finite and their standard parts are not all zero, the j th being 1; reducing the homogeneous equations gives a point of $\mathcal{Y}(\mathbb{C})$. Another coordinate of maximal modulus differs by a finite unit, whose nonzero standard part rescales all reduced coordinates by one scalar, so the reduction is well defined. A regular map near the reduction point can be written in affine coordinates using denominators nonzero at that ordinary point. Their evaluations are finite units, so taking standard parts commutes with these expressions. This proves compatibility with regular morphisms, and hence independence of the projective embedding.

For the valuative interpretation, take the set-sized field E generated over $\mathbb{C}$ by the finitely many homogeneous coordinates and use its valuation ring $E \cap \mathcal{O}_{\mathbb{C}}$, with residue field $\mathbb{C}$. The properness criterion [20, Tag 0BX5] applies in that set-sized setting; no scheme with a proper-class coefficient field is needed for this interpretation.

At a smooth point of a curve, choose an affine presentation, with denominators nonzero at p , in which the equations for the coordinates other than σ have an invertible ordinary Jacobian minor. Translate their ordinary values to zero and write them as a tuple Z . The formal map consisting of σ and these equations has invertible linear part. Its formal inverse gives the other coordinates as ordinary-coefficient series in σ . Evaluating at any $\varepsilon \in \mathfrak{m}_{\mathbb{C}}$ gives a point reducing to p ; all defining identities are preserved by Lemma 29.5, and every chart denominator remains a unit.

For uniqueness, suppose two tuples Z, Z' have the same value of σ and the same reduction. Subtract the polynomial equations, or first clear their unit denominators. Their difference factors as

$$0 = M(\varepsilon, Z, Z')(Z - Z'), \quad M(\varepsilon, Z, Z') = M_0 + \Delta,$$

where M_0 is the invertible ordinary Jacobian minor and every entry of Δ is infinitesimal. Thus $\text{st}(\det M) = \det M_0 \neq 0$, so M is invertible and $Z = Z'$. This proves the claimed bijection without completeness of an ambient valuation topology. $\square$

Lemma 30.3 (A nonconstant trajectory reduces to a zero). *If $\mathbf{y}$ is a nonconstant solution and $p = \text{st}(\mathbf{y})$, then $k \geq 1$ in equation (30.2).*

Proof. Put $u = \sigma(\mathbf{y})$. Reduction gives $u \in \mathfrak{m}_{\mathbb{C}}$. The parameter value 0 corresponds to the ordinary point p under Lemma 30.2; since $\mathbf{y}$ is nonconstant, $u \neq 0$. Hence $\partial u \in \mathfrak{m}_{\mathbb{C}}$ by Lemma 29.4. The local expression gives $\partial u = f(u) = c_k u^k(1 + \varepsilon)$ with $\varepsilon \in \mathfrak{m}_{\mathbb{C}}$. For $k = 0$ the right side has nonzero standard part, and for $k < 0$ it is infinite; both contradict the infinitesimal left side. $\square$

The lemma includes the poles of ξ , and because projective reduction also controls the infinite affine solutions it is stronger than the statement that a bounded affine solution tends to an equilibrium. Nothing here treats an ordinary path on a Riemann surface as a fine-continuous path in $\mathbf{No}[i]$.

30.3 Simple zeros

Lemma 30.4 (Constant-coefficient eigenvectors). *If $\lambda \in \mathbb{C}$ and $u \in \mathbf{No}[i]^{\times}$ satisfy $\partial u = \lambda u$, then $\lambda \in \mathbb{R}$ and $u = c e^{\lambda \omega}$ with $c \in \mathbb{C}^{\times}$; conversely these solve the equation for real λ . Such a solution is infinitesimal exactly when $\lambda < 0$.*

Proof. Write $\lambda = \alpha + i\beta$, $\rho = |u|$ and $\zeta = u/\rho$. Conjugation gives $\partial(u\bar{u}) = 2\alpha u\bar{u}$, so $\partial\rho/\rho = \alpha$, $\partial\zeta = i\beta\zeta$ and $\zeta\bar{\zeta} = 1$. The coordinates of ζ are finite, so $\partial\zeta$ is infinitesimal (Lemma 29.4), while $\text{st}\zeta$ has modulus one; reducing the equation forces $\beta = 0$. Then $\partial(ue^{-\alpha\omega}) = 0$ and the constant field is $\mathbb{C}$. The converse uses $\partial\omega = 1$, and $e^{\alpha\omega}$ is infinitesimal, ordinary or infinite according as α is negative, zero or positive. $\square$

For $\partial = \partial$ this is the first-order part of Corollary 8.8, and the proof is its second proof; it uses only a finite unit and its standard part, and assumes no infinite imaginary exponential. The new use is inside an exhaustive nonlinear classification.

Proposition 30.5 (Linearization and all simple-zero solutions). *Suppose $f(\sigma) = \lambda\sigma + O(\sigma^2)$ with $\lambda \neq 0$. There is a unique $h_p(\sigma) = \sigma + O(\sigma^2) \in \mathbb{C}[[\sigma]]$ with $h'_p(\sigma)f(\sigma) = \lambda h_p(\sigma)$. The nonzero infinitesimal solutions of $\partial u = f(u)$ are exactly $u = h_p^{-1}(c e^{\lambda\omega})$ with $c \in \mathbb{C}^{\times}$ when $\lambda \in \mathbb{R}_{<0}$, and there are none otherwise.*

Proof. Write $f(\sigma) = \lambda\sigma(1 + \sigma q(\sigma))$. Then $\lambda/f(\sigma) - 1/\sigma$ is in $\mathbb{C}[[\sigma]]$, because $(1 + \sigma q)^{-1} - 1$ is divisible by σ . Let $H(\sigma)$ be its formal primitive with constant term zero and put

$$h_p(\sigma) = \sigma \exp H(\sigma), \quad H(\sigma) = \int_0^{\sigma} (\lambda/f(s) - 1/s) ds.$$

The integral divides the coefficient of s^j by the nonzero integer $j + 1$, and the exponential is a formal series in the zero-constant H . Thus $h_p = \sigma + O(\sigma^2)$. Its logarithmic derivative is λ/f , proving the required identity. Conversely, if $h_p = \sigma U$ with $U(0) = 1$, the identity forces $U'/U = \lambda/f - 1/\sigma = H'$. The derivative of $U \exp(-H)$ vanishes, so this series equals its constant term 1. This proves uniqueness; the linear coefficient 1 gives the formal inverse. For a nonzero infinitesimal solution the chain rule gives $\partial h_p(u) = \lambda h_p(u)$ with $h_p(u) \sim u \neq 0$, and Lemma 30.4 gives necessity and the form. Conversely each displayed input is infinitesimal when $\lambda < 0$, all evaluations exist, and differentiating the inverse identity gives $\partial u = f(u)$; c is unique by injectivity of the inverse formal map. $\square$

For the quadratic local field $f(\sigma) = \lambda\sigma + a\sigma^2$, the normalized maps are explicitly

$$h_p(\sigma) = \frac{\sigma}{1 + (a/\lambda)\sigma}, \quad h_p^{-1}(v) = \frac{v}{1 - (a/\lambda)v}.$$

Their denominators are units on infinitesimal inputs, and $h'_p f = \lambda h_p$ follows directly. This includes the local linearization used in the logistic example (Example 32.3).

The formal linearization is classical [18, Lemma 6.1(i), arXiv v1]. The exclusion of every nonreal λ , including those with negative real part, is the specifically surcomplex restriction: a classical spiralling decay at $\lambda = -1 + i$ does not become a nonzero element of $\mathbf{No}[i]$ satisfying that eigenvalue equation.

30.4 Multiple zeros: complete power–logarithm families

Suppose $f(\sigma) = c_{m+1}\sigma^{m+1} + O(\sigma^{m+2})$ with $m \geq 1$. Integrating the Laurent series $1/f$ termwise gives the *time primitive*

$$\mathbf{t}(\sigma) = c_* \sigma^{-m} + \sum_{r=1}^{m-1} e_r \sigma^{-r} + \varrho \operatorname{Log} \sigma + g(\sigma), \quad c_* = -\frac{1}{m c_{m+1}}, \quad g \in \sigma \mathbb{C}[[\sigma]], \quad (30.4)$$

where ϱ is the coefficient of σ^{-1} in $1/f$, the sum is empty for $m = 1$, and all other integration constants are fixed to zero (changing them only relabels the parameter C below). For every nonzero infinitesimal u , Lemmas 29.5 and 29.7 give $\mathfrak{D}\mathbf{t}(u) = \mathfrak{D}u/f(u)$, the denominator being $c_{m+1}u^{m+1}(1 + \mathfrak{m}_{\mathbb{C}}) \neq 0$. So a solution of $\mathfrak{D}u = f(u)$ satisfies

$$\mathbf{t}(u) = \omega + C \quad \text{for exactly one } C \in \mathbb{C}, \quad (30.5)$$

because the constant field is $\mathbb{C}$. Here uniqueness of C is for the chosen strip logarithm in $\mathbf{t}$; changing that choice can relabel C . The scale argument below is independent of the choice.

Lemma 30.6 (The leading scale is forced). *Every nonzero infinitesimal solution satisfies $u \sim \beta \omega^{-1/m}$ for exactly one $\beta \in \mathbb{C}^\times$ with $\beta^m = c_*$.*

Proof. Divide the remaining terms of equation (30.4) by u^{-m} . The negative powers give multiples of u^{m-r} with $m-r > 0$; the regular part gives $u^m g(u)$; all are infinitesimal. The logarithmic term is infinitesimal after this division by Lemma 29.7 with $q = m$. Thus $\mathbf{t}(u) = c_* u^{-m}(1 + \eta)$ for some $\eta \in \mathfrak{m}_{\mathbb{C}}$. Since $(\omega + C)/\omega = 1 + C/\omega$ also has residue 1, equation (30.5) implies $c_* u^{-m}/\omega \in 1 + \mathfrak{m}_{\mathbb{C}}$. Inverting this unit gives

$$(u/x_m)^m \in c_*(1 + \mathfrak{m}_{\mathbb{C}}), \quad x_m = \omega^{-1/m}.$$

Writing $v = u/x_m$, the positive number $|v|^m$ has ordinary residue $|c_*| > 0$. Hence $|v|$ is bounded above and below by positive ordinary real numbers: an infinite or infinitesimal $|v|$ would make $|v|^m$ infinite or infinitesimal. Therefore v is a finite unit. Its unique standard part $\beta \in \mathbb{C}^\times$ satisfies $\beta^m = c_*$, and $v/\beta \in 1 + \mathfrak{m}_{\mathbb{C}}$, proving the claim. $\square$

Fix $\beta^m = c_*$ and an ordinary logarithm $\log \beta$, and put $x_m = \omega^{-1/m}$ and $y_\omega = (\log \omega)/\omega$. Both are infinitesimal, though not treated as algebraically independent after evaluation. Writing $u = \beta x_m W$ with $W \in 1 + \mathfrak{m}_{\mathbb{C}}$, dividing equation (30.5) by $\omega = x_m^{-m}$ and using the last formula of Lemma 29.7 gives

$$0 = W^{-m} - 1 + \sum_{r=1}^{m-1} e_r \beta^{-r} x_m^{m-r} W^{-r} - \frac{\varrho}{m} y_\omega + x_m^m (\varrho \log \beta + \varrho L(W) + g(\beta x_m W) - C), \quad (30.6)$$

all powers and logarithms of W being expanded at $W = 1$. In particular, the logarithmic contribution is

$$x_m^m \varrho \operatorname{Log}(\beta x_m W) = -\frac{\varrho}{m} y_\omega + x_m^m \varrho (\log \beta + L(W)),$$

which fixes the sign and the factor $1/m$ in equation (30.6).

Proposition 30.7 (Existence, uniqueness and completeness at a multiple zero). *For every $\beta^m = c_*$ and $C \in \mathbb{C}$ there is a unique $w_{\beta,C}(X, Y) \in 1 + (X, Y)\mathbb{C}[[X, Y]]$ satisfying equation (30.6) formally with (x_m, y_ω) replaced by (X, Y) . Then $u_{\beta,C} = \beta x_m w_{\beta,C}(x_m, y_\omega)$ is a nonzero infinitesimal solution of $\partial u = f(u)$, and once the logarithm of each β is fixed every such solution equals exactly one $u_{\beta,C}$.*

Proof. Put $W = 1 + U$. The negative powers of $1 + U$ and $L(1 + U)$ are ordinary formal series in U , and the argument $\beta X(1 + U)$ of g has zero constant term. Thus the right side of equation (30.6) defines a series in (X, Y, U) . Every term except $(1 + U)^{-m} - 1$ contains X or Y , since $m - r \geq 1$ and $m \geq 1$. Its constant term is zero and its U -derivative at the origin is exactly $-m \neq 0$. The formal implicit lemma gives a unique zero-constant $U_{\beta,C}(X, Y)$, hence $w_{\beta,C} = 1 + U_{\beta,C}$.

Both x_m and y_ω are infinitesimal, so the evaluation lemma gives a root in $1 + \mathfrak{m}_{\mathbb{C}}$. Reversing the algebra yields $\mathfrak{t}(u_{\beta,C}) = \omega + C$. Differentiation gives $\partial u_{\beta,C}/f(u_{\beta,C}) = 1$; the denominator is nonzero and the leading term of $u_{\beta,C}$ is βx_m , so this is a nonzero solution of $\partial u = f(u)$.

Conversely, for any nonzero infinitesimal solution the leading-scale lemma fixes β , and the time identity fixes C after the logarithm of that β has been fixed. Its quotient $u/(\beta x_m)$ solves the same evaluated implicit equation. Uniqueness in the entire monad of 1 identifies it with the constructed value, regardless of its support. This step uses the unit divided difference from Lemma 29.6; it does not assume that formal evaluation is injective. Distinct β have different leading coefficients, and distinct C have different values of $\mathfrak{t}$. Finally $c_* \neq 0$ and characteristic zero give exactly m distinct roots of $\beta^m = c_*$ in $\mathbb{C}$. $\square$

The time parameter also has an exact first-order separation. On one fixed branch, let $C \neq C'$ and set $W_C = u_{\beta,C}/(\beta x_m)$ and $W_{C'} = u_{\beta,C'}/(\beta x_m)$. Subtracting the two implicit equations gives

$$(-m + \eta)(W_C - W_{C'}) = (C - C')x_m^m, \quad \eta \in \mathfrak{m}_{\mathbb{C}}.$$

Indeed, only the term $-Cx_m^m$ depends explicitly on C , and the divided difference in W has residue $-m$. Consequently

$$u_{\beta,C} - u_{\beta,C'} \sim -\frac{\beta}{m}(C - C')\omega^{-1-1/m}.$$

Thus the constant distinguishes solutions at a specified next scale.

When $f(\sigma) = c_{m+1}\sigma^{m+1}$ exactly, all e_r , ϱ and g vanish, and the implicit solution has the explicit form

$$w_{\beta,C}(X, Y) = (1 + CX^m)^{-1/m}, \quad u_{\beta,C} = \beta\omega^{-1/m}(1 + C/\omega)^{-1/m}.$$

The fractional power is the formal binomial series with constant term 1, evaluated at the infinitesimal C/ω . Its inverse m th power is $1 + C/\omega$, so it gives the time identity $c_* u_{\beta,C}^{-m} = \omega + C$ directly, for all $m \geq 1$.

Remark 30.8 (No hidden solutions at another scale). A formal family alone would not prove the theorem. Completeness has two steps: every solution satisfies $\mathfrak{t}(u) = \omega + C$ because the constant field is exactly $\mathbb{C}$, and that identity forces the scale $\omega^{-1/m}$. Uniqueness in the actual monad then excludes corrections with supports outside the displayed two-variable series; in particular there is no additional beyond-all-orders parameter.

Remark 30.9 (Coordinate and branch choices). A change of local parameter transports the solutions by an invertible ordinary formal series. If $\tilde{\sigma} = a\sigma + O(\sigma^2)$, $a \in \mathbb{C}^\times$, then $\tilde{c}_{m+1} = a^{-m}c_{m+1}$, so $\tilde{c}_* = a^m c_*$ and the corresponding leading coefficient is $\tilde{\beta} = a\beta$. The number m of branches is unchanged.

On a fixed coordinate branch, replacing $\log \beta$ by $\log \beta + 2\pi i k$ adds $2\pi i k \varrho$ to $\mathfrak{t}$ and relabels the same solution by $C' = C + 2\pi i k \varrho$. The combination $\varrho \log \beta - C$ in the implicit equation is unchanged. Thus the parametrizations depend on the choices, while the family of points and the existence criterion do not; in particular the leading coefficient β is not a coordinate invariant.

Proof of Theorem 30.1. Every nonconstant solution reduces to an ordinary point (Lemma 30.2), which is a zero of ξ by Lemma 30.3; Propositions 30.5 and 30.7 give exactly the local lists. Conversely, evaluate the completed local coordinate map of $\mathcal{X}$ at each listed nonzero infinitesimal: the point is nonconstant, hence generic, and $\bar{\partial}\sigma = f(\sigma)$ with the chain rule and the rule for inverses gives $\bar{\partial}g(\mathbf{y}) = (\xi g)(\mathbf{y})$ for every rational function g . To check injectivity directly, a nonzero local Laurent expansion factors as $a\sigma^r(1 + \sigma q(\sigma))$ with $a \in \mathbb{C}^\times$, $r \in \mathbb{Z}$ and $q \in \mathbb{C}[[\sigma]]$. At a nonzero infinitesimal u its value is $au^r(1 + uq(u)) \neq 0$, since the last factor has residue 1. Thus no nonzero rational function is killed, and the construction gives a field embedding, not only an assignment of coordinates. A constant point has zero derivative and solves the equation exactly when ξ is regular and vanishes there. $\square$

31 One finite-scale Hahn field, and derivation independence

31.1 All solutions in one set-sized field

Let $p_1, \dots, p_r$ be the multiple zeros of ξ , of orders $m_j + 1$, and $\lambda_1, \dots, \lambda_s$ the negative real coefficients at simple zeros. Put $M = \text{lcm}(m_1, \dots, m_r)$ (the empty least common multiple being 1) and let

$$\mathfrak{G}_\xi = \langle \omega^{1/M}, \log \omega, e^{\lambda_1 \omega}, \dots, e^{\lambda_s \omega} \rangle \quad (31.1)$$

be the multiplicative group they generate. These are positive Conway monomials: ω is log-atomic, so $\log \omega$ is a monomial; positive rational powers of ω are monomials; and $\lambda_j \omega$ is purely infinite, whose exponential is a monomial [1, §3.2, Definition 5.1 and Remark 5.18]. Thus $\mathfrak{G}_\xi$ is a finitely generated ordered monomial group, relations among the generators being allowed. Let $K_\xi = \mathbb{C}((\mathfrak{G}_\xi)) \subseteq \mathbf{No}[i]$ be the field of series with complex coefficients and reverse well-ordered support in $\mathfrak{G}_\xi$. In the additive notation of Section 20 it is K_{Γ_ξ} for the finitely generated group Γ_ξ of exponents of its monomials, a group not contained in $\mathbb{R}$. Indeed, $\log \omega$ is positive infinite but $(\log \omega)^n \prec \omega$ for every positive integer n ; the ordered group already has distinct logarithmic and power scales.

Theorem 31.1 (Uniform finite-scale localization). *For every normalized derivation $\bar{\partial}$:*

- (i) K_ξ is a set-sized differential subfield of $\mathbf{No}[i]$ of cardinality $2^{\aleph_0}$, and $\bar{\partial}|_{K_\xi}$ does not depend on $\bar{\partial}$;
- (ii) every constant or nonconstant solution of $(\mathcal{X}, \xi)$ lies in $\mathcal{X}(K_\xi)$;
- (iii) the same K_ξ serves simultaneously for every ordinary parameter of the families of Theorem 30.1.

Proof. A Hahn field over an ordered group is a field: after extracting a leading term, invert a unit $1 + \varepsilon$ by its geometric series (Lemma 2.6). Every monomial in that inverse remains in the same group. The finitely generated group $\mathfrak{G}_\xi$ is countable, so its series form a set of cardinality at most $|\mathbb{C}|^{\aleph_0} = 2^{\aleph_0}$; this bound includes every permitted support and coefficient assignment. The constant series give the reverse inequality, and ordinary coefficients are constants of every $\bar{\partial}$.

For a monomial $\mathbf{g} = \omega^{a/M}(\log \omega)^b \prod_j e^{n_j \lambda_j \omega}$ with $a, b, n_j \in \mathbb{Z}$, exponential compatibility, the product rule and $\partial \omega = 1$ give the *three-shift formula*

$$\frac{\partial \mathbf{g}}{\mathbf{g}} = \frac{a}{M\omega} + \frac{b}{\omega \log \omega} + \sum_{j=1}^s n_j \lambda_j. \quad (31.2)$$

In particular $\partial \log \omega = 1/\omega$, and the same rules apply to negative integer powers by differentiating inverses. Relations among the generators cause no ambiguity: two product representations are equal in $\mathbf{No}[i]$, hence their differentiated values are equal for each ∂ . The displayed formula is independent of that derivation.

For a reverse well-ordered support $S \subseteq \mathfrak{G}_\xi$, all resulting derivative supports lie in

$$S \cup \omega^{-1}S \cup (\omega \log \omega)^{-1}S.$$

Each translate is reverse well-ordered, and a finite union has the same property. For a fixed output monomial, each translate has at most one preimage in S , so at most three input monomials contribute. Thus the termwise derivative family is Hahn-summable, including possible collisions or cancellations between translates. Strong additivity now gives $\partial K_\xi \subseteq K_\xi$, and the coefficient formula proves that all normalized derivations have the same restriction to the whole Hahn field, not just to its finite sums.

At a multiple zero, $m_j \mid M$, so $x_{m_j} = (\omega^{1/M})^{-M/m_j}$ and $y_\omega = (\log \omega)\omega^{-1}$ belong to $\mathfrak{G}_\xi$. Every evaluated term of equation (30.3) has its monomial in that group, and the evaluation lemma supplies summability. At a stable simple zero, the inverse linearizer is an ordinary zero-constant series in $ce^{\lambda_j \omega}$, so its value is also in K_ξ . Varying c , C or β changes only ordinary coefficients and does not enlarge the group.

Finally choose an affine chart about the ordinary reduction point. Its coordinate functions have ordinary formal expansions in the local parameter. Evaluation uses only monomials from $\mathfrak{G}_\xi$, by the same support lemma, and remains in K_ξ ; rational functions may also have a finite Laurent principal part, handled by inversion in the field. Hence the entire point lies in $\mathcal{X}(K_\xi)$. Constant solutions already have coordinates in $\mathbb{C}$. This proves all three assertions, including cases where either list of zeros is empty. $\square$

Remark 31.2 (An instance of the monomial criterion, and a multi-scale workspace). Equation (31.2) says that $\mathfrak{G}_\xi$ is differentially closed on monomials in the sense of Definition 20.1, so for $\partial = \partial$ the stability of K_ξ is an instance of Lemma 20.2. The field K_ξ is an explicit *multi-scale* ∂ -stable Hahn workspace, with power, logarithmic and exponential scales. The open continuation “classify logarithmic derivatives in multi-scale ∂ -stable Hahn workspaces” (Section 51.13) is adjacent: member 11 supplies such workspaces but does not classify logarithmic derivatives on them, so the continuation stands.

Corollary 31.3 (A continuum of independent solutions inside two scales). *The Hahn field $\mathbb{C}((\langle \omega, \log \omega \rangle))$ contains $2^{\aleph_0}$ distinct solutions of $\partial u = u^3 - u^2$ that are algebraically independent over $\mathbb{C}$ as a family; the field they generate has transcendence degree $2^{\aleph_0}$ over $\mathbb{C}$.*

Proof. At $u = 0$ the vector field has a double zero with leading coefficient -1 . Thus $m = 1$, $c_* = 1$, and Proposition 30.7 gives a nonzero infinitesimal solution $u_C \sim \omega^{-1}$ for every $C \in \mathbb{C}$, in the two-scale field $K = \mathbb{C}((\langle \omega, \log \omega \rangle))$. Fixing the logarithm of $\beta = 1$, distinct C give distinct time values and therefore distinct solutions. They are nonconstant: a nonzero infinitesimal is not an ordinary complex number.

The three-shift proof makes K a set-sized differential field, and its constant field is exactly $\mathbb{C}$, since $\mathbb{C} \subseteq K \subseteq \mathbf{No}[i]$ and $\ker \partial = \mathbb{C}$ on $\mathbf{No}[i]$. Rosenlicht's algebraic-independence theorem, in the form of [18, Lemma 7.2 and Proposition 7.3, arXiv v1], applies to any finite set of distinct nonconstant solutions in this differential field. Every polynomial relation uses finitely many variables, so the whole family $(u_C)_{C \in \mathbb{C}}$ is algebraically independent. It has cardinality $|\mathbb{C}| = 2^{\aleph_0}$ and generates a field of exactly that transcendence degree. This uses the classical theorem as an import, entirely within a set-sized field. $\square$

A finitely generated *monomial group* allows arbitrary well-based infinite sums, which is why Corollary 31.3 is consistent with Theorem 31.1: finite monomial generation is not finite transcendence degree. The notion is algebraic independence of differentially *algebraic* solutions; it is a different notion from the differential transcendence proved in the companion tail-spans report [29], and there is no conflict.

31.2 Independence of the normalized derivation

Theorem 31.4 (Derivation-independent autonomous sector). *Let ∂_1, ∂_2 be normalized derivations. For every smooth connected projective curve $\mathcal{X}/\mathbb{C}$ and nonzero meromorphic ξ , the differential embeddings $(\mathbb{C}(\mathcal{X}), \xi) \hookrightarrow (\mathbf{No}[i], \partial_1)$ and $(\mathbb{C}(\mathcal{X}), \xi) \hookrightarrow (\mathbf{No}[i], \partial_2)$ are the same embeddings of underlying fields. For every $F \in \mathbb{C}[Y, Z]$,*

$$\{y \in \mathbf{No}[i] : F(y, \partial_1 y) = 0\} = \{y \in \mathbf{No}[i] : F(y, \partial_2 y) = 0\}, \quad (31.3)$$

and if $F \neq 0$ and y lies in either set then $\partial_1 y = \partial_2 y$. No equality of the two derivations on all of $\mathbf{No}[i]$ is asserted.

Proof. For curves, Theorem 31.1 places every solution for either derivation in the same K_ξ , where the derivations agree. The corresponding field embedding is therefore differential for one exactly when it is differential for the other.

For the scalar assertion, $F = 0$ gives the whole class $\mathbf{No}[i]$ for both solution collections; no derivative equality is claimed in that case. Now let $F \neq 0$. A nonzero constant has no solutions. Otherwise factor F over $\mathbb{C}$ and consider a solution for ∂_1 ; some irreducible factor R vanishes at $(y, \partial_1 y)$. If $y \in \mathbb{C}$, both derivatives vanish and there is nothing to prove. A factor involving only Y forces this case. A factor involving only Z is $Z - c$ up to a nonzero scalar, and $\partial_1 y = c$ implies $\partial_1(y - c\omega) = 0$, hence $y = c\omega + d$ with $d \in \mathbb{C}$. Both derivations agree on that value.

It remains to treat $y \notin \mathbb{C}$ with R involving both variables. The evaluation map from the one-dimensional integral algebra $A = \mathbb{C}[Y, Z]/(R)$ to $\mathbf{No}[i]$ is injective. Indeed, a nonzero kernel would be a nonzero prime of this affine curve, hence maximal; its residue field would be $\mathbb{C}$ by the weak Nullstellensatz, forcing $y \in \mathbb{C}$. Thus evaluation extends to an embedding of $L = \text{Frac}(A)$. Here Y is transcendental and $L/\mathbb{C}(Y)$ is finite separable. Irreducibility in characteristic zero gives $R_Z \neq 0$ in L : otherwise R would divide its nonzero partial derivative of smaller Z -degree.

On the smooth projective curve with function field L , define

$$\xi Y = Z, \quad \xi Z = -\frac{R_Y}{R_Z} Z. \quad (31.4)$$

These values annihilate R and define the unique derivation on L with $\xi Y = Z$, by separability. It is nonzero, since $Z = 0$ in A would force the irreducible R to be associated to Z , contrary to its involving both variables.

Write $z = \partial_1 y$. Differentiating the evaluated relation gives

$$R_Y(y, z)z + R_Z(y, z)\partial_1 z = 0.$$

The denominator $R_Z(y, z)$ is nonzero by injectivity, so this identity proves that evaluation intertwines the derivations on both generators, hence on all of L . The curve assertion then makes the same embedding differential for ∂_2 , giving $\partial_2 y = z = \partial_1 y$ and $F(y, \partial_2 y) = 0$. Interchanging the derivations proves the equality of the two solution collections. $\square$

For example, take $F(Y, Z) = Z^2 - Y$. On its rational curve the induced vector field has $\xi Y = Z$ and $\xi Z = 1/2$. If y is a nonconstant solution, put $z = \partial y \neq 0$; differentiating $z^2 = y$ gives $2z\partial z = z$, hence $z = (\omega + C)/2$ and

$$y = \frac{(\omega + C)^2}{4}, \quad C \in \mathbb{C}.$$

Every displayed value solves the equation. There is also the scalar constant solution $y = 0$, but the point $(Y, Z) = (0, 0)$ is not a constant solution of the induced vector field, since $\xi Z = 1/2$ there. This is why constant scalar solutions are separated before passing to an embedding of a curve's function field.

Corollary 31.5 (Uniform scalar bound). *For every nonzero $F \in \mathbb{C}[Y, Z]$ there is one Hahn subfield $K_F = \mathbb{C}((\mathfrak{G}_F)) \subseteq \mathbf{No}[i]$ with a finitely generated monomial group and cardinality $2^{\aleph_0}$ containing all solutions of $F(y, \partial y) = 0$ for every normalized derivation simultaneously. In particular that solution collection is a set.*

Proof. For each irreducible factor R involving both variables, form the curve and vector field of equation (31.4) and take its finitely generated group $\mathfrak{G}_R$ from Theorem 31.1. Let $\mathfrak{G}_F$ be generated by the union of these finitely many groups together with ω and $\log \omega$. It is finitely generated even if there are no such factors, and the construction depends only on F .

The inclusion of each monomial group includes its entire Hahn field, so all nonconstant solutions belonging to a mixed factor are in K_F . Factors in Y alone contribute constants, and factors $Z - c$ contribute $c\omega + d$; these too lie in K_F . This covers every factor and every normalized derivation at once. The three-shift and countability arguments apply to the union of generators, proving differential stability, a common derivative restriction and cardinality $2^{\aleph_0}$. A nonzero constant F has no solutions and is covered by the same choice of the two base generators. $\square$

The hypothesis $F \neq 0$ is necessary for a set-sized bound: if $F = 0$, its solution collection is all of $\mathbf{No}[i]$.

Corollary 31.6 (Real autonomous equations). *For $F \in \mathbb{R}[Y, Z]$ the set of solutions in $\mathbf{No}$ does not depend on the normalized derivation, and on every solution of a nonzero such equation all normalized derivations take the same value.*

Proof. Intersect equation (31.3) with $\mathbf{No}$ and use the pointwise equality of derivatives. $\square$

Corollary 31.7 (Two earlier corollaries hold for every normalized derivation). *For every normalized derivation ∂ : the equation $\partial y = \lambda y$ with $\lambda \in \mathbb{C}$ has a nonzero solution in $\mathbf{No}[i]$ exactly when $\lambda \in \mathbb{R}$, the solutions then being $ce^{\lambda\omega}$, $c \in \mathbb{C}$; and the only solutions in $\mathbf{No}[i]$ of $\partial u = 1 + u^2$ are $u = \pm i$. These are the first-order statements of Corollary 8.8 and Corollary 8.11. The second-order statements of Corollary 8.8, such as the absence of a nonzero solution of $\partial^2 y + y = 0$, are not first-order autonomous equations in one unknown, are not covered by Theorem 31.4, and are not upgraded.*

Proof. The first statement is Lemma 30.4. The second is Example 32.4, or Theorem 31.4 with $F = Z - 1 - Y^2$ together with Corollary 8.11 for ∂ . $\square$

Remark 31.8 (What the independence theorem does not say). It does not say that **No** has only one normalized derivation, nor that every surreal is first-order autonomous differential-algebraic over $\mathbb{C}$; it concerns a sharply specified common sector. Coefficients in $\mathbf{No}[i] \setminus \mathbb{C}$, higher differential order and simultaneous systems are outside its statement, and their coefficients or solutions may probe scales on which normalized derivations differ. Which higher-order sectors distinguish normalized derivations is left open, and nothing here implies a uniform finite-scale bound for arbitrary higher-order equations.

32 Real branches and worked equations

Suppose $\mathcal{X}, \xi, p, \sigma$ are defined over $\mathbb{R}$ and $f(\sigma) = c\sigma^{m+1} + O(\sigma^{m+2})$ with $c \in \mathbb{R}^\times$. The real version of the time primitive uses $\log|\sigma|$ in place of $\text{Log } \sigma$, and on a branch $\sigma \sim \beta\omega^{-1/m}$ with real β one uses $\log|\beta|$ in equation (30.6); all coefficients are then real for real C .

Corollary 32.1 (Exact real branch count). *At such a multiple zero the real nonconstant solutions are exactly the families of Proposition 30.7 with real β , real C and the real logarithm convention. There is one real branch when m is odd; when m is even there are two if $c < 0$ and none if $c > 0$; each branch is in bijection with $\mathbb{R}$. At a simple zero, real nonconstant solutions exist exactly when $\lambda < 0$, with real nonzero amplitude in the normalized linearizer.*

Proof. For nonzero real u , the sign of u is fixed and $\partial \log |u| = \partial u/u$, whether $u > 0$ or $u < 0$. The real time primitive therefore satisfies $\mathbf{t}_{\mathbb{R}}(u) = \omega + C$ with $C \in \mathbb{R}$. The leading-scale argument gives a real $\beta = \text{st}(u/\omega^{-1/m}) \neq 0$, and $W = u/(\beta\omega^{-1/m}) \in 1 + \mathfrak{m}$ is positive. Thus the real implicit equation uses $\log|\beta|$ and $\log W$. Conversely, with real β and C that formal equation has real coefficients. Its unique solution has real coefficients, either by the coefficient recursion or by uniqueness after complex conjugation, and evaluates to a real solution. Distinct time constants give distinct solutions as before.

If m is odd, $\beta^m = -1/(mc)$ has exactly one real root. If m is even, it has two real roots precisely when $c < 0$, and none when $c > 0$. These roots are nonzero and index the stated families. At a real simple zero the normalized linearizer and its inverse have real coefficients. Therefore $h_p^{-1}(ae^{\lambda\omega})$ is real exactly when its amplitude a is real; nonconstant infinitesimal solutions require $\lambda < 0$ and $a \neq 0$. $\square$

The sign restrictions reflect the orientation $\partial\omega = 1$, not a choice of complex branch cut (compare Warning 23.4). For example, if $\beta < 0$ and the complex convention chooses $\log \beta = \log |\beta| + i\pi$, then $\mathbf{t}_{\mathbb{C}} = \mathbf{t}_{\mathbb{R}} + i\pi\varrho$ and $C_{\mathbb{C}} = C_{\mathbb{R}} + i\pi\varrho$ on that branch. Thus a real solution need not have a real time constant in an arbitrary complex-logarithm convention. The real primitive removes this fixed imaginary shift.

Example 32.2 (Growth can be a stable branch at infinity). $\partial u = u$ has $u = 0$ and $u = ce^{\omega}$, $c \in \mathbb{C}^\times$. At the affine zero the eigenvalue is $+1$, so no nonconstant solution reduces to 0; at infinity the parameter $\sigma = 1/u$ has $\partial\sigma = -\sigma$, eigenvalue -1 , and every nonconstant solution reduces to infinity. “Negative simple eigenvalue” is a projective statement, not a restriction to bounded affine solutions. Directly, $\partial(ue^{-\omega}) = 0$ forces $u = ce^{\omega}$, so the family is exhaustive. For $c \neq 0$ its local coordinate is $\sigma = c^{-1}e^{-\omega} \in \mathbf{m}_{\mathbb{C}}$, displaying the stable branch in the chart at infinity.

Example 32.3 (The logistic equation). For $\partial u = -u + u^2$ the equilibria 0 and 1 have eigenvalues -1 and $+1$, and at infinity $\sigma = 1/u$ satisfies $\partial\sigma = \sigma - 1$, nonzero at $\sigma = 0$. Every nonconstant solution is therefore

$$u = \frac{ce^{-\omega}}{1 + ce^{-\omega}}, \quad c \in \mathbb{C}^\times.$$

Indeed, for $u \neq 1$ put $v = u/(1 - u)$. The quotient rule gives $\partial v = -v$, hence $v = ce^{-\omega}$ by Lemma 30.4. Conversely $1 + ce^{-\omega}$ is a finite unit of residue 1, so the formula is always defined and solves the logistic equation. The value $c = 0$ gives the equilibrium $u = 0$; the other equilibrium $u = 1$ is separate and is never obtained by an ordinary c . Since $\text{st}(u) = 0$ for every member of the nonconstant family, none has been lost at the other affine zero or at infinity. Real solutions correspond exactly to real c , together with the separate equilibrium 1.

Example 32.4 (Riccati without a tangent trajectory). For $\partial u = 1 + u^2$ the zeros $\pm i$ are simple with eigenvalues $\pm 2i$, and at infinity $\partial(1/u) = -(1 + (1/u)^2)$ is nonzero at $1/u = 0$. By Theorem 30.1 the only solutions in $\mathbf{No}[i]$ are $u = i$ and $u = -i$. This reproves Corollary 8.11 for every normalized derivation by a second, projective route, and it rules out a surcomplex element playing the differential role of $\tan \omega$, without any assertion about a separately defined symbolic trigonometric function.

Example 32.5 (An explicit logarithmic correction). Consider the branch at $u = 0$ of $\partial u = -u^2(1 + u)$. A time primitive is $\mathfrak{t}(u) = 1/u + \text{Log } u - L(1 + u)$, with $\mathfrak{t}'(u) = -1/(u^2(1 + u))$; here $m = 1$, $\beta = 1$, $\varrho = 1$, and we choose $\log \beta = 0$. Use formal variables X, Y before evaluating them at $x_1 = 1/\omega$ and $y_\omega = (\log \omega)/\omega$. The formal implicit equation for the unit W is

$$W^{-1} + XL(W) - Y - XL(1 + XW) - CX - 1 = 0,$$

and with $b = Y + CX$ its first homogeneous terms are $W = 1 - b + (b^2 - Xb - X^2) + O((X, Y)^3)$. Thus, with $L = \log \omega$,

$$u_C = \frac{1}{\omega} - \frac{L + C}{\omega^2} + \frac{(L + C)^2 - (L + C) - 1}{\omega^3} + \dots,$$

the remainder being the evaluation of a series in $X(X, Y)^3\mathbb{C}[[X, Y]]$. This is a formal total-degree statement; it does not assert convergence of truncations in the fine topology. The formula describes all *nonconstant* solutions reducing to 0; the constant solution $u = 0$ must also be included.

There is a second family at $u = -1$. In the parameter $v = u + 1$ the equation is $\partial v = -v(1 - v)^2$, with eigenvalue -1 . Its normalized linearizer is explicitly

$$h(v) = \frac{v}{1 - v} \exp\left(\frac{v}{1 - v}\right), \quad h(v) = v + O(v^2).$$

Here the exponential is the ordinary formal series, evaluated only at an infinitesimal argument. Its logarithmic derivative is $h'(v)/h(v) = 1/(v(1 - v)^2)$, so $h'(v)(-v(1 - v)^2) = -h(v)$. Thus the nonconstant solutions reducing to -1 are $u = -1 + h^{-1}(ae^{-\omega})$, $a \in \mathbb{C}^\times$. At infinity, $\sigma = 1/u$ satisfies $\partial\sigma = 1 + 1/\sigma$, which has a pole at zero, so there are no nonconstant solutions reducing to infinity. Together the two families and the equilibria 0, -1 give the complete list. Both families are real exactly for real time constant C or real amplitude a , respectively.

33 Holomorphic differentials and abelian varieties

33.1 Holomorphic differentials forbid constant-speed compact flows

Proposition 33.1 (Infinitesimal contraction). *Let $\mathcal{Y}/\mathbb{C}$ be smooth and projective, of any finite dimension, and let φ be a global regular 1-form on $\mathcal{Y}$. For every $\mathbf{y} \in \mathcal{Y}(\mathbf{No}[i])$, $\varphi_{\mathbf{y}}(\partial \mathbf{y}) \in \mathfrak{m}_{\mathbb{C}}$.*

Proof. Reduce $\mathbf{y}$ to $p \in \mathcal{Y}(\mathbb{C})$ by Lemma 30.2. On an affine neighbourhood of p , choose finitely many regular coordinate functions $t_1, \dots, t_N$. After shrinking around p if necessary, write $\varphi = \sum_j a_j dt_j$ with each a_j regular there. The coordinates $t_j(\mathbf{y})$ and coefficients $a_j(\mathbf{y})$ are finite: regular functions are ratios with denominators nonzero at p , whose evaluations are units with ordinary nonzero residue. By Lemma 29.4, each $\partial t_j(\mathbf{y})$ is infinitesimal. Consequently

$$\varphi_{\mathbf{y}}(\partial \mathbf{y}) = \sum_j a_j(\mathbf{y}) \partial t_j(\mathbf{y}) \in \mathfrak{m}_{\mathbb{C}},$$

since $\mathfrak{m}_{\mathbb{C}}$ is an ideal of the finite-element ring. The algebraic chain rule makes this contraction invariant under changing the chosen coordinates. No analytic integration or choice of a global coordinate system is used. $\square$

Corollary 33.2 (Holomorphic dual obstruction). *Let $\mathcal{X}/\mathbb{C}$ be a smooth projective curve, $0 \neq \varphi \in H^0(\mathcal{X}, \Omega^1)$, and ξ the meromorphic vector field with $\varphi(\xi) = 1$. There is no nonconstant differential embedding $(\mathbb{C}(\mathcal{X}), \xi) \hookrightarrow (\mathbf{No}[i], \partial)$.*

Proof. A nonconstant solution would give $1 = \varphi_{\mathbf{y}}(\xi(\mathbf{y})) = \varphi_{\mathbf{y}}(\partial \mathbf{y}) \in \mathfrak{m}_{\mathbb{C}}$. $\square$

The same obstruction is visible in the divisor of ξ . Locally write $\varphi = \sigma^r a(\sigma) d\sigma$ with $r \geq 0$ and $a(0) \neq 0$. Then $\xi = \sigma^{-r} a(\sigma)^{-1} \partial/\partial \sigma$ has order $-r \leq 0$: it is nonzero where φ is nonzero and has a pole at a zero of φ . Thus it has no zeros anywhere, in agreement with Theorem 30.1.

Theorem 33.3 (Squarefree hyperelliptic obstruction). *Let $Q \in \mathbb{C}[Y]$ be squarefree with $\deg Q \geq 3$. Then the solutions in $\mathbf{No}[i]$ of*

$$(\partial y)^2 = Q(y) \tag{33.1}$$

are exactly the constants $y = c \in \mathbb{C}$ with $Q(c) = 0$.

Proof. Every constant root of Q solves the equation. Suppose instead that y is nonconstant and put $z = \partial y \neq 0$. Squarefreeness makes Q a nonsquare in $\mathbb{C}(Y)$, since each root has odd order one. Hence $Z^2 - Q(Y)$ is irreducible, and the scalar-to-curve reduction in Theorem 31.4 identifies (y, z) with a nonconstant differential point on its smooth projective model.

We check directly that $\varphi = dY/Z$ is a global regular 1-form on this model. At finite points with $Z \neq 0$ this is immediate. At a root a of Q , squarefreeness gives $Q'(a) \neq 0$, so Z is a local parameter and the identity $2Z dZ = Q'(Y) dY$ gives

$$\frac{dY}{Z} = \frac{2 dZ}{Q'(Y)}.$$

The right side is regular at $(a, 0)$, with unit denominator. These cases cover the affine curve, which is smooth because Q and Q' have no common root.

At infinity put $w = 1/Y$, $v = Zw^{g+1}$ and let $q(w) = w^{\deg Q} Q(1/w)$, so $q(0) \neq 0$ is the leading coefficient of Q . If $\deg Q = 2g + 2$, $g \geq 1$, the local equation is $v^2 = q(w)$. There are two points over infinity, with $v(0) = \pm \sqrt{q(0)} \neq 0$; at each, w is a local parameter and

$$\frac{dY}{Z} = -\frac{w^{g-1}}{v} dw$$

is regular. If $\deg Q = 2g + 1$, the equation is $v^2 = wq(w)$. Its unique point at infinity has parameter v , since the derivative in w at $(0, 0)$ is $-q(0) \neq 0$. Solving formally gives $w = q(0)^{-1}v^2 + O(v^4)$. The

same formula $dY/Z = -w^{g-1}dw/v$ is then a unit multiple of $v^{2g-2}dv$, again regular. The exponents are nonnegative precisely in the degrees under consideration.

Thus Proposition 33.1 applies to φ . But its contraction at the nonconstant point is $\varphi_{(y,z)}(\partial(y,z)) = \partial y/z = 1$, whereas that proposition makes it infinitesimal. This contradiction excludes every nonconstant solution. $\square$

Remark 33.4 (A necessary distinction at branch points). Differentiating $z^2 = Q(y)$ with $\partial y = z \neq 0$ gives $\partial z = Q'(y)/2$; the cancellation is valid for a nonconstant solution, not for a constant solution at $z = 0$. So equation (33.1) and the vector system $\partial y = z$, $\partial z = Q'(y)/2$ have different constant-point behaviour: a simple root c gives the constant solution $y = c$ of the scalar equation, but $(c, 0)$ is not a stationary point of the vector system. In particular an elliptic translation system has no points in $\mathbf{No}[i]$ solving that vector equation, although the squared scalar equation keeps its branch-point constants.

Example 33.5 (Squarefreeness and degree are real hypotheses). For $Q(Y) = Y^3$, which is not squarefree, $y = 4/(\omega + C)^2$ with $C \in \mathbb{C}$ is nonconstant and satisfies $(\partial y)^2 = y^3$. For degree two, $y = (e^\omega + e^{-\omega})/2$ satisfies $(\partial y)^2 = y^2 - 1$. Indeed, the first derivative is $-8/(\omega + C)^3$, whose square is $64/(\omega + C)^6 = y^3$; in the second case $\partial y = (e^\omega - e^{-\omega})/2$, and subtracting the two squares gives 1. Both solutions are nonconstant for every normalized derivation. Neither hypothesis of Theorem 33.3 can simply be deleted.

33.2 The formal logarithm and the identity monad

This subsection concerns a *constant* abelian variety $\mathbf{A}/\mathbb{C}$ of dimension d , not one whose coefficients vary in the differential field. Let $\mathbf{A}_1(\mathbf{No}[i])$ be the kernel of reduction $\mathbf{A}(\mathbf{No}[i]) \rightarrow \mathbf{A}(\mathbb{C})$, a group homomorphism by Lemma 30.2; the constant inclusion is a section. The formal completion of $\mathbf{A}$ at its identity carries a commutative formal group law $\mathcal{F}(T, S) \in \mathbb{C}[[T_1, \dots, T_d, S_1, \dots, S_d]]^d$, $\mathcal{F}(T, S) = T + S + \text{higher terms}$. The geometric setting of abelian varieties and tangent spaces is standard [19, Chapter I, §1 and Theorem 6.4]; the characteristic-zero logarithm fact is proved here.

Lemma 33.6 (Normalized formal-group logarithm). *For a commutative d -dimensional formal group over $\mathbb{C}$ there is a unique formal map $\log_{\mathcal{F}}(T) = T + O(T^2)$ with $\log_{\mathcal{F}}(\mathcal{F}(T, S)) = \log_{\mathcal{F}}(T) + \log_{\mathcal{F}}(S)$. It has a formal inverse $\exp_{\mathcal{F}}(T) = T + O(T^2)$, and its differential is the normalized invariant differential matrix of the formal group.*

Proof. Put $J(T) = \partial_S \mathcal{F}(T, 0)$. Its constant matrix is the identity, so it is invertible over $\mathbb{C}[[T]]$. Translating the cotangent basis at zero gives the column of invariant formal forms

$$(\varphi_1, \dots, \varphi_d)^t = J(T)^{-1} dT.$$

The dual invariant tangent fields commute: their infinitesimal translations commute because $\mathcal{F}$ is commutative. For such fields v, w , the identity $d\varphi_j(v, w) = v(\varphi_j(w)) - w(\varphi_j(v)) - \varphi_j([v, w])$ has all three terms zero. These fields form a basis over $\mathbb{C}[[T]]$, so each φ_j is closed.

Here is the coefficientwise integration step. If $\alpha_n = \sum_i a_i(T) dT_i$ is closed with all a_i homogeneous of degree n , define

$$H_{n+1}(T) = \frac{1}{n+1} \sum_i T_i a_i(T).$$

Closedness gives $\partial_j a_i = \partial_i a_j$; Euler's homogeneous identity then gives $\partial_j H_{n+1} = (a_j + \sum_i T_i \partial_i a_j)/(n+1) = a_j$. Integrating each homogeneous part in this way produces the unique zero-constant primitive of each φ_j . The resulting vector $\ell(T)$ has identity linear part.

Associativity, and commutativity for the first variable, give

$$J(\mathcal{F}(T, S))^{-1} d\mathcal{F}(T, S) = J(T)^{-1} dT + J(S)^{-1} dS.$$

Thus the total differential in (T, S) of $\ell(\mathcal{F}(T, S)) - \ell(T) - \ell(S)$ vanishes. In characteristic zero a formal series with all partial derivatives zero is constant; here its constant term is zero. This proves the homomorphism identity and defines $\log_{\mathcal{F}} = \ell$. The identity linear part gives the formal inverse $\exp_{\mathcal{F}}$ by degree recursion.

For uniqueness, compose another normalized logarithm with $\exp_{\mathcal{F}}$. The result is an additive formal map with identity linear part. For any homogeneous term H_n of degree $n > 1$, additivity gives $2^n H_n(T) = H_n(2T) = 2H_n(T)$, so $H_n = 0$. The map is the identity, as required. $\square$

By the smooth-coordinate argument of Lemma 30.2, now with d free parameters and an invertible complementary Jacobian, the formal coordinates give a bijection between $\mathbf{A}_1(\mathbf{No}[i])$ and $\mathbf{m}_{\mathbb{C}}^d$. They describe every point of the monad, not only points already presented by a chosen series. All coordinates of $\mathcal{F}$, $\log_{\mathcal{F}}$ and $\exp_{\mathcal{F}}$ have zero constant term. Their evaluations therefore remain infinitesimal, and the composition clause of Lemma 29.5 evaluates both inverse identities and the group identity on every infinitesimal tuple.

For a local check, the one-dimensional multiplicative formal group $\mathcal{F}(T, S) = T + S + TS$ has $J(T) = 1 + T$ and

$$\log_{\mathcal{F}}(T) = \log(1 + T), \quad \exp_{\mathcal{F}}(U) = e^U - 1.$$

These are ordinary formal series at zero, evaluated only on the infinitesimal domain. This example illustrates the local lemma; the later global splitting uses properness of an abelian variety.

A formal coordinate change $\tilde{T} = H(T)$ with $H(0) = 0$ and ordinary invertible linear part B satisfies

$$\log_{\tilde{\mathcal{F}}}(H(T)) = B \log_{\mathcal{F}}(T).$$

Indeed, composition with $\exp_{\mathcal{F}}$ is an additive formal map with linear part B , and the same homogeneous argument kills every higher term. Thus the logarithm is intrinsically Lie-algebra valued; its coordinate vectors change by the ordinary linear tangent map, not by a nonlinear reparametrization.

Proposition 33.7 (Global splitting through the identity monad). *In a fixed ordinary basis of $\text{Lie}(\mathbf{A})$ there is a group isomorphism*

$$\mathbf{A}(\mathbf{No}[i]) \longrightarrow \mathbf{A}(\mathbb{C}) \oplus \mathbf{m}_{\mathbb{C}}^d, \quad \mathbf{y} \longmapsto (\text{st}(\mathbf{y}), \log_{\mathcal{F}}(\mathbf{y} - \text{st}(\mathbf{y}))), \quad (33.2)$$

with inverse $(c, u) \mapsto c + \exp_{\mathcal{F}}(u)$. Intrinsically the second summand is $\text{Lie}(\mathbf{A})(\mathbf{m}_{\mathbb{C}})$, independent of the basis.

Proof. Reduction is a homomorphism because the ordinary group law commutes with reduction, and the inclusion of ordinary points is a section. Every point has the unique decomposition $\mathbf{y} = c + \mathbf{q}$ with $c = \text{st}(\mathbf{y}) \in \mathbf{A}(\mathbb{C})$ and $\mathbf{q} = \mathbf{y} - c \in \mathbf{A}_1(\mathbf{No}[i])$. Uniqueness also follows from $\mathbf{A}(\mathbb{C}) \cap \mathbf{A}_1(\mathbf{No}[i]) = \{0\}$. Commutativity makes this a direct-product decomposition of groups.

The evaluated formal logarithm identifies the second factor with the additive group $\mathbf{m}_{\mathbb{C}}^d$. Its inverse sends u to $\exp_{\mathcal{F}}(u)$, which has reduction zero. Consequently $(c, u) \mapsto c + \exp_{\mathcal{F}}(u)$ is inverse to the displayed map in both directions. The coordinate-change identity above identifies its second factor canonically with $\text{Lie}(\mathbf{A})(\mathbf{m}_{\mathbb{C}})$. $\square$

No complex analytic uniformization, no period lattice and no exponential defined on every surcomplex Lie-algebra element is used; $\exp_{\mathcal{F}}$ is evaluated only on $\mathfrak{m}_{\mathbb{C}}^d$. In the terms of the companion Tate report this is the formal-group treatment of a *constant* good-reduction variety over $\mathbf{No}[i]$, with reduction by standard part (Remark 34.6).

33.3 The exact logarithmic-derivative image

For $\mathbf{y} \in \mathbf{A}(\mathbf{No}[i])$ its *algebraic logarithmic derivative* is

$$\mathrm{dlog}_{\partial} \mathbf{y} = d(\cdot - \mathbf{y})_{\mathbf{y}}(\partial \mathbf{y}) \in \mathrm{Lie}(\mathbf{A})(\mathbf{No}[i]), \quad (33.3)$$

the differential of translation by $-\mathbf{y}$ applied to $\partial \mathbf{y}$; equivalently its coordinates are the contractions of the normalized invariant differentials with $\partial \mathbf{y}$. It is a tangent vector, not the argument of a complex number. For $\mathbf{A}$ replaced by the multiplicative group it would be $z \mapsto z^{\dagger}$. The class $\partial \mathfrak{m}_{\mathbb{C}} = \partial \mathcal{O}_{\mathbb{C}}$ is a $\mathbb{C}$ -linear subspace of $\mathfrak{m}_{\mathbb{C}}$ (Lemma 29.4), and $\partial : \mathfrak{m}_{\mathbb{C}} \rightarrow \partial \mathfrak{m}_{\mathbb{C}}$ is a bijection because $\mathfrak{m}_{\mathbb{C}} \cap \mathbb{C} = \{0\}$.

Theorem 33.8 (Abelian logarithmic-derivative image). *For every normalized derivation and every constant abelian variety $\mathbf{A}/\mathbb{C}$ of dimension d ,*

$$\mathrm{dlog}_{\partial} \mathbf{y} = \partial \log_{\mathcal{F}}(\mathbf{y} - \mathrm{st}(\mathbf{y})), \quad \mathrm{dlog}_{\partial}(\mathbf{A}(\mathbf{No}[i])) = \mathrm{Lie}(\mathbf{A})(\partial \mathfrak{m}_{\mathbb{C}}). \quad (33.4)$$

The kernel is exactly $\mathbf{A}(\mathbb{C})$, and in any ordinary Lie basis

$$0 \longrightarrow \mathbf{A}(\mathbb{C}) \longrightarrow \mathbf{A}(\mathbf{No}[i]) \xrightarrow{\mathrm{dlog}_{\partial}} (\partial \mathfrak{m}_{\mathbb{C}})^d \longrightarrow 0 \quad (33.5)$$

is split exact, with splitting $a \mapsto \exp_{\mathcal{F}}((\partial|_{\mathfrak{m}_{\mathbb{C}}})^{-1}a)$.

Proof. For each fixed point $\mathbf{y}$, the point $c = \mathrm{st}(\mathbf{y})$ is ordinary and has zero derivative. Translation by $-c$ therefore preserves the logarithmic derivative and gives $\mathbf{q} = \mathbf{y} - c$ in the identity monad. Writing its formal coordinate tuple as $T(\mathbf{q})$, the evaluated chain rule and the differential formula from Lemma 33.6 give

$$\partial \log_{\mathcal{F}}(\mathbf{q}) = J(T(\mathbf{q}))^{-1} \partial T(\mathbf{q}) = \mathrm{dlog}_{\partial} \mathbf{y}.$$

This differentiates the coordinate tuple at the fixed point; it does not require differentiating a standard-part map.

Under Proposition 33.7, the logarithmic derivative is exactly the homomorphism $(c, u) \mapsto \partial u$. The restriction $\partial : \mathfrak{m}_{\mathbb{C}} \rightarrow \partial \mathfrak{m}_{\mathbb{C}}$ is a $\mathbb{C}$ -linear bijection: surjectivity onto that image is its definition, and its kernel is $\mathfrak{m}_{\mathbb{C}} \cap \mathbb{C} = \{0\}$. Applied coordinatewise, it proves the asserted image and shows that the kernel consists precisely of the points with $u = 0$, namely $\mathbf{A}(\mathbb{C})$.

Let P_{∂} denote this restriction's inverse. Then $a \mapsto \exp_{\mathcal{F}}(P_{\partial}a)$ is a homomorphism, has logarithmic derivative a , and reduces to the identity point. It is the unique section with image in the identity monad. In particular, all points over one value a form its translate by the kernel $\mathbf{A}(\mathbb{C})$. No surjectivity of ∂ on all of $\mathbf{No}[i]$ was used. $\square$

Remark 33.9 (Agreement with, and contrast to, the multiplicative group). For $\partial = \partial$, $\partial \mathfrak{m}_{\mathbb{C}} = \mathcal{A} + i\mathcal{A} = \mathcal{A}_{\mathbb{C}}$ in the notation of Lemma 16.2, so the image is $\mathrm{Lie}(\mathbf{A})(\mathcal{A}_{\mathbb{C}})$. The multiplicative group is not proper, and for it Theorem 8.1 constrains only the imaginary part: $\{z^{\dagger}\} = \mathbf{No} + i\mathcal{A}$. For an abelian variety the *whole* Lie coordinate is constrained, real and imaginary parts alike. The theorem is not a consequence of the algebraic closedness of $\mathbf{No}[i]$: it specifies a generally proper differential image.

33.4 A primitive obstruction when primitives exist

For the obstruction construction and finite-primitive criterion, assume $\bar{\partial}$ is surjective, as ∂ is. By equation (2.3), applied coordinatewise, every $z \in \mathbf{No}[i]$ is $z = \text{pp}(z) + \text{ct}(z) + \varepsilon$ with $\text{pp}(z) \in \mathcal{P}_{\mathbb{C}}$, $\text{ct}(z) \in \mathbb{C}$ and $\varepsilon \in \mathfrak{m}_{\mathbb{C}}$. For $a \in \mathbf{No}[i]$ choose $\mathfrak{b}$ with $\bar{\partial}\mathfrak{b} = a$ and put

$$\mathfrak{o}_{\bar{\partial}}^{\mathbb{C}}(a) = \text{pp}(\mathfrak{b}) \in \mathcal{P}_{\mathbb{C}}. \quad (33.6)$$

Two primitives differ by an ordinary complex constant, so this is well defined; it is $\mathbb{C}$ -linear and surjective, with $\mathfrak{o}_{\bar{\partial}}^{\mathbb{C}}(\bar{\partial}P) = P$ for $P \in \mathcal{P}_{\mathbb{C}}$. For $\bar{\partial} = \partial$ it is the componentwise obstruction $\mathfrak{o}_{\partial}^{\mathbb{C}}(a) = \mathfrak{o}(\text{Re } a) + i\mathfrak{o}(\text{Im } a)$. It is *not* the surcomplex phase obstruction $\Phi(a) = \mathfrak{o}(\text{Im } a)$ of Definition 7.4, which ignores the real part, and the two must not be conflated: Remark 33.9 is the reason.

Corollary 33.10 (Finite-primitive criterion for abelian motion; surjective $\bar{\partial}$). *Let $\bar{\partial}$ be surjective and $a \in \text{Lie}(\mathbf{A})(\mathbf{No}[i])$. The equation $\text{dlog}_{\bar{\partial}} \mathbf{y} = a$ is solvable in $\mathbf{A}(\mathbf{No}[i])$ if and only if $\mathfrak{o}_{\bar{\partial}}^{\mathbb{C}}(a) = 0$ componentwise in one, equivalently every, ordinary Lie basis. Then, for a finite primitive vector $\mathfrak{b}$ with $\bar{\partial}\mathfrak{b} = a$, all solutions are*

$$\mathbf{y} = c + \exp_{\mathcal{F}}(\mathfrak{b} - \text{st}(\mathfrak{b})), \quad c \in \mathbf{A}(\mathbb{C}). \quad (33.7)$$

Moreover $\ker \mathfrak{o}_{\bar{\partial}}^{\mathbb{C}} = \bar{\partial}\mathfrak{m}_{\mathbb{C}}$.

Proof. Choose any primitive vector $\mathfrak{b}$. Vanishing of its purely infinite part means precisely that it is finite; the remaining parts are an ordinary constant vector and an infinitesimal vector. For a finite primitive, put $u = \mathfrak{b} - \text{st}(\mathfrak{b}) \in \mathfrak{m}_{\mathbb{C}}^d$. It still satisfies $\bar{\partial}u = a$. If $\mathfrak{b}'$ is another primitive, then $\mathfrak{b}' - \mathfrak{b} \in \mathbb{C}^d$; if one is finite, so is the other, and subtracting standard parts gives the same u . This is the unique infinitesimal primitive. Conversely an infinitesimal primitive has zero purely infinite part. This proves $\ker \mathfrak{o}_{\bar{\partial}}^{\mathbb{C}} = \bar{\partial}\mathfrak{m}_{\mathbb{C}}$ coordinatewise, and Theorem 33.8 gives solvability and the full translate of its kernel displayed above.

A change of ordinary Lie basis is an invertible matrix over $\mathbb{C}$. It commutes with $\bar{\partial}$, with the $\mathbb{C}$ -linear obstruction and, on finite vectors, with standard part. Thus vanishing in one basis is equivalent to vanishing in every basis, and the solution formula represents the same points. $\square$

The standard part in equation (33.7) is taken only after the finiteness of $\mathfrak{b}$ has been established; “the standard part” of an arbitrary infinite primitive would not be a valid substitute. Surjectivity is used to define the obstruction on every Lie-algebra input. For an input with an explicit primitive, the image theorem tests finiteness directly, without a surjectivity assumption.

Corollary 33.11 (Sharp one-power speed threshold; every normalized derivation). *Fix $0 \neq v \in \text{Lie}(\mathbf{A})(\mathbb{C})$ and $p \in \mathbb{R}$. For every normalized derivation $\bar{\partial}$, the equation $\text{dlog}_{\bar{\partial}} \mathbf{y} = \omega^{-p}v$ is solvable exactly when $p > 1$. In that case all solutions are*

$$\mathbf{y} = c + \exp_{\mathcal{F}}\left(\frac{\omega^{1-p}}{1-p}v\right), \quad c \in \mathbf{A}(\mathbb{C}).$$

In particular no nonzero constant speed and no speed $\omega^{-1}v$ is realizable.

Proof. Exponential compatibility and $\bar{\partial}\omega = 1$ give $\bar{\partial}\omega^r = r\omega^{r-1}$ for every real r , and $\bar{\partial}\log \omega = 1/\omega$. Thus an explicit primitive vector is $\omega^{1-p}v/(1-p)$ for $p \neq 1$, and $(\log \omega)v$ for $p = 1$, for every normalized derivation.

When $p > 1$, the first vector is infinitesimal, so the image theorem and its splitting give exactly the displayed solutions. When $p < 1$, any coordinate with $v_j \neq 0$ is infinite in modulus; the same is true of $(\log \omega)v_j$ when $p = 1$. Every other primitive differs by an ordinary constant vector, which cannot make that coordinate finite. There is therefore no infinitesimal primitive, and Theorem 33.8 excludes a solution. This argument uses neither surjectivity nor a globally defined primitive obstruction. $\square$

This is the abelian analogue of Corollary 10.2. The integration is elementary; its force is that it is an exact obstruction for every constant abelian variety in every finite dimension, and that it constrains a real speed $\omega^{-p}v$ as well as an imaginary one, which the multiplicative group does not.

34 What the autonomous theory settles, and what it does not

Remark 34.1 (A finite geometric decision criterion). For a presented smooth constant projective curve and a nonzero meromorphic vector field, the divisor of that field has finite support. Thus Theorem 30.1 reduces existence of a nonconstant surcomplex trajectory to a finite test: find the zeros, compute their orders, and at each simple zero compute the linear coefficient. A multiple zero contributes solutions; a simple zero contributes solutions exactly when its coefficient is real and negative. The classification then specifies all solutions as Hahn evaluations.

To call this an effective procedure, the presentation must supply algorithms for that zero divisor, the local orders and leading coefficients, and exact tests for vanishing imaginary part and real sign. Exact coefficient-field arithmetic alone does not supply these geometric operations. Algebraic complex presentations equipped with polynomial factorization and local-expansion algorithms provide a sufficient setting. These requirements are a contract for a procedure, not an implementation in the repository.

The output is a finite list of local families with their ordinary parameters and formal recursions. An infinite series is represented by that defining data, not by listing every coefficient. A chosen parameter must itself have a representation if coefficients are to be computed effectively. No unconditional computability claim is made for arbitrary unencoded complex constants. In this qualified sense the criterion lifts item N25 for the stated class; beyond it N25 stands.

Remark 34.2 (Why ordinary formal initial-value results are not enough). At an ordinary finite time a vector field that is nonzero at a point has a nonconstant local solution through it. In $\mathbf{No}[i]$ every projective point has an ordinary reduction and its finite coordinates have infinitesimal derivatives, so a nonzero constant velocity at the reduction is impossible: admissible trajectories are centred at zeros, and their rate of approach is constrained by the differential structure. This explains the apparent reversal between the local formal picture and Theorem 30.1 without conflating their time parameters, and it is the nonlinear form of Remark 11.8. For example, the ordinary formal equation $dy/ds = 1$ has the solution $y = s$ centred at zero. The intrinsic equation $\partial y = 1$ in $\mathbf{No}[i]$ instead has exactly $y = \omega + C$, $C \in \mathbb{C}$. These points reduce to infinity in $\mathbb{P}^1$. There the local parameter $\sigma = 1/y$ satisfies $\partial \sigma = -\sigma^2$, a double zero of the projective vector field, just as the classification requires. Substituting an infinitesimal for the ordinary time s cannot preserve $ds/ds = 1$ under ∂ , since its intrinsic derivative is infinitesimal.

Remark 34.3 (Established inputs and proposed contributions). The published construction of a surreal derivation is an input. The theory of smooth curves, formal implicit equations, invariant differentials and characteristic-zero formal-group logarithms is classical; the local formal normal forms are [18, Lemma 6.1], and Rosenlicht's independence theorem is imported from [18, Lemma 7.2, Proposition 7.3], whose numbered statements are cited from the consulted arXiv version 1 and are

not assumed to match the journal numbering. The scalar imaginary obstruction overlaps Part II. Member 11 proposes as new only the exhaustive ambient realization theorem with its finite-scale support bound, the derivation independence of the constant first-order solution sets, the projective differential-form obstruction and the exact abelian logarithmic-derivative image; historical novelty is a separate question (item N93).

Remark 34.4 (Boundaries). The curve theorem does not classify higher-dimensional autonomous systems: Proposition 33.1 survives in every dimension, but near a singular point several eigendirections interact, and integer relations among the eigenvalues of the linear part together with noncommuting normal-form terms prevent the one-coordinate time primitive from proving a general classification. The abelian theorem assumes a constant variety; nonconstant abelian varieties can acquire a connection term after coordinate changes, which the constant-section splitting does not address, and noncommutative algebraic groups need another argument because their invariant forms need not be closed, which is where the proof of Lemma 33.6 deliberately stops. For instance, the affine group with law $(a, b)(a', b') = (aa', b + ab')$ has left-invariant forms da/a and db/a . Near its identity put $a = 1 + x$, $b = y$. The second form is $dy/(1 + x)$, and

$$d\left(\frac{dy}{1+x}\right) = -\frac{dx \wedge dy}{(1+x)^2} \neq 0.$$

Thus its invariant forms cannot all be differentials of formal logarithm coordinates with additive group law. This pinpoints the commutativity step needed above; it is not a classification of noncommutative surcomplex differential equations.

Remark 34.5 (Foundational scope and a formalization route). $\mathcal{X}(\mathbf{No}[i])$, $\mathbf{A}(\mathbf{No}[i])$ and the exact sequences over them are class-level statements, read in a conservative class theory such as NBG with the usual pointwise algebraic definitions (Section 2.1); no collection of all surreal supports is used as a set. Each formal evaluation uses a set-sized support, Theorem 31.1 supplies one set-sized field for the whole solution set of a fixed equation, and the imported independence theorem is applied in that field; for the abelian theorem each formal computation uses finitely many input series and the global sequence is stated pointwise. A dependency-ordered formalization uses four interfaces:

- (i) a valued differential field with ordinary residue section, constant field $\mathbb{C}$, small derivative on finite elements and a specified element of derivative 1, together with the exponential, order and strong-additivity laws of Definition 29.1;
- (ii) finite-variable strong formal evaluation, zero-constant composition and the chain rule for that intrinsic derivation;
- (iii) smooth algebraic local coordinates identifying the entire residue monad with infinitesimal tuples, with projective reduction;
- (iv) the one-variable time primitive, forced scale and constant-field uniqueness argument, followed by commutative formal groups and invariant differentials for the abelian extension.

A Hahn identity alone supplies neither the geometric coordinates nor the summability bridge needed to evaluate them.

The current Lean library already proves actual real and complex finite-variable evaluation, strong summability, the ring laws and zero-constant composition. The live ledger `docs/FORMALIZATION.md` records this reusable algebraic portion of Lemma 29.5. It does not yet prove that lemma's chain rule for ∂ , the full projective-monad statement Lemma 30.2, or the autonomous classification. Checked fine-differentiation rules concern a different derivative and cannot fill the intrinsic-derivation interface

by a change of notation. The source package supplied no Lean file; that historical fact is distinct from current partial library coverage (Warning 49.4).

Remark 34.6 (Relation to other reports). The companion report on Hahn–Tate uniformization notes, among its limitations, that curves with nonnegative valuation of j need a different treatment, such as formal groups around a good-reduction model and a residue elliptic curve [30]. Proposition 33.7 does this for *constant* abelian varieties over $\mathbf{No}[i]$, with reduction by standard part; it does not do it over that report’s Hahn fields with a coarse valuation, so this is a cross-link and not an answer. The Tate construction and the present one use different residue maps; identifying their formal-group vocabulary does not identify their hypotheses or fields. The Tate result itself imposes no Berarducci–Mantova differential equation, whereas this part determines an intrinsic logarithmic-derivative image. Likewise, “autonomous flow” here means a solution of an intrinsic differential equation. The dynamics report’s embedding of a near-identity map into a formal flow is a statement about formal iteration and its infinitesimal generator. These are different constructions, even when they use some of the same formal calculus.

Table 8: Proof dependencies of Part VI, after member 11’s audit.

Conclusion	Necessary inputs, and the check against overreach
Finite derivative (Lemma 29.4)	$\partial\omega = 1$, the sign rule and finite coordinates. No surjectivity.
Formal evaluation (Lemma 29.5)	Strong Hahn summability and the ordered-word lemma. No fine-topological limit.
Projective reduction (Lemma 30.2)	Finitely many homogeneous coordinates, standard part, smooth local implicit equations; poles checked in local coordinates.
Simple zeros (Proposition 30.5)	Formal linearization, constant field $\mathbb{C}$, Lemma 30.4. A negative real part alone is insufficient.
Multiple zeros (Proposition 30.7)	A time primitive, the constant-field identity $\mathbf{t}(u) - \omega \in \mathbb{C}$, a forced leading scale, implicit unit uniqueness. Formal construction alone is insufficient.
Finite-scale field (Theorem 31.1)	All local families and the three-shift formula. Finite monomial generation is not finite transcendence degree.
Derivation independence (Theorem 31.4)	Constant coefficients and the normalization. Variable coefficients and higher order are not covered.
Hyperelliptic exclusion (Theorem 33.3)	Squarefreeness, degree at least three, holomorphy of dY/Z . Branch-point constants are retained.
Abelian image (Theorem 33.8)	A constant abelian variety, its residue section, the commutative formal logarithm on the identity monad. No global uniformization.
Primitive obstruction (Corollary 33.10)	In addition, surjectivity of ∂ . Standard part only of finite primitive vectors.
Power-speed threshold (Corollary 33.11)	Explicit power or logarithmic primitives and the abelian image theorem; no surjectivity assumption.

Part VII

Transfinite critical potentials: second-order equations at every logarithmic depth

This part is the contribution of member 13 (appendix C). It treats the scalar second-order equations

$$\partial^2 y + (Q_\alpha + c(\ell_\alpha^\dagger)^2)y = 0, \quad Q_\alpha = \frac{1}{4} \sum_{\beta < \alpha} (\ell_\beta^\dagger)^2,$$

where α is an arbitrary ordinal, c is an ordinary real number, $\ell_\alpha = \omega^{\omega^{-\alpha}}$ is the log-atomic tower that continues the iterated logarithms of equation (3.2) through every ordinal, and $\ell^\dagger = \partial\ell/\ell$. The unknown ranges over all of **No** or **No**[i]. For every ordinal the solution space is two-dimensional when $c \leq \frac{1}{4}$ and zero when $c > \frac{1}{4}$ (Theorem 38.1); no set of earlier logarithmic scales decides which case occurs (Theorem 39.2); and at every nonzero limit ordinal three nested, ∂ -stable, real closed Hahn fields that contain the same potential have solution dimensions 0, 1 and 2, the solution field over the first having the upper triangular Borel subgroup of $\mathrm{SL}_2(\mathbb{C})$ as differential Galois group (Theorems 40.4 and 41.2). Four relations to the rest of the report fix its place.

- *The missing oscillator.* The supercritical half is the second-order statement of Corollary 8.8 read at another scale. Remark 37.5 makes this precise: one gauge identity carries $\partial^2 y + b^2 y = 0$ to the Euler operator at the scale e^ω , and the family to the same Euler operator at the scale ℓ_α ; in both cases the obstruction is an infinite accumulated phase, $|b|\omega$ in the first and $\sqrt{c - \frac{1}{4}}\ell_{\alpha+1}$ in the second. Neither statement contains the other.
- *First-order thresholds and Part V.* The coefficient $\ell_n^\dagger$ is the boundary case $p = 1$ of Corollary 10.3, and Corollary 37.1 carries that obstructed boundary to every ordinal depth. Stage 0 is a power-Hahn regular-singular system, and its classification in Part V agrees with the one here. For $\alpha \geq 1$ the potential lies outside $K_{\mathbb{R}}$ except at $(\alpha, c) = (1, 0)$, which is the previous stage's critical endpoint $Q_{1,0} = Q_{0,1/4}$. The Conway exponents outside $\mathbb{R}$ in the other potentials put them beyond that coefficient field (Remark 38.4).
- *Part III.* For a real coefficient, Theorem 17.6 excludes a one-dimensional solution space in **No**. The family confirms this without that part's import, exhibits the one-dimensional case inside a set-sized subfield exactly where a primitive is missing (Theorem 40.4 and Proposition 40.5), and has an explicit canonical amplitude and phase (Remark 38.5).
- *Workspaces.* The fields of Section 40 are full Hahn fields K_Γ in the sense of Section 20, of higher rank and with logarithmic monomials. Like Examples 18.2 and 20.7 they separate closure properties, here with algebraic closure, derivative stability and closure under logarithms all present while an exponential and then a primitive are missing (Remark 40.6).

Every statement of this part is relative to the one fixed ∂ (Warning 3.3). The main construction uses the clauses of Theorem 3.1 other than (BM6), because the primitives it needs are supplied by the tower itself; surjectivity enters only the real Riccati remark (Remark 38.2) and the comparison with Part III (Remark 38.5), which also uses that part's import. The part adds one input that Theorem 3.1 does not list: the ordinal log-atomic formulas of [3], imported with their limit cases in Proposition 35.3.

35 The log-atomic tower, and the alphabet of this part

Convention 35.1 (The alphabet of this part). Throughout Part VII:

- Ordinals are written α, β, ζ , and λ always denotes a *nonzero limit ordinal*; in this part λ is never an eigenvalue. An ordinal is regarded as a surreal number, and ω^x is the Conway monomial map of equation (2.1), with Warning 2.2 in force. Exponent-group elements are written γ . Member 13's ordinal index η is ζ here.
- $\ell_\alpha = \omega^{\omega^{-\alpha}}$, $\varpi_\alpha = \sum_{\beta < \alpha} \omega^{-\beta}$, $\ell_{<\alpha} = \omega^{\varpi_\alpha}$ and $u_\alpha = \omega^{\varpi_\alpha/2}$. Member 13 writes p_α , s_α , P_α , U_α for $\omega^{-\alpha}$, ϖ_α , $\ell_{<\alpha}$, u_α : here p is the real threshold exponent of Corollaries 10.2 and 10.3, s is a scale as in ∂_s , the P_j are phases, and U is an ordinary complex domain.
- Logarithmic derivatives are written with $\dagger$. Member 13's a_α , A_α and h_α are $\ell_\alpha^\dagger$, $\ell_{<\alpha}^\dagger$ and $u_\alpha^\dagger = \frac{1}{2}\ell_{<\alpha}^\dagger$ (Lemma 36.1); a is a real coefficient and A a matrix here.
- Potentials are written Q : Q_α as above (Lemma 36.1), $Q_{\alpha,c} = Q_\alpha + c(\ell_\alpha^\dagger)^2$, and a general potential $Q \in \mathbf{No}[i]$. Member 13 writes $q_{\alpha,c}$ and q ; in this report q is the rank-one coefficient of Theorem 8.4.
- A scale is written s , with the rescaled derivation ∂_s of Table 1; the Euler operator is $s\partial_s$, which is $\partial_{\log s}$ when $s > 0$. Member 13 writes T , D_T and E_T ; the sans-serif T is an adjoined generator here, never a number. Its partner U with $U^2 T' = 1$ is u here.
- $\text{Wr}(y_1, y_2) = y_1 \partial y_2 - y_2 \partial y_1$, as in equation (17.10), and Wr_s is the same expression for ∂_s ; member 13 writes W , which is a unitary matrix here. For a differential subfield $F \subseteq \mathbf{No}[i]$ and $Q \in F$, $\text{Sol}_F(Q) = \{y \in F : \partial^2 y + Qy = 0\}$.
- At a limit ordinal the exponent groups are $\Delta_0 \subset \Delta_1 \subset \Delta_2$ and the fields are $H_{\Delta_j} = \mathbb{R}((t^{\Delta_j}))$ and $K_{\Delta_j} = \mathbb{C}((t^{\Delta_j}))$, as in Table 13. Member 13 writes Γ_j , K_j and k_j ; the report's K_0 is $\mathbb{C}((t))$, and Γ_n are the stages of Theorem 20.3.
- Y is a fundamental matrix, $\psi_{a,b}$ a differential automorphism, $\text{Bor}_2(\mathbb{C})$ the upper triangular Borel subgroup of $\text{SL}_2(\mathbb{C})$, and $\text{Sch}(s)$ the Schwarzian. Member 13 writes Φ , $\sigma_{a,b}$, $B(\mathbb{C})$ and $\mathcal{S}_\partial$; here Φ is the phase map, σ the local parameter of Part VI, B an accumulated phase and $\mathcal{S}$ the strip. The word “Borel” in this part names that algebraic subgroup; it is unrelated to the Borel summation disclaimed in item N37.
- Member 13's real infinitesimal primitive θ of an angular rate is the infinitesimal phase residue η of Convention 5.1(b), and its Riccati solution b is w here.
- For $g \neq 0$, $f \prec g$ means $f/g \in \mathfrak{m}_{\mathbb{C}}$, and $f \sim g$ means $f/g - 1 \in \mathfrak{m}_{\mathbb{C}}$, as in Part VI.

Warning 35.2 (Member 13's $\mathfrak{m}$ is not this report's $\mathfrak{m}$). Member 13 writes $\mathfrak{m}(x) = \omega^x$ for the Conway monomial map, and $\mathfrak{i}$ and $\mathcal{O}$ for the infinitesimals and the finite surreals. In this report $\mathfrak{m}$ is the ideal of infinitesimals and never takes an argument, $\mathcal{O}$ is the class of finite surreals, and the monomial map is written ω^x or, in increasing orientation, t^{-x} (equation (2.2)). Every formula of the member has been transcribed accordingly. A reader moving between the two texts should translate $\mathfrak{m}(\cdot)$ before reading anything else.

Proposition 35.3 (The ordinal log-atomic tower: imported). *For every ordinal α the families $(\omega^{-\beta})_{\beta < \alpha}$ and $(\ell_{\beta+1})_{\beta < \alpha}$ are summable, ℓ_α is positive infinite and log-atomic, and*

$$\log \ell_\alpha = \ell_{\alpha+1}, \quad \partial \ell_\alpha = \ell_{<\alpha}^{-1}, \quad \ell_{<\alpha} = \omega^{\varpi_\alpha} = \exp\left(\sum_{\beta < \alpha} \ell_{\beta+1}\right). \quad (35.1)$$

For every real r , $\ell_\alpha^r = \exp(r\ell_{\alpha+1}) = \omega^{r\omega^{-\alpha}}$ and $\ell_{<\alpha}^r = \omega^{r\varpi_\alpha}$.

These are imported, not proved here: [3, Lemma 2.3, Proposition 2.5, Lemmas 2.6–2.8 and the pre-derivation formula on p. 10], where the notation is $\lambda_{-\alpha} = \log_\alpha \omega$, and for the real powers the real-linear exponent formula in the proof of [3, Lemma 2.3]; the numbering is that of arXiv version 3, checked for this review. For finite n , ℓ_n is the n -fold logarithm of ω and $\ell_{<n} = \ell_0 \cdots \ell_{n-1}$, so equation (35.1) returns equation (3.2); $\ell_0 = \omega$ and $\ell_1 = \log \omega$. At an infinite limit λ , ℓ_λ is the simplest positive infinite surreal below every ℓ_β with $\beta < \lambda$. The monomial $\ell_{<\alpha}$ is the monomial product $\prod_{\beta < \alpha} \ell_\beta$, and its two readings — sum the exponents $\omega^{-\beta}$ in normal form, or exponentiate the sum of the logarithms $\ell_{\beta+1}$ — agree by the monomial-product logarithm identity of [3, Lemma 2.3]. At a limit ordinal that identity is an imported fact; replacing it by ordinary convergence of finite products would be invalid (Warning 2.5). Likewise the limit-stage derivative formula is a fact about this derivation, not the result of formally differentiating a finite iterate, and the normalization $\partial\omega = 1$ alone does not determine it (Remark 42.4). The displayed real-power identities do not define ω^z for non-real z , or a complex power of a positive infinite scale; Corollary 37.1 will give the relevant differential obstruction. The real powers rest on two facts from [3, Section 2]: the logarithm of a monomial is purely infinite, and the correspondence between the exponent of a monomial and its logarithm is real-linear. Since the monomial map is multiplicative, $u_\alpha^2 = \ell_{<\alpha}$. The empty-index conventions are $\varpi_0 = 0$, $\ell_{<0} = u_0 = 1$ and $\partial\ell_0 = 1$. Every displayed family is indexed by the set of ordinals below one fixed α ; none is summed over the class of all ordinals.

Remark 35.4 (What the notation does not assert). The symbols ℓ_α designate particular log-atomic numbers. They do not assert a globally defined transfinite iteration or composition of $\log$ on **No** (Remark 4.7), and Proposition 39.5 shows that the tower has no final stage inside **No**. The remark after Corollary 10.3 — “no transfinite logarithm tower is used” — remains true of Part II; this part imports that tower.

Lemma 35.5 (Separation of scales). *The families $(\omega^{-\beta})_{\beta < \alpha}$ and $(\ell_{\beta+1})_{\beta < \alpha}$ are strictly decreasing, and $\varpi_\beta < \varpi_\zeta$ for $\beta < \zeta$. Consequently $\ell_\beta^\dagger = \omega^{-\varpi_{\beta+1}} = \ell_{<\beta+1}^{-1} = \partial\ell_{\beta+1}$,*

$$\frac{\ell_{\beta+1}^\dagger}{\ell_\beta^\dagger} = \frac{1}{\ell_{\beta+1}}, \quad \ell_\zeta^\dagger \prec \ell_\beta^\dagger \quad (\beta < \zeta), \quad (35.2)$$

and at a limit λ , $\ell_\lambda \prec \ell_\beta$ for every $\beta < \lambda$.

Proof. The monomial map is strictly increasing and $-\zeta < -\beta$, which gives the first assertion, including the decrease of $\ell_{\beta+1} = \omega^{\omega^{-(\beta+1)}}$. For $\beta < \zeta$ the difference $\varpi_\zeta - \varpi_\beta = \sum_{\beta \leq \xi < \zeta} \omega^{-\xi}$ has leading term $\omega^{-\beta}$ with coefficient 1; hence it is positive and ϖ increases. By equation (35.1), $\ell_\beta^\dagger = \ell_{<\beta}^{-1} \ell_\beta^{-1} = \omega^{-\varpi_\beta - \omega^{-\beta}} = \omega^{-\varpi_{\beta+1}}$, and $\partial\ell_{\beta+1} = \partial\log \ell_\beta = \ell_\beta^\dagger$. Distinct monomials are Archimedean separated, and $\varpi_{\beta+2} - \varpi_{\beta+1} = \omega^{-(\beta+1)}$ gives the ratio. Finally $\ell_\lambda/\ell_\beta = \omega^{\omega^{-\lambda} - \omega^{-\beta}}$ has strictly negative Conway exponent, so it is infinitesimal. The same argument works for any $\beta < \zeta$, not only at a limit. $\square$

For finite n , $\ell_n^\dagger = 1/(\ell_0 \ell_1 \cdots \ell_n)$ is the coefficient $b_{n,1}$ of Corollary 10.3: the obstructed boundary case $p = 1$ at depth n .

36 The critical identity at every ordinal

Lemma 36.1 (Summation and triangular differentiation). *For every ordinal α the families $(\ell_\beta^\dagger)_{\beta < \alpha}$, $((\ell_\beta^\dagger)^2)_{\beta < \alpha}$ and $(\ell_\beta^\dagger \ell_\zeta^\dagger)_{\beta, \zeta < \alpha}$ are summable, and*

$$\ell_{<\alpha}^\dagger = \sum_{\beta < \alpha} \ell_\beta^\dagger, \quad (36.1)$$

$$\partial(\ell_\beta^\dagger) = -\ell_\beta^\dagger \ell_{<\beta+1}^\dagger \quad \text{for every ordinal } \beta, \quad (36.2)$$

$$\partial(\ell_{<\alpha}^\dagger) = -\frac{1}{2} \left((\ell_{<\alpha}^\dagger)^2 + \sum_{\beta < \alpha} (\ell_\beta^\dagger)^2 \right). \quad (36.3)$$

In particular Q_α is a well-defined surreal, $Q_0 = 0$, and $Q_\alpha > 0$ for $\alpha \geq 1$.

Proof. In the orientation of equation (2.2), $\ell_\beta^\dagger = t^{\varpi_{\beta+1}}$ with exponents strictly increasing in β (Lemma 35.5); an ordinal-indexed increasing family of exponents is well ordered and each exponent occurs once, so the first two families are summable (Definition 2.4); doubling the exponents preserves their strict increase. Summability of $(\ell_{\beta+1})_{\beta < \alpha}$ was supplied by Proposition 35.3.

Here the double family's support and finite incidence can also be seen directly. For $\beta \leq \zeta$ its exponent is

$$\varpi_{\beta+1} + \varpi_{\zeta+1} = 2 \sum_{\xi \leq \beta} \omega^{-\xi} + \sum_{\beta < \xi \leq \zeta} \omega^{-\xi}.$$

The first place where these coefficient strings differ shows that these exponents increase in the lexicographic order on (β, ζ) with $\beta \leq \zeta < \alpha$. This is a well-order: choose the least first coordinate and then the least second coordinate in any nonempty subset. The coefficient string also recovers the two indices uniquely. Thus each exponent receives one ordered pair on the diagonal and two off it. This proves both conditions for summability and licenses restrictions to either triangle, multiplication of the two sums and regrouping. It is the Hahn product theorem in this particular support [1, Section 2.4].

Since $\ell_{<\alpha} = \exp(\sum_{\beta < \alpha} \ell_{\beta+1})$, clauses (BM4) and (BM3) give $\ell_{<\alpha}^\dagger = \sum_{\beta < \alpha} \partial \ell_{\beta+1} = \sum_{\beta < \alpha} \ell_\beta^\dagger$, which is equation (36.1). As $\ell_\beta^\dagger = \ell_{<\beta+1}^{-1}$, this gives equation (36.2). Differentiating equation (36.1) by (BM3),

$$\partial(\ell_{<\alpha}^\dagger) = - \sum_{\beta < \alpha} \sum_{\zeta \leq \beta} \ell_\beta^\dagger \ell_\zeta^\dagger,$$

a subfamily of the summable double family. The square $(\ell_{<\alpha}^\dagger)^2$ counts each off-diagonal unordered pair twice and each diagonal pair once, so twice the triangular sum is $(\ell_{<\alpha}^\dagger)^2 + \sum_{\beta < \alpha} (\ell_\beta^\dagger)^2$; all regroupings are licensed by the joint summability just proved (Warning 2.5). For $\alpha \geq 1$ the leading term of Q_α is $\frac{1}{4}(\ell_0^\dagger)^2 = \frac{1}{4}\omega^{-2} > 0$. $\square$

Theorem 36.2 (Exact critical solutions and Wronskian). *For every ordinal α ,*

$$u_\alpha^\dagger = \frac{1}{2} \ell_{<\alpha}^\dagger, \quad \partial^2 u_\alpha + Q_\alpha u_\alpha = 0, \quad u_\alpha^2 \partial \ell_\alpha = 1. \quad (36.4)$$

Both u_α and $u_\alpha \ell_\alpha$ solve $\partial^2 y + Q_\alpha y = 0$, and $\text{Wr}(u_\alpha, u_\alpha \ell_\alpha) = 1$.

Proof. The first identity is the derivative of $\log u_\alpha = \frac{1}{2} \sum_{\beta < \alpha} \ell_{\beta+1}$. Hence, by equation (36.3),

$$\frac{\partial^2 u_\alpha}{u_\alpha} = \partial(u_\alpha^\dagger) + (u_\alpha^\dagger)^2 = \frac{1}{2} \partial(\ell_{<\alpha}^\dagger) + \frac{1}{4} (\ell_{<\alpha}^\dagger)^2 = -\frac{1}{4} \sum_{\beta < \alpha} (\ell_\beta^\dagger)^2 = -Q_\alpha.$$

The third identity is $\ell_{<\alpha} \partial \ell_\alpha = 1$. Write $u = u_\alpha$ and $s = \ell_\alpha$. Differentiating $u^2 \partial s = 1$ gives $2\partial u \partial s + u \partial^2 s = 0$, so $\partial^2(us) + Q_\alpha us = s(\partial^2 u + Q_\alpha u) + 2\partial u \partial s + u \partial^2 s = 0$, and $\text{Wr}(u, us) = u^2 \partial s = 1$. $\square$

Remark 36.3 (A limit stage is not a missing induction step). At a successor, $\ell_{<\alpha+1} = \ell_{<\alpha} \ell_\alpha$ and $Q_{\alpha+1} = Q_\alpha + \frac{1}{4} (\ell_\alpha^\dagger)^2$. At a limit, the same identity for $\partial^2 u_\alpha$ comes from the single summable triangular calculation of Lemma 36.1. No derivative of a nonconvergent net and no interchange of an uncontrolled double series occurs, and Q_λ is defined by the Hahn sum of its increments, without assuming topological convergence of the partial sums Q_β . Summing those partial sums would be a different, invalid operation: at every nonzero limit λ , the family $(Q_\beta)_{0 < \beta < \lambda}$ has infinitely many contributions $\frac{1}{4} \omega^{-2}$ at the same exponent and is not summable.

Lemma 36.4 (A fundamental pair spans every solution). *Let F be a differential field of characteristic zero with constants C_F , let $Q \in F$, and let $y_1, y_2 \in F$ solve $\partial^2 y + Qy = 0$ with $\text{Wr}(y_1, y_2) \neq 0$. Then $\text{Wr}(y_1, y_2) \in C_F$, and every solution in F has a unique expression in $C_F y_1 + C_F y_2$. Thus this solution space has dimension two over C_F .*

Proof. For any two solutions v, w , $\partial \text{Wr}(v, w) = v \partial^2 w - w \partial^2 v = -Qvw + Qwv = 0$. In particular $\mathcal{W} = \text{Wr}(y_1, y_2)$ is a nonzero constant. For a third solution y , solve the two linear equations

$$\begin{pmatrix} y_1 & y_2 \\ \partial y_1 & \partial y_2 \end{pmatrix} \begin{pmatrix} a \\ b \end{pmatrix} = \begin{pmatrix} y \\ \partial y \end{pmatrix}.$$

Their determinant is $\mathcal{W}$, so their unique coefficients are $a = \text{Wr}(y, y_2)/\mathcal{W}$ and $b = \text{Wr}(y_1, y)/\mathcal{W}$. The numerator and denominator in each quotient have derivative zero, hence $a, b \in C_F$. Conversely, constant linear combinations remain solutions. If such a combination is zero, its derivative is zero too, and the same invertible matrix makes both coefficients zero. This proves spanning and independence. $\square$

Applied to Theorem 36.2, these formulas give exactly $y = u_\alpha(a + b\ell_\alpha)$, with unique ordinary constants a, b (real in $\mathbf{No}$, complex in $\mathbf{No}[i]$). The first two stages make the normalization concrete: for $\alpha = 0$ the equation is $\partial^2 y = 0$ and the pair is $(1, \omega)$; for $\alpha = 1$ the potential is $\frac{1}{4} \omega^{-2}$ and the pair is $(\sqrt{\omega}, \sqrt{\omega} \log \omega)$. Both Wronskians are 1.

With $F = \mathbf{No}[i]$ the solution collection is first a class; the spanning formulas of Theorem 38.1 show that it is a set, isomorphic to $\mathbb{C}^2$ or to 0, so no dimension theory of a proper-class vector space is used, as in Remark 25.6. All Hahn and Picard–Vessiot fields of Sections 40 and 41 are set-sized from the outset.

37 The Euler threshold at an infinite scale, and the missing oscillator

Corollary 37.1 (No nonreal powers of an infinite scale). *Let $s \in \mathbf{No}$ with $s > \mathbb{R}$, so that $\partial s > 0$ by (BM7), and let $\nu \in \mathbb{C} \setminus \mathbb{R}$. Then*

$$\partial z = \nu s^\dagger z, \quad \text{equivalently} \quad s \partial_s z = \nu z, \quad (37.1)$$

has no nonzero solution in $\mathbf{No}[i]$. For real ν its solutions are exactly $\mathbb{C} s^\nu$, with $s^\nu = \exp(\nu \log s)$.

Proof. Write $\nu = a + ib$ with $a, b \in \mathbb{R}$ and $b \neq 0$. The imaginary coefficient $b s^\dagger$ has the explicit real primitive $b \log s$. This primitive is infinite in modulus: $\log s$ exceeds every real number, since an ordinary upper bound would exponentiate to an ordinary upper bound on s . If a nonzero solution existed, the direction (i) $\Rightarrow$ (ii) of Theorem 8.4, or its direct polar proof, would supply an infinitesimal real η with $\partial\eta = b s^\dagger$. Then $\partial(\eta - b \log s) = 0$, so $\eta - b \log s \in \mathbb{R}$ by the constant-field property. An ordinary constant cannot turn an infinite primitive into a finite one, a contradiction. This uses the displayed primitive and no surjectivity of ∂ .

For real ν , $s^\nu = \exp(\nu \log s)$ is nonzero and has logarithmic derivative $\nu s^\dagger$. If z is any solution, $\partial(z/s^\nu) = 0$, so $z/s^\nu \in \mathbb{C}$; conversely each such constant multiple solves the equation, including zero. $\square$

Member 13 proves this from an infinitesimal real primitive of the angular rate of a unit, obtained from the strong Hahn logarithm; that primitive is the infinitesimal phase residue η of Theorem 6.3, and its argument is the direction (i) $\Rightarrow$ (iii) of Theorem 8.4. The route is recorded as agreement and not printed twice. The forbidden coefficient $\nu s^\dagger$ may be very small — at $s = \ell_\alpha$ it lies below $\ell_\beta^\dagger$ for every $\beta < \alpha$ — and the obstruction is the infinite primitive $\log s$, not the size of the coefficient (Warning 8.5). Three earlier statements are instances: $s = e^\omega$ gives the first-order statement of Corollary 8.8; $s = \ell_n$ gives the case $p = 1$ of Corollary 10.3, multiplied by $\text{Im } \nu$; and $s = \ell_1$, $\nu = -i$ gives Proposition 28.4, since $\partial_\tau y = iy / \log \omega$ is $\partial y = -i \ell_1^\dagger y$.

Lemma 37.2 (Liouville transformation as an operator identity). *Let F be a differential field of characteristic zero, and let $s, u, Q \in F$ with $s \neq 0$, $\partial s \neq 0$, $u \neq 0$, $\partial^2 u + Qu = 0$ and $u^2 \partial s = 1$. Define $\partial_s = (\partial s)^{-1} \partial$ and $s^\dagger = \partial s / s$. Then for every $v, c \in F$,*

$$\left(\partial^2 + Q + c(s^\dagger)^2 \right)(uv) = u(s^\dagger)^2 \left((s \partial_s)^2 - s \partial_s + c \right) v. \quad (37.2)$$

Proof. Differentiating $u^2 \partial s = 1$ gives $2\partial u \partial s + u \partial^2 s = 0$. Since $\partial v = (\partial s) \partial_s v$ and $\partial^2 v = (\partial^2 s) \partial_s v + (\partial s)^2 \partial_s^2 v$, the finite product rule gives

$$\partial^2(uv) + Quv = (\partial^2 u + Qu)v + (2\partial u \partial s + u \partial^2 s) \partial_s v + u(\partial s)^2 \partial_s^2 v = u(\partial s)^2 \partial_s^2 v.$$

Since $\partial_s s = 1$, $(s \partial_s)^2 - s \partial_s = s^2 \partial_s^2$; adding the term $c(s^\dagger)^2 uv$ gives equation (37.2). $\square$

The rescaled operator ∂_s is a derivation and has the same constant field as ∂ , since its multiplying scalar is nonzero. The identity does not differentiate c , so c need not be constant in this lemma; the Euler classification below does require an ordinary constant. It gives a bijection between the two kernels by $v \mapsto uv$, because both u and $u(s^\dagger)^2$ are nonzero. It also transports Wronskians exactly:

$$\text{Wr}(uv_1, uv_2) = u^2 (\partial s) \text{Wr}_s(v_1, v_2) = \text{Wr}_s(v_1, v_2).$$

These are pointwise field identities, so they also apply in $\mathbf{No}[i]$. They treat v as an arbitrary element of the field and do not assert that every surreal is a composite $v(s)$ and invoke no substitution operator; such a distinction is essential in view of the global composition obstruction of [2, Theorem 8.4] (Remark 4.7).

Proposition 37.3 (Euler classification at a positive infinite scale). *Let $s \in \mathbf{No}$ with $s > \mathbb{R}$ (so $\partial s > 0$), let $c \in \mathbb{R}$, and put $r_{\pm} = \frac{1}{2}(1 \pm \sqrt{1 - 4c})$ when $c \leq \frac{1}{4}$. Then*

$$\ker_{\mathbf{No}[i]}((s\partial_s)^2 - s\partial_s + c) = \begin{cases} \mathbb{C}s^{r_+} + \mathbb{C}s^{r_-}, & c < \frac{1}{4}, \\ \mathbb{C}s^{1/2} + \mathbb{C}s^{1/2}\log s, & c = \frac{1}{4}, \\ 0, & c > \frac{1}{4}, \end{cases}$$

and the real kernel is the same with $\mathbb{R}$ in place of $\mathbb{C}$.

Proof. The operator is $s^2(\partial_s^2 + cs^{-2})$, and ∂_s is a derivation of $\mathbf{No}[i]$ with constants $\mathbb{C}$. For real r , $s\partial_s(s^r) = rs^r$. If $c < \frac{1}{4}$ the two distinct real roots of $r^2 - r + c$ give solutions with $\text{Wr}_s(s^{r_+}, s^{r_-}) = (r_- - r_+)s^{r_+ + r_- - 1} = r_- - r_+ \neq 0$, and Lemma 36.4 for ∂_s gives completeness. If $c = \frac{1}{4}$, put $v_0 = s^{1/2}$ and $v_1 = s^{1/2}\log s$. Then

$$(s\partial_s - \tfrac{1}{2})v_0 = 0, \quad (s\partial_s - \tfrac{1}{2})v_1 = v_0.$$

Both therefore lie in the squared factor's kernel, and $\text{Wr}_s(v_0, v_1) = v_0^2\partial_s(\log s) = 1$. The same spanning lemma proves completeness at the repeated root. If $c > \frac{1}{4}$ the roots are $\nu_{\pm} = \frac{1}{2} \pm i\sqrt{c - \frac{1}{4}}$, the operator factors as $(s\partial_s - \nu_+)(s\partial_s - \nu_-)$ with commuting constant-coefficient factors. If their product kills v , first apply Corollary 37.1 to $(s\partial_s - \nu_-)v$: it must be zero. Apply that corollary again to obtain $v = 0$. Thus the conclusion does not assume that a candidate solution has a power or exponential form. For the real statement, the displayed bases are real and independent over $\mathbb{C}$. Taking imaginary parts of a real solution's unique complex expansion forces both coefficients to be real. $\square$

The number $\frac{1}{4}$ and the repeated-root logarithm are the classical Euler phenomenon. The nonreal-root argument excludes *every* nonzero element of $\mathbf{No}[i]$, not merely a particular trial formula.

Remark 37.4 (The Euler operator is a constant-coefficient operator at a rescaled scale). This remark is an observation of this merge; member 13 does not state it. For $s > 0$ with $\partial s \neq 0$, one has $s\partial_s = \partial_{\log s}$ (Proposition 4.2), and $\log s$ is positive infinite when $s > \mathbb{R}$. So Proposition 37.3 is the order-two case of the constant-coefficient classification of Theorem 13.2, with $P(X) = X^2 - X + c$, read for $\partial_{\log s}$ and $\log s$ in place of ∂ and ω : the roots are real exactly when $c \leq \frac{1}{4}$, and $\exp(r \log s) = s^r$ for each real root r . Theorem 13.2 is stated for ∂ only; its proof goes through for $\partial_{\log s}$ with Corollary 37.1 in place of Corollary 8.8, but nothing below uses that, because member 13's direct proof is complete.

Remark 37.5 (The missing oscillator is the Euler problem at the scale e^ω). This remark is an observation of this merge, with its proof; member 13 does not state it. Take $s = e^\omega$. Then $s^\dagger = 1$, $s\partial_s = \partial$, and $u = e^{-\omega/2}$ satisfies $u^2\partial s = 1$ and $\partial^2 u - \frac{1}{4}u = 0$, so Lemma 37.2 with $Q = -\frac{1}{4}$ reads

$$(\partial^2 + c - \tfrac{1}{4})(e^{-\omega/2}v) = e^{-\omega/2}(\partial^2 - \partial + c)v.$$

For $c = \frac{1}{4} + b^2$ with $b \in \mathbb{R} \setminus \{0\}$ the left operator is $\partial^2 + b^2$, and the case $c > \frac{1}{4}$ of Proposition 37.3 is the second-order statement of Corollary 8.8; for $c = \frac{1}{4} - \kappa^2$ with $\kappa > 0$ real it returns exactly $Ae^{\kappa\omega} + Be^{-\kappa\omega}$, $A, B \in \mathbb{C}$, solving $\partial^2 y - \kappa^2 y = 0$. At the repeated endpoint $c = \frac{1}{4}$ it instead gives $A + B\omega$, as Theorem 13.2 does. At the scale $s = \omega$ the same lemma, with $u = 1$ and $Q = 0$, gives stage 0 of the family, and at the scale $s = \ell_\alpha$, with $u = u_\alpha$ and $Q = Q_\alpha$, it gives Theorem 38.1. So the constant-potential oscillator, stage 0 and stage α are one Euler threshold read at the scales

$\log s = \omega$, ℓ_1 and $\ell_{\alpha+1}$, and in every supercritical case the obstruction is the infinite accumulated phase $\sqrt{c - \frac{1}{4}} \log s$ of Corollary 37.1; the constant-potential threshold $b^2 > 0$ is the Euler threshold $c > \frac{1}{4}$ shifted by the gauge potential $-\frac{1}{4}$.

This is the precise sense, and the only sense, in which the supercritical half of Theorem 38.1 generalizes the missing oscillator: it moves the obstructed accumulated phase from ω down through every stage of the log-atomic tower, where the forbidden potential $(c - \frac{1}{4})(\ell_\alpha^\dagger)^2$ lies below every earlier scale (Theorem 39.2). The oscillator is not a member of the family: its potential is an ordinary constant, its gauge potential $-\frac{1}{4}$ is negative, and its scale e^ω lies above the whole tower, since $\log e^\omega = \ell_0$. Conversely the family is not a corollary of Corollary 8.8: it needs the imported tower, the critical identity Theorem 36.2 and the criterion at the scale $\ell_{\alpha+1}$. Neither statement says anything about a general potential (Remark 38.3).

38 The transfinite classification

Theorem 38.1 (Transfinite critical classification). *For every ordinal α and every $c \in \mathbb{R}$, the potential $Q_{\alpha,c} = Q_\alpha + c(\ell_\alpha^\dagger)^2$ is a well-defined surreal, and the solution space of*

$$\partial^2 y + Q_{\alpha,c} y = 0 \quad (38.1)$$

in $\mathbf{No}$ has dimension two over $\mathbb{R}$ if $c \leq \frac{1}{4}$ and dimension zero if $c > \frac{1}{4}$. The identical statement holds in $\mathbf{No}[i]$ over $\mathbb{C}$. For $c < \frac{1}{4}$ a basis is

$$u_\alpha \ell_\alpha^{r+}, \quad u_\alpha \ell_\alpha^{r-}, \quad r_\pm = \frac{1}{2}(1 \pm \sqrt{1 - 4c}), \quad (38.2)$$

and for $c = \frac{1}{4}$ a basis is

$$u_{\alpha+1}, \quad u_{\alpha+1} \ell_{\alpha+1}. \quad (38.3)$$

Every displayed basis element is a single Conway monomial, and $Q_{\alpha,1/4} = Q_{\alpha+1}$.

Proof. The coefficients exist by Lemma 36.1. Apply Lemma 37.2 in $F = \mathbf{No}[i]$ with $s = \ell_\alpha$, $u = u_\alpha$ and $Q = Q_\alpha$; its hypotheses are Theorem 36.2, and $s^\dagger = \ell_\alpha^\dagger$. Multiplication by u_α is a bijection of $\mathbf{No}[i]$ and of $\mathbf{No}$, and the factor $u(s^\dagger)^2$ is nonzero, so equation (38.1) is equivalent to $((s\partial_s)^2 - s\partial_s + c)v = 0$, and Proposition 37.3 gives the dimensions and bases. At $c = \frac{1}{4}$ use $u_\alpha \ell_\alpha^{1/2} = \omega^{\varpi_\alpha/2 + \omega^{-\alpha}/2} = u_{\alpha+1}$ and $\log \ell_\alpha = \ell_{\alpha+1}$. The monomial assertion is $u_\alpha \ell_\alpha^r = \omega^{\varpi_\alpha/2 + r\omega^{-\alpha}}$, by Proposition 35.3. $\square$

The theorem is proved at limit ordinals by the same argument as at successors; it is not an induction that silently skips limits. The critical endpoint is exactly the next stage, including after a limit. In Conway normal form,

$$Q_{\alpha,c} = \frac{1}{4} \sum_{\beta < \alpha} \omega^{-2\varpi_{\beta+1}} + c \omega^{-2\varpi_{\alpha+1}}, \quad (38.4)$$

$$y_\pm = \omega^{\varpi_\alpha/2 + r_\pm \omega^{-\alpha}} \quad (c < \frac{1}{4}), \quad (38.5)$$

$$y_0 = \omega^{\varpi_{\alpha+1}/2}, \quad y_1 = \omega^{\varpi_{\alpha+1}/2 + \omega^{-(\alpha+1)}} \quad (c = \frac{1}{4}), \quad (38.6)$$

with no unspecified primitive, inverse operator or integration constant; for $c = 0$ the last term of equation (38.4) is omitted. The Wronskian transport following Lemma 37.2 gives, in the printed order, $\text{Wr}(y_+, y_-) = -\sqrt{1 - 4c}$ and $\text{Wr}(y_0, y_1) = 1$. Consequently every solution is a unique constant linear combination of its displayed pair; the constants are the Wronskian ratios of Lemma 36.4.

Remark 38.2 (A Riccati form). For a real root r of $r^2 - r + c$, the logarithmic derivative of $u_\alpha \ell_\alpha^r$ is

$$w = \frac{1}{2} \ell_{<\alpha}^\dagger + r \ell_\alpha^\dagger, \quad \partial w + w^2 + Q_{\alpha,c} = 0. \quad (38.7)$$

Indeed $\partial \ell_\alpha^\dagger = -\ell_\alpha^\dagger (\ell_{<\alpha}^\dagger + \ell_\alpha^\dagger)$ by equation (36.2), and the coefficient of $(\ell_\alpha^\dagger)^2$ in the residual is $r^2 - r + c$. When $c > \frac{1}{4}$ there is no *real* Riccati solution: if such a w existed, surjectivity would supply $H \in \mathbf{No}$ with $\partial H = w$. Then $y = \exp H \neq 0$ would satisfy $\partial^2 y + Q_{\alpha,c} y = (\partial w + w^2 + Q_{\alpha,c}) y = 0$, contradicting Theorem 38.1. This implication uses (BM6), together with the real exponential law; the second-order nonexistence itself does not.

Remark 38.3 (Not a classification of arbitrary potentials). The theorem is exact for the family equation (38.4). It does not say that every surreal potential has this form, that a comparison of leading coefficients settles arbitrary perturbations, or that every $\varepsilon \prec (\ell_\alpha^\dagger)^2$ preserves subcritical or supercritical behaviour of $Q_{\alpha,c} + \varepsilon$; that would need further differential asymptotic estimates, which the gauge identity alone does not give. At the critical endpoint a smaller perturbation can change the answer: for any real d ,

$$Q_{\alpha,1/4} + d(\ell_{\alpha+1}^\dagger)^2 = Q_{\alpha+1,d}.$$

The added term is $\prec (\ell_\alpha^\dagger)^2$ when $d \neq 0$, yet $d = 0$ gives dimension two and $d = 1$ gives dimension zero. No comparison theorem for general surreal Riccati equations is used.

Remark 38.4 (Stage zero is a regular-singular system; the finite stages are first-order thresholds). At $\alpha = 0$: $\ell_0 = \omega$, $u_0 = 1$, $Q_0 = 0$ and $\ell_0^\dagger = t$, so equation (38.1) is $\partial^2 y + ct^2 y = 0$. With $\partial_\tau = -\omega \partial$ one has $\omega^2 \partial^2 = \partial_\tau^2 + \partial_\tau$, so the equation is $\partial_\tau^2 y + \partial_\tau y + cy = 0$, the power-Hahn regular-singular system $\partial_\tau Y = A_0 Y$ of Definition 23.5 for $Y = (y, \partial_\tau y)$, with $A_0 = \begin{pmatrix} 0 & 1 \\ -c & -1 \end{pmatrix}$ and $A_+ = 0$. Its eigenvalues μ solve $\mu^2 + \mu + c = 0$ and are real exactly when $c \leq \frac{1}{4}$, so Theorem 25.3 gives dimension 2 or 0, in agreement; the bases correspond through $t^\mu = \omega^r$ with $\mu = -r$, and at $c = \frac{1}{4}$ the Jordan block gives $\omega^{1/2}$ and $\omega^{1/2} \log \omega$. This comparison is made by this merge and is recorded as agreement.

There is one overlap at the next stage: $Q_{1,0} = Q_1 = \frac{1}{4} \omega^{-2} = Q_{0,1/4}$ belongs to $K_{\mathbb{R}}$ and is the repeated-root case just described. For $\alpha = 1$ with $c \neq 0$, the term $c\omega^{-2\varpi_2} = c\omega^{-2-2\omega^{-1}}$ has Conway exponent $-2 - 2\omega^{-1} \notin \mathbb{R}$, so $Q_{1,c} \notin K_{\mathbb{R}}$. For every $\alpha \geq 2$ the same monomial occurs with coefficient $\frac{1}{4}$ in Q_α ; its exponent differs from all later terms, including the perturbation, so it cannot cancel. Thus these potentials also lie outside $K_{\mathbb{R}}$. This exact support test, rather than merely the appearance of a logarithm in a formula, establishes the boundary of Part V. For finite n the supercritical obstruction is the boundary case $p = 1$ of Corollary 10.3 at depth n , and Corollary 37.1 extends it to every ordinal.

Remark 38.5 (Amplitude and phase of a supercritical member). This remark is an observation of this merge, with its proof; member 13 does not state it. Let $c > \frac{1}{4}$ and put $\rho = (c - \frac{1}{4})^{-1/4} u_{\alpha+1} > 0$. Then

$$\partial^2 \rho + Q_{\alpha,c} \rho = \rho^{-3},$$

because $\partial^2 u_{\alpha+1} = -Q_{\alpha+1} u_{\alpha+1}$ by Theorem 36.2, $Q_{\alpha,c} - Q_{\alpha+1} = (c - \frac{1}{4})(\ell_\alpha^\dagger)^2$, and $(\ell_\alpha^\dagger)^2 u_{\alpha+1}^4 = 1$ since $u_{\alpha+1}^2 = \ell_{<\alpha+1} = (\ell_\alpha^\dagger)^{-1}$. Moreover $\rho^{-2} = \sqrt{c - \frac{1}{4}} \partial \ell_{\alpha+1}$, whose normalized primitive is $\sqrt{c - \frac{1}{4}} \ell_{\alpha+1}$, a purely infinite monomial with zero infinitesimal part. By Theorem 38.1 the equation is intrinsically oscillatory in the sense of Theorem 17.7, so Theorem 17.6(i) makes ρ the *unique* positive Pinney solution, Theorem 17.7 has canonical phase $P = \sqrt{c - \frac{1}{4}} \ell_{\alpha+1}$ with $\eta = 0$, and the companion system has differential phase spectrum $\{P, -P\}$. The Pinney identity is direct; the uniqueness, the canonical phase and the spectrum use Part III, hence the import Theorem 15.6 (item N53) and (BM6). At the scale e^ω of Remark 37.5 the same formula gives $\rho = |b|^{-1/2}$ and $P = |b| \omega$ for $\partial^2 y + b^2 y = 0$, which for $b = 1$ is the value $\rho = 1$ recorded after Theorem 17.6.

39 Invisible perturbations, and no final stage

Proposition 39.1 (The exact earlier-stage asymptotics). *For $\beta < \alpha$ and $c, d \in \mathbb{R}$,*

$$Q_{\alpha,c} - Q_\beta \sim \frac{1}{4}(\ell_\beta^\dagger)^2, \quad Q_{\alpha,c} - Q_{\alpha,d} = (c-d)(\ell_\alpha^\dagger)^2 \prec (\ell_\beta^\dagger)^2 \quad (c \neq d). \quad (39.1)$$

Proof. After subtracting Q_β , the largest monomial left in equation (38.4) is $(\ell_\beta^\dagger)^2$, with coefficient $\frac{1}{4}$; every other term is strictly smaller by Lemma 35.5. More explicitly, dividing the remainder after that leading term by $(\ell_\beta^\dagger)^2$ shifts its entire Hahn support strictly below the monomial 1. The remainder is therefore infinitesimal, including at limit α ; this uses the well-ordered support of the whole sum, not a limit of finite tails. For the second assertion, the common initial sum cancels exactly, and $(\ell_\alpha^\dagger/\ell_\beta^\dagger)^2$ is infinitesimal. Multiplication by the nonzero ordinary constant $c-d$ preserves that property. $\square$

Theorem 39.2 (No set of earlier scales decides solvability). *For every set $\mathcal{B}$ of ordinals there are positive surreals Q_-, Q_+ with*

$$Q_+ - Q_- \prec (\ell_\beta^\dagger)^2 \quad \text{and} \quad Q_\pm - Q_\beta \sim \frac{1}{4}(\ell_\beta^\dagger)^2 \quad (\beta \in \mathcal{B}),$$

such that $\partial^2 y + Q_- y = 0$ has two independent surcomplex solutions and $\partial^2 y + Q_+ y = 0$ has only $y = 0$ in $\mathbf{No}[i]$.

Proof. Take the ordinal $\alpha = \sup\{\beta + 1 : \beta \in \mathcal{B}\} + 1$. This is a set ordinal, is at least 1, and exceeds every member of $\mathcal{B}$; for the empty set it gives $\alpha = 1$. Now take $Q_- = Q_{\alpha,0} = Q_\alpha$ and $Q_+ = Q_{\alpha,1}$. Both are positive, with leading term $\frac{1}{4}\omega^{-2}$; Theorem 38.1 gives dimensions two and zero, and Proposition 39.1 gives the two precision assertions. $\square$

Example 39.3 (The first infinite stage). At $\alpha = \omega$, with $\ell_n = \log_n \omega$,

$$Q_\omega = \frac{1}{4\omega^2} + \frac{1}{4\omega^2(\log \omega)^2} + \frac{1}{4\omega^2(\log \omega)^2(\log_2 \omega)^2} + \cdots \quad (39.2)$$

The coefficient $(\ell_\omega^\dagger)^2$ is smaller than every displayed monomial, yet adding it changes the dimension from two to zero, and $Q_\omega + c(\ell_\omega^\dagger)^2$ has the exact threshold $c = \frac{1}{4}$. The exact Hahn sum in equation (39.2) does determine Q_ω uniquely. What fails to determine solvability is prescribing its coefficients at all the displayed finite logarithmic scales while allowing a remainder smaller than every one of them. The two potentials agree in that data, but differ at the additional monomial $(\ell_\omega^\dagger)^2$ in their full normal forms.

Warning 39.4 (What the invisibility statement is about). Theorem 39.2 concerns the explicitly named precision hierarchy $(\ell_\beta^\dagger)^2$, not every conceivable observation or algorithm. For a fixed set $\mathcal{B}$, put

$$\mathcal{E}_\mathcal{B} = \{e \in \mathbf{No} : e \prec (\ell_\beta^\dagger)^2 \text{ for every } \beta \in \mathcal{B}\}.$$

It is closed under addition and multiplication by ordinary real constants, since each normalized error is infinitesimal. The theorem gives $Q_+ - Q_- \in \mathcal{E}_\mathcal{B}$, so solvability does not descend to these additive equivalence classes. For nonempty $\mathcal{B}$, this error class is nonzero and proper: the constructed difference belongs to it, whereas 1 does not. It is therefore not an ideal of the surreal field, since multiplying a nonzero member by its inverse would give 1. For empty $\mathcal{B}$ it is all of $\mathbf{No}$, as the vacuous condition requires. No discontinuity in the fine surreal topology is asserted (Warning 2.5).

Proposition 39.5 (There is no all-ordinal critical potential). *There is no surreal Q with $Q - Q_\beta \sim \frac{1}{4}(\ell_\beta^\dagger)^2$ for every ordinal β . In particular $\frac{1}{4} \sum_{\beta \in \mathbf{On}} (\ell_\beta^\dagger)^2$ is not a Hahn sum in $\mathbf{No}$.*

Proof. The relation at β forces the coefficient of $(\ell_\beta^\dagger)^2$ in the normal form of Q to be $\frac{1}{4}$, since the support of Q_β consists of earlier monomials. The support of Q would then contain the proper class of distinct monomials $(\ell_\beta^\dagger)^2$, contradicting the set-sized support of a surreal. $\square$

This does not conflict with the existence of Q_α for every set ordinal, uncountable ones included: every initial segment is a set, their union is not. The absence of a final stage is a boundary of the construction, not a convergence problem to be repaired by a summation convention (Section 2.1).

Remark 39.6 (Relation to the valuation-insufficiency pair). Example 10.4 and Warning 10.5 show that for first-order equations a real-exponent valuation cannot see the slower logarithmic scale that decides solvability. Theorem 39.2 is the second-order, transfinite counterpart: within this family no set of logarithmic scales below the relevant stage decides, while the single coefficient c at the stage α does. It concerns this family only (Remark 38.3).

40 Three differential Hahn fields at a limit ordinal

Fix a nonzero limit ordinal λ and put

$$\Delta_0 = \text{span}_{\mathbb{R}}\{\varpi_\beta : \beta < \lambda\}, \quad \Delta_1 = \Delta_0 + \mathbb{R} \varpi_\lambda, \quad \Delta_2 = \Delta_1 + \mathbb{R} \omega^{-\lambda}, \quad (40.1)$$

set-sized ordered real vector subgroups of the additive surreals. Let $H_{\Delta_j} = \mathbb{R}((t^{\Delta_j}))$ and $K_{\Delta_j} = H_{\Delta_j}[i] = \mathbb{C}((t^{\Delta_j}))$ be the full Hahn fields, viewed in $\mathbf{No}$ and $\mathbf{No}[i]$; since each Δ_j is a group, $t^{\Delta_j} = \omega^{\Delta_j}$, and membership of an exponent does not depend on the orientation.

Warning 40.1 (Two levels of support). $\text{span}_{\mathbb{R}}$ means *finite* real linear combinations. It is not closure under Hahn sums inside the exponent group: an element of Δ_0 has bounded *inner* support, while an element of H_{Δ_0} may have arbitrary well-ordered *outer* support of t -exponents in that group (equivalently reverse well-ordered Conway exponents). The outer sum defining Q_λ is allowed; the inner sum ϖ_λ is not an allowed exponent of H_{Δ_0} . Everything below depends on this distinction, and confusing the two levels would destroy the example.

Lemma 40.2 (Bounded support in the exponent group). *Every $\gamma \in \Delta_0$ has Conway support in $\{\omega^{-\zeta} : \zeta < \beta\}$ for some $\beta < \lambda$. Consequently $\varpi_\lambda \notin \Delta_0$, $\omega^{-\lambda} \notin \Delta_1$, and $\varpi_\lambda, \omega^{-\lambda}$ are linearly independent over $\mathbb{R}$ modulo Δ_0 .*

Proof. Write $\gamma = \sum_{j=1}^m r_j \varpi_{\beta_j}$ with $\beta_j < \lambda$. For a nonempty list choose $\beta = \max_j \beta_j < \lambda$; for the empty list take $\beta = 0$. Every summand, and therefore their sum after any cancellations, has zero coefficient at $\omega^{-\zeta}$ for $\zeta \geq \beta$. This proves the support bound; it does not assert the converse that every bounded inner support lies in Δ_0 .

The coefficient of $\omega^{-\zeta}$ in ϖ_λ is 1 at every $\zeta < \lambda$, so no such bound is possible. Every element of Δ_1 has zero coefficient at $\omega^{-\lambda}$, since all its generators do. If $a\varpi_\lambda + b\omega^{-\lambda} \in \Delta_0$, compare the coefficient at $\omega^{-\lambda}$ to obtain $b = 0$. Next choose an inner-support bound $\beta < \lambda$ for that element and compare its coefficient at $\omega^{-\beta}$; this gives $a = 0$. This proves independence modulo Δ_0 , not just separate nonmembership of the two generators. $\square$

All three groups are set-sized: they are images of sets of finite lists of generators and real coefficients. They are divisible because they are real vector spaces. The Hahn real-closedness theorem therefore makes every H_{Δ_j} real closed, and adjoining i makes every K_{Δ_j} algebraically closed [1, Section 2.3], with the same hypotheses as Theorem 20.3. The complex Hahn field is exactly this quadratic extension: split each coefficient into its real and imaginary parts; both supports remain subsets of the original well-ordered support. The monomials u_λ and ℓ_λ witness the two strict inclusions respectively, by Lemma 40.2.

Proposition 40.3 (Derivative stability is a support assertion). *Each H_{Δ_j} and K_{Δ_j} is ∂ -stable and contains Q_λ ; their constants are $\mathbb{R}$ and $\mathbb{C}$. More precisely, the logarithmic derivative of every monomial ω^γ with $\gamma \in \Delta_1$ lies in H_{Δ_0} , and with $\gamma \in \Delta_2$ in H_{Δ_2} .*

Proof. If $\beta < \lambda$ then $\beta + 1 < \lambda$, so $\varpi_{\beta+1} \in \Delta_0$ and $\ell_\beta^\dagger = \omega^{-\varpi_{\beta+1}} \in H_{\Delta_0}$; hence $\ell_{<\beta}^\dagger$ for $\beta \leq \lambda$, and Q_λ , have support in Δ_0 and lie in the full Hahn field H_{Δ_0} . For $\gamma = \sum_{j=1}^m r_j \varpi_{\beta_j} \in \Delta_0$ with $\beta_j < \lambda$, $\omega^\gamma = \prod_j \ell_{<\beta_j}^{r_j}$ by Proposition 35.3 and the multiplicativity of the monomial map; so $(\omega^\gamma)^\dagger = \sum_j r_j \ell_{<\beta_j}^\dagger \in H_{\Delta_0}$. A further term $r\varpi_\lambda$ adds $r\ell_{<\lambda}^\dagger \in H_{\Delta_0}$, and a further $r\omega^{-\lambda}$ adds $r\ell_\lambda^\dagger = r\omega^{-\varpi_\lambda - \omega^{-\lambda}} \in H_{\Delta_2}$. So each Δ_j is differentially closed on monomials (Definition 20.1), and Lemma 20.2 gives ∂ -stability of H_{Δ_j} and K_{Δ_j} ; that step uses strong additivity (BM3), not continuity. Explicitly, $\partial(\omega^\gamma) = \omega^\gamma(\omega^\gamma)^\dagger$ shifts the latter support by γ , which stays in the same additive group. For an arbitrary Hahn series, strong additivity gives a summable family of these termwise derivatives; their union of supports still lies in the group, so their sum lies in the full Hahn field. A constant of a subfield is an ambient constant, and the fields contain $\mathbb{R}$, respectively $\mathbb{C}$, giving equality of constant fields. $\square$

Theorem 40.4 (Solution dimensions zero, one and two). *For $j = 0, 1, 2$,*

$$\dim_{\mathbb{R}} \text{Sol}_{H_{\Delta_j}}(Q_\lambda) = \dim_{\mathbb{C}} \text{Sol}_{K_{\Delta_j}}(Q_\lambda) = j, \quad (40.2)$$

and the complex solution spaces are 0, $\mathbb{C}u_\lambda$ and $\mathbb{C}u_\lambda + \mathbb{C}u_\lambda\ell_\lambda$.

Proof. By Theorem 36.2 and Lemma 36.4 every ambient solution is $au_\lambda + bu_\lambda\ell_\lambda$ with $a, b \in \mathbb{C}$, a combination of the two distinct monomials with exponents $\varpi_\lambda/2$ and $\varpi_\lambda/2 + \omega^{-\lambda}$. By Lemma 40.2 neither lies in Δ_0 , only the first lies in Δ_1 , and both lie in Δ_2 . A nonzero constant combination of distinct monomials cannot cancel either support position: its coefficient at the first exponent is exactly a , and at the second exactly b . Thus membership in K_{Δ_0} forces $a = b = 0$; membership in K_{Δ_1} forces only $b = 0$; and K_{Δ_2} permits both. The real case has the identical support test with $a, b \in \mathbb{R}$. This rules out hidden solutions with more complicated supports, since the ambient spanning theorem was applied before restricting to any of the three fields. $\square$

Proposition 40.5 (An exponential gap followed by a primitive gap). *H_{Δ_0} and H_{Δ_1} are closed under log on positive elements. Nevertheless*

$$\log u_\lambda \in H_{\Delta_0}, \quad u_\lambda \notin H_{\Delta_0}; \quad u_\lambda^{-2} \in H_{\Delta_1}, \quad \text{no primitive of } u_\lambda^{-2} \text{ lies in } H_{\Delta_1}.$$

The larger field H_{Δ_2} is not closed under log: $\ell_\lambda \in H_{\Delta_2}$ but $\log \ell_\lambda = \ell_{\lambda+1} \notin H_{\Delta_2}$.

Proof. For $\beta \leq \lambda$, $\log \ell_{<\beta} = \sum_{\zeta < \beta} \ell_{\zeta+1}$, and the exponent of $\ell_{\zeta+1}$ is $\omega^{-(\zeta+1)} = \varpi_{\zeta+2} - \varpi_{\zeta+1} \in \Delta_0$ because $\zeta + 2 < \lambda$ for $\zeta < \lambda$; so $\log \ell_{<\beta} \in H_{\Delta_0}$, and the logarithm of any monomial with exponent in Δ_1 is a finite real combination of these, hence in H_{Δ_0} . A positive element of H_{Δ_j} is $r\omega^\gamma(1 + \varepsilon)$

with $r > 0$ real and $\varepsilon \prec 1$ in H_{Δ_j} , and $\log(1 + \varepsilon)$ is a summable Taylor series in ε with support in Δ_j (Lemma 2.6); the case $\varepsilon = 0$ gives the zero series. Consequently

$$\log(r\omega^\gamma(1 + \varepsilon)) = \log r + \log \omega^\gamma + \sum_{n \geq 1} \frac{(-1)^{n+1}}{n} \varepsilon^n \in H_{\Delta_j} \quad (j = 0, 1).$$

The series is strongly summable inside the fixed group, since its exponents are finite sums of exponents of ε . This proves the claimed closure without assuming it in advance. Now $\log u_\lambda = \frac{1}{2} \log \ell_{<\lambda} \in H_{\Delta_0}$ while $\varpi_\lambda/2 \notin \Delta_0$. Also $u_\lambda^{-2} = \omega^{-\varpi_\lambda} \in H_{\Delta_1}$ and $\partial \ell_\lambda = u_\lambda^{-2}$; every real primitive is $\ell_\lambda + r$ with $r \in \mathbb{R}$, and its coefficient at exponent $\omega^{-\lambda} \notin \Delta_1$ is still 1, so no such primitive lies in H_{Δ_1} .

Finally, every element of Δ_2 has inner support contained in $\{\omega^{-\zeta} : \zeta \leq \lambda\}$. Hence $\omega^{-(\lambda+1)} \notin \Delta_2$. The positive monomial ℓ_λ lies in H_{Δ_2} , but its logarithm $\ell_{\lambda+1} = \omega^{\omega^{-(\lambda+1)}}$ does not. This proves the failure of logarithm closure in the third field. $\square$

Remark 40.6 (Which closure is missing, and the relation to Parts III and IV). Full Hahn summation, real closedness, derivative stability and closure under logarithms are all present in H_{Δ_0} , and still $\text{Sol}_{H_{\Delta_0}}(Q_\lambda) = 0$: the first missing operation is an exponential, the second a primitive, which is the phenomenon of Example 18.2 at a limit stage. The third field contains both fundamental solutions, but loses logarithm closure: Proposition 40.5 supplies the explicit witness. Thus solving this equation and being closed under logarithms are separate properties. Δ_0 is divisible and differentially closed on monomials, so it is its own differential Hahn hull (Theorem 20.3); the example is a further witness that the hull is not solution closure (Warning 20.5). Over the whole of **No**, Theorem 17.6 excludes a one-dimensional solution space for a real potential, because a primitive of u^{-2} produces a second solution from one ((BM6)); H_{Δ_1} lacks exactly that primitive, which is why the one-dimensional case occurs there and nowhere in **No**. The three complex fields are algebraically closed, so no step of the jump is an algebraic root; compare Remark 8.12.

41 A Borel Picard–Vessiot group at every limit stage

Keep λ and take the set-sized differential field $k = K_{\Delta_0}$, with constants $\mathbb{C}$. By Theorem 36.2,

$$\partial u_\lambda = h u_\lambda, \quad h = \frac{1}{2} \ell_{<\lambda}^\dagger \in k, \quad \partial \ell_\lambda = u_\lambda^{-2}. \quad (41.1)$$

The calculation below needs neither an abstract existence theorem for Picard–Vessiot extensions nor a black-box Galois computation, in keeping with Section 21.

Lemma 41.1 (Algebraic independence of the two generators). *u_λ and ℓ_λ are algebraically independent over k .*

Proof. Their exponents $\varpi_\lambda/2$ and $\omega^{-\lambda}$ are linearly independent over $\mathbb{Q}$ modulo Δ_0 (Lemma 40.2). In a finite relation $\sum_{(n,m)} f_{nm} u_\lambda^n \ell_\lambda^m = 0$ with $f_{nm} \in k$, a nonzero term has support in the coset $\Delta_0 + n\varpi_\lambda/2 + m\omega^{-\lambda}$, distinct pairs give disjoint cosets, and terms in disjoint cosets cannot cancel. $\square$

Theorem 41.2 (Borel differential Galois group). *The field $L = k(u_\lambda, \ell_\lambda) \subseteq K_{\Delta_2}$ is a Picard–Vessiot field for $\partial^2 y + Q_\lambda y = 0$ over k . Its constants are $\mathbb{C}$, its transcendence degree over k is two, and its differential k -automorphisms are exactly*

$$\psi_{a,b}(u_\lambda) = a u_\lambda, \quad \psi_{a,b}(\ell_\lambda) = a^{-2} \ell_\lambda + b, \quad a \in \mathbb{C}^\times, \quad b \in \mathbb{C}. \quad (41.2)$$

On the ordered solution basis $(u_\lambda, u_\lambda \ell_\lambda)$ the group is

$$\text{Bor}_2(\mathbb{C}) = \left\{ \begin{pmatrix} a & d \\ 0 & a^{-1} \end{pmatrix} : a \in \mathbb{C}^\times, d \in \mathbb{C} \right\} \subseteq \text{SL}_2(\mathbb{C}). \quad (41.3)$$

Proof. By equation (41.1), L is a differential field. It contains the fundamental matrix

$$Y = \begin{pmatrix} u_\lambda & u_\lambda \ell_\lambda \\ hu_\lambda & hu_\lambda \ell_\lambda + u_\lambda^{-1} \end{pmatrix}, \quad \det Y = 1,$$

whose entries generate u_λ and $\ell_\lambda = (u_\lambda \ell_\lambda)/u_\lambda$ over k . No new constant occurs, because $L \subseteq \mathbf{No}[i]$, whose constants are $\mathbb{C} \subseteq k$; these are the Picard–Vessiot requirements. The transcendence degree is Lemma 41.1. For a differential k -automorphism ψ , $\psi(u_\lambda)^\dagger = h = u_\lambda^\dagger$, so $\psi(u_\lambda) = au_\lambda$ with $a \in \mathbb{C}^\times$; then $\partial\psi(\ell_\lambda) = a^{-2}u_\lambda^{-2}$, so $\psi(\ell_\lambda) - a^{-2}\ell_\lambda = b \in \mathbb{C}$. Conversely, Lemma 41.1 makes equation (41.2) a well-defined k -automorphism of the rational function field $k(u_\lambda, \ell_\lambda)$; equation (41.1) shows that it commutes with ∂ on the generators, hence on L . It sends $u_\lambda \ell_\lambda$ to $abu_\lambda + a^{-1}u_\lambda \ell_\lambda$, so its matrix is $\begin{pmatrix} a & ab \\ 0 & a^{-1} \end{pmatrix}$, and $d = ab$ is arbitrary. The parameters and the action are algebraic, so the algebraic group is the Borel subgroup itself, not only an abstract list of automorphisms. $\square$

Corollary 41.3 (After the first Hahn enlargement). *Over K_{Δ_1} the Picard–Vessiot field is $K_{\Delta_1}(\ell_\lambda)$, with differential Galois group $\ell_\lambda \mapsto \ell_\lambda + b$, $b \in \mathbb{C}$.*

Proof. K_{Δ_1} contains u_λ and not ℓ_λ , which is therefore transcendental over it, K_{Δ_1} being algebraically closed. Its derivative lies in K_{Δ_1} , an automorphism fixing u_λ has $a = 1$, and the same fundamental matrix and no-new-constants argument apply. $\square$

Remark 41.4 (Scope of the Galois computation). “Minimal” refers to the solution-generated field L , not to the much larger K_{Δ_2} : every fundamental system is a constant invertible combination of $(u_\lambda, u_\lambda \ell_\lambda)$ and generates L . The intermediate Hahn fields are explicit workspaces, and no minimality is claimed for them among arbitrary fields. The other differential Galois groups computed in this report are commutative — $\mathbf{G}_m(\mathbb{C})$ in Theorem 21.1, the tori of Theorem 21.12 and Corollary 17.10, and $(\mathbb{C}, +)$ in Theorem 26.3 and Corollary 41.3 — so $\text{Bor}_2(\mathbb{C})$ is the only non-commutative one; it reflects the two successive missing operations of Proposition 40.5, an exponential and a primitive.

42 Schwarzian form, the first limit stage, and what this part does not claim

42.1 Schwarzian coordinates

For $s \in \mathbf{No}[i]$ with $\partial s \neq 0$ put

$$\text{Sch}(s) = \frac{\partial^3 s}{\partial s} - \frac{3}{2} \left(\frac{\partial^2 s}{\partial s} \right)^2, \quad (42.1)$$

a finite differential rational expression; no topology and no composition is involved.

Lemma 42.1 (Schwarzian and a normalized fundamental pair). *For $Q \in \mathbf{No}[i]$ the following are equivalent: some $s \in \mathbf{No}[i]$ has $\partial s \neq 0$ and $\text{Sch}(s) = 2Q$; the equation $\partial^2 y + Qy = 0$ has a fundamental pair in $\mathbf{No}[i]$. If y_1, y_2 is such a pair, every such s is a constant fractional linear transform of y_2/y_1 .*

Proof. Given s , algebraic closedness supplies $v = (\partial s)^{-1/2} \neq 0$, for either root. Then $v^\dagger = -\frac{1}{2}\partial^2 s/\partial s$ and $\partial^2 v/v = -\frac{1}{2}\partial^3 s/\partial s + \frac{3}{4}(\partial^2 s/\partial s)^2 = -\frac{1}{2}\text{Sch}(s)$, so v and vs are solutions with $\text{Wr}(v, vs) = v^2\partial s = 1$. Conversely, for a fundamental pair with constant $\text{Wr} \neq 0$, the quotient $s = y_2/y_1$ has $\partial s = \text{Wr}/y_1^2$, and the same differentiation gives $\text{Sch}(s) = -2\partial^2 y_1/y_1 = 2Q$. If s and $s_0 = y_2/y_1$ both have Schwarzian $2Q$, their reconstructed fundamental pairs differ by an invertible constant matrix (m_{jk}) (Lemma 36.4), so $s = (m_{11}s_0 + m_{12})/(m_{21}s_0 + m_{22})$, the denominator being nonzero because $\partial s_0 \neq 0$ rules out $s_0 \in \mathbb{C}$; conversely every such quotient comes from another fundamental pair. $\square$

Corollary 42.2 (Transfinite Schwarzian potentials). *For every ordinal α , $\text{Sch}(\ell_\alpha) = 2Q_\alpha$. The s with $\partial s \neq 0$ and $\text{Sch}(s) = 2Q_{\alpha,c}$ are the constant fractional linear transforms of $\ell_\alpha^{\sqrt{1-4c}}$ if $c < \frac{1}{4}$ and of $\ell_{\alpha+1}$ if $c = \frac{1}{4}$; if $c > \frac{1}{4}$ there is no such s in $\mathbf{No}[i]$.*

Proof. Use the pair $u_\alpha, u_\alpha \ell_\alpha$ for Q_α , the quotient of the pair equation (38.2) for $c < \frac{1}{4}$, and of equation (38.3) for $c = \frac{1}{4}$; then apply Lemma 42.1 and Theorem 38.1. $\square$

So the invisible perturbation of Theorem 39.2 also crosses the image of the scalar Schwarzian map. This is an exact projective reformulation of Theorem 38.1, not an independent result, and the classical Schwarzian correspondence is reproved only to make its application to surcomplex scalars unambiguous.

42.2 The first limit stage in normal form

Example 42.3 ($\lambda = \omega$). The inner exponent sum is the strong geometric series $\varpi_\omega = \sum_{n \geq 0} \omega^{-n} = \omega/(\omega - 1)$; the identity does not turn an outer monomial into a variable-base exponential. Then

$$u_\omega = \omega^{\omega/(2(\omega-1))}, \quad \ell_\omega = \omega^{\omega^{-\omega}}, \quad \ell_\omega^\dagger = \omega^{-\omega/(\omega-1)-\omega^{-\omega}}, \quad \partial \ell_\omega = \omega^{-\omega/(\omega-1)} = u_\omega^{-2}.$$

ℓ_ω is a positive infinite number below every $\log_n \omega$; its role as a primitive of the infinite inverse logarithmic product is recorded in the proof of [2, Theorem 8.4], and its existence is not a new assertion. The whole surcomplex solution space of $\partial^2 y + Q_\omega y = 0$ is

$$\mathbb{C} \omega^{\omega/(2(\omega-1))} + \mathbb{C} \omega^{\omega/(2(\omega-1))+\omega^{-\omega}},$$

$Q_\omega + (\ell_\omega^\dagger)^2$ has only the zero solution, and at the boundary $Q_\omega + \frac{1}{4}(\ell_\omega^\dagger)^2 = Q_{\omega+1}$ the basis is $\omega^{\varpi_{\omega+1}/2}$, $\omega^{\varpi_{\omega+1}/2+\omega^{-(\omega+1)}}$. All of this is exact. The groups of equation (40.1) are

$$\Delta_0 = \bigoplus_{n \geq 0} \mathbb{R} \omega^{-n}, \quad \Delta_1 = \Delta_0 + \mathbb{R} \frac{\omega}{\omega - 1}, \quad \Delta_2 = \Delta_1 + \mathbb{R} \omega^{-\omega};$$

the direct sum is inside the additive surreals, its members have finitely many terms, and it is not the field of Laurent series in ω^{-1} . Every term of Q_ω has exponent $-2(1 + \omega^{-1} + \dots + \omega^{-n}) \in \Delta_0$, so H_{Δ_0} contains the potential; adjoining the direction $\omega/(\omega - 1)$ admits the first solution monomial, and adjoining $\omega^{-\omega}$ the second. At higher limits “finite inner support” is replaced by “support bounded below λ ” (Lemma 40.2); no countability assumption is used there or in Theorem 41.2.

42.3 Inputs, boundaries and continuations

Remark 42.4 (Which assumptions matter, and the derivation). The normal-form and strong-additivity inputs justify every limit stage. The ordinal derivative formula of Proposition 35.3 fixes

the coefficients and the bases; the normalization $\partial\omega = 1$ alone does not determine it. The exact constant field is essential both to the angular obstruction and to the spanning argument, and the complex argument needs only infinitesimal logarithms, not an exponential at arbitrary imaginary arguments. For another surreal derivation the finite identities may still hold, but the limit-stage derivatives need not be the Berarducci–Mantova ones; no derivation-independent version is claimed, and in particular nothing here is asserted for the class of normalized derivations of Part VI. These are not statements about coordinate differentiation in $\mathcal{O}(U)((t^\Gamma))$ (Part VIII) or about fine-differentiable class functions (Warning 4.1).

Remark 42.5 (Established inputs and proposed contributions). The finite critical hierarchy is classical Kneser–Weber–Hartman–Hille theory, with a modern treatment and the positive-solution reduction in [33, Section 2], whose Schrödinger potential has the opposite sign to Q here; neither the finite threshold nor reduction of order is new. The derivation is Berarducci and Mantova’s [1], the ordinal log-atomic formulas are [3, Section 2], Hahn real closedness and Neumann’s lemma are standard, and the finite-primitive obstruction is Part II. The general logical theory of such solvability is not new either: [3, p. 17] discusses the predicate $\Omega(a)$ for the existence of a nonzero solution of $4y'' + ay = 0$, and the asymptotic differential algebra of [34] is neither replaced nor claimed. Mantova’s monotonicity and Taylor results for transseries [35] concern functional composition and are not used or superseded. The main proofs use no model completeness, no differential closure and no finite-system splitting theorem. Member 13 proposes as its contributions only the transfinite classification (Theorem 38.1), the invisibility theorem (Theorem 39.2), and especially the limit-stage filtration and its Galois group (Theorems 40.4 and 41.2); the gauge and Schwarzian mechanisms are standard, and no isolated elementary identity is promoted. Priority is provisional (item N103).

Remark 42.6 (A formalization interface). The proofs split into an imported surreal interface — Conway normal forms, Hahn summability, the log-atomic derivative identities and the strongly additive derivation — and explicit finite differential algebra: the operator identity, the Euler factorization, Wronskian spanning, exponent-coset independence and the automorphism calculation. Member 13 recommends isolating that interface rather than trying to infer the whole derivation from $\partial\omega = 1$. No Lean file is supplied (Warning 49.4).

Table 9: Proof dependencies of Part VII.

Conclusion	Necessary inputs, and the check against overreach
Tower (Proposition 35.3)	Imported from [3], limit cases included. Not derived from $\partial\omega = 1$.
Critical identity (Theorem 36.2)	Strong additivity, the exponential clause, joint summability of the double family. No limit of partial sums.
Nonreal powers (Corollary 37.1)	Theorem 8.4, direction (i) $\Rightarrow$ (iii); constant field $\mathbb{C}$. No primitive.
Euler classification (Proposition 37.3)	Wronskian spanning for ∂_s , real powers, Corollary 37.1.
Classification (Theorem 38.1)	The gauge identity with $s = \ell_\alpha$. Not a classification of arbitrary potentials.
Invisibility (Theorem 39.2)	Scale separation; a set of ordinals has a strict upper bound. The named hierarchy only.
Field jump (Theorem 40.4)	Bounded inner support, the monomial criterion, Hahn real closedness. Finite spans, not Hahn closure, of exponents.
Borel group (Theorem 41.2)	Coset independence, constants of $\mathbf{No}[i]$. No abstract Picard–Vessiot existence theorem.

Part VIII

Coordinate differential equations in common-domain Hahn rings

We now change operators. Throughout Part VIII the derivative is D_z of Table 1: it differentiates holomorphic coefficient *functions* in an ordinary complex coordinate z and holds every Hahn monomial and every scalar parameter constant. No scalar derivation on the coefficient field is involved in Sections 43 to 47; Section 48 then puts the two operators side by side and proves the compatibility statements that do hold.

The two halves of this report have different constant fields. The general solution of a coherent linear equation is parametrized by *Hahn* constants; the general solution of a scalar equation for ∂ is parametrized, when it exists at all, by *ordinary complex* constants. A number can be a coefficient constant without being a differential-field constant — $D_z\omega = 0$ while $\partial\omega = 1$ — and that difference is the whole reason the two theories coexist.

43 The fixed-common-domain ring

Let Γ be a set-sized ordered abelian subgroup of the additive surreals, interpret $t^\gamma = \omega^{-\gamma}$ as in equation (2.2), and put

$$K_\Gamma = \mathbb{C}((t^\Gamma)) = \left\{ \sum_{\gamma \in S} c_\gamma t^\gamma : S \subseteq \Gamma \text{ well ordered, } c_\gamma \in \mathbb{C} \right\},$$

with valuation v the least exponent and $v(0) = +\infty$; so $v(t^\gamma) > 0$ means the monomial is infinitesimal. Here Γ may have higher rank than $\mathbb{R}$, and *no divisibility assumption on Γ is needed* for any construction in this part. Let $U \subseteq \mathbb{C}$ be an ordinary *nonempty connected* open set, let $\mathcal{O}(U)$ denote the ordinary holomorphic functions on U , and define

$$\mathcal{H}_\Gamma(U) = \mathcal{O}(U)((t^\Gamma)) = \left\{ F(z) = \sum_{\gamma \in S} f_\gamma(z) t^\gamma : S \text{ well ordered, } f_\gamma \in \mathcal{O}(U) \right\}. \quad (43.1)$$

Every coefficient is holomorphic on the *same* U . Differentiation is coefficientwise,

$$D_z F = \sum_{\gamma \in S} f'_\gamma(z) t^\gamma,$$

a derivation of the ring that creates no new exponents; its support is a subset of the original support. For vectors and matrices, support means the union of the supports of the finitely many entries. A “nonzero coefficient function” means one not identically zero, so the global support is not redefined each time a coefficient happens to vanish at a particular z .

Warning 43.1 (Two neighbours this ring is not, and one notation clash). $\mathcal{H}_\Gamma(U)$ is *not* the larger ring with an unrelated convergent germ of shrinking radius for each exponent and no common domain, and it is *not* the formal power-series ring $\mathbb{C}[[z]]((t^\Gamma))$ or $K_\Gamma[[z]]$. These distinctions are already emphasized in the repository’s analytic and foundational reports [24, 26, 22], and Example 44.7 and Section 48.1 exhibit the separating witnesses. There is no requirement that the family (f_γ) be uniformly bounded in γ , and no numerical series over exponents need converge: holomorphicity and summability impose different conditions and are used at different steps. Notation: in equation (43.1) the symbol $\mathcal{O}(U)$ denotes ordinary holomorphic functions; it is *not* the finite-element valuation class $\mathcal{O}$, which never carries an argument.

Evaluation at an ordinary $z_0 \in U$ is a ring homomorphism to K_Γ , coefficientwise, and

$$\ker(D_z : \mathcal{H}_\Gamma(U) \rightarrow \mathcal{H}_\Gamma(U)) = K_\Gamma, \quad (43.2)$$

because an ordinary holomorphic function with zero derivative on a connected U is constant. If U is simply connected, define the normalized primitive coefficientwise,

$$I_{z_0} F = \sum_{\gamma} \left(\int_{z_0}^z f_{\gamma}(\zeta) d\zeta \right) t^{\gamma},$$

the ordinary integrals being path independent and holomorphic; they do not enlarge the support.

A section also has a value at a finite-halo point, after enlarging the exponent workspace if necessary: for $z_0 \in U$ and infinitesimal h ,

$$F^{\sharp}(z_0 + h) = \sum_{\gamma \in S} \sum_{n \geq 0} \frac{f_{\gamma}^{(n)}(z_0)}{n!} t^{\gamma} h^n. \quad (43.3)$$

The union of supports lies in $S + (\text{supp } h)^*$, and the two-support lemma together with the ordered-word count of Lemma 2.6 and appendix A gives strong summability of the double family. No uniform bound on the ordinary Taylor coefficients is required. This is *infinitesimal Taylor evaluation only*, not evaluation at an arbitrary infinite coordinate.

44 Linear coherent systems with positive perturbation

Theorem 44.1 (Positive-perturbation fundamental matrix). *Let U be simply connected, $z_0 \in U$, and*

$$A = A_0 + E \in \text{Mat}_d \mathcal{H}_\Gamma(U), \quad A_0 \in \text{Mat}_d \mathcal{O}(U), \quad S = \text{supp } E \subseteq \Gamma_{>0}.$$

Then there is a unique Hahn-coherent matrix X with

$$D_z X = AX, \quad X(z_0) = I_d. \quad (44.1)$$

It is invertible in $\text{Mat}_d \mathcal{H}_\Gamma(U)$, its support is contained in S^ , and its coefficient at exponent 0 is the ordinary normalized fundamental matrix Υ_0 of $D_z \Upsilon_0 = A_0 \Upsilon_0$, $\Upsilon_0(z_0) = I_d$. Uniqueness holds among all Hahn-coherent matrices, not only those whose support was prescribed in advance.*

Proof. Ordinary holomorphic linear theory gives Υ_0 on U , normalized at z_0 , with everywhere nonzero determinant: local existence and uniqueness, analytic continuation along ordinary paths, and simple connectedness [12, Ch. 4]. Put $G = \Upsilon_0^{-1} E \Upsilon_0$, whose support lies in S , and $TH = I_{z_0}(GH)$. For each n the support of $T^n I_d$ lies in the set of sums of n members of S , and at a fixed exponent only finitely many ordered words in S contribute, by Lemma 2.6. Hence

$$V = \sum_{n \geq 0} T^n I_d \quad (44.2)$$

is strongly summable with support in S^* and satisfies $V = I_d + TV$, so $D_z V = GV$ and $V(z_0) = I_d$; then $X = \Upsilon_0 V$ solves equation (44.1). Writing $V = I_d + W$ with W of positive support, the geometric matrix series $\sum_{n \geq 0} (-W)^n$ is strongly summable by the same lemma and is a two-sided inverse, so X is invertible with the claimed exponent-0 coefficient.

For uniqueness let Z be the difference of two coherent solutions and put $H = \Upsilon_0^{-1}Z$. If $H \neq 0$, take the least exponent γ of its global support. The equation $D_z H = GH$ forces $H'_\gamma = 0$: any contribution on the right would need a coefficient of H at $\gamma - s < \gamma$ for some $s \in S$, $s > 0$. The initial condition gives $H_\gamma(z_0) = 0$, so $H_\gamma = 0$ — a contradiction. Note that this argument never used a prescribed support monoid. $\square$

The proof yields an explicit recursion, which is a *support certificate* as well as an existence proof: with $V_0 = I_d$ and all unlisted coefficients zero,

$$V'_\gamma = \sum_{s+\delta=\gamma} G_s V_\delta, \quad V_\gamma(z_0) = 0 \quad (\gamma > 0), \quad (44.3)$$

where every dependency satisfies $\delta < \gamma$ and every displayed sum is finite.

Warning 44.2 (Not an imported convergence proof). The familiar iterated-integral form equation (44.2) does *not* import an ordinary convergence proof into the surreal fine topology. The primitive is an ordinary coefficientwise primitive and the infinite sum is a strongly summable Hahn family. At higher rank the successive support levels need not become cofinal in the value group, and no such cofinality is claimed or used.

Corollary 44.3 (Inhomogeneous coherent systems). *Under the hypotheses of Theorem 44.1, for $f \in \mathcal{H}_\Gamma(U)^d$ and $c \in K_\Gamma^d$ the unique solution of $D_z y = Ay + f$, $y(z_0) = c$, is*

$$y = X(c + I_{z_0}(X^{-1}f)),$$

with support contained in $(\text{supp } c + S^) \cup (\text{supp } f + S^*)$.*

Proof. The ordinary product rule holds coefficientwise because each Hahn coefficient is a finite sum, which verifies the formula; the support follows from the supports of X and X^{-1} and from $S^* + S^* = S^*$; uniqueness follows from Theorem 44.1 applied to differences. $\square$

Corollary 44.4 (Liouville determinant identity). *With the same normalization,*

$$\det X = \det \Upsilon_0 \cdot E(I_{z_0}(\text{tr } E)).$$

Here E is the formal exponential of a positive-support coherent section, not an unrestricted scalar complex exponential.

Proof. Finite matrix algebra gives Jacobi's formula $D_z \det X = (\text{tr } A) \det X$. The displayed expression satisfies the same scalar equation and equals 1 at z_0 , so the $d = 1$ case of Theorem 44.1 identifies them. The exponent has positive support, so Lemma 2.6 admits its exponential. $\square$

Proposition 44.5 (A negative leading exponent forbids a coherent scalar exponential). *If $a \in \mathcal{H}_\Gamma(U)$ has $v(a) < 0$, then $D_z Y = aY$ has no nonzero solution in $\mathcal{H}_\Gamma(U)$.*

Proof. Differentiation can delete coefficient functions but cannot create smaller exponents, so $v(D_z Y) \geq v(Y)$. Since U is connected, $\mathcal{O}(U)$ is an integral domain and $v(aY) = v(a) + v(Y) < v(Y)$ for $Y \neq 0$. The leading exponents cannot agree. $\square$

In particular $D_z Y = i\omega Y$ has no nonzero common-domain Hahn-coherent solution on *any* connected ordinary domain. The argument excludes not merely one series representation but every nonzero coherent solution in the stated ring.

Example 44.6 (Positive support is not necessary for matrix systems). If $N \in \text{Mat}_d(\mathbb{C})$ satisfies $N^2 = 0$ and $1 \in \Gamma$, then

$$D_z X = t^{-1} N X, \quad X(0) = I_d, \quad X = I_d + z t^{-1} N$$

is a coherent *polynomial* matrix despite its negative exponent. Nilpotence permits a finite algebraic cancellation that the scalar argument of Proposition 44.5 excludes. So the positive-support hypothesis is sufficient, not necessary, and negative-exponent matrix systems need finer analysis; this example bounds Proposition 44.5 deliberately.

Example 44.7 (A scale obstruction, and how rescaling resolves it). Consider

$$D_z Y = t^{-1} Y, \quad Y(0) = 1, \quad (44.4)$$

with $1 \in \Gamma$. It has no nonzero solution in $\mathcal{H}_\Gamma(U)$ on any ordinary nonempty domain, by Proposition 44.5: $v(D_z Y) \geq v(Y)$ while $v(t^{-1} Y) = v(Y) - 1$. The formal z -series $\sum_{n \geq 0} t^{-n} z^n / n!$ *does* exist as an element of $K_\Gamma[[z]]$, but its coefficient-family exponents are unbounded below, so it is not an element of the fixed-domain ring equation (43.1): formal existence and coherent existence have separated. Now rescale $z = tw$; since t is D_z -constant the equation becomes $D_w Y = Y$, $Y(0) = 1$, whose solution on an ordinary w -domain is the ordinary holomorphic e^w . That describes a *microscopic* z -domain, not a common ordinary one: at $v(z) > 1$ the original Taylor family is also strongly summable, because z/t is infinitesimal, whereas at $z = tw$ with nonzero ordinary w one uses the ordinary coefficient value e^w and *not* a Hahn sum of infinitely many exponent-zero terms $w^n/n!$. This is a domain-and-scale obstruction, not a failure of the formal differential equation, and it is why the positive-support hypothesis of Theorem 45.2 cannot simply be dropped.

45 A polynomially nonlinear initial-value theorem

A nonlinear theorem needs both a domain on which an ordinary background solution exists and a reason why every new Hahn coefficient is determined by earlier ones. Positivity of the support supplies the latter. The source manuscripts prove this theorem twice, in two notations, with two different admissibility hypotheses on the nonlinearity: polynomial dependence on the dependent variables, or holomorphic dependence on a neighbourhood of the graph of the background solution with Taylor evaluation in the vector variable. Both versions are recorded; the second needs the following admissibility lemma.

Lemma 45.1 (Admissibility of nonlinear evaluation). *Let $d \geq 1$, let $y_0 : U \rightarrow \mathbb{C}^d$ be holomorphic, let $W \subseteq \mathbb{C} \times \mathbb{C}^d$ be an ordinary open neighbourhood of its graph, let $S \subseteq \Gamma_{>0}$ be well ordered, and let F_0 and each F_s , $s \in S$, be holomorphic maps $W \rightarrow \mathbb{C}^d$. Write formally*

$$F(z, Y) = F_0(z, Y) + \sum_{s \in S} t^s F_s(z, Y). \quad (45.1)$$

If $u \in \mathcal{H}_\Gamma(U)^d$ has positive support, the Taylor evaluation of equation (45.1) at $y_0 + u$ is well defined and every coefficient is holomorphic on U . If $\text{supp } u \subseteq M$ and $M = E^$ for a positive well-ordered $E \supseteq S$, then the result has support in M .*

Proof. Let T be the finite union of the supports of the components of u ; then $S \cup T$ is well ordered and positive. A term in a Taylor coefficient has exponent $s + \beta_1 + \cdots + \beta_k$ with $s \in S \cup \{0\}$ and each $\beta_j \in T$. Lemma 2.6 gives a well-ordered union and only finitely many ordered decompositions

at each exponent, and there are finitely many vector-component choices at each fixed length k . So each coefficient is a finite sum of products of holomorphic derivatives of the F_s along y_0 with coefficients of u , hence holomorphic on U . The support containment follows from closure of M under addition. $\square$

No ordinary analytic convergence theorem is applied to a sum indexed by surreal exponents: the ordinary Taylor coefficients supply holomorphic functions, and Lemma 2.6 supplies the summation permission.

Theorem 45.2 (Support-certified positive-support initial-value problem). *In the setting of Lemma 45.1, assume U is simply connected, $z_0 \in U$, and $y'_0(z) = F_0(z, y_0(z))$. Let $\mathbf{v} \in K_\Gamma^d$ have positive support, and put*

$$E = S \cup \text{supp } \mathbf{v}, \quad M = E^*.$$

Then there exists a solution of the form

$$Y = y_0 + u, \quad u \in \mathcal{H}_\Gamma(U)^d \text{ of positive support,} \quad (45.2)$$

to

$$D_z Y = F(z, Y), \quad Y(z_0) = y_0(z_0) + \mathbf{v}, \quad (45.3)$$

and it is unique. Its support satisfies the explicit certificate

$$\text{supp } u \subseteq M \setminus \{0\}. \quad (45.4)$$

Convention 45.3 (The uniqueness class, stated exactly). Uniqueness in Theorem 45.2 is uniqueness among coherent solutions of the form $y_0 + (\text{positive-support vector})$ on U — that is, among all positive-support coherent solutions with the specified ordinary reduction y_0 . The support containment equation (45.4) is a *conclusion*, not a hypothesis: no support was prescribed, and uniqueness therefore holds among all positive-support coherent solutions and not merely among those whose support was assumed to lie in M . Nothing wider is claimed. In the *linear* case the least-exponent argument delivers more, and Theorem 44.1 states that stronger conclusion separately: uniqueness there is among all Hahn-coherent matrices.

Proof. Put $A(z) = D_Y F_0(z, y_0(z))$, the Jacobian along the background solution. Ordinary complex linear theory gives an invertible holomorphic fundamental matrix Υ on U with $\Upsilon' = A\Upsilon$, $\Upsilon(z_0) = I_d$; simple connectedness removes monodromy [12, Ch. 3–4].

M is well ordered by Lemma 2.6. Perform recursion along $M_{>0}$. For a proposed positive-support u , expansion gives $F_0(z, y_0 + u) = F_0(z, y_0) + Au + R_0(z, u)$ where every monomial of R_0 has degree at least two in u , and each perturbation term $t^s F_s(z, y_0 + u)$ carries the additional positive exponent s . Hence at a fixed exponent γ the equation reads

$$u'_\gamma = A(z)u_\gamma + R_\gamma(z), \quad u_\gamma(z_0) = \mathbf{v}_\gamma, \quad (45.5)$$

where R_γ depends only on coefficients u_β with $0 < \beta < \gamma$: a contribution from R_0 has at least two positive exponents summing to γ , so each is smaller, and a contribution from a perturbation has $s > 0$ before any factor of u , so every remaining exponent is smaller. The only term containing u_γ is the isolated linear term Au_γ .

At a fixed γ only finitely many contributions occur: expand each previously admitted exponent as a finite word in E ; all contributions then determine ordered words in E with sum γ , of which Lemma 2.6 permits only finitely many, and there are finitely many vector indices at each fixed

length. Each F_s has finitely many polynomial monomials, or finitely many relevant Taylor terms at each order by Lemma 45.1, and only finitely many s can participate. So unbounded degrees across different s do not introduce an infinite coefficient sum, and *no common bound on the polynomial degrees of the F_s is required*.

By induction R_γ is an ordinary holomorphic vector on U , and the ordinary linear inhomogeneous problem equation (45.5) has the unique holomorphic solution

$$u_\gamma(z) = \Upsilon(z) \left(\mathbf{v}_\gamma + \int_{z_0}^z \Upsilon(\zeta)^{-1} R_\gamma(\zeta) d\zeta \right) \quad (45.6)$$

on U : the integral is an *ordinary* complex integral of a holomorphic integrand on a simply connected set, so it is path independent and holomorphic there. This completes the recursion; the recursion is set-sized and no limit value at an accumulation exponent is needed, each coefficient being assigned by its own equation. The resulting support lies in $M_{>0}$, hence is well ordered, and Lemma 45.1 makes the substitution summable, so equation (45.3) holds coefficientwise.

For uniqueness in the sense of Convention 45.3, let u and $\tilde{u}$ be two positive-support coherent solutions with nonzero difference. The finite union of their supports is well ordered, so the difference has a least exponent γ , below which the coefficients agree. At γ all nonlinear and perturbation difference contributions therefore cancel, leaving $(u_\gamma - \tilde{u}_\gamma)' = A(z)(u_\gamma - \tilde{u}_\gamma)$ with zero initial value; ordinary linear uniqueness makes that coefficient identically zero, a contradiction. $\square$

Warning 45.4 (What the proof does not assume, and what the theorem does not promise). There is no Banach-space norm on the collection of coefficients, no contraction in the surreal fine topology, no convergence of Picard iterates, and no shrinking domain per exponent: every R_γ and every u_γ in equation (45.6) is holomorphic on the *original* U . The recursion uses well-founded support dependence and ordinary linear uniqueness, and nothing else.

The ordinary background solution is a *hypothesis*. The theorem does not promise to continue it past its ordinary singularities; its existence on an arbitrary prescribed domain is not asserted, and it may encounter poles or other obstructions beyond U . Simple connectedness is required. Negative-valued perturbations and arbitrary radius-free coefficient germs are not covered (Examples 44.6 and 44.7), and unrestricted substitution into an arbitrary holomorphic function of Y without specifying its domain and support conditions is out of scope. This is a differential-equation specialization of the support-recursion viewpoint already present in the repository, not a claim to have invented positive-support recursion; its additional content is the explicit initial-value contract, the common-domain argument through a single fundamental matrix, and the uniqueness clause of Convention 45.3.

Remark 45.5 (The linear case, and one practical rule). For $F_0(z, Y) = A_0(z)Y + b_0(z)$ with perturbations affine in Y , the nonlinear terms vanish and every higher coefficient needs one linear inhomogeneous solve with the *same* fundamental matrix Υ : the expensive ordinary homogeneous problem is solved once, not once per Hahn exponent. An effective implementation still needs a way to enumerate the finitely many decompositions of a requested exponent. A well-ordered support is a *semantic* certificate, not automatically an effective presentation; for a finite positive grid, finite truncations often reduce the requirement to integer-composition enumeration, but for an arbitrary program-described high-rank support no general terminating enumeration algorithm is claimed.

45.1 Two Riccati witnesses

Example 45.6 (A coherent entire solution with a pole at an infinite coordinate). Take $d = 1$, $F_0(Y) = Y^2$, $y_0 = 0$, no coefficient perturbation, and initial value $\mathbf{v} = t$. Then

$$y = \frac{t}{1 - tz} = \sum_{n \geq 0} t^{n+1} z^n \quad (45.7)$$

satisfies $D_z y = y^2$, $y(0) = t$. Every coefficient is a *polynomial* in z , so this is a coherent section on the whole ordinary complex plane. Yet the rational expression has a pole at the infinite point $z = t^{-1} = \omega$, which is not in the finite halo of $\mathbb{C}$: evaluating the Hahn series there would produce infinitely many copies of t and is not a strongly summable evaluation. So being coherent on the whole ordinary plane does not mean being regular at every surcomplex coordinate. The location of the pole explains, rather than contradicts, the exact geometric-series identity.

Example 45.7 (An entire coefficient family with an exact recursion). Take $F_0(z, Y) = Y^2$, $y_0 = 0$, $U = \mathbb{C}$, $S = \{1\}$, so

$$D_z Y = Y^2 + t, \quad Y(0) = 0. \quad (45.8)$$

Theorem 45.2 produces a unique positive-support coherent solution in $\mathcal{O}(\mathbb{C})((t^{\mathbb{Z}}))$; writing $Y = \sum_{n \geq 1} a_n z^{2n-1} t^n$, the coefficient equations give

$$a_1 = 1, \quad (2n - 1)a_n = \sum_{j=1}^{n-1} a_j a_{n-j} \quad (n \geq 2), \quad (45.9)$$

so

$$Y = tz + \frac{t^2 z^3}{3} + \frac{2t^3 z^5}{15} + \frac{17t^4 z^7}{315} + \frac{62t^5 z^9}{2835} + \cdots, \quad (45.10)$$

every coefficient an entire polynomial. Formally this is $\sqrt{t} \tan(\sqrt{t} z)$; that notation denotes the displayed Taylor identity, and the final series has integer exponents even when the chosen workspace does not contain $\sqrt{t}$. For finite surcomplex z the argument $\sqrt{t} z$ is infinitesimal and a small-argument tangent interpretation is legitimate in an enlarged workspace; but substituting $z = \omega = t^{-1}$ gives exponents $1 - n$, unbounded below, and the family is not summable. Entire coefficient functions on the ordinary plane do not justify evaluation at every infinite surcomplex input. This is an exact coherent solution on a common ordinary domain, not an all-scale surcomplex entire function, and it therefore does not contradict an all-scale rigidity theorem stated in a different function class.

46 Monodromy stays an ordinary-domain construction

On a non-simply-connected ordinary domain U , lift the coefficient system to its ordinary universal covering surface and choose a lift of the base point. The proof of Theorem 44.1 applies there: ordinary holomorphic primitives of coefficient one-forms are path independent on the cover, and the support calculation is unchanged.

Theorem 46.1 (Monodromy valued in $\mathrm{GL}_d(K_\Gamma)$). *Let U be a nonempty connected ordinary domain with base point z_0 , and let $A = A_0 + E$ satisfy the holomorphic and positive-support hypotheses of Theorem 44.1, with perturbation support S , without assuming that U is simply connected. Analytic continuation of the normalized fundamental matrix around an ordinary based loop ℓ yields another fundamental matrix X_ℓ , and their ratio has zero coordinate derivative, so*

$$X_\ell = X M_\ell, \quad M_\ell \in \mathrm{GL}_d(K_\Gamma). \quad (46.1)$$

The coefficient of M_ℓ at exponent 0 is the ordinary monodromy matrix of A_0 , and all its other exponents lie in $S^* \setminus \{0\}$. Under a fixed convention for composition of based loops these matrices form the usual monodromy representation; reversing the loop-composition convention reverses the multiplication order.

Proof. Immediate from equation (43.2) applied on the cover, together with $X(z_0) = I_d$, which lets M_ℓ be read off at the lifted endpoint; the support statement is Theorem 44.1. $\square$

No fine-continuous surcomplex path is asserted: the loop is an ordinary complex path, continuation is coefficientwise, and the result is a matrix of Hahn constants. This is the same domain separation that makes coefficientwise contour theory meaningful.

Example 46.2 (A residue invisible to ordinary monodromy). On a branch domain for the ordinary logarithm containing 1, consider

$$D_z Y = \frac{\lambda + \varepsilon}{z} Y, \quad Y(1) = 1, \quad \lambda \in \mathbb{C}, \quad v(\varepsilon) > 0. \quad (46.2)$$

The coherent solution is $Y = z^\lambda E(\varepsilon \log z)$, with $z^\lambda = \exp(\lambda \log z)$ in ordinary complex analysis. A positive loop about 0 changes $\log z$ by $2\pi i$ and gives

$$M = e^{2\pi i \lambda} E(2\pi i \varepsilon). \quad (46.3)$$

If $\lambda \in \mathbb{Z}$ the *ordinary* monodromy is 1, while the full monodromy is not 1 when $\varepsilon \neq 0$; by equation (6.3) its leading change is exactly $2\pi i \varepsilon$. So reduction to the ordinary coefficient can erase a genuine global obstruction to a single-valued coherent solution.

Warning 46.3 (The cross-category pair). In equation (46.2) the element ε is constant for D_z but need not be constant for the scalar ∂ . It is *not* an infinitesimal phase residue in the sense of Convention 5.1(b); it is a Hahn coefficient of a coordinate problem. The finite-primitive criterion of Theorem 8.4 is therefore not the criterion for solving equation (46.2). Concretely:

$$\varepsilon = \omega^{-1} \quad \text{is permitted in the coordinate problem } D_z Y = \frac{\lambda + \varepsilon}{z} Y,$$

$$\varepsilon = \omega^{-1} \quad \text{is prohibited as the imaginary coefficient of } \partial y = i\omega^{-1}y.$$

This is the sharpest available illustration that the two halves of this report are not competing answers to one problem. They are answers to precisely typed problems that happen to share differential notation.

47 A genuinely two-scale, noncommuting example

Let $\Gamma = \mathbb{Z}^2$ with the lexicographic order and use monomials with

$$v(t) = (0, 1), \quad v(u) = (1, 0), \quad \text{so } u \prec t^n \text{ for every ordinary } n \geq 1.$$

This is a Hahn workspace for *coordinate* differentiation: no formula for ∂u is assumed, and none is needed. Set

$$N = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}, \quad M = \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix}, \quad A(z) = tN + uzM.$$

Both matrices are nilpotent and $NM \neq MN$. The normalized fundamental matrix has the expansion $X(z) = \sum_{p,q \geq 0} t^p u^q X_{pq}(z)$ with $X_{00} = I_2$, and the recursion

$$X'_{pq} = NX_{p-1,q} + zMX_{p,q-1}, \quad X_{pq}(0) = 0 \quad (p+q > 0), \quad (47.1)$$

a negative index meaning zero. Ordinary polynomial integration gives

$$X_{10} = zN, \quad X_{01} = \frac{z^2}{2}M, \quad X_{11} = z^3 \left(\frac{NM}{6} + \frac{MN}{3} \right) = \begin{pmatrix} z^3/6 & 0 \\ 0 & z^3/3 \end{pmatrix},$$

and since $\text{tr } A = 0$, Corollary 44.4 gives $\det X = 1$ exactly.

Warning 47.1 (The naive exponential is wrong, and total degree is not a valuation cutoff). It would be incorrect to replace X by $E(tzN + uz^2M/2)$: that expression's mixed coefficient is $z^3(NM + MN)/4$, not X_{11} , and the difference at tu is

$$\frac{z^3}{12}(MN - NM) \neq 0.$$

The order of multiplication inside iterated integrals matters even though every individual coefficient calculation is finite. This is exactly why appendix A counts *ordered* words: an unordered or multiset version of the support count would not carry Theorems 44.1 and 45.2.

Separately: the delivered verification program computes all X_{pq} with $p+q \leq 6$, checks the coefficient equations and initial conditions, and verifies $\det X = 1$ to the same total degree. *Total degree is not a valuation cutoff here*: t^{100} has higher total degree but lower valuation than u . The support theorem justifies the complete series; the finite jet is a different, explicitly declared representation of part of it.

48 Formal, microscopic, and coherent: three different promises

48.1 Formal solvability is the weakest of the three

Proposition 48.1 (Formal existence over any characteristic-zero field). *Let K be any field of characteristic zero and $A(Z) = \sum_{n \geq 0} A_n Z^n \in \text{Mat}_d(K[[Z]])$. The formal problem $\partial_Z X = A(Z)X$, $X(0) = I_d$ has a unique solution $X = \sum_{n \geq 0} X_n Z^n$, given by*

$$X_0 = I_d, \quad (n+1)X_{n+1} = \sum_{k=0}^n A_k X_{n-k}. \quad (48.1)$$

Division by $n+1$ is permitted and the right side uses only known coefficients; no topology and no analytic estimate is involved. In particular the formal problem is always solvable for $K = \mathbf{No}[i]$.

Warning 48.2 (Three promises, worked through $A = i\omega$). Proposition 48.1 does not assert that the solution can be *evaluated* at a given surcomplex number. For $A = i\omega$ it gives

$$X(Z) = \sum_{n \geq 0} \frac{(i\omega Z)^n}{n!}.$$

At any nonzero *ordinary* z the valuations of the terms are unbounded below, so this family is not strongly summable (Remark 11.9). At an infinitesimal h with ωh infinitesimal it *is* summable, and equals $E(i\omega h)$. And common-domain coherence is excluded outright by Proposition 44.5, which

rules out not just this series representation but every nonzero coherent solution of $D_z Y = i\omega Y$ in the stated ring. So

$$\text{formal solvability} \neq \text{microscopic evaluability} \neq \text{common-domain coherence},$$

and the three must never be substituted for one another.

48.2 The coefficient derivation on a common-domain family

The one place where the two operators of this report legitimately meet is the following. Fix a common-domain family $F(z) = \sum_{\gamma \in S} f_\gamma(z) t^\gamma$ as in equation (43.1) and write, for each monomial,

$$\partial t^\gamma = \sum_{\delta} d_{\delta\gamma} t^\delta, \quad d_{\delta\gamma} \in \mathbb{R}. \quad (48.2)$$

These scalars form a support-controlled matrix; only set-sized columns are ever used at a time, and it is not a matrix indexed by all surreals as a set.

Lemma 48.3 (Finite incoming contributions). *For every well-ordered exponent set S , the union $T_S = \bigcup_{\gamma \in S} \text{supp } \partial t^\gamma$ is well ordered, and for each δ only finitely many $\gamma \in S$ have $d_{\delta\gamma} \neq 0$.*

Proof. The family $(t^\gamma)_{\gamma \in S}$ is summable. Apply (BM3), including its clause that the *derived* family is summable; the two conclusions are exactly the two clauses of Definition 2.4. $\square$

Theorem 48.4 (Differentiation preserves common-domain coherence). *The rule*

$$\partial_H F = \sum_{\delta \in T_S} \left(\sum_{\gamma \in S} d_{\delta\gamma} f_\gamma \right) t^\delta \quad \left(\text{equivalently } \partial_H F = \sum_{\gamma \in S} f_\gamma(z) \partial t^\gamma \right) \quad (48.3)$$

defines a $\mathbb{C}$ -linear derivation with values in the common-domain Hahn algebra over the differential hull of Γ . Every inner sum is finite; each new coefficient is holomorphic on the same ordinary domain U , so differentiating the scalar coefficients neither shrinks nor destroys the common domain; and

$$\partial_H D_z F = D_z \partial_H F. \quad (48.4)$$

If the exponent group Γ is differentially stable (Definition 20.1) the values stay in $\mathcal{H}_\Gamma(U)$; for an arbitrary initial Γ the same assertions hold after passing to its differential Hahn hull (Theorem 20.3).

Proof. Lemma 48.3 gives the well-ordered output support and the finiteness of each inner sum, so each new coefficient is a finite real-linear combination of functions holomorphic on the same U , hence holomorphic there. Linearity is immediate. For the product rule, expand the Hahn product of two families and use $\partial(t^{\gamma+\eta}) = t^\gamma \partial t^\eta + t^\eta \partial t^\gamma$; standard product summability and (BM3) make the expanded families summable, and coefficient comparison then uses only finite sums. Finally ordinary differentiation is $\mathbb{C}$ -linear and commutes with each finite inner sum in equation (48.3), which gives equation (48.4). Stability of Γ , or passage to the hull, keeps the supports inside the workspace. $\square$

Warning 48.5 (The derivative may need a larger monomial workspace). ∂_H need *not* preserve a previously chosen smaller monomial group: the support of ∂t^γ can contain monomials outside a preselected subgroup (Example 20.7). Its representing matrix $(d_{\delta\gamma})$ is only row-finite on each specified summable input support; no claim is made that an individual derivative has finite support or that the matrix has a finite global representation. A derivative routine should therefore return both a value *and* the parent in which that value is represented.

Theorem 48.6 (Total intrinsic derivative of an evaluation). *Let F be a common-domain family, let $c \in U$ and $h \in \mathfrak{m}_{\mathbb{C}}$, and let $F^{\sharp}(c+h)$ be the finite-halo evaluation equation (43.3). Then*

$$\partial(F^{\sharp}(c+h)) = (\partial_{\mathbb{H}}F)^{\sharp}(c+h) + \partial h \cdot (D_z F)^{\sharp}(c+h). \quad (48.5)$$

Equivalently, for $z \in \mathcal{O}_{\mathbb{C}}$ with $c = \text{st}(z) \in U$ and $h = z - c$, $\partial(\text{ev}_z F) = \text{ev}_z(\partial_{\mathbb{H}}F) + \text{ev}_z(D_z F) \partial z$. All terms may be placed in one set-sized differentially stable Hahn workspace after adjoining $\text{supp } h$ and taking the hull (Theorem 20.3).

Proof. Apply (BM3) to the summable double family equation (43.3). Since c is an ordinary complex point, all the numbers $f_{\gamma}^{(n)}(c)/n!$ are constants of ∂ , and $\partial h = \partial z$. The product rule splits the derivative of each term into

$$(\partial t^{\gamma}) \frac{f_{\gamma}^{(n)}(c)}{n!} h^n \quad \text{and} \quad t^{\gamma} \frac{f_{\gamma}^{(n)}(c)}{(n-1)!} h^{n-1} \partial h \quad (n \geq 1),$$

the second omitted for $n = 0$. The first family is summable by Lemma 48.3 followed by the same positive-support Taylor argument; the second is the corresponding Taylor family for $D_z F$ multiplied by the fixed scalar ∂h . So the split and the rearrangement are legitimate (Warning 2.7 is satisfied), and reindexing gives the two terms of equation (48.5). $\square$

An ordinary holomorphic lift $f^{\sharp}$ has ∂ -constant coefficients, so equation (48.5) reduces to $\partial(f^{\sharp}(z)) = (f')^{\sharp}(z) \partial z$. A general coherent family carries the additional coefficient term, and this distinction is independent of whether the evaluated value has a short expression.

Example 48.7 (Two worked total derivatives). (i) $F(z) = t^2 e^z$ at $z = t$: here $\partial_{\mathbb{H}} F = -2t^3 e^z$, $D_z F = t^2 e^z$, $\partial t = -t^2$, and

$$\partial(t^2 E(t)) = (-2t^3 - t^4) E(t),$$

the extra $-t^4 E(t)$ being the moving-argument term. Differentiating only the scalar prefactor, or only the analytic argument, would miss part of the answer. (ii) $F(Z) = \omega Z^2 + tZ$ at $z = 1+t$: here $\partial_{\mathbb{H}} F = Z^2 - t^2 Z$, $D_z F = 2\omega Z + t$, $\partial z = -t^2$, and

$$F(z) = \omega + 2 + 2t + t^2, \quad \partial(F(z)) = 1 - 2t^2 - 2t^3,$$

which equation (48.5) reproduces exactly; keeping only the argument derivative would lose the leading constant 1. The family $F(Z) = \omega$ already disproves that omission.

48.3 Fixed contours, fixed centres, and moving poles

Corollary 48.8 (Coefficientwise linear naturality). *Any fixed $\mathbb{C}$ -linear operation on ordinary coefficient functions commutes with $\partial_{\mathbb{H}}$ wherever that operation is defined coefficientwise. In particular, for a fixed ordinary piecewise smooth contour $\gamma_c \subseteq U$,*

$$\partial \left(\int_{\gamma_c} F(z) dz \right) = \int_{\gamma_c} \partial_{\mathbb{H}} F(z) dz, \quad \int_{\gamma_c} F dz := \sum_{\gamma} \left(\int_{\gamma_c} f_{\gamma} dz \right) t^{\gamma}, \quad (48.6)$$

and the analogue holds for a coefficientwise Laurent residue at a fixed ordinary centre, whenever all coefficient functions belong to the stated punctured-domain class.

Proof. Apply the fixed linear operation to the finite inner sums of equation (48.3): both orders of application give the coefficient $\sum_{\gamma} d_{\delta\gamma} L(f_{\gamma})$ at δ . $\square$

No surcomplex-valued Riemann integral is introduced: each side of equation (48.6) is a coefficient-wise functional. For example, on a punctured ordinary disk about $a \in \mathbb{C}$ the family $F(z) = t/(z - a)$ has contour integral $2\pi it$ around a positive simple loop, and both sides of equation (48.6) equal $-2\pi it^2$; this is an exact Hahn identity.

Warning 48.9 (A moving contour or a moving pole is a different statement). The contour in Corollary 48.8 is fixed and ordinary. A moving contour, moving endpoints, or a residue with a different centre convention requires additional terms or a different statement; none is hidden in the corollary. At an *actual moving pole* the coefficient term cannot be dropped. Use the formal rational operators $\tilde{\partial}$ and ∂_Z of Proposition 4.3: for $a \in \mathbf{No}[i]$ and $F(Z) = 1/(Z - a)$,

$$\tilde{\partial}F = \frac{\partial a}{(Z - a)^2}, \quad \partial_Z F = -\frac{1}{(Z - a)^2}, \quad \partial \frac{1}{z - a} = -\frac{\partial z - \partial a}{(z - a)^2}, \quad (48.7)$$

these rational identities holding whenever the denominators are nonzero and needing no coherent expansion. A fixed-centre residue must not be silently replaced by an actual residue at a . The repository already distinguishes those residues; intrinsic differentiation supplies another reason to keep them apart. Likewise, the formal Hahn residue reading of Warning 18.3 is not an identification with a cluster residue or an actual-point contour residue.

Remark 48.10 (A pointwise-only evaluation identity). For a *fixed* ordinary $z \in U$, strong additivity does give

$$\partial(F(z)) = \sum_{\gamma} f_{\gamma}(z) \partial(t^{\gamma}),$$

a pointwise scalar identity. Turning it into a statement about a common-domain coherent family, differentiating again in z , or integrating it along a path requires separate control of the coefficient-family supports and of the domain — which is precisely what Theorems 48.4 and 48.6 and Corollary 48.8 supply, under their hypotheses. The distinct coefficient rings documented in the repository remain distinct after ∂ is added.

48.4 Formal power series need their own admissibility condition

Let K be a differential subfield of $\mathbf{No}[i]$ (or take $K = \mathbf{No}[i]$). On $K[[Z]]$ the formal coefficient and variable derivatives commute:

$$\tilde{\partial}F = \sum_n (\partial c_n) Z^n, \quad \partial_Z F = \sum_{n \geq 1} n c_n Z^{n-1}, \quad \tilde{\partial} \partial_Z F = \partial_Z \tilde{\partial} F.$$

Evaluation at a scalar a requires summability. If the families defining $F(a)$, $(\tilde{\partial}F)(a)$ and $(\partial_Z F)(a)$ are all summable, then the same computation gives

$$\partial(F(a)) = (\tilde{\partial}F)(a) + \partial a \cdot (\partial_Z F)(a). \quad (48.8)$$

Proposition 4.5 is the sharper version, which assumes only the summability of the evaluated family and *derives* the other two.

Warning 48.11 (Do not apply this to an unevaluable germ). Equation (48.8) must not be applied to an arbitrary unevaluable formal germ. The finite-halo theorem Theorem 48.6 supplies its own summability proof; an unrestricted formal coefficient sequence does not. The limiting case is Warning 6.2(iii): the evaluation rule $f(h)$ of Lemma 6.1 requires *ordinary constant* coefficients, and $\sum_{n \geq 0} \omega^{n^2} X^n$ is not summable at $X = \omega^{-1}$ because its monomials ω^{n^2-n} are not reverse well

ordered. The infinitesimal chain rule must not be exported to arbitrary $\mathbf{No}[i]$ -coefficient streams without a joint support certificate. On the specific ring $\mathcal{O}(U)((t))$ one may define $\partial_H = -t^2 D_t$ and D_z coefficientwise, and they commute because both sides have identical coefficient formulas; that statement need not hold as a meaningful endomorphism assertion on an arbitrarily chosen K_Γ , since the scalar derivative can require monomials outside that workspace (Warning 48.5).

48.5 The four categories, side by side

Table 10: Four typed categories that share differential notation. They are not competing answers to one problem.

Category	Constants	Constructive positive result	Boundary
Scalar $(\mathbf{No}[i], \partial)$	$\mathbb{C}$	Finite-primitive criterion, finite-phase solution formula, gauge normal form, constant-coefficient classification	$\partial y = iy$ has no nonzero solution; no global compatible exponential
Real-exponent Hahn scalar K_Γ , $\Gamma \subseteq \mathbb{R}$, $1 \in \Gamma$	$\mathbb{C}$	$\partial - \lambda$ bijective for $\lambda \in \mathbb{C}^\times$; $P(\partial)$ bijective on $\mathbb{C}((t))$ when $P(0) \neq 0$; exact residuals	t has no primitive there; the real-exponent hypothesis is substantive
Formal $K[[Z]]$	K	Linear coefficient recursion, always solvable	Evaluation may not be summable
Common-domain $\mathcal{H}_\Gamma(U)$	K_Γ	Positive-perturbation linear and polynomially nonlinear initial-value theorems, support certificates, monodromy in $\mathrm{GL}_d(K_\Gamma)$	A negative scalar leading exponent prevents a coherent exponential; nilpotence is an exception

Part IX

Computation, formalization, and the register of non-claims

49 Certificate-oriented computational contracts

49.1 Theorem-driven outputs, not a universal solver

The exact criterion suggests a usable symbolic interface. Given a represented coefficient $q = a + ib$, a backend may try to produce primitives A, B with $\partial A = a$ and $\partial B = b$ together with a certified decomposition $B = P + r + \eta$ with $P \in \mathcal{P}$, $r \in \mathbb{R}$, $\eta \in \mathfrak{m}$. It need not use the normalization $\mathcal{I}_0$ to do so, since $\mathrm{pp}(B) = \mathfrak{o}(b)$ is unchanged by the real integration constant (Remark 5.7). The logical workflow is

$$q = a + ib \mapsto B \mapsto \mathrm{pp}(B) \mapsto \begin{cases} \text{a base-field solution,} & \mathrm{pp}(B) = 0, \\ \text{a certified obstruction and an extension symbol,} & \mathrm{pp}(B) \neq 0. \end{cases} \quad (49.1)$$

Warning 49.1 (This is a specification, not a terminating algorithm). Equation (49.1) is a mathematical specification. For arbitrary program-described surreal normal forms, integration, exact support access and zero testing may all be unavailable. A failed search for a primitive is not a proof that none exists — in the full field every element has one — and a failed representation of an infinitesimal exponential is not a proof that no mathematical solution exists. A backend should return a well-defined symbolic primitive and state that the purely infinite part has not been decided; it must not report “no solution” merely because its representation cannot compute that part. The repository’s capability-tier distinctions are therefore especially important at this differential boundary [25].

Three logically distinct outcomes arise for the ambient question, and a fourth once a workspace is declared.

Table 11: Result contracts. The last row is not a nonexistence theorem.

Result	Certificate required	Mathematical conclusion
Workspace solution	Rational exponent r , an element $k \in \mathcal{H}$ with positive support and $-rt + \partial k = q$	A certified nonzero solution $t^r E(k)$ lies in the declared Hahn field (Theorem 19.3)
Ambient-only solution	A verified primitive A of a , a <i>finite</i> accumulated phase B of b with verified $\text{pp}(B) = 0$, an admissible infinitesimal-residue expression, and certified failure of the declared workspace solvability test	A nonzero solution $\exp(A)E(i\eta)$ exists in $\mathbf{No}[i]$, but the workspace obstruction is nonzero (Theorem 8.4)
Ambient phase obstruction	A verified accumulated phase B of b with certified nonzero $\text{pp}(B)$	No nonzero surcomplex solution exists; the gauge normal form returns P and the removable factor g (Theorem 9.1)
Undetermined representation	A required primitive, decomposition, equality or support certificate is unavailable	No existence or nonexistence conclusion

For the finite exact core — finite rational Hahn polynomials with Gaussian-rational coefficients, represented as a finite map $q \in \mathbb{Q} \mapsto c_q \in \mathbb{Q}(i)$ with zero coefficients deleted — the first three outcomes are decidable. Proposition 18.1 gives a terminating derivative; additive integration inside $\mathcal{H}$ succeeds exactly when the exponent-1 coefficient vanishes, and otherwise equation (18.4) gives an ambient primitive using one explicit $\log \omega$ constructor. Two tests must be offered *separately*: the *workspace* test rejects every nonzero term below exponent 1 and rejects a nonreal exponent-1 coefficient (with Gaussian-rational coefficients a real exponent-1 coefficient is automatically rational), and the *ambient* test examines the imaginary part b and succeeds precisely when $b = 0$ or every exponent of $\text{supp } b$ exceeds 1. For arbitrary computably described Hahn supports none of this is advertised as a total decision procedure, and the semantic formula $\Phi(q) = \text{pp}(\mathcal{I}_0 \text{Im } q)$ remains exact without promising an algorithm for $\mathcal{I}_0$ or for equality of arbitrary represented scalars.

Warning 49.2 (What a certificate must actually establish). A solution certificate consists of exact derivative identities plus support and size evidence. For a finite expression the identities may reduce to field algebra plus the real exponential and logarithm rules. For an infinitesimal Taylor expression the certificate must identify positive support and justify the summations used in the derivative rule. For a *nonsolvability* certificate it is not enough to report a large leading term numerically: one must prove that an accumulated phase has nonzero purely infinite part. The output $t^r E(k)$ is a finite exact expression denoting an infinite Hahn series, not a finite list of its coefficients.

A finite-phase certificate need not compute the whole normalized primitive: it suffices to exhibit a finite B and prove $\partial B = \text{Im } q$. This is useful when semantic existence is known but a general symbolic integration procedure is unavailable.

The two inputs $q_1 = i\omega^{-2}$ and $q_2 = i\omega^{-1}$ are distinguished by the exact accumulated phases $-\omega^{-1}$ and $\log \omega$. A numerical specialization $\omega := M$ for a large real M would instead give ordinary complex equations with nonzero oscillatory solutions in *both* cases. Such a specialization is not an embedding of differential structures and cannot certify the surreal answer.

49.2 A derivation-aware representation interface

Every exact parent should record both its ambient embedding and its closure capabilities. A Laurent parent may advertise: closed under ∂ ; closed under $(\partial - \lambda)^{-1}$ for $\lambda \neq 0$; *not* closed under primitives. A log-exponential extension can support more primitives without thereby supporting arbitrary global phases, and adding a constructor named **Exp** is not enough: its domain and its derivative contract must be stated.

Table 12: Operations, required certificates, and result boundaries.

Object or request	Required certificate	Result boundary
Intrinsic derivative	A selected derivation and a parent field, with a supported image in a known parent	May require a larger parent: a monomial derivative need not lie in the old one (Warning 48.5)
Coordinate derivative	One common coefficient domain	Every Hahn monomial and scalar parameter stays constant
Hahn evaluation	A joint reverse-well-ordered support with finite multiplicity	A formal expression may not denote a sum at the input (Remark 11.9)
Coherent total derivative	One common coefficient domain and <i>both</i> derivatives	Dropping ∂_H changes the answer (Example 48.7)
Finite-phase integrating factor	A real accumulated phase B with verified $\text{pp}(B) = 0$	A nonzero solution in $\mathbf{No}[i]$
Phase obstruction	A verified nonzero purely infinite part of an accumulated phase	Nonexistence of a nonzero homogeneous solution
Hahn resolvent	A real-exponent group with $1 \in \Gamma$, and a support-shift certificate; a lower exponent bound for a prefix	An exact Hahn solution <i>in that workspace</i> ; prefix computation is not stream equality
Coherent initial-value solver	Base solution, common domain, positive support, finite decomposition access	A certified coefficient recursion
Global phase	An explicit phase convention with its supported domain	A group law does not imply compatibility with ∂ (Example 11.7)
Oscillatory extension	A distinct parent and the rule $\partial T = iT$	Not a surcomplex scalar (Corollary 21.8)

Warning 49.3 (A truncation must record the *type* of omitted information). A truncation of Y_H in equation (18.10) can carry the exact residual equation (18.11) and a real-power support tail. That certificate does *not* rule out an exponentially small homogeneous correction in the full field (Proposition 18.12). A computation that returns “unique solution” must say whether uniqueness is

in K_{Γ} , in the full $\mathbf{No}[i]$, in an oscillatory extension, or among coherent sections of a named ring. Likewise, returning $\exp(\int A)$ is not a certificate for a complex integrating factor: the software must either verify the finite-phase criterion, select a differential extension, or report that the required operation is not implemented. An unevaluated expression and a proven element of the requested field are different outputs. A matrix jet specified by total degree, a Laurent jet specified by valuation, and an exact recursively defined series must remain three different output types (Warning 47.1). Branch continuation and monodromy should use ordinary-domain path data rather than an implied fine-continuous path in the surreal class (Theorem 46.1). For several extension symbols, the integer relation lattice of equation (21.6) is additional required data.

A realistic first implementation can stay inside an explicit differential algebra: rational expressions in a finite collection of real-power and logarithmic scales; finite constant-coefficient operators; formal Laurent series represented by a recurrence or by coefficient access. At each extension a derivative routine should return both a value and its parent, and an integration routine should distinguish a proved primitive, a proved obstruction *in the selected parent*, and a failure of the available algorithm to decide. The missing t -coefficient of equation (18.3) is a proved parent obstruction; failure to recognize a complicated primitive in the full surreal field is not.

49.3 Formalization boundaries

A proof-assistant development can begin from a set-sized ordered field with a derivation, a chosen infinite scale w with $\partial w = 1$, real constants, and specified local exponential and logarithm operations with their support theorems. The bounded-derivative argument (Lemma 3.8), the complexification (Proposition 3.5), the phase decomposition (Theorem 6.3) and the first-order criterion (Theorem 8.4) then become lemmas over explicit hypotheses. Surjectivity of the derivation is a *genuine additional hypothesis*, not something inferred from real closedness.

Four concrete proof obligations are worth naming, in increasing order of difficulty.

- (1) The explicit rational monomial derivative formula equation (18.1) on $\mathcal{H} = \mathbb{C}((t^{\mathbb{Q}}))$. The primitive-image and logarithmic-image theorems (Theorems 19.2 and 19.3) then reduce to support translation, leading-term factorization and the infinitesimal exponential theorem. This is a tractable independent target.
- (2) The implication “monomial support closure $\Rightarrow$ closure of the full small-support Hahn type under ∂ ”, which is the content of Lemma 20.2 and Theorem 20.3. It complements, rather than duplicates, a generic Hahn convolution development.
- (3) Lemma 48.3: for each admissible input support, each output coefficient receives only finitely many monomial-derivative contributions. Once available, the common-domain coefficient interface becomes finite linear algebra at every output exponent.
- (4) The represented quotient $\mathcal{P}$ of Warning 2.1, so that the rank-one classification does not accidentally require a type whose elements are proper classes.

The interfaces should keep ∂ , ∂_Z , $\tilde{\partial}$, ∂_H , D_z and fine differentiability separate. Evaluation is a map with a support certificate, and Proposition 4.5 is a compatibility theorem for that map; it is not a coercion identifying operators. An exponential operation should carry the domain predicate $\text{Im } z \in \mathcal{O}$ whenever it promises the intrinsic differential law. The difficult Berarducci–Mantova existence construction can remain an explicit imported interface until separately formalized; it is a much larger dependency and is not needed to check the logical consequences of a stated interface.

Warning 49.4 (Formalization plan and present scope). The report-specific differential theorems remain pending in the repository ledger `docs/FORMALIZATION.md`. Existing checked prerequisites there must be distinguished from a formalization of the Berarducci–Mantova derivation, its phase obstruction, the finite-matrix classification or the coherent differential-equation package. The preceding paragraphs specify further proof obligations and a possible order of development; they do not assert that the concrete surreal carrier has already been connected to every required Hahn, real-closed-field, exponential or derivation interface.

50 Relation to the literature and to the companion reports

The proof has three layers. At the foundation are the established surreal field and exponential constructions and the chosen strongly additive, surjective derivation [1, 7]. The normal-form splitting and the infinitesimal formal calculus connect that structure to the repository’s finite-angle geometry [23]. Above them are the explicitly proved differential consequences: the finite-primitive criterion, the gauge normal form, the maximal strip, the scalar and constant-coefficient classifications, the workspace images, and the coordinate initial-value theorems.

The universal H -field results [3] explain why the differential viewpoint is a major part of surreal theory rather than an ad hoc extra operation; through Part IV they are cited for context and are not a substitute for an unstated linear-solvability theorem. In Part III that changes: the elementary embedding of [3] is used as a transfer principle, and the splitting and surjectivity facts of [4, Lemma 2.4.19] are used as premises (Theorem 15.6, item N53). The amplitude–phase results of [5] over Hardy fields, and the classical Pinney parametrization [6], are the antecedents of Theorems 17.6 and 17.7 and are kept visible there; no solution of the Boshernitzan conjecture named in [5] is claimed. The ω -series composition results [2] show that an external function interpretation is valid on a substantial specified class while warning against unrestricted composition (Warning 22.2). The global exponential construction [8] supplies the crucial contrasting structure: algebraic global exponentiation and number-derivation compatibility are different requirements (Warning 11.6). Bournez and Guilmant [9] study closure properties of selected surreal fields far beyond the elementary constructions of Section 20, and no theorem of theirs is used to justify them. Ordinary complex initial-value theory [12] supplies the analytic ingredient of Part VIII, and [13, 14] supply the Picard–Vessiot and differential-module terminology in their standard sense. Neumann [10] and Higman [11] are the published sources for appendix A.

The repository documents are treated as unrefereed project manuscripts that motivate the gap, not as replacements for the published differential foundations; none of their analytic-geometry or contour theorems is needed for the main proofs here. The following cross-links matter.

- **Trigonometry.** That report supplies the finite-angle phase group and explicitly declines to choose a derivation [23]. Theorem 6.5 reproves the part needed here and adds the derivative identity; Warning 11.6 states precisely what the present no-go theorems do and do not say about its global phase extensions. Its existing constructions need not be removed or renamed as false; their contracts need only be distinguished from the intrinsic differential one, and the derivation should be named wherever it is used.
- **Analysis.** That report’s derivation-obstruction theorem, for one chosen canonical global surcomplex exponential, is *prior coverage* [24]. Corollary 11.4 strengthens it in phase-independence and does not contradict it; the two are compatible. Part VIII reuses its common-domain coherent-section language and refines the germ-versus-common-domain-versus-formal distinction into the explicit separations of Table 10.

- **Computer algebra and foundations.** The capability taxonomy and the subsection “Differential equations must name their category” [25], and the four-interface discipline of [26], are answered directly by Tables 1, 11 and 12 and by the tiering contract of Section 49.1.
- **Spectral theory.** The companion report *Surcomplex Spectral Theory* [27] supplies the finite algebraic spectral vocabulary that Section 16.1 uses, and its “spectrum” is the algebraic eigenvalue and singular-value list. That is *not* the differential phase spectrum of Convention 5.1(e), and Warning 5.2 states the distinction in the terms both reports need. Theorem 16.5 is the bridge between them for Hermitian coefficients; Example 16.10 shows there is no bridge in general. **Cross-link both ways.**
- **Hahn–Tate uniformization.** That report records that curves with nonnegative valuation of j need formal groups around a good-reduction model [30], and it makes no assertion about the Berarducci–Mantova derivation. Proposition 33.7 treats constant good-reduction abelian varieties over $\mathbf{No}[i]$ by their formal groups, and Theorem 33.8 is the collection’s first contact between surreal derivations and abelian varieties; it is a cross-link, not an answer to that report’s question over Hahn fields with a coarse valuation (Remark 34.6).
- **Tail-spans.** The Euler derivation ∂_τ of Part V is the operator that report uses at the scale $\log t$ [29]. Corollary 31.3 concerns algebraic independence of differentially *algebraic* solutions and is a different notion from that report’s differential transcendence; there is no conflict.
- **Classical regular-singular and autonomous theory.** Levelt’s Jordan decomposition [15], its exposition for formal connections [16], and generalized-series operator methods [17] are the antecedents of Part V; the geometric formalism of autonomous first-order equations, their local normal forms and Rosenlicht’s theorem [18] are the antecedents of Part VI, with [19, 20] for the algebraic geometry.
- **Classical oscillation theory and the log-atomic tower.** The finite critical hierarchy of Part VII is classical Kneser–Weber–Hartman–Hille theory, treated in [33]; the ordinal log-atomic formulas are [3, Section 2]; the solvability predicate Ω of [3, p. 17] and the asymptotic differential algebra of [34] are the general theory the family instantiates; and Mantova’s composition results [35] are neither used nor superseded (Remark 42.5).

51 The register of non-claims

This section is part of the result. Each item below was stated explicitly by at least one of the eleven source manuscripts, or is an artifact-level finding of the audit of a source package and is marked as such; all of them survive the merge. Several are also stated in place above, and the cross-references say where. Items N1–N52 come from the first seven manuscripts, N53–N64 from the eighth, N65–N79 from member 12, N80–N94 from member 11 and N95–N106 from member 13; where a later member changed the status of an earlier item, the item is amended in place and its original wording is kept.

51.1 Recovered on audit

An audit of this merge against its seven sources found two scope disclaimers stated in a source and missing here. They are restored, since each limits a claim the report otherwise appears to make.

- R1. **The interface table proposes contracts.** One source explicitly distinguishes its certificate and interface specifications from implemented functions. Accordingly, Table 12 is a proposed contract, not evidence that a complete differential solver exists. The companion computer-algebra report supplies the capability taxonomy; current checked prerequisites and pending obligations are tracked separately in `docs/FORMALIZATION.md`. This historical source disclaimer is not an audit asserting that no related implementation exists anywhere in the repository.
- R2. **A recorded run is not a reproduction promise.** One source notes that a local environment’s Python version and elapsed time need not match its bundled run. Appendix D reproduces the delivered records and states the versions they were recorded under; it does not assert that a rerun elsewhere will reproduce those versions or timings, and a mismatch in either is not evidence of a failed check.

51.2 Provenance of the mathematics

- N1. The Berarducci–Mantova derivation is *imported*, not constructed and not reproved (Theorem 3.1). Its construction via log-atomic numbers, a pre-derivation there, path derivatives and nested truncation ranks is not reproduced. The summability and surjectivity proofs are substantial imported inputs, not consequences of the product rule alone. Assigning a derivative to a few familiar monomials is not a construction of a strongly additive derivation on every surreal, and this report uses the published conclusions rather than a pretended implementation.
- N2. No uniqueness characterization of ∂ is offered (Warning 3.3). Other surreal derivations exist; none of them is classified here; the listed structural properties do not characterize every possible surreal derivation; and the word “canonical”, where it survives, is a naming convention for this distinguished construction. *Amended with member 11*: Part VI does not classify them either. It proves, for the class of normalized derivations of Definition 29.1, that one sector — first-order autonomous equations with constant coefficients — is the same for all of them; that is a statement about a class of derivations, not a classification of its members, and no uniqueness of a normalized derivation is claimed (item N80).
- N3. Every intrinsic statement is relative to this one fixed ∂ with the normalization $\partial\omega = 1$, which is indispensable for discussing a named equation such as $\partial y = iy$. The results are *not* derivation-agnostic. *Amended with member 11*: this holds for every part except Part VI, whose statements are proved for every normalized derivation and flag the two that need surjectivity (Convention 29.2). No theorem of another part is upgraded; the first-order statements of Corollaries 8.8 and 8.11 are reproved for every normalized derivation there (Corollary 31.7), and the second-order statements are not. Member 12’s hypotheses (Remark 23.9) are clauses of the normalized-derivation definition, but Part V is stated, and here kept, for the fixed ∂ only; see Section 51.13.
- N4. Smallness is *derived* here from the H -field positivity clause (Lemma 3.8), and is therefore not advertised as part of the imported theorem. Two of the source manuscripts instead import it, one of them flagging it as specific to the chosen derivation and not a general fact about derivations of non-Archimedean fields; that route is recorded in Remark 3.9. Either way, smallness is a structural hypothesis about this derivation.

- N5. The positive-support summability lemma is imported in its standard additive-exponent form (Lemma 2.6) and is additionally proved in appendix A. One source audit records that the Neumann paper was *not retrieved in full* and that the publisher page did not provide accessible full text in that session; the article therefore imports the standard lemma rather than claiming to reprove or re-audit the original paper, and appendix A exists so that nothing depends on an uninspected passage. Higman’s publication metadata was checked against the publisher. Gonshor’s book is used as a standard reference for normal forms and the surreal exponential; only its publisher catalogue was identified, and the whole book was not downloaded or audited.
- N6. The restricted composition, Taylor and global-incompatibility results for ω -series are *separately* imported (Warning 22.2); they do not follow from Theorem 3.1 and are not hypotheses hidden inside the phase criterion. The composition obstruction is quoted with its hypotheses intact, for this particular derivation, and is *not* strengthened into an assertion about every conceivable notion of composition or about every other choice of derivation. Nothing here settles alternative global composition structures. The finite-phase obstruction of this report is a direct differential-algebraic argument, not a restatement or strengthening of that composition theorem.
- N7. The universal H -field results are cited for context, not as a substitute for an unstated linear-solvability theorem, and the elementary embedding of transseries is a theorem about a specified ordered differential language which must not be upgraded to a statement about every simultaneously added analytic, compositional or complex exponential operation (Remark 22.3). Bournez–Guilmant is cited as contextual work only; no theorem of it justifies Theorems 20.3 and 20.8.
- N8. The class-theoretic framing is an *assumption* (Section 2.1): definable classes, or a class-theoretic presentation with enough replacement for the stated constructions. The exact sequences are elementwise and are not the formation of a set of proper-class cosets (Warning 9.3).

51.3 Novelty, refereeing, verification

- N9. No priority or novelty claim is made, and no claim to resolve a named published open problem. These are proved deductions from explicitly stated hypotheses; some may admit formulations already familiar in the general theory of H -fields, and the deductions are *not* asserted to be absent from every published treatment of H -fields, transseries or differential algebra. Ordinary constant-coefficient differential algebra and formal resolvents have a long history; their specialization here is exposition.
- N10. No independent refereeing and no peer review is claimed. The source manuscripts were research expositions prepared with AI assistance; this merge is of the same character.
- N11. No proof-assistant verification, no Lean source, no machine-checked proof object (Warning 49.4). The general proofs remain human-readable mathematical arguments.
- N12. The literature consultation was targeted, not an exhaustive novelty or priority search, and the bibliography is correspondingly targeted.
- N13. The seven repository audits were *targeted coverage audits*, pinned to one revision, and read-only. None was a line-by-line verification of the fifteen reports or of their archived

sources; none asserts that the word “derivation”, or Berarducci–Mantova, or transseries, is absent from every source manuscript; none assesses the correctness of the repository’s other claims. The reading scopes recorded in the sources include the documentation catalogue and manifest; the analysis, trigonometry, foundations and computer-algebra READMEs; the opening foundational section of the analysis source; selected computer-algebra source ranges; and lines 1–220 of the trigonometry article. Several of the audits record that code-search or connector calls returned no usable matches or were truncated, and that those empty or partial responses were *not* used as evidence of absence and not treated as a complete inventory. The assessments are revision-specific: no claim is made that the identified gap remains unfilled on a later commit or branch. The repository’s existing phase decomposition and chosen-exponential obstruction are acknowledged as prior coverage, not presented as missing results. No repository file was modified by any of the seven source packages, and the placement suggestions they made were suggestions. The later, equally scoped audits of members 11 and 12, pinned to `a3124af79f66b8b9c196d76b4cbc5ac3938907c4`, are items N93 and N77, and that of member 13, pinned to `048b72cf7cbfc8ab246e4f73788c10460cb3f6e0`, is item N104.

51.4 Effectivity: what is exact but not computable

- N14. $\mathcal{I}_0$ is a normalization defined by a uniqueness property, not an algorithm (Warning 7.2). It is not asserted to be multiplicative, not asserted to preserve arbitrary Hahn sums, and not asserted to be effective for arbitrary surreal inputs or computable from a supplied symbolic representation. No implementation of it on arbitrary program-described surreals, no universal zero test, and no full differential-algebraic solver is supplied.
- N15. $\mathfrak{o}$ and Φ are exact but not presented as computable functions on arbitrary finitely described surreal inputs. The semantic formula $\Phi(q) = \text{pp}(\mathcal{I}_0 \text{Im } q)$ is exact without promising an algorithm for $\mathcal{I}_0$ or for equality of arbitrary represented scalars. An unevaluated primitive, or an undecided finiteness test, is *not* a negative solvability proof (Warning 49.1).
- N16. Equation (49.1) is a specification, not a terminating algorithm. Only the first three outcomes of Table 11 are claimed decidable, and only for the finite exact core; for arbitrary computably described Hahn supports they are not advertised as total decision procedures.
- N17. The rational decision procedure is restricted to $\mathbb{Q}(X)$ although Theorem 10.6 is stated over $\mathbb{R}(X)$, and it *raises* on unsupported input rather than reporting mathematical nonexistence (Warning 10.7). No algorithm for an arbitrary real coefficient is asserted.
- N18. A well-ordered support is a *semantic* certificate, not automatically an effective presentation (Remark 45.5). No general terminating enumeration algorithm for arbitrary high-rank program-described supports is claimed. A closed formula for a coefficient stream is not a finite list of every coefficient, and the coefficient recursion of equation (18.16) does not make equality of arbitrary program-generated streams decidable.
- N19. ∂_H is represented by a matrix that is only *row-finite on each specified summable input support*; no claim is made that an individual monomial derivative has finite support, or that the matrix has a finite global representation, or that ∂_H preserves a previously chosen smaller monomial group (Warning 48.5).

- N20. Theorem 20.3 is a semantic existence construction, not an effective algorithm for computing $\partial(t^\gamma)$ at arbitrary surreal exponents, and its countably many stages do not constitute a terminating algorithm for every input representation (Warning 20.5).

51.5 Scope of the scalar theory

- N21. The criterion classifies *scalar homogeneous first-order* multiplicative equations, and nothing more. It is not a claim that all nonlinear equations have solutions, that $\mathbf{No}[i]$ is differentially closed — it is not (Remark 8.12) — that arbitrary nonlinear scalar equations or all higher-order operators on the full class are classified, or that primitives are computable from finite input descriptions. *Amended with member 11*: one nonlinear class is now classified. Theorems 30.1 and 31.4 describe every solution in $\mathbf{No}[i]$ of every first-order autonomous equation $F(y, y') = 0$ with $F \in \mathbb{C}[Y, Z]$, and more generally of every constant meromorphic vector field on a smooth projective curve, for every normalized derivation. The item stands for everything beyond that class: variable coefficients in $\mathbf{No}[i] \setminus \mathbb{C}$, higher order, simultaneous systems and higher-dimensional autonomous systems (items N81–N82), and $\mathbf{No}[i]$ remains not differentially closed. *Amended with member 13*: one explicit family of linear second-order equations with variable coefficients is classified at every ordinal depth, $\partial^2 y + (Q_\alpha + c(\ell_\alpha^\dagger)^2)y = 0$ (Theorem 38.1). The item stands for every other higher-order operator, and in particular for arbitrary perturbations of that family (item N99).
- N22. No general inhomogeneous existence theorem is claimed *by Part II*. Outside the admissible locus $\Phi(q) = 0$ its only assertion is *uniqueness* (Theorem 12.1 and Warning 12.2); the absence of a nonzero homogeneous solution does not imply inhomogeneous nonexistence, and the additional differential-field structure that a general inhomogeneous existence theorem would need is not invoked there. Nor is it claimed that such broader solvability questions are open in the literature. Theorem 12.1 proves surjectivity of $\partial - q$ only in the admissible case, and no claim is made *by that scalar argument* about surjectivity of all linear differential operators on $\mathbf{No}[i]$: a trivial kernel is explicitly distinguished from a failure of surjectivity, and $\partial - i$ on $\mathbb{C}((t))$ is the witness (Remark 18.9). Part III does prove first-order inhomogeneous existence (equation (15.2)) and, from it, existence for every forcing of every finite system (Corollary 15.11); that conclusion rests *entirely* on the separately imported Theorem 15.6 and not on the rank-one criterion, as Warning 15.12 states. It leaves the present item standing for the scalar theory as such, it does not make any homogeneous equation solvable, and it changes none of the witnesses of Warning 12.2: what it adds is that their uniqueness is now known to be accompanied by existence. No effectivity follows (item N57). *Amended with member 12*: for power-Hahn regular-singular systems and forcing in $K_{\mathbb{R}}^n$, inhomogeneous existence holds *without* that import, with a solution in $K_{\mathbb{R}}[\log t]^n$ (Theorem 26.5 and Remark 26.6); for arbitrary surcomplex forcing nothing is inferred there (item N68).
- N23. The constant-coefficient theorem and its matrix corollary concern *ordinary complex constant* coefficients and matrices only, not arbitrary surcomplex-coefficient systems (Remark 13.4). The kernel theorems classify *homogeneous* solutions in $\mathbf{No}[i]$ only; a full kernel does not settle an inhomogeneous equation over a smaller subfield. *Amended with member 12*: the item stands as written. The variable-coefficient class of Part V is a different statement, for the Euler derivation ∂_τ and power-Hahn coefficients $A_0 + A_+$, and it does not extend the constant-coefficient theorem to arbitrary surcomplex coefficients; its own limits are items N65–N79. *Amended with member 13*: the item still stands. Proposition 37.3 is the order-two

constant-coefficient classification read for a rescaled derivation $\partial_{\log s}$ (Remark 37.4); it does not extend Theorem 13.2 to arbitrary coefficients, and its limits are items N95–N106.

- N24. Theorem 15.9 gives a complete phase normal form for every ordinary finite system, including non-Hermitian ones, and Theorem 16.5 computes its invariant from ordinary algebraic eigenvalues *when the coefficient is iH with H Hermitian*. This formula does not extend to arbitrary coefficients: Example 16.10 gives a nonnormal coefficient with instantaneous eigenvalues $\pm i$ whose system nevertheless has a fundamental matrix in $\mathrm{GL}_2(\mathbf{No}[i])$, so that integrating imaginary parts of eigenvalues would predict $\{\omega, -\omega\}$ instead of the correct $\{0, 0\}$. Within the scalar theory of Part II taken alone, none of this is available: there the trace condition is only the first invariant (Proposition 14.2), the torus theorem treats *diagonal* systems only (Theorem 21.12), constant matrices are settled while arbitrary matrices with surreal entries are not, and no general matrix normal form follows from the scalar result — the normal form of Part III needs the separate import of Theorem 15.6. The normal form covers every finite rank over $\mathbf{No}[i]$, but does not compute the normalizing gauges or phase obstructions on arbitrary input descriptions. Effective procedures on specified input languages require additional information beyond instantaneous eigenvalues (Warnings 16.11 and 17.13).

For power-Hahn regular-singular systems, $\mathrm{Ph}(-tA) = \{(\mathrm{Im} \lambda_j) \log t\}$, with the λ_j the eigenvalues of the ordinary residual matrix. The normalizing gauge is explicit and lies in $\mathrm{GL}_n(K_{\mathbb{R}}[\log t])$ (Proposition 25.12); within this class the module splits into rank-one factors with obstructions read off from that matrix (Remark 28.6). Example 16.10 lies outside the class. Explicit computation for broader input languages remains open.

- N25. No differential closedness, no universal Picard–Vessiot closure, no classification of all differential extensions, and no decision procedure for arbitrary nonlinear differential equations is claimed. The Picard–Vessiot constructions of Section 21 are explicit; the set-sized ones deliberately avoid any blanket application of Picard–Vessiot existence theory to a proper-class base, and the proper-class ones are proved by hand for exactly that reason. The extension $\mathbb{C}(\omega)(\mathcal{E})$ of Corollary 21.8 is used only to exclude a differential embedding, and no assertion whatever is made about its full constant field. *Amended with member 11:* for presented constant curves and meromorphic vector fields with exact algebraic coefficients, Remark 34.1 gives a finite geometric decision criterion for the existence of nonconstant solutions, conditional on exact arithmetic and decidable sign tests (item N90). No decision procedure for arbitrary nonlinear differential equations is claimed.
- N26. Theorem 11.2’s maximality does not claim that the differential law alone uniquely specifies every normalization on $\mathcal{S}$; ratios of admissible values are differential constants.
- N27. All nonexistence results concern surcomplex *number* solutions under ∂ (Warning 8.9). They do not deny ordinary oscillatory functions, do not say that $\sin z$ and $\cos z$ fail their ordinary differential equations, and do not deny algebraically defined global surcomplex phase extensions or the Ehrlich–Kaplan canonical global surcomplex exponentials, which satisfy different structural contracts and should not be described as nonexistent (Warning 11.6). The obstruction is only to *also* satisfying pointwise differential compatibility for this ∂ .
- N28. No general composition law on $\mathbf{No}$ is constructed and unrestricted substitution “replace ω by x in every surreal” is never used (Remark 4.7).

- N29. Splitting a differentiated infinite sum into a coefficient-derivative part and a variable-derivative part requires *both* separated families to be summable independently (Warning 2.7); summability of the sums does not imply it.
- N30. ct is not a multiplicative standard part (Warning 2.3), and the real-power rule $\partial\omega^r = r\omega^{r-1}$ is asserted for ordinary real r only and must not be extended by silently replacing r with an arbitrary surreal exponent (Warning 2.2).
- N31. The infinitesimal exponential's injectivity is *local* to $\mathfrak{m}_{\mathbb{C}}$ and is not a global injectivity claim; the infinitesimal evaluation rule requires *ordinary constant* coefficients and must not be exported to arbitrary $\mathbf{No}[z]$ -coefficient streams without a joint support certificate (Warnings 6.2 and 48.11).
- N32. Strong summability is not valuation convergence and is not fine-topological convergence (Warning 2.5). No cofinal growth of $n\sigma$ in a higher-rank value group is established, and no argument uses one. The truncation residual identities certify a formal remainder at a stated order, not convergence of the truncation sequence in the fine surreal topology, and the factorial Hahn series is not an ordinary analytic function germ and not a fine-topology limit of its truncations (Warning 18.14). Numerical substitution is declined as a surrogate for a Hahn identity.

51.6 Scope of the workspaces

- N33. Neither localization theorem gives closure under *all* set-indexed summable families (Warning 20.9): an arbitrary-support closure is a different and stronger demand, a localized field need not contain every Hahn series over its own monomials, and there is no Hahn completeness, no finite data representation, no algorithm for the adjoined maps, no birthday or cardinal bound, no computable enumeration and no termination claim for symbolic integration. Arbitrary future coherent families are not automatically covered. Theorem 20.8 is not proved to produce a full Hahn field with a fixed exponent group, and the lighter variant of Remark 20.10 is proved to be neither.
- N34. Theorem 20.3's minimality holds only among full Hahn fields indexed by divisible exponent groups, not among all differential subfields; the hull group itself need not be countable; and the theorem is not primitive closure, nor closure under real exponentiation, arbitrary logarithms, or solutions of all first-order linear equations (Warning 20.5 and Example 18.2).
- N35. The sharp support threshold is deliberately restricted to real-exponent Hahn coefficients, and the criterion provably does *not* globalize to $\mathbf{No}$ by valuation alone (Warning 10.5). The valuation test of Corollary 18.4 is not a valuation-only classification of arbitrary surreal coefficients. The threshold must not be merged into the general resolvent.
- N36. Theorem 18.6's real-exponent hypothesis is substantive: in a higher-rank group a fixed increment 1 need not dominate all relevant scales, and no claim replaces the support argument by an assertion that successive valuation improvements converge (Warning 18.7). Proposition 18.8 is confined to the rank-one Laurent field with $P(0) \neq 0$, and its residual bound is a certificate for a finite truncation: it does not assert that the truncation equals the full solution.
- N37. The factorial examples assert no Borel summation, no Stokes multiplier and no analytic continuation rule (Warning 18.14 and Proposition 18.12); the formal Hahn residue reading of

Warning 18.3 is not an identification with a cluster residue or an actual-point contour residue from the repository's analytic reports.

- N38. The finite-rank classification over $\mathbf{No}[i]$ does not automatically descend to a prescribed smaller workspace: a phase-normalizing gauge or a required inhomogeneous solution can leave that workspace. A classification over such restricted bases is not supplied here.

51.7 Scope of the coordinate theory

- N39. The coherent theory requires *positive* support of the perturbation. That hypothesis is sufficient, not necessary: the nilpotent example Example 44.6 shows a coherent solution with a negative exponent, and negative-exponent matrix systems need finer analysis (Warning 45.4).
- N40. Theorem 45.2 requires either polynomial dependence on the dependent variables or holomorphic dependence with the Taylor admissibility of Lemma 45.1, together with a *supplied* ordinary background solution on U and U simply connected. The background solution's existence on an arbitrary prescribed domain is not asserted, the theorem does not continue it past its ordinary singularities, negative-valued perturbations and arbitrary radius-free coefficient germs are not covered, and unrestricted substitution into an arbitrary holomorphic function of Y without specifying its domain and support conditions is out of scope. Uniqueness is exactly the class of Convention 45.3 and nothing wider.
- N41. No general irregular-singularity classification, no Stokes classification and no Riemann–Hilbert correspondence over arbitrary Hahn fields is offered.
- N42. Monodromy (Theorem 46.1) is an *ordinary*-domain construction on the ordinary universal cover with coefficientwise continuation; no fine-continuous surcomplex path is asserted. Monodromy for the *nonlinear* coherent problem on non-simply-connected domains remains future work.
- N43. The finite-halo evaluation equation (43.3) is infinitesimal Taylor evaluation only, not evaluation at an arbitrary infinite coordinate (Examples 45.6 and 45.7).
- N44. The analytic interface is stated for a *fixed* ordinary open domain and a *fixed* ordinary contour or residue centre. Moving contours, moving endpoints, moving poles and other residue conventions are explicitly outside it (Warning 48.9), and statements are deliberately not transported between arbitrary formal germs and fixed-domain holomorphic families.
- N45. Formal existence, microscopic evaluability and common-domain coherence are three different promises and are never substituted for one another (Warning 48.2). The formal recursion asserts nothing about evaluability at a given surcomplex number.
- N46. The Riccati example of Example 45.7 gives an exact coherent solution on a common ordinary domain, not an all-scale surcomplex entire function; it does not contradict an all-scale rigidity theorem stated in a different function class.
- N47. Theorem 44.1 assumes no norm on coefficients, no contraction, no convergence of Picard iterates and no shrinking domain per exponent (Warning 45.4); nothing here claims to have invented positive-support recursion.

51.8 Scope of the executable checks

- N48. The first eight delivered verification suites check *finite exact symbolic identities and finite jets*, and nothing else. In total they record $1,395 + 115 + 149 + 258 + 291 + 150 + 259 + 62$ passing checks (appendix D); a matrix identity or a whole checked coefficient prefix counts as one check in two of the suites. The suites of members 11 and 12 record 10 and 8 passing checks and carry their own scope items, N94 and N78–N79, and the suite of member 13 records 70, with scope items N105–N106; every limitation stated in the rest of this item applies to them as well. They do not construct the surreal numbers, do not implement Conway normal forms or the Berarducci–Mantova path construction, do not verify arbitrary Hahn summability or strong additivity for arbitrary supports, do not verify class-size or proper-class statements, do not verify the Hahn–Neumann support lemma or Higman’s lemma, do not verify transfinite recursion, and do not verify any of the general nonexistence or infinite-support theorems. They are regression suites for examples, not proof assistants and not implementations of a general surreal differential solver.
- N49. The suites model ∂ by ordinary symbolic differentiation: as $-t^2 d/dt$ on t -expressions, as d/dx on expressions in an ordinary variable standing in for ω , and as $f \mapsto \text{cancel}(iT \partial f / \partial T)$ on the oscillatory extension. The primitive tests replace ω by an ordinary positive variable and check finite algebraic differential identities; they do not infer that an ordinary exponential is a scalar-compatible surcomplex exponential. A finite Taylor check is a formal coefficient check up to its stated order.
- N50. Declared cutoffs are part of the claim, not incidental: one suite uses a formal-series cutoff of 16 with a deterministic seed; another uses matrix *total degree* 6 — explicitly not a valuation cutoff, since t^{100} has higher total degree and lower valuation than u — a Laurent reciprocal-series order of 12, and nonlinear checks through t^{12} ; another checks resolvent residuals for λ in a listed finite set and N in a listed finite range; another checks factorial residuals to $N = 16$. A successful check of a residual at finitely many orders is not a proof of its general formula, and symbolic agreement of finite Taylor coefficients is not a proof of Hahn summability.
- N51. The restricted rational classifier accepts exact Gaussian-rational or rational coefficients and rejects floats, strings, non-Gaussian-rational coefficients, unbound parameters, and non-rational functions by *raising*; those rejections are counted as passing checks in one suite, and the script is explicit that it is not an expression parser (Warning 10.7).
- N52. Two artifact-level discrepancies in the source packages are recorded rather than silently repaired: one package’s PowerShell build script was supplied but not executed on Windows in the preparation environment, and one package’s README lists a checksum file that is not present among the delivered files. Neither affects any mathematical statement.

51.9 Scope of the matrix theory

The eighth manuscript states twelve explicit non-claims of its own. All survive here, together with the audit findings recorded against its package.

- N53. Theorem 15.6 is an *import*, not a result of this report. The splitting of monic scalar operators into first-order factors and the surjectivity of nonzero scalar operators are taken from [4, Lemma 2.4.19] and transferred along the elementary embedding of [3]; neither is reproved. The Berarducci–Mantova derivation and that elementary embedding are likewise imported

wholesale (items N1 and N7). “Linearly closed” in the cited source means the splitting property and *not* that every homogeneous equation has a full solution set in the base field; reading it the second way would contradict Corollary 8.8 (Warning 15.7).

- N54. The rank-one phase obstruction, the bounded-primitive ideal, the diagonal Galois tori and the universal exponential algebra used in Part III are existing repository or literature theory, reproved or specialized where a proof was wanted, and are not discoveries of the eighth manuscript; its own README and its comparison table say so (Remarks 15.1 and 15.2 and Section 17.6). The classical Pinney parametrization equation (17.10) and the abstract Schwarz-closed-field existence principle behind it are likewise disclaimed as antecedents, not contributions [6, 4, 5].
- N55. No named published open problem is claimed solved by Part III, and in particular no new solution of the Boshernitzan conjecture discussed in [5] is claimed. Its own novelty assessment is provisional and is narrowed to three statements — the arbitrary-dimensional Hermitian spectral-integral formula equation (16.3), the sharp entrywise perturbation criterion Corollary 16.7 with Proposition 16.9, and the normalized non-Abelian integration theorem Theorem 17.1 — on the basis of a *targeted* literature search that did not locate an exact prior statement. That is evidence for treating them as candidate new results, not a priority determination, and this report makes no novelty claim of its own (item N9).
- N56. No proof-assistant verification, no Lean formalization and no independent refereeing of Part III is claimed; its strongest claims require independent mathematical review. This repeats items N11 and N12 for the eighth member, whose own text states it in those words.
- N57. Non-effectivity. The obstruction map $\mathfrak{o}$ and the normalizing gauges G and W are exact *semantic* operations, as $\mathcal{I}_0$ is (Warning 7.2). Even for finite matrices, arbitrary surreal input carries no assumed finite computable encoding, and no factorization algorithm follows from the first-order transfer in Theorem 15.6. Identifying restricted, support-certified input languages for which eigenvalue primitives, their purely infinite parts and a normalizing unitary gauge admit certified effective computation is posed as an open problem, not solved; the exact recurrences of Example 17.3 and equation (17.16) are instances, not a solution. *Amended with member 12:* one such language is now supplied for one class. For power-Hahn regular-singular systems with finite rational supports and algebraic coefficients (or the effective coefficient operations of Section 28.3), that procedure computes the Hahn invariants by a finite calculation, and the phase spectrum is read off from the residual eigenvalues (Proposition 25.12). Its gauge is not unitary and is not finitely listed, and for indirectly supplied supports it is not an oracle algorithm (item N69); the question as posed, for unitary gauges and general inputs, stays open.
- N58. Every statement of Part III is relative to the one fixed intrinsic ∂ of Theorem 3.1 with $\partial\omega = 1$. None is a claim about every surreal derivation, about derivatives of a class function of a surreal variable, or about any of the coordinate operators of Table 1. Infinite-dimensional matrices, topological initial-value problems on $\mathbf{No}$, Stokes data over smaller analytic fields and a global surcomplex exponential are all outside its assertions (Warning 17.13).
- N59. All dimensions in Part III are ordinary finite integers, and every displayed Hahn sum there — equation (17.3), equation (17.14), equation (17.19) — is a support-certified formal sum, not a limit in the fine topology on $\mathbf{No}[i]$. This is item N32 applied to the matrix part.

- N60. The metric theorems make no Lyapunov and no uniform-equivalence claim. Warning 15.17 exhibits a positive horizontal metric of scale $e^{-2\omega}$ for printed coefficients whose ordinary solutions grow, and Theorem 15.16 has no hypothesis relating a horizontal metric to any fixed norm.
- N61. Theorem 16.5 does not remove the connection term $W^*\partial W$ of equation (16.5); that term is present and is merely invisible in the ordered additive quotient by $\mathcal{A}$. It is not the assertion that diagonalization commutes with differentiation, and the classification is not the assertion that every phase grade has a horizontal vector in $\mathbf{No}[z]^n$ — by Corollary 15.11 only the zero grade does (Warning 16.6).
- N62. The proper-class handling of Remark 17.14 is a size justification for the proofs, not an algorithm. It does not say that any fixed Hahn workspace already contains the needed primitives or solutions (Example 18.2), and it gives no effective presentation, no birthday or cardinal bound, and no minimal witness-closed subfield.
- N63. The eighth suite’s own scope statement is narrow by its authors’ wording: its checks do not verify the arbitrary-support Hahn arguments, the published model-theoretic inputs, the existence of a gauge for an arbitrary surreal matrix, or the general theorems. Its deliberate counterexamples are asserted *as* counterexamples — the nonnormal coefficient is required to have eigenvalues $\pm i$ together with a base-field fundamental matrix, and the noncommuting exponential defect is required to be nonzero — so a suite that passed while those assertions failed would be reporting the opposite of what it appears to report.
- N64. The eighth package’s verification program *rewrites its own delivered evidence*. Its closing lines compute the package root as the parent of the script’s directory, create `data/` there, and write `data/verification.json`; its README documents the behaviour as regeneration. Appendix D therefore reproduces the record *as delivered* and does not re-run it, and appendix E repeats the instruction to run any source suite on a copy. A byte-identity check made after such a run compares two equally modified copies and establishes nothing. This is an artifact-level hazard, like the two in item N52; it affects no mathematical statement. The delivered records remain historical; the later review rerun on isolated copies is recorded separately at the end of appendix D.

51.10 Scope of the regular-singular theory

Member 12 states fifteen explicit limitations of its own, by this merge’s count. All survive here, together with the audit findings recorded against its package (item N79).

- N65. *The class.* The residual matrix must have ordinary complex entries, and the positive perturbation must be power-Hahn with well-ordered real exponents (Definition 23.5); only the unknown is unrestricted. The theorems are not a principle that the standard part of every finite surreal coefficient controls all solutions: Proposition 28.4 is an infinitesimal coefficient with zero residue and no solution. That counterexample does not refute a classification based on richer logarithmic or differential phase data (Remark 28.5).
- N66. *No higher rank.* The proofs do not extend unchanged to a general higher-rank exponent group: the positive real lower bound of S bounds word lengths, and the shear uses a diagonal Euler derivation with real weights. General Hahn support principles may survive at higher rank, but an intrinsic Berarducci–Mantova derivation on arbitrary monomials need not act

by multiplication by their Hahn exponents, and merely replacing $\mathbb{R}$ by an ordered subgroup of $\mathbf{No}$ in the formulas is not legitimate.

- N67. *No global objects.* No global surcomplex exponential, no trigonometric function at an infinite argument, no fine-topological initial-value theorem and no extension of all ordinary complex local solutions is constructed; the nonexistence of an imaginary Euler mode is part of the conclusion. Adjoining $\log t$ does not manufacture a surcomplex fundamental matrix on a nonzero-frequency summand (Remark 26.4).
- N68. *Forcing.* Existence after adjoining $\log t$ is proved only for forcing in $K_{\mathbb{R}}^n$; no theorem for arbitrary surcomplex forcing is inferred from the coefficient calculation. “Fredholm” and “index zero” are algebraic: no Banach space, closed-range argument, compact operator or topology on $\mathbf{No}[i]$ is involved (Remark 26.6).
- N69. *Finite dependence is not an algorithm.* The critical set S_{crit} is sufficient, not minimal: matrix products can vanish and word contributions can cancel (Example 27.2). A finite certificate for the invariants is not a finite listing of the gauge $G_{\mathfrak{h}}$, which may have infinite support. For a support supplied only indirectly, finding the exact word sums that hit a resonance need not be effective, finitely many black-box coefficient queries cannot certify that an unseen resonant coefficient is absent, and iterating through all exponents below $\max \text{Lev}(A_0)$ need not terminate (Section 28.3).
- N70. *The Hahn classification is for $\Gamma = \mathbb{R}$ only.* The complete invariants of Theorem 25.7 are for $K_{\mathbb{R}}$; over K_{Γ} real parts need not be removable, even rank one retains the real exponent modulo Γ , and transferring the classification to $K_{\mathbb{Q}}$ would identify $\partial_{\tau}y = \sqrt{2}y$ with the trivial equation (Remark 28.3).
- N71. *Which smallness.* The invariance of Corollary 25.4 is under perturbations of positive Hahn order, not under numerically small changes of the residual matrix (Example 27.6).
- N72. *Non-canonical entries.* Only the block similarity invariants of the nilpotent matrix N are independent of choices; its displayed entries are not asserted to be canonical across Jordan bases.
- N73. *Sets, not classes.* The ambient solution collection is identified with a set, and no dimension theory for a proper-class vector space is used (Remark 25.6).
- N74. *Not a coordinate derivative.* ∂_{τ} is the intrinsic derivation rescaled; t names the surreal ω^{-1} , equation (23.3) is an identity for the intrinsic derivation, and no differentiability in the fine topology is assumed; symbolic checks with a formal variable verify algebraic identities realized by the embedding (Remark 23.3).
- N75. *Antecedents.* Regular-singular normal forms, nilpotent logarithmic factors and the classification of unipotent formal differential modules are classical [15, 16]; generalized-series operator methods are van der Hoeven’s [17]; and the scalar nonoscillation phenomenon already exists in this report (Lemma 25.2). None is relabelled as new. Member 12 proposes as its contribution only the finite-resonance support theorem, the constructive normal form with support and finite-monomial certificates, and the exact comparison of Hahn and full-surcomplex solutions and morphisms; general theorems about generalized-series differential modules may imply parts of the construction in other terminology.

- N76. *Novelty and verification.* No independent refereeing and no Lean verification is claimed. No named published open conjecture is claimed resolved. Priority is provisional: the searches identified the direct antecedents [1, 3, 15, 16, 17] and this report, and did not locate in them the specific finite-resonance statement together with the full ambient spectral selection; that is a bounded assessment, not evidence that no equivalent result exists, and a specialist should check priority before any publication claim.
- N77. *The repository comparison.* Member 12 inspected the repository on 22 September 2026 at `a3124af79f66b8b9c196d76b4cbc5ac3938907c4`: the root README, the documentation catalogue and this report's README, together with neighbouring analytic-geometry coverage. The comparison is to the material returned in that review and is not an assertion of permanent absence from later revisions; the whole repository was not searched and its other drafts were not audited; and its README distinguishes the pinned revision from every README read later in the review. The repository was read, not modified.
- N78. *The checks.* The program handles finite positive rational supports, a specified rational cutoff and an explicitly supplied generalized-eigenvalue block decomposition. It is not a parser or decision procedure for arbitrary surreal numbers or indirectly described infinite supports. Its eight checks detect sign errors, missing matrix-order terms and incorrect nilpotent inverse bounds; they do not replace the support lemma, the proof excluding arbitrary ambient solutions, or an independent review, and the infinite-support theorems are proved in the text, not by the tests.
- N79. *Audit findings on package 12* (artifact level; no mathematical statement is affected). The program writes its `-output` file, by default `verification.json` in the current working directory, and prints the same JSON, and the package's `verify` target overwrites both delivered records; placed here as `code/12-regular-singular-Makefile`, that file still names the package's original file names, not the placed ones. Run it only on a copy. The field `ambient_dimension: 1` in the mixed-frequency record is a label written by the script, not a computed quantity. The tail-invariance check compares resonant coefficients only through weight 4. The two delivered records carry the same JSON content, and there is no checksum manifest. The record states SymPy 1.14.0; the package README says the run used Python 3 without a minor version, and the program's docstring requires Python 3.10 or later.

51.11 Scope of the autonomous theory

Member 11 states fifteen explicit limitations of its own, by this merge's count. All survive here, together with the audit findings recorded against its package (item N94).

- N80. *Derivations.* Every theorem of Part VI holds for every normalized derivation, but no equality of two normalized derivations on $\mathbf{No}[i]$ is asserted: they agree on the solutions of every nonzero $F(y, y') = 0$ and on the fields K_ξ , and nowhere else is claimed. No claim is made that $\mathbf{No}$ has a unique normalized derivation, or that every surreal is first-order autonomous differential-algebraic over $\mathbb{C}$, and no normalized derivation is classified (Convention 29.2 and Remark 31.8).
- N81. *Coverage.* The classification is complete for first-order equations on curves only. Coefficients in $\mathbf{No}[i] \setminus \mathbb{C}$, higher differential order and simultaneous systems are outside it and may probe scales on which normalized derivations differ; which higher-order sectors distinguish

normalized derivations is left open, and no uniform finite-scale bound for arbitrary higher-order equations is implied.

- N82. *No higher-dimensional classification.* Proposition 33.1 survives in every dimension, but higher-dimensional autonomous systems are not classified: near a singular point several eigendirections interact, and integer relations among eigenvalues and noncommuting normal-form terms defeat the one-coordinate time primitive (Remark 34.4).
- N83. *Differential-algebraic only.* Nothing is asserted about derivatives of arbitrary fine-topological class functions, global complex exponentiation or topological convergence of surreal series; the formal implicit uniqueness is not a convergence statement about Newton iteration; and no ordinary path on a Riemann surface is treated as a fine-continuous path in $\mathbf{No}[i]$.
- N84. *Monomial generation is not transcendence degree.* The finite monomial generation of K_ξ is not a finite transcendence-degree bound: Corollary 31.3 places $2^{\aleph_0}$ algebraically independent solutions inside a two-generator monomial group.
- N85. *The strip logarithm is pointwise.* Log is used pointwise or on a specified leading-term class; it is not a global additive logarithm and defines no global surcomplex exponential (Lemma 29.7).
- N86. *Classical and imported inputs.* The local formal normal forms are classical [18, Lemma 6.1], Rosenlicht's algebraic-independence theorem is imported [18, Lemma 7.2, Proposition 7.3], and the curve-and-derivation formalism is that of [18], whose numbered statements are cited from the consulted arXiv version 1 and are not assumed to match the journal numbering. The construction of surreal derivations, formal implicit equations, invariant differentials and characteristic-zero formal-group logarithms are inputs, not contributions; [19] supplies the standard setting only, and the formal-logarithm argument is proved in full (Lemma 33.6).
- N87. *Non-canonical parametrizations.* Parametrizations depend on the local parameter and on the choice of $\log \beta$, and the leading coefficient β is coordinate-dependent; only the number of branches, the family of points and the existence criterion are canonical (Remark 30.9).
- N88. *Constant commutative groups only.* The abelian results hold for constant abelian varieties over $\mathbb{C}$ only. Nonconstant abelian varieties, which can acquire a connection term, and noncommutative algebraic groups, whose invariant forms need not be closed, are excluded. No complex analytic uniformization, no period lattice and no exponential on all surcomplex Lie-algebra elements is used; $\exp_{\mathcal{F}}$ is evaluated only on $\mathfrak{m}_{\mathbb{C}}^d$.
- N89. *Surjectivity and standard part.* Surjectivity of the derivation is used to define $\mathfrak{o}_{\mathcal{F}}^{\mathbb{C}}$ on every input and in Corollary 33.10. The explicit primitives in Corollary 33.11 establish that threshold for every normalized derivation without surjectivity. A standard part is taken only after finiteness of the primitive has been established.
- N90. *Effectivity.* The finite geometric decision criterion is an effective procedure only with effective divisor and local-expansion data, as well as exact coefficient operations, equality and sign tests. Algebraic complex presentations with the stated geometric algorithms suffice. It is not an unconditional computability claim about arbitrary unencoded complex constants, and producing an entire formal solution as a finite string requires its defining implicit equation and coefficient recursion (Remark 34.1).
- N91. *Classes.* $\mathcal{X}(\mathbf{No}[i])$, $\mathbf{A}(\mathbf{No}[i])$ and the exact sequences over them are class-level statements read NBG-style and pointwise; no collection of all surreal supports is used as a set (Remark 34.5).

- N92. *No trigonometry.* The Riccati example asserts nothing about separately defined symbolic trigonometric functions (Example 32.4).
- N93. *Novelty, verification and the repository comparison.* No peer review or Lean verification of the autonomous theorems is claimed. The source supplied no Lean file. Current library results do cover the algebraic finite-variable evaluation and composition clauses; the intrinsic chain rule and geometric reduction bridge remain pending, as distinguished in Remark 34.5. No named published open problem is claimed solved and no exhaustive priority certificate exists: the results are candidate new theorems with proofs, and a specialist may identify earlier equivalents. The repository comparison was scoped to the root README, the documentation inventory and this report's README at `a3124af79f66b8b9c196d76b4cbc5ac3938907c4`, with scoped keyword searches; irrelevant keyword results were not treated as evidence of absence, not every repository manuscript or publication was read, and the repository was read, not modified. The source's review log is kept in this directory as `11-autonomous-dynamics-sources.md`.
- N94. *The checks, and audit findings on package 11.* The program's ten exact finite checks — the logistic solution, the two counterexamples of Example 33.5, the time primitive and a total-degree truncation of Example 32.5 — catch sign, normalization and coefficient errors in concrete instances; they do not establish the support lemma, the algebraic-geometric reduction, the completeness of the solution list, or novelty. At the artifact level: the program itself writes only to standard output and accepts `-degree` from 2 to 7 (default 4), but the package's build script, placed as `code/11-autonomous-dynamics-build.sh`, and its README's rebuild instructions redirect that output onto the delivered `verification.txt`; run them only on a copy. There is no checksum manifest.

51.12 Scope of the transfinite critical potentials

Member 13 states twenty-one explicit limitations of its own, by this merge's count. All survive here, together with the audit findings recorded against its package (item N106); its two open directions, which it does not claim to have solved, are among the continuations of Section 51.13.

- N95. *Scalar unknowns.* The unknown of Part VII is a scalar of a specified differential field, not a function on an ordinary interval (Convention 3.6); “no solution in $\mathbf{No}[i]$ ” is not a claim that an ordinary complex initial-value problem has no analytic solution, and nothing is asserted about coordinate differentiation in $\mathcal{O}(U)((t^\Gamma))$ or about fine-differentiable class functions (Remark 42.4).
- N96. *One derivation, and an imported tower.* Every statement is relative to the fixed ∂ (item N3). The ordinal log-atomic formulas, limit cases included, are imported from [3, Section 2]; they are not obtained by differentiating finite iterates, and the normalization $\partial\omega = 1$ alone does not determine them. No derivation-independent version is claimed: for another derivation the finite identities may still hold, but the limit-stage derivatives need not be the Berarducci–Mantova ones, and nothing is asserted for the class of normalized derivations of Part VI (Proposition 35.3 and Remark 42.4).
- N97. *No transfinite iteration or composition.* The ℓ_α are particular numbers; no globally defined transfinite iteration or composition of log, no substitution $v(s)$ and no global composition is used (Remark 35.4 and Lemma 37.2, [2, Theorem 8.4]). The Conway monomial map is not $x \mapsto \exp(x \log \omega)$, and no variable-exponent power rule is used (Warning 2.2).

- N98. *Sums and classes.* Every sum is a Hahn sum of a set-indexed summable family; none is a limit of partial sums in the fine topology, Q_λ is the sum of its increments and not of the Q_β , the imported product identity is not replaced by convergence of finite products, and no proper-class family is summed. There is no all-ordinal potential (Proposition 39.5), and that is a boundary of the construction, not a convergence problem. Solution collections are sets; no dimension theory of a proper-class vector space is used.
- N99. *This family only.* No classification of arbitrary potentials, and no comparison of leading coefficients, is asserted; no comparison theorem for surreal Riccati equations is used; that every $\varepsilon \prec (\ell_\alpha^\dagger)^2$ preserves subcritical or supercritical behaviour is not proved and would need further differential asymptotic estimates, and at $c = \frac{1}{4}$ a smaller perturbation can matter (Remark 38.3).
- N100. *What invisibility means.* Theorem 39.2 concerns the named hierarchy $(\ell_\beta^\dagger)^2$, not every conceivable observation or algorithm. It is a statement about a quotient by an additive class of errors, not a discontinuity in the fine topology, and that class is not called an ideal of the field (Warning 39.4).
- N101. *Field-theoretic scope.* $\text{span}_\mathbb{R}$ means finite combinations, not Hahn closure inside the exponent group, and at $\lambda = \omega$ the group Δ_0 is not a Laurent-series field (Warning 40.1 and Example 42.3). All three fields are algebraically closed after adjoining i and stable under the derivation. The first two real fields are closed under logarithms; the third is not, as witnessed in Proposition 40.5. No exponential or general solution closure is asserted for them; “minimal” refers to the solution-generated field only, and no minimality of the intermediate Hahn fields among arbitrary fields is claimed (Remarks 41.4 and 40.6).
- N102. *Inputs and antecedents.* The finite critical hierarchy and reduction of order are classical [33], whose potential has the opposite sign; the derivation [1], the log-atomic formulas [3], Hahn real closedness and the positive-support lemma are imported; the obstruction for nonreal powers is this report’s criterion, reproved there in a special case; the gauge and Schwarzian mechanisms are standard, and the Schwarzian form is not an independent result. The general theory of such solvability, including the predicate Ω of [3, p. 17], is not new; [34] is neither replaced nor claimed; [35] is neither used nor superseded. The main proofs use no model completeness, no differential closure and no finite-system splitting theorem, and surjectivity only in the real Riccati remark (Remarks 38.2 and 42.5).
- N103. *Novelty and verification.* Member 13 proposes as new only the transfinite classification, the invisibility theorem, and the limit-stage filtration with its Borel group; no isolated elementary identity is promoted. Priority is provisional: the literature search was targeted and did not certify exhaustive coverage, an expert may recognize a conclusion as a known consequence of a broader theorem, and the arguments do not depend on the novelty assessment. No named published conjecture is claimed resolved. No independent refereeing, no proof-assistant verification and no Lean implementation is claimed. This report itself makes no novelty claim (item N9).
- N104. *The repository comparison.* Member 13 read the repository at 048b72cf7cbfc8ab246e4f73788c10460cb3f6e0: its tree, root README, the catalogue docs/README.md, the listing of docs/new/, the introductions of the entire-functions and Gamma-functions reports, and source ranges of this article. Connector searches for “Kneser” and “iterated-logarithm” returned no hits, including none for wording visible in an inspected source, and were not

treated as evidence of absence. The comparison is bounded and is not a proof of absence from the repository, its history or later commits; the repository was read, not modified. *Recorded by this merge:* the article it read already contained the first-order counterpart Corollary 10.3, titled “Iterated-logarithm thresholds at every finite depth”, which the member does not cite; Lemma 35.5 and Corollary 37.1 now connect the two.

- N105. *The checks.* The 70 exact symbolic checks — the critical and Riccati identities at lengths 0 to 12 (39), Euler powers, bases and Wronskians (8), the gauge identity (4), Schwarzian identities (4) and the fundamental matrix and Borel action (15) — test finite identities, with the Euler checks modelling the scale by an ordinary positive variable. They do not construct the surreal numbers or the derivation, do not prove the imported derivative formula, do not certify transfinite Hahn summability or exponent-field membership, and do not establish novelty; a count is not a measure of proof strength, and no finite list of checks certifies a theorem for every ordinal.
- N106. *Audit findings on package 13* (artifact level; no mathematical statement is affected). The program writes to standard output and, with `-output`, to the named file; the package README’s `python verify.py -output verification.txt` would overwrite a record of that name, so run it only on a copy. The placed manifest `data/13-critical-potential s-manifest.json` lists five files with SHA-256 digests; all five match the delivered package, but only the program and its record are shipped here, as `code/13-critical-potential s-verify.py` and `data/13-critical-potentials-verification.txt`, and the package’s `article.tex`, 21-page `article.pdf` and `README.md` are not. The record states Python 3.13.5 and SymPy 1.14.0.

51.13 Open continuations, hedged

The following are directions for further development. They are *not* assertions that the corresponding questions are unresolved in the literature.

Two items that earlier editions of this list posed have since been answered inside the report, by Part III, and are recorded here as answered rather than deleted; two more are answered in part by Part V, and a third is adjacent to Part VI; all three are re-scoped below.

- *Which variable-coefficient matrix equations keep their fundamental matrices in $\mathbf{No}[i]$.* Corollary 15.11 answers this for every finite system: a fundamental matrix exists in $\mathrm{GL}_n(\mathbf{No}[i])$ exactly when every member of $\mathrm{Ph}(A)$ is zero, and $\dim_{\mathbb{C}} \ker(\partial - A) = m_0(A)$ in general. For Hermitian coefficients Corollary 16.8 makes the test explicit in terms of ordinary algebraic eigenvalues: the dimension is the number of eigenvalues lying in $\mathcal{A}$. The distinction the item asked for — genuine phase obstruction against workspace obstruction — is exactly the distinction between $\mathrm{Ph}(A) \neq \{0, \dots, 0\}$ and the workspace phenomena of Part IV. For $A = iH$ with H Hermitian, Corollary 16.7 shows that the entire phase multiset is unchanged by Hermitian entrywise perturbations of H in $\mathcal{A}_{\mathbb{C}}$.

What the item also asked — which nontriangular differential modules admit a filtration whose rank-one factors have *computable* obstructions — remains open, and for the reason the item gave: that needs genuine module-theoretic information, not the eigenvalues of an instantaneous coefficient matrix. Example 16.10 confirms the reason rather than removing it. The remaining question here is effective computation on a specified input language; finite-rank classification over $\mathbf{No}[i]$ itself is already supplied by Theorem 15.9.

- *Connecting the phase invariant with the strongest available linear-solvability theorems for this H -field.* This is done in Theorem 15.6 and Corollaries 16.8 and 15.11, and it separates existence from dimension exactly as the item asked: existence for *every* forcing holds unconditionally, while the dimension of the homogeneous solution space is the multiplicity $m_0(A)$ of the phase zero. The separation is not free — existence comes from the imported Theorem 15.6 and not from the rank-one criterion (Warning 15.12, item N53) — and no effective version of either half is claimed (item N57).

Three of the remaining continuations are touched by Parts V and VI: two are answered *in part*, and one only has an adjacent result. They are re-scoped here rather than deleted.

- *Find explicit formulas for the already defined phase spectrum of further classes of non-Hermitian coefficients,* the obstruction being that a nonnormal similarity gauge need not be entrywise finite (Warning 16.11). Answered for one class: for power-Hahn regular-singular systems $\text{Ph}(-tA) = \{(\text{Im } \lambda_j) \log t\}$, with an explicit gauge and no import (Proposition 25.12, member 12). A general explicit formula outside the currently computed classes remains open; Example 16.10 rules out the instantaneous-eigenvalue formula.
- *Identify restricted, support-certified input languages on which the phase map, the purely infinite parts of eigenvalue primitives and a normalizing unitary gauge become effectively computable.* Answered in part for one class: finite rational supports with algebraic coefficients for power-Hahn regular-singular systems (Section 28.3 and Remark 28.6). The gauge there is not unitary and not finitely listed, and the continuation stands for unitary gauges and for every other class.
- *Classify logarithmic derivatives in multi-scale ∂ -stable Hahn workspaces where support shifts lack the uniform rational form of equation (18.1).* Not answered. Member 11 supplies explicit multi-scale ∂ -stable Hahn fields K_ξ , with power, logarithmic and exponential scales (Remark 31.2), but does not classify logarithmic derivatives on them; member 13 supplies the higher-rank logarithmic fields H_{Δ_j} of Section 40, with an explicit logarithmic-derivative formula for their monomials (Proposition 40.3), and does not classify them either.

Members 11 and 12 add continuations of their own. From member 12: a *layered* analogue of the regular-singular theory for coefficient fields containing $\log \omega$, $\log \log \omega$ and their real powers, where Proposition 28.4 shows residual data at the first power scale to be insufficient and a finite sequence of scale-dependent residual matrices might decide existence for a restricted logarithmic class — any answer must separate derivative effects from monomial valuation and justify every support interchange anew; which representations of semialgebraically or recursively specified well-ordered supports make exact resonance-word extraction decidable; and a modular formalization whose Hahn part needs only finite-dimensional linear algebra, strong summation and the positive-support lemma, and whose ambient part needs only the derivation clauses, complexification and the finite-element estimate. From member 11: autonomous systems in higher dimension; nonconstant abelian varieties and noncommutative algebraic groups; and which higher-order differential-algebraic sectors distinguish normalized derivations. One question is recorded by this merge rather than by either source: every hypothesis member 12 lists (Remark 23.9) is a clause of the definition of a normalized derivation, and on $K_{\mathbb{R}}$ the operator $-\omega \partial$ acts by $t^a \mapsto at^a$ for every normalized ∂ , so Part V appears to hold for every normalized derivation; that has not been audited line by line, and no statement of Part V is upgraded.

Member 13 adds two directions and does not claim either solved: characterize the ordered exponent groups Γ for which Q_λ lies in $\mathbb{R}((t^\Gamma))$ and ∂ restricts, and classify the solution dimension there by the two basis exponents; and study controlled nonconstant perturbations of c after the

exact reduction to $\partial_s^2 + cs^{-2}$. Its part is adjacent to member 12's layered continuation above without answering it: within its one family the coefficient at the last scale decides existence and no set of earlier scales does (Theorem 39.2 and Remark 39.6), but no restricted logarithmic class of regular-singular type is classified. It is likewise adjacent to the extension of the workspace algorithms to logarithmic and higher-rank parents below: the fields H_{Δ_j} are explicit such parents, with no algorithm.

The remaining continuations stand. Build explicitly typed analytic and coefficient interfaces that prove exchange rules with contour functionals under their correct common-domain and summability hypotheses, always fixing whether centres and contours move. Extend the workspace algorithms from real-exponent Hahn fields to logarithmic and higher-rank parents with explicit support-change certificates, preserving the distinction between a representation of a primitive and a decision about its infinite part. Separate negative-exponent coherent systems into finite nilpotent cases, cases reducible by a gauge transformation, and genuinely non-coherent cases, each proposed reduction carrying its own support proof. Develop monodromy for coherent coordinate equations on non-simply-connected domains in the nonlinear case.

52 Conclusion

The internal derivative does not turn a surcomplex number into an ordinary complex-valued function at infinity. Real exponential integration is extremely rich; unit-circle phases are severely constrained, because only a constant ordinary phase and an infinitesimal residue already live in the base field. The exact consequence is

$$\{y^\dagger : y \in \mathbf{No}[i]^\times\} = \mathbf{No} + i\partial\mathbf{m} = \mathbf{No} + i\partial\mathcal{O}, \quad \partial\mathbf{m} \subsetneq \mathbf{m}.$$

The purely infinite part of an accumulated phase of the imaginary coefficient is therefore the complete rank-one obstruction. It supplies a canonical normal form, distinguishes algebraic closure from differential solution closure, explains the missing oscillator, determines the maximal domain of any derivation-compatible exponential, removes every nonreal characteristic root from a constant-coefficient solution space, and fixes how many formal phase generators a diagonal Picard–Vessiot extension needs. At the workspace level two sharper identities are available,

$$\partial(\mathcal{H}) = \{f \in \mathcal{H} : [t]f = 0\}, \quad \{y^\dagger : y \in \mathcal{H}^\times\} = \mathbb{Q}t + \{g : g = 0 \text{ or } v(g) > 1\},$$

giving exact auditable tests rather than heuristic simplification rules, and showing why derivative closure, primitive closure and exponential closure must be separate capabilities.

The same obstruction survives the passage to finite matrices, and there it becomes a multiset. Every finite system over $\mathbf{No}[i]$ has a differential phase spectrum $\text{Ph}(A) \subset \mathcal{P}$, complete for gauge equivalence, of which the trace test equation (14.3) sees only the sum; and when the coefficient is iH with H Hermitian the spectrum is computed by integrating the ordinary algebraic eigenvalues of H and keeping the purely infinite parts of their primitives. The connection term left behind by pointwise diagonalization is neither ignored nor approximated: by Lemma 16.3 a unitary gauge is entrywise finite, so that term lands in $\text{Mat}_n(\mathcal{A}_\mathbb{C})$, which is exactly the kernel the invariant annihilates. That is why the perturbation theorem needs no spectral gap. What the argument needs instead is the strictness $\mathcal{A} \subsetneq \mathbf{m}$ that Warning 5.4 insists on, and an additional imported input. The eigenvalue computation needs Hermitian symmetry, which Example 16.10 shows cannot be dropped from that formula.

Two further strata show the same obstruction from other sides. For regular-singular systems $\partial y = -tAy$ with an ordinary residual matrix and a positive real-power perturbation, finitely many resonance-reaching coefficients determine the Hahn invariants, real residual parts are sheared away by real monomials, nilpotent Levelt data are removed by one actual logarithm, and what survives in $\mathbf{No}[i]$ is exactly the imaginary parts of the residual eigenvalues: $\text{Ph}(-tA) = \{(\text{Im } \lambda_j) \log t\}$, a non-Hermitian sector computed without the matrix part's imported input. And for first-order autonomous equations with constant coefficients — constant vector fields on curves — every surcomplex trajectory is centred at a zero of the field, a simple zero contributes exactly when its eigenvalue is negative real, a multiple zero contributes finitely many power-logarithm families, and the whole answer is the same for every normalized surreal derivation. Constant abelian varieties obey the same rule with their whole Lie coordinate: dlog lands exactly in $\text{Lie}(\mathbf{A})(\partial \mathbf{m}_{\mathbb{C}})$.

The same obstruction also runs down the log-atomic tower. For the second-order potentials $Q_\alpha + c(\ell_\alpha^\dagger)^2$ one gauge identity reduces the equation to the Euler operator at the scale ℓ_α , and the threshold $c = \frac{1}{4}$ is exact at every ordinal depth: the missing oscillator is the same Euler problem at the scale e^ω , and in every supercritical case the obstruction is an infinite accumulated phase, now $\sqrt{c - \frac{1}{4}} \ell_{\alpha+1}$. No set of earlier logarithmic scales decides solvability, and at every limit stage the solutions appear one at a time in differential Hahn fields that are already real closed and closed under logarithms — first after an exponential, then after a primitive — with the Borel subgroup of $\text{SL}_2(\mathbb{C})$ as the Galois group of the two steps together.

A distinct theory of coordinate initial-value problems remains available and useful. On a common ordinary complex domain, positive-support Hahn perturbations of a holomorphic problem are solved uniquely by support recursion through ordinary linear equations, with explicit support certificates, a Liouville determinant identity, and monodromy valued in $\text{GL}_d(K_\Gamma)$. The two theories coexist without conflict once the derivative, the parent field, the coefficient domain and the summability certificate are part of every statement.

The resulting picture is coherent rather than paradoxical: $\mathbf{No}[i]$ is algebraically closed; ∂ is surjective; real exponential integrals exist; finite angular motion is exact; and a sustained nonzero ordinary angular velocity is nevertheless absent. These are different closure properties, and the obstruction sequence equation (9.3) records their precise boundary. The boundary worth remembering is this:

Every surcomplex direction has a finite angle, but a differential evolution prescribing a purely infinite *accumulated* phase need not have a surcomplex value. A phase convention may assign a value; it cannot force that value to satisfy the incompatible number-derivation law.

And the distinctions are not a notation policy. Their compatibility and incompatibility are theorems: an intrinsic derivative of a scalar, an external derivative of a function, and the substitution that might connect them are three pieces of structure, not one.

Part X

Appendices

A The positive-support calculus, proved

This appendix supplies the combinatorial facts used throughout. It works in a set-sized ordered abelian group and uses no completeness assumption on the group. The finite-word principle is the well-ordered-alphabet case of Higman’s lemma [11]; the support consequence is the classical Hahn–Neumann fact [10]. Both are cited as published sources, and a proof is included as well, so that nothing in this report depends on an uninspected passage. The published citations are *not* replaced by this appendix, and this appendix is not replaced by them.

A.1 Two elementary facts about well-ordered supports

First, every sequence in a well-ordered set has an infinite nondecreasing subsequence: choose an occurrence of the least value in the whole tail, then an occurrence of the least value in the tail strictly after that index, and continue; the selected indices increase and the selected values do not decrease.

Lemma A.1 (Sum of two supports). *If A, B are well-ordered subsets of an ordered abelian group, then $A + B$ is well ordered, and for each γ only finitely many pairs $(a, b) \in A \times B$ satisfy $a + b = \gamma$. A finite union of well-ordered subsets is well ordered.*

Proof. If $a_n + b_n$ were strictly decreasing, select a nondecreasing subsequence of the a_n ; the corresponding b_n would then be strictly decreasing, contradicting well ordering of B . In a linear order, the absence of an infinite strictly decreasing sequence implies well ordering in the classical set-theoretic setting used here: a nonempty subset with no least element would produce such a sequence by recursive choice.

If infinitely many distinct pairs had a fixed sum, select a sequence of distinct pairs. Their a entries are distinct, so a nondecreasing subsequence of them is strictly increasing, and the corresponding b entries strictly decrease — impossible. Finally, an infinite decreasing sequence in a finite union would have an infinite subsequence inside one member of the union. $\square$

A.2 Finite words over a well-ordered alphabet

A word $a_1 \cdots a_m$ *embeds* in a word $b_1 \cdots b_n$ if there are indices $1 \leq j_1 < \cdots < j_m \leq n$ with $a_k \leq b_{j_k}$ for every k . The empty word embeds in every word.

Lemma A.2 (Finite-word lemma). *Every infinite sequence of finite words over a well-ordered alphabet contains two words, in their original order, such that the earlier embeds in the later.*

Proof. Call a sequence *bad* if it contains no such pair, and suppose a bad sequence exists. Choose one minimally in the following sense: after a fixed prefix $w_0, \dots, w_{n-1}$, choose w_n of smallest possible length among words that permit continuation to an infinite bad sequence. Natural-number lengths make each choice possible, and no w_n is empty, since the empty word embeds in everything.

Write $w_n = v_n a_n$ with a_n the last letter, and select indices $n_0 < n_1 < \dots$ for which the letters a_{n_j} are nondecreasing. Consider the sequence

$$w_0, \dots, w_{n_0-1}, v_{n_0}, v_{n_1}, v_{n_2}, \dots$$

It cannot be bad, because v_{n_0} is shorter than the minimal choice w_{n_0} after the same prefix. So it has a good pair. That pair cannot lie wholly inside the prefix, since the original sequence was bad. If it runs from a prefix word to some v_{n_j} , then the prefix word also embeds in w_{n_j} , again contradicting badness. And if it runs from v_{n_i} to v_{n_j} with $i < j$, append the nondecreasing last letters to the embedding: $w_{n_i} = v_{n_i} a_{n_i}$ embeds in $v_{n_j} a_{n_j} = w_{n_j}$, once more a contradiction. Hence no bad sequence exists. $\square$

A.3 From words to Hahn sums

Let $S \subseteq \Gamma_{>0}$ be well ordered. If a word in S embeds in another, the sum of its letters is at most the sum of the other's; and equality forces the words to be identical, because any unmatched letter is strictly positive and any strict comparison of matched letters contributes a strictly positive difference, so a finite sum of such contributions cannot vanish.

An infinite strictly decreasing sequence of word sums would therefore be a bad word sequence, contradicting Lemma A.2; hence S^* is well ordered. If infinitely many distinct words had the same sum, an embedded pair among them — which Lemma A.2 supplies — would force two of them to be identical. So each exponent has only finitely many word representations. This proves Lemma 2.6.

Remark A.3 (Ordered words, and what this appendix does *not* prove). The count is over *ordered* words, not over unordered multisets. That is what makes the lemma suffice for noncommuting matrix products: each word retains the order in which its coefficient matrices were multiplied, and finite matrix dimensions introduce only finitely many intermediate indices for a word of fixed length. Warning 47.1 is the concrete witness that the ordering matters. Combined with Lemma A.1, this justifies multiplying an arbitrary Hahn series by a positive-support power series, the Taylor-evaluation families of equations (6.1) and (43.3), and every coefficient recursion in this report.

It does *not* establish cofinal growth of $n\sigma$ in a higher-rank value group, and no assertion of that kind is used anywhere (Warnings 44.2 and 2.5).

B Notation, and a concordance with the source manuscripts

Table 13: Notation used in this report.

Symbol	Meaning
$\mathbf{No}, \mathbf{No}[i]$	The real surreal class field and its complexification $\mathbf{No}[i]$; class-sized carriers, with all supports and summed families sets.
∂, x'	The fixed simplest Berarducci–Mantova derivation, normalized by $\partial\omega = 1$, extended to $\mathbf{No}[i]$ by $\partial i = 0$.
∂_s	The rescaled derivation $\partial f / \partial s$, with $\partial_s s = 1$.
$\partial_Z, \tilde{\partial}$	Formal differentiation of the indeterminate on $\mathbf{No}[i][[Z]]$, and coefficientwise intrinsic differentiation there.

Symbol	Meaning
D_z	Coordinate differentiation of holomorphic coefficient functions in an ordinary complex z ; every monomial and scalar is held constant.
∂_H	Intrinsic differentiation of the Hahn monomials of a common-domain family, holding z fixed.
D_t	Formal differentiation in a Hahn parameter treated as an indeterminate.
F'_{fine}	The pointwise fine order-field derivative of a class function; not a field derivation.
$\mathcal{O}, \mathfrak{m}$	The finite real surreals and the real infinitesimals, with $\mathcal{O} = \mathbb{R} \oplus \mathfrak{m}$.
$\mathcal{O}_{\mathbb{C}}, \mathfrak{m}_{\mathbb{C}}$	$\mathcal{O} + i\mathcal{O}$ and $\mathfrak{m} + i\mathfrak{m}$.
$\mathcal{P}$	The additive class of purely infinite normal forms, including 0; <i>not</i> the class of all infinite surreals.
pp, ct, fin, st	The purely infinite projection; the coefficient of ω^0 , defined on all of $\mathbf{No}$; ct + infinitesimal part; and the standard part, defined on $\mathcal{O}$ only. ct and st agree only on $\mathcal{O}$.
t^γ, t	The Conway monomial $\omega^{-\gamma}$, and $t = \omega^{-1}$; not a generic exponential power.
E, L	The infinitesimal Hahn exponential on $\mathfrak{m}_{\mathbb{C}}$ and logarithm on $1 + \mathfrak{m}_{\mathbb{C}}$.
cis	The finite-angle phase map on $\mathcal{O}$, with kernel $2\pi\mathbb{Z}$.
θ	A finite angle : the argument of cis, always in $\mathcal{O}$.
η	An infinitesimal phase residue : the unique element of $\mathfrak{m}$ left after factoring out the constant ordinary unit c .
B	An accumulated phase : any primitive of the imaginary coefficient. Not an angle; obstructed exactly when $\text{pp}(B) \neq 0$.
$z^\dagger$	The logarithmic derivative $\partial z/z$ for $z \neq 0$; presumes no logarithm.
$\mathcal{I}_0$	The unique primitive with $\text{ct } \mathcal{I}_0(b) = 0$; a normalization, not an algorithm. $\mathcal{I}_0 _{\mathcal{A}}$ is the unique infinitesimal primitive.
$\mathcal{A}$	The bounded-primitive ideal $\partial\mathfrak{m} = \partial\mathcal{O}$; a convex ideal of $\mathcal{O}$, strictly inside $\mathfrak{m}$.
$\mathfrak{o}, \Phi$	$\mathfrak{o}(b) = \text{pp } \mathcal{I}_0(b) \in \mathcal{P}$, and $\Phi(q) = \mathfrak{o}(\text{Im } q)$.
$\mathcal{S}, E_{\mathcal{S}}$	The additive strip $\mathbf{No} + i\mathcal{O}$, not a subring, and its derivation-compatible exponential.
$\mathbf{T}(\mathbf{No}), \mathbf{T}(\mathbb{C})$	$\{u \in \mathbf{No}[i] : u\bar{u} = 1\}$ and the ordinary complex unit circle.
H_Γ, K_Γ	$\mathbb{R}((t^\Gamma))$ and $\mathbb{C}((t^\Gamma))$; $K_0 = \mathbb{C}((t))$, on which $\partial = -t^2 d/dt$.
$\mathcal{H}$	The rational-exponent workspace $\mathbb{C}((t^{\mathbb{Q}}))$, a full Hahn field.
$\mathcal{H}_\Gamma(U)$	The fixed-common-domain ring $\mathcal{O}(U)((t^\Gamma))$, all coefficients holomorphic on one ordinary U .
$\mathcal{O}(U)$	Ordinary holomorphic functions on U ; <i>not</i> $\mathcal{O}$.
$\mathcal{W}$	The proper-class ω -series field with its restricted composition theory.
$\mathbf{T}$	A transcendental differential generator with $\partial\mathbf{T} = i\mathbf{T}$ (or $= a\mathbf{T}$); never a surcomplex number.
$\mathbf{C}, \mathbf{S}, \mathbf{v}$	The recovered oscillator elements and the conjugation-fixed Cayley coordinate in the extension.
$\Lambda, \mathbf{G}_m(\mathbb{C})$	The integer phase-relation lattice, and the multiplicative algebraic group.
τ, ∂_τ	In Part V: $\tau = \log t = -\log \omega$, and the Euler derivation $\partial_\tau = -\omega\partial$, the rescaled ∂_s at $s = \tau$.
$K_{\mathbb{R}}, K_{\Gamma}^{\geq 0}, K_{\Gamma}^{> 0}$	$\mathbb{C}((t^{\mathbb{R}}))$, and the elements of K_Γ of nonnegative, respectively positive, valuation; in Part V $\Gamma \subseteq \mathbb{R}$ need not contain 1.

Symbol	Meaning
$\text{Eig}(A_0)$, $\text{Lev}(A_0)$	The ordinary eigenvalues of a residual matrix, and its Levelt resonances $\lambda - \mu \in \mathbb{R}_{>0}$.
A_0° , $A^\natural$, $A^\flat$	A Jordan form of the residual matrix; the normalized Hahn coefficient; the constant form $iD + N$.
$G_\natural$, F , G_τ	The normalizing Hahn gauge, the Hahn gauge to the constant form, and the gauge in $\text{GL}_n(K_{\mathbb{R}}[\tau])$ to iD .
hob_A	The Hahn obstruction of a regular-singular system; not $\mathfrak{o}$.
$\bar{\partial}$	In Part VI: an arbitrary normalized derivation; ∂ is one of them.
$\mathcal{X}$, ξ , σ	A smooth projective curve, a meromorphic vector field on it, and a local parameter.
$\mathbf{A}$, $\log_{\mathcal{F}}$, $\exp_{\mathcal{F}}$	A constant abelian variety and its formal logarithm and exponential.
$\text{dlog}_{\bar{\partial}}$, $\mathfrak{o}_{\bar{\partial}}^{\mathbb{C}}$	The abelian logarithmic derivative, and the complex primitive obstruction $\text{pp}(\mathfrak{b})$ with $\bar{\partial}\mathfrak{b} = a$; not Φ .
Log , $\mathfrak{t}$	A strip logarithm, used pointwise, and a time primitive.
ℓ_α , ϖ_α	In Part VII: the log-atomic number $\omega^{\omega^{-\alpha}}$ for an ordinal α , extending the iterated logarithms ℓ_n ; and the exponent sum $\sum_{\beta < \alpha} \omega^{-\beta}$.
$\ell_{<\alpha}$, u_α	The monomials ω^{ϖ_α} (the monomial product of the ℓ_β , $\beta < \alpha$) and $\omega^{\varpi_\alpha/2}$.
Q_α , $Q_{\alpha,c}$	The critical potential $\frac{1}{4} \sum_{\beta < \alpha} (\ell_\beta^\dagger)^2$ and $Q_\alpha + c(\ell_\alpha^\dagger)^2$; $Q_{\alpha,1/4} = Q_{\alpha+1}$.
λ , Δ_j	In Sections 40 and 41: a nonzero limit ordinal, never an eigenvalue; and the exponent groups of equation (40.1).
Wr , Sch	The Wronskian $y_1 \partial y_2 - y_2 \partial y_1$, and the Schwarzian of equation (42.1).
$\text{Bor}_2(\mathbb{C})$, $\psi_{a,b}$	The upper triangular Borel subgroup of $\text{SL}_2(\mathbb{C})$, and the automorphisms of equation (41.2); unrelated to Borel summation.

Warning B.1 (The concordance, and three symbols that must not be reused). Across the source manuscripts the derivation is written ∂ , $\bar{\partial}$, D , δ , der , ∂_{BM} and with a prime; two of them write ∂ for the intrinsic derivation and D_z for the coordinate one, and one writes D for a Picard–Vessiot derivation on a wholly different field. This report fixes ∂ for the intrinsic derivation and keeps Table 1 as the reference. The letter D never names a second derivation of $\mathbf{No}$ or $\mathbf{No}[i]$. As an operator it occurs only where it is named: as the abbreviation $D := \partial|_{K_0} = -t^2 d/dt$ in Proposition 18.1, which is the fixed derivation restricted to $\mathbb{C}((t))$ and not another one; as ordinary polynomial differentiation on $\mathbb{C}[x]$ in Proposition 12.3, where nothing surreal is differentiated; and, with a subscript, as the coordinate operators D_z and D_t of Table 1 and the Jacobian $D_Y F_0$ of Part VIII. Otherwise D , with or without a subscript P or Q , is a diagonal matrix, in Parts III and V, and the calligraphic $\mathcal{D}_A$ of Section 17.5 is the first-integral derivation of a polynomial ring. (An earlier edition of this warning said that D appears only as polynomial differentiation; that was not accurate. The correction changes no label and no statement.) Member 11 writes D for an arbitrary normalized derivation, which is $\bar{\partial}$ here (Convention 29.2); member 12 writes $\delta = -\omega\partial$, which is the rescaled derivation $\partial_\tau = \partial_s$ at $s = \tau = \log t$ here (Remark 23.2), because δ is already a source notation for the infinitesimal phase residue in Table 2.

The letter J is not used in this report (Convention 5.6). In the sources it denotes the normalized intrinsic primitive (here $\mathcal{I}_0$), the representative class of purely infinite surreals (here $\mathcal{P}$), the inverse of $\partial|_{\mathfrak{m}}$ (here $\mathcal{I}_0|_{\mathcal{A}}$), and — in the companion *rank-one Berkovich* report, which remains standalone — the zero-constant *coordinate* primitive of a convergent power series, whose “complete primitive

obstruction” is an annular Laurent residue and is unrelated to the phase obstruction. Any cross-link between the two reports must name the operator, not the letter.

The letter U is used in this report for an ordinary complex domain (Part VIII), and once more, with V , as a pair of formal placeholders in the statement of the exponential and logarithm identities (Lemma 6.1), where they stand for nothing at all. One source manuscript flagged its own collision between U as a transcendental differential generator and U as an ordinary domain; the generator is T here, and neither use ever denotes a surreal number or a matrix over one. The eighth manuscript writes U for a unitary matrix; this report writes W , and Convention 15.5 fixes the rest of that part’s alphabet: H for a Hermitian matrix — never the hyphenated adjective in “ H -field” and never the coefficient field H_Γ — G for a gauge matrix, the upper-case upgrade of the scalar gauge g of equation (9.1); $\mathcal{M}$ for a horizontal metric, where the eighth manuscript writes G ; $\mathcal{R}$ for the rotation generator, where it writes the forbidden J ; and Π for the additive phase lattice inside $\mathcal{P}$, because Λ is already the integer relation lattice of equation (21.6), which is the object that manuscript calls L . Its Ψ is this report’s $\mathfrak{o}$ (Convention 5.1(e)) and is *not* the Ψ of Corollary 11.4, which is a placeholder for a putative derivation-compatible exponential and is the letter’s only occurrence here; its $\mathcal{I}$ is $\mathcal{A}$. In Part III the letter W denotes a unitary matrix and M a finite differential module; Part VIII uses the same two letters for a positive-support matrix, an ordinary neighbourhood and a well-ordered exponent set, and the two parts never share a statement. The unitary group itself is $U_n(\mathbf{No}[i])$, printed upright so that it is not read as a domain.

Member 12’s letters are renamed in Convention 23.6: its Jordan form J and nilpotent parts J_λ are A_0° and N_λ° (the letter J stays retired); $\ell = \log t$ is τ and δ is ∂_τ ; K is $K_\mathbb{R}$, and $\mathcal{O}_K$, $\mathfrak{m}_K$ are $K_\mathbb{R}^{\geq 0}$, $K_\mathbb{R}^{> 0}$; its P , H , B are G_0 , $G_\natural$, $A^\natural$; its M and $\mathcal{R}$ are S^* and $\text{Lev}(A_0)$; its Π_γ , L_γ , $K_{\lambda\mu}$ are pr_γ , $\mathcal{H}_\gamma$, $\mathcal{N}_{\lambda\mu}$; its T , R_0 , C , Q are $t^{D_{\text{re}}}$, D_{re} , $A^\flat$, G_τ ; its set of labels B is $\text{Im Eig}(A_0)$; its obstruction ob_A is the Hahn obstruction hob_A ; its operator $\mathcal{D}_A$ is written $\partial_\tau - A$; a matrix it calls U is X ; a real weight it calls η is γ' ; and its “phase labels” are residual frequencies (Warning 23.7). It keeps D for a real diagonal matrix, as Part III does, and V, V_λ, V_b, V_0 for $\mathbb{C}^n$ and its generalized-eigenspace and frequency subspaces.

Member 11’s letters are renamed in Convention 29.3: its derivation D is $\bar{\partial}$ and its field $K = \mathbf{No}(i)$ is $\mathbf{No}[i]$; its curve parameter t is σ , so that t remains the number ω^{-1} (Warning 4.1(1)); its curve X , vector field V , points P and reductions $\bar{P}$ are $\mathcal{X}$, ξ , $\mathbf{y}$ and $\text{st}(\mathbf{y})$; its 1-forms η, θ_j are φ, φ_j , since η and θ are the phase residue and the finite angle here; its abelian variety A is $\mathbf{A}$ and its constant $A = -1/(mc_{m+1})$ is c_* ; its branch logarithm $L(z)$ is the strip logarithm $\text{Log } z$, and its formal-group L, E are $\log_{\mathcal{F}}, \exp_{\mathcal{F}}$; its time primitive T is $\mathfrak{t}$, with B_r, ρ, H renamed e_r, ϱ, g ; its ℓ_D is $\text{dlog}_{\bar{\partial}}$, its $\mathfrak{B}_D = D\mathfrak{m}_K$ is $\bar{\partial}\mathfrak{m}_\mathbb{C}$ (which is $\mathcal{A}_\mathbb{C}$ for $\bar{\partial} = \partial$), and its Obs_D is $\mathfrak{o}_{\bar{\partial}}^\mathbb{C}$, which is *not* Φ , and its complex primitive vector B is $\mathfrak{b}$, since B is an accumulated phase here; its infinitesimal error δ is an ε , its formal series $F_{p,\beta,C}$ is $w_{p,\beta,C}$, and its evaluation points x, y are x_m, y_ω .

Member 13’s letters are renamed in Convention 35.1, and its most dangerous letter first (Warning 35.2): its $\mathfrak{m}(x)$ is the Conway monomial map ω^x , and its $\mathfrak{i}$ and $\mathcal{O}$ are $\mathfrak{m}$ and $\mathcal{O}$. Its p_α , s_α , P_α , U_α are $\omega^{-\alpha}$, ϖ_α , $\ell_{<\alpha}$, u_α ; its a_α , A_α , h_α are $\ell_\alpha^\dagger$, $\ell_{<\alpha}^\dagger$, $u_\alpha^\dagger$; its potentials $q_{\alpha,c}$ and q are $Q_{\alpha,c}$ and Q , since q is the rank-one coefficient here; its coordinate T with D_T , E_T and partner U is a scale s with ∂_s , $s\partial_s$ and u ; its Wronskian W is Wr ; its Γ_j , K_j , k_j are Δ_j , H_{Δ_j} , K_{Δ_j} ; its fundamental matrix Φ , automorphisms $\sigma_{a,b}$, Borel group $B(\mathbb{C})$ and Schwarzian $\mathcal{S}_\partial$ are Y , $\psi_{a,b}$, $\text{Bor}_2(\mathbb{C})$ and Sch ; its infinitesimal primitive θ of an angular rate is η ; its Riccati solution b is w ; and its ordinal index η is ζ .

Finally, $\mathcal{O}$ never takes an argument. Where a source wrote $\mathcal{O}(V)$ for holomorphic functions, this report writes $\mathcal{O}(V)$.

Dependency chain

Theorem 3.1 (imported) $\implies$ Proposition 3.5, Lemma 6.1 $\implies$ Theorem 6.3, Corollary 6.7
 $\implies$ Theorem 8.1, Theorem 8.4 $\implies$ Theorem 9.1, Theorem 11.2, Theorem 13.2.

The ideal-structure theorem Theorem 7.5 needs only the constant field, surjectivity, the H -field positivity clause and $\partial\omega = 1$; the polar decomposition additionally needs real closedness, standard part and the infinitesimal exponential; the image theorem additionally needs real exponential compatibility. The bounded-derivative lemma alone gives a quick independent proof that $\partial y = iy$ forces $y = 0$ (Corollary 8.8, second proof). The constant-coefficient theorem needs only the nonreal-eigenvalue obstruction, the constant field and the real exponential. The Laurent resolvents need strong additivity and $\partial t = -t^2$. The extension theorems need the constant-coefficient obstruction and a formal Laurent expansion in a new indeterminate. Strong additivity is essential for the infinitesimal chain rule and for Theorem 48.4; the Laurent identities can be verified inside formal series algebra independently of any construction of all surreals, the embedding being what identifies their operator with ∂ . Neither the composition theorem nor the Picard–Vessiot background is a dependency of the phase criterion. So the most striking negative conclusions do not depend on delicate analytic-geometry results or on a full model-theoretic classification.

C Provenance: what each archive contributed

This report merges eleven source manuscripts. The manuscripts themselves are not distributed with this report; the disposition of each source result is recorded in the provenance material below and in the package README. The identifiers below are the merge identifiers; among the first seven the original archive names collided, three pairs of them differing only by a trailing marker, so the mapping is recorded here and in the package README because it is not recoverable from the filenames. Members 01–09 are pinned to `e260237db9b71da8b74a0c13c8e6355119091100`; member 10 joined later and is pinned to `aa846271b4dcae2c055b216126a87210292ec19b`; members 11 and 12 joined together after it, from the archives `surreal_autonomous_dynamics` and `surcomplex_regular_singular`, and are pinned to `a3124af79f66b8b9c196d76b4cbc5ac3938907c4`. The first of those archive names is misleading on two counts: its field is **No** $[i]$ with real corollaries, not **No** alone, and its “dynamics” are flows of autonomous differential equations, not iteration of maps. Member 13 joined after them, from the archive `surreal_transfinite_critical_potentials`, and is pinned to `048b72cf7cbfc8ab246e4f73788c10460cb3f6e0`; its code and data are placed under the prefix `13-critical-potentials`.

Table 14: Provenance by member.

Member	Distinctive contribution retained in this report
01 <i>thresholds and Galois tori</i>	The exact image as the boxed organizing identity (Theorem 8.1); the obstruction as an $\mathbb{R}$ -linear surjection with the gauge <i>uniqueness</i> theorem and the rank-one tensor corollary (Theorem 9.1 and Corollary 9.4); the sharp support threshold over a real-exponent Hahn field with strictness at exponent 1 and the valuation-insufficiency counterexample (Theorem 10.1 and Example 10.4); the upper-triangular criterion with its descending recursion, the Jacobi trace test and the $\text{diag}(i, -i)$ counterexample (Theorem 14.1 and Proposition 14.2); the rational and diagonal Picard–Vessiot extensions with from-scratch no-new-constants proofs, the saturated integer lattice, the torus, and the $(i, 2i)$ versus $(i, \sqrt{2}i)$ examples (Theorems 21.1 and 21.12 and Example 21.13); the formal oscillator in complex and real rational form with the order-incompatibility argument (equation (21.4) and Theorem 21.10); the non-summability of the exponential series at ω (Remark 11.9); the phase-independent no-global-exponential corollary (Corollary 11.4); the “at most one, not none” warning with its two witnesses (Warning 12.2); the class-hygiene discipline (Warning 2.1); the coefficient-domain tiering contract (Section 49.1).
02 <i>word lemma and nonlinear IVP</i>	The entire coordinate half: the common-domain ring with the explicit disowning of both the shrinking-radius germ ring and the full formal ring (equation (43.1) and Warning 43.1); the linear fundamental-matrix theorem with uniqueness among <i>all</i> Hahn-coherent matrices, the support certificate and the explicit recursion (Theorem 44.1); the inhomogeneous corollary, the Liouville determinant identity, the negative-leading-exponent proposition and the nilpotent counterexample bounding it (Corollaries 44.3 and 44.4, Proposition 44.5, and Example 44.6); the nonlinear deformation theorem with no common degree bound (Theorem 45.2); monodromy valued in $\text{GL}_d(K_\Gamma)$ with the example where ordinary monodromy is trivial and full monodromy is not (Theorem 46.1 and Example 46.2); the two-scale noncommuting computation and the refutation of the naive exponential (Section 47); the coherent entire solution poled at an infinite coordinate (Example 45.6); the cross-category pair (Warning 46.3); the three-promise separation worked through $A = i\omega$ (Warning 48.2); the strong Laurent resolvent with its residual certificate and the exact factorial solution (Proposition 18.8 and Example 18.10); the four-category separation table (Table 10); and the from-scratch proof of the ordered-word lemma (appendix A).

Member	Distinctive contribution retained in this report
05 <i>logarithmic- derivative image</i>	The exact decision criterion for rational coefficients and its raising-rather-than-reporting contract (Theorem 10.6 and Warning 10.7); the order-theoretic structure and convexity of the bounded-primitive ideal (Theorem 7.5(iii)); the countable-stage differential Hahn hull with its monomial criterion, its minimality caveat, the real-group criterion, the $\mathbb{Q}\sqrt{2}$ counterexample and the “differentiation closure is not integration closure” example (Lemma 20.2, Theorem 20.3, Proposition 20.6, and Examples 18.2 and 20.7); the analytic coefficient interface with coefficientwise linear naturality, the total-derivative chain rule and the separate admissibility condition for formal power series (Theorems 48.4 and 48.6, Corollary 48.8, and Section 48.4); and the algebraic unit-modulus example (Example 10.8).
06 <i>finite primitive and oscillation</i>	The base skeleton, and in particular: the strip theorem with <i>pointwise</i> maximality (Theorems 11.1 and 11.2); the character-independent differential defect (Proposition 11.5); the products, duals and no-finite-order corollary (Corollary 9.4); the direct-sum description (equation (7.5)); the summary table of distinguishable situations (Table 3); the factorially divergent forced oscillator with its closed residual and anti-topological disclaimer (Theorem 18.13 and Warning 18.14); and the theorem that oscillation cannot retain the H -field rule (Theorem 21.10).
07 <i>exact sequence and rational exponents</i>	The explicit $\mathbb{C}((t^{\mathbb{Q}}))$ workspace chapter, with the exact derivative image, the exact logarithmic-derivative image, the single resonant coefficient, the resonant normal form and the workspace-versus-ambient discussion (Theorems 19.2 and 19.3, Proposition 19.4, Remark 19.5, and Table 5); the derivative taxonomy including the fine order-field derivative and the proof that a fine derivative does not determine an intrinsic one (Table 1 and Warning 4.1); the change-of-scale observation (Proposition 4.2); the two-derivative chain rule with <i>derived</i> summability (Proposition 4.5 and Remark 4.6); the integration-by-parts correction for normalized primitives (Proposition 7.3); the restricted ω -series composition bridge and the precise sense in which two derivatives agree (Section 22 and Remark 22.1); the terminating finite inverse of $\partial - i$ on $\mathbb{C}[\omega]$ (equation (12.3)); the union-of-fields anti-lemma (Warning 20.4); and the four-outcome contract that returns a reason rather than a failure flag (Table 11).
08 <i>coherent IVP and general resolvent</i>	The resolvent over a general real-exponent K_{Γ} with its substantive hypothesis (Theorem 18.6 and Warning 18.7); the ambient-versus-workspace uniqueness of the factorial solution, with its exponentially small full-field ambiguity (Proposition 18.12); the valuation test for finite phase (Corollary 18.4); the negative-valued coefficient that destroys common-domain coherence, with the rescaling that resolves it (Example 44.7); “the same printed equation under different derivatives” (Warning 4.1(7)); the Riccati example with an entire coefficient family (Example 45.7); the oscillatory extension proved Picard–Vessiot via the absence of proper differential ideals in the Laurent ring (Theorem 21.2); the anti-Banach disclaimer (Warning 45.4); the certificate table (Table 12); and the rule that a truncation must record the <i>type</i> of omitted information (Warning 49.3).

Member	Distinctive contribution retained in this report
09 <i>BM bridge and derived smallness</i>	The derivation of smallness and of strict order reversal on the infinitesimals from the H -field positivity axiom — the provenance this report uses (Lemma 3.8); the theorem that intrinsic differentiation preserves common-domain coherence, with the total intrinsic derivative and the fixed-contour and moving-pole discussion (Theorems 48.4 and 48.6, Corollary 48.8, and Warning 48.9); the bounded-primitive ideal as a named object with its full structure theory, including the cut description and the induced order-preserving quotient isomorphism (Theorem 7.5); the increasingly-slow-scale examples (Corollary 10.3 and Table 3); the resolvent with its support certificate and the whole-prefix coefficient recurrence with its exact guarantee (Theorem 18.6 and Remark 18.15); the constant-coefficient coefficient stream and the $\sum \omega^{n^2} X^n$ support counterexample (Warning 48.11); and the derivation-aware representation interface (Table 12).
10 <i>Hermitian phase spectral integral</i>	The whole of Part III: matrix gauge equivalence and the complete phase normal form for every finite system, with the differential phase spectrum as its complete invariant (Theorem 15.9 and Convention 5.1(e), Warning 5.2); the imported splitting and first-order surjectivity input with its transfer (Theorem 15.6 and Warning 15.7); existence for every forcing and the exact kernel dimension (Corollary 15.11 and Warning 15.12); the canonical phase projectors and the refined trace identity (Proposition 15.13 and Corollary 15.14); the complete horizontal metric cone with its anti-Lyapunov example and the unitary normal form (Theorems 15.16 and 15.18 and Warning 15.17); the attained min–max, the entrywise Weyl bound and the bounded-gauge estimate (Lemmas 16.1 to 16.3 and Warning 16.4); the spectral–integral theorem itself with gap-free perturbation invariance, the exact kernel count and its sharpness (Theorem 16.5, Corollaries 16.7 and 16.8, and Proposition 16.9); the two deliberate boundary counterexamples and the coupled unitary example (Examples 16.10 and 16.12 and Warning 16.11); normalized non-Abelian integration with its cocycle law and the exact noncommuting Hahn recurrence (Theorem 17.1, Warning 17.2, and Example 17.3); the real rotation-block descent and the real metric cone with its parity (Theorems 17.4 and 17.5); the positive Ermakov–Pinney dichotomy and the canonical amplitude–phase reduction (Theorems 17.6 and 17.7); the exact Airy Hahn expansion with its Wronskian, amplitude jet and truncation residual (Section 17.4); the resonance algebra with its Dickson finiteness and the singular three-phase invariant algebra (Theorem 17.8 and Example 17.9); the phase lattice as a Galois torus, its group algebra and real form (Corollary 17.10, Proposition 17.11, and Warning 17.12); and the set-sized witness-closed reading of the proper-class arguments (Remark 17.14).

Member	Distinctive contribution retained in this report
11 <i>autonomous dynamics</i>	The whole of Part VI: the class of normalized derivations and the scope amendment it forces (Definition 29.1 and Convention 29.2); formal evaluation in several variables, formal implicit uniqueness in a monad, and strip logarithms for every normalized derivation (Lemmas 29.5 to 29.7); projective reduction and the reduction-to-a-zero lemma (Lemmas 30.2 and 30.3); the realization classification with its simple-zero and multiple-zero families and the no-hidden-solution argument (Theorem 30.1, Propositions 30.5 and 30.7, and Remark 30.8); uniform finite-scale localization with the three-shift formula, and the continuum of algebraically independent solutions inside two scales (Theorem 31.1 and Corollary 31.3); derivation independence with its scalar and real corollaries (Theorem 31.4 and Corollaries 31.5 and 31.6); the exact real branch count and the worked equations (Corollary 32.1 and Example 32.5); infinitesimal contraction, the holomorphic dual obstruction and the squarefree hyperelliptic exclusion with the two counterexamples bounding it (Proposition 33.1, Corollary 33.2, Theorem 33.3, and Example 33.5); the normalized formal-group logarithm, the identity-monad splitting, the abelian logarithmic-derivative image, the finite-primitive abelian criterion and the speed threshold (Lemma 33.6, Proposition 33.7, Theorem 33.8, and Corollaries 33.10 and 33.11); the finite geometric decision criterion (Remark 34.1); and the review log <code>11-autonomous-dynamics-sources.md</code> .
12 <i>regular- singular spectral selection</i>	The whole of Part V: the Euler derivation on arbitrary real-exponent Hahn fields (Lemma 23.1); the power-Hahn regular-singular class; the resonance-reaching support and the finite-dependence lemma (Definition 24.3 and Lemma 24.4); the normalized Hahn–Levelt form with support control, and the finite resonance theorem with its degree bound and finite coefficient replacement (Theorems 24.5 and 24.7 and Corollaries 24.8 and 24.9); the real-monomial shear (Theorem 25.1); spectral selection with the ambient dimension $m_{\mathbb{R}}(A_0)$, perturbation invariance and real descent (Theorem 25.3 and Corollaries 25.4 and 25.5); the classifications over $K_{\mathbb{R}}$ and over $\mathbf{No}[i]$ with exact morphism dimensions and tensor dimensions (Theorems 25.7 and 25.9 and Corollaries 25.8 and 25.11); the Hahn solution space, the transcendence of $\log t$, minimal logarithmic repair with Galois group $(\mathbb{C}, +)$, and the exact Hahn obstruction sequence (Theorems 26.1, 26.3 and 26.5 and Lemma 26.2); the six worked examples (Section 27); the finite-monomial, one-logarithm envelope (Theorem 28.1); the power-Hahn boundary (Proposition 28.4); and the exact finite procedure with its limits (Section 28.3).

Member	Distinctive contribution retained in this report
13 <i>transfinite critical potentials</i>	The whole of Part VII: the imported ordinal log-atomic tower with its limit-stage caveats and scale separation (Proposition 35.3 and Lemma 35.5); the summable triangular differentiation and the exact critical solutions with Wronskian one at every ordinal (Lemma 36.1 and Theorem 36.2); the Wronskian spanning lemma, the Liouville gauge identity and the Euler classification at a positive infinite scale (Lemmas 37.2 and 36.4 and Proposition 37.3); the transfinite classification with its Conway normal forms, Wronskians and Riccati form (Theorem 38.1 and Remark 38.2); the earlier-stage asymptotics, the invisibility theorem, the first infinite stage and the absence of an all-ordinal potential (Propositions 39.1 and 39.5, Theorem 39.2, and Example 39.3); the three Hahn fields at a limit ordinal with bounded inner support, derivative stability, the 0, 1, 2 jump and the exponential-then-primitive gap (Lemma 40.2, Propositions 40.3 and 40.5, and Theorem 40.4); the Borel Picard–Vessiot group and its additive remainder (Lemma 41.1, Theorem 41.2, and Corollary 41.3); the Schwarzian form (Lemma 42.1 and Corollary 42.2); the first limit stage in normal form (Example 42.3); and its boundaries, inputs and formalization interface (Section 42.3).

Remark C.1 (How the merge was performed, and the count of non-claims). A result that all seven prove is printed once. Where two members proved the same statement by genuinely different routes, both routes are kept and marked: Lemma 3.8 against Remark 3.9 for smallness; the three proofs of Corollary 8.8; the two constant-field proofs in Theorems 21.1 and 21.2; appendix A against the published citation; and the two localization constructions of Section 20, which prove different things. Where two members stated the same theorem with *different* uniqueness qualifiers, the qualifier printed is the one its proof supports and no wider (Convention 45.3), with the stronger linear statement kept separately (Theorem 44.1). Where one name covered different coefficient rings, the rings are named (Theorems 10.1 and 18.6 and Proposition 18.8). Nothing is strengthened beyond what a source proves.

The first seven manuscripts state, between them, $13 + 16 + 18 + 13 + 24 + 21 + 13 = 118$ explicit non-claims, with heavy overlap, and the eighth states a further 12, giving 130. Members 11 and 12 state fifteen each, by this merge’s count, giving 160, and member 13 states twenty-one, giving 181. Section 51 consolidates them into 106 numbered items, organized by theme; every one of the 181 has a home there, and many items carry several sources’ phrasings at once. Two of the 118 are artifact-level discrepancies in the source packages rather than mathematical limitations, and they are recorded as such in item N52; a third, the eighth package’s self-rewriting verification script, is item N64; and the audit findings on the packages of members 12, 11 and 13 are items N79, N94 and N106.

Integrating member 13 followed the same rules. Its reproof of the obstruction for nonreal powers of an infinite scale, through an infinitesimal primitive of the angular rate, is Theorem 6.3 with the direction (i) $\Rightarrow$ (iii) of Theorem 8.4, and is recorded as agreement (Corollary 37.1); its derivative-stability argument is an instance of Lemma 20.2 and is cited as such (Proposition 40.3); and its stage 0 agrees with Theorem 25.3 (Remark 38.4). Its own proofs are kept: they use neither surjectivity nor the import of Theorem 15.6. Three observations that the member does not state are recorded as such, with proofs: the Euler operator as a constant-coefficient operator at a rescaled scale (Remark 37.4), the missing oscillator as the Euler problem at the scale e^ω (Remark 37.5),

and the amplitude and phase of a supercritical member, which use Part III and flag its import (Remark 38.5); the comparison of stage 0 with Part V is likewise made by this merge. Hedges were amended in place and not removed: items N13, N21, N23 and N48, the remark after Corollary 10.3, Remark 28.5, the opening of Section 21, the remark after Theorem 17.6, Warning 5.3, and the continuation on multi-scale Hahn workspaces. One omission in the member’s repository comparison is recorded, with its pin kept (item N104).

Integrating members 11 and 12 followed the same rules. Their reproofs of results already in the report are recorded as agreements and cited: smallness (Lemma 29.4 and Lemma 25.2(i), against Lemma 3.8), the missing oscillator (Lemma 30.4 and Lemma 25.2(ii), against the second proof of Corollary 8.8), the ordered-word lemma (Lemma 24.1 against Lemma 2.6, with member 12’s Higman-free real-exponent route kept as a marked second proof), the strip logarithm (Lemma 29.7 against Theorem 11.1), the logarithmic separator (Proposition 28.4 against Example 10.4), the missing logarithmic primitive (Lemma 26.2 against Example 18.2), and the monomial adjunction example, printed only as the second row of Table 5. Where a reproof holds under weaker hypotheses — every normalized derivation, or no surjectivity and no import — the statement is kept in its new part with that scope, and no statement of an older part is widened. Genuinely different routes are kept and marked: Example 32.4 beside Corollary 8.11, and Proposition 25.12 beside the existence half of Theorem 15.9. Hedges that the new parts affect were amended in place and not removed: items N2, N3, N21, N22, N23, N24, N25, N48 and N57, the remark after Corollary 8.11, Warnings 3.2, 3.3, 16.11 and 17.13, and three open continuations. One observation that neither source states is recorded as such (Remark 25.13), and one question is recorded rather than claimed among the open continuations (Section 51.13).

Integrating the eighth member followed the same rules. Its reproofs of results already in the report — the convexity and strictness of $\mathcal{A}$, the exact logarithmic-derivative image, the saturation of the relation lattice — are recorded as agreements and cited, not printed again (Remarks 15.1 and 15.2 and Section 17.6). Its two genuinely different routes are kept and marked: the rank- r differential simplicity of Proposition 17.11(i) beside the rank-one argument of Theorem 21.2, and the metric proof of the Pinney dichotomy beside the classical parametrization equation (17.10). Where it strengthened a statement this report had deliberately hedged, the hedge was amended rather than removed and the new hypothesis was named: item N22 now records that first-order inhomogeneous existence holds but only by the import of Theorem 15.6. Item N24 distinguishes the complete finite-system phase classification from the Hermitian eigenvalue formula; the non-Hermitian counterexample limits the latter. The maintained review corrects earlier summaries that had conflated those two scopes.

D Verification record, exactly as delivered

Each source package shipped a deterministic exact-symbolic program together with its *recorded* run. This appendix reproduces those records. It does not re-run them and does not merge their counts into a single suite: eleven suites with eleven scopes remain eleven suites.

Table 15: The eleven recorded verification runs.

Member	Recorded	Scope and declared cutoffs
01	1,395 checks passed	Standard-library Python only, exact Gaussian rationals, deterministic seed 20260921, formal-series cutoff 16. Categories: Leibniz rule 250; additivity 250; primitive identity including the logarithmic term 250; conjugation compatibility 250; power-threshold regressions 49; positive-phase primitives 100; small-phase differential equation as a jet 100; unit-circle exponential identity as a jet 100; Cayley logarithmic derivative as a jet 1; exact polynomial-forcing inverses 36; repeated real-root amplitudes 9. Not a surreal-number implementation and not a proof checker, by its own docstring.
02	115 checks passed	Python with SymPy, exact symbolic only. Groups: power primitives 7; finite-phase formula identities 3; iterated-logarithm primitives 15; Laurent resolvent residuals 10; noncommuting matrix coefficient equations and initial values 56; matrix ODE and determinant jets 5; nonlinear rational solution and jets 15; infinitesimal monodromy jets 2; total chain-rule identities 2. Matrix cutoff is <i>total degree</i> 6, not a valuation cutoff; Laurent reciprocal order 12; nonlinear checks through t^{12} .
05	149 checks passed	Python with SymPy. Exposes a restricted rational-phase classifier as an importable API that type-checks its arguments, rejects nan, complex infinity and infinities, rejects free symbols other than the declared variable, coerces to polynomials over $\mathbb{Q}$, and <i>raises</i> outside $\mathbb{Q}(\text{variable})$. Check families include the real power rule over 22 rationals; power and logarithmic primitives to depth 3; twelve hand-listed rational-phase examples; a 6×6 degree sweep; eight rejection tests; the algebraic unit $(1 + i\omega)/\sqrt{1 + \omega^2}$; truncated $E(-it)$ residuals for $N = 0..12$ against equation (10.5); the total chain rule with its moving-argument term; commutation of the two coefficient operations; a fixed-contour sample; and a constant-coefficient example.
06	258 checks passed	Python with SymPy, models ∂ as $-t^2 d/dt$ on t -expressions and the extension derivation by $f \mapsto \text{cancel}(iT \partial f / \partial T)$; uses an ordinary variable as a stand-in for ω . Categories: total chain rule 7; scale primitives 23; formal exponential/logarithm 45; phase and gauge algebra 5; oscillatory extension 8; constant-coefficient modes 18; Laurent resolvent residuals 60; factorial oscillator 35; polynomial resolvent cutoffs 21; Laurent valuation bounds 36. Covers real-power primitives at $p \in \{-2, 0, \frac{1}{2}, 1, \frac{3}{2}, 2, 3\}$, the log-scale ladder at depths 0..3 and $p \in \{0, 1, 2, 3\}$, and truncated infinitesimal identities at orders 1..9. Its own trailing scope note reads: exact finite symbolic identities only; these checks do not prove the infinite-support, class-size, or general differential-field theorems.

Member	Recorded	Scope and declared cutoffs
07	291 checks passed	Python with SymPy plus a finite Hahn core implemented as a validating map from rational exponents to Gaussian-rational coefficients, with a terminating derivative, a primitive that rejects a nonzero resonant $[t^1]$, an ambient primitive pair returning (p, c) for $p + c \log \omega$, a classifier returning <i>workspace</i> / <i>ambient-only</i> / <i>phase-obstructed</i> , and a workspace certificate (r, k) asserting that k has positive support. The normalizer <i>rejects</i> floats, strings and non-Gaussian-rational coefficients, and four such rejections are counted among the passing checks; the script states that it is not an expression parser. Thirteen groups, including rational monomials 87 and the finite product rule 81.
08	150 checks passed	Python with SymPy, deterministic, with no numerical tolerance — an assertion requires exact simplification to zero. Covers monomial derivatives and primitives at nine rational exponents $\{-\frac{5}{2}, -1, -\frac{1}{3}, 0, \frac{1}{4}, 1, \frac{4}{3}, 2, \frac{7}{2}\}$; the logarithmic primitive; sixteen product-rule checks on four sparse expressions with complex coefficients; twenty-four resolvent-truncation residuals for $\lambda \in \{1, 2, i, 1 + i\}$ and $N = 1..6$, each required to equal exactly the first omitted derivative with the alternating sign; sixteen factorial residuals for $N = 1..16$; the Riccati coefficients $a_1..a_{10}$; constant-coefficient examples; the Cayley phase identity; and the oscillator identities in the adjoined extension.
09	259 checks passed	Python with SymPy over formal symbols, with $\partial(\cdot) = \text{expand}(-t^2 \partial/\partial t)$; a matrix identity or a whole checked coefficient prefix counts as <i>one</i> check. Groups: derivation identities 41 (including $\partial(t^n)$ for $n = -12..12$ and $\partial^n t$ for $n = 0..12$ against $(-1)^n n! t^{n+1}$); primitive identities 15; formal small-series identities 24 at orders 4, 6, 8, 10; coherent total-derivative examples 8, including the moving coefficient value $1 - 2/\omega^2 - 2/\omega^3$ and a moving-pole identity; general resolvent residuals 81 for $\lambda \in \{i, 2, 1 + i\}$ and $N = 0..8$; factorial-series residuals 13; whole-prefix coefficient recurrences 9 through t^{10} with lower bounds 1, -2 , -5 ; constant-coefficient operator examples 33; matrix identities 35.

Member	Recorded	Scope and declared cutoffs
10	62 checks passed	Python with SymPy, exact symbolic and rational arithmetic throughout, with no numerical sampling. Groups: the nonnormal counterexample 3 — unimodular gauge, $\partial G = AG$, and characteristic polynomial $\zeta^2 + 1$, asserted <i>as</i> a counterexample (Example 16.10); the coupled unitary example 4 (orthogonality, skew-Hermiticity, the exact phase gauge, the instantaneous determinant, Example 16.12); the Jordan metric 2 (Warning 15.17); the three-phase system 6, including its invariant Hermitian metric, its gauge identity, three polynomial first integrals verified in the <i>original</i> coupled coordinates, and the quadratic relation of equation (17.23); the noncommuting recurrence 12 with 12 unitarity-coefficient checks, one naive-exponential defect identity and one assertion that the commutator defect is nonzero (Example 17.3); the Airy recurrence 16 with the first amplitude correction and the exact truncation residual at $N = 16$ (equations (17.16) and (17.20)); and three Pinney checks (determinant one, invariant metric, symplectic amplitude reduction) carried out on arbitrary symbolic jets under the substitution for the second derivative (equations (17.9) and (17.11)). The record also carries the degree-8 amplitude jet of equation (17.19) and the 3×3 coefficient matrix of equation (17.22) as strings, both matching the displayed forms. Its own trailing scope note reads: finite exact symbolic checks; not a proof of general or infinite-support theorems.
11	10 checks passed	Python 3.13.5 with SymPy 1.14.0, exact symbolic arithmetic in a total-degree quotient of $\mathbb{Q}[C][X, Y]$, at the default truncation -degree 4 (the option accepts 2 to 7, and the number of checks depends on it). Checks: the logistic solution, with $e^{-\omega}$ modelled by a symbol q with derivative $-q$; the repeated-root counterexample $(\partial y)^2 = y^3$; the degree-two counterexample, with e^{ω} modelled by q with derivative q; the time primitive of $-t^2(1+t)$; the implicit equation of Example 32.5 through total degrees 1, 2, 3 and 4; the printed homogeneous corrections F_1, F_2; and the differential residual through X^6, with the derivation modelled as $-X^2\partial_X + X\partial_L$ after $X = 1/\omega$, $Y = XL$, $L = \log \omega$. The record also prints $F_1, \dots, F_4$ and reports 7 as the first possibly nonzero degree of the truncation residual. Its trailing note: these tests do not verify the general proofs, formalization or novelty.

Member	Recorded	Scope and declared cutoffs
12	8 checks passed	Python with SymPy 1.14.0 (no Python version recorded), exact rational and Gaussian-rational arithmetic, a finite-cutoff normalizer for finite rational supports with explicitly supplied generalized-eigenvalue blocks, and a residual check of $\partial_\tau G - AG + GA^\dagger$ made coefficientwise and independently of the recursion. Checks: the interaction-generated resonance $A_2^\dagger = 2E_{13}$ of Example 27.1, with residual through weight 6; the logarithmic fundamental matrix, $\partial_\tau Y = AY$ identically and $\det Y = t^3$, with $x = t^{1/2}$ and ∂_τ modelled as $\frac{x}{2}\partial_x + \partial_\tau$; the nilpotent residual block of Example 27.3 through weight 8; the mixed-frequency system of Example 27.4 through weight 7; tail invariance after adding a weight $\frac{7}{3}$ coefficient, compared through weight 4; the accumulating identity $((1 - \frac{1}{j}) - 1)(-j) = 1$ for $2 \leq j \leq 100$; the forced nilpotent example of Example 27.5 with a failed constant-image test; and twelve seeded random cases (seed 20260922) through weight 4, with and without Jordan blocks. Its scope field reads: finite algebraic checks, not a formal verification of the theorems.
13	70 checks passed	Python 3.13.5 with SymPy 1.14.0, exact symbolic expressions and no floating-point evaluation; a nonzero residual raises and gives a nonzero exit status. Suites: the finite hierarchy 39 — for lengths $n = 0, \dots, 12$, with generators a_j and the derivation $a_j \mapsto -a_j \sum_{k \leq j} a_k$ modelling $\ell_j^\dagger$, the triangular identity equation (36.3), the critical potential identity of Theorem 36.2 and the Riccati residual equation (38.7); Euler identities 8 — real-power eigenvalue, the Euler polynomial, the distinct-root basis with Wronskian $-2d$ for roots $\frac{1}{2} \pm d$ and the repeated-root basis with Wronskian 1, in an ordinary positive variable T with d/dT ; the gauge identity equation (37.2) with its normalization, first-derivative cancellation and Wronskian 4; the Schwarzian 4, including that the ratio of two solutions has Schwarzian $2Q$; and the Picard–Vessiot suite 15 — $\det Y = 1$, the four entries of the fundamental-matrix equation, derivative compatibility of the automorphism on both generators, the four entries of the Borel action and the four entries of the group law. Its trailing note: finite identities only, not a proof-assistant verification; transfinite summability, the imported derivation, exponent-field membership and novelty need the mathematical arguments.

The first seven runs were recorded under Python 3.13.5, six of them under SymPy 1.14.0 and the seventh using the standard library only. The eighth records SymPy 1.14.0 and a status of PASS with 62 checks, but records no Python version; its package pins `sympy==1.14.0` and its README states Python 3.10 or later. The ninth record (member 11) states Python 3.13.5 and SymPy 1.14.0; the tenth (member 12) states SymPy 1.14.0, its README says Python 3, and its program requires Python 3.10 or later; the eleventh (member 13) states Python 3.13.5 and SymPy 1.14.0, and its program requires Python 3.10 or later. All eleven record zero failures. The consolidated scope limits are item N48 to item N52, item N63 to item N64, items N78–N79, item N94 and items N105–N106

of Section 51. Artifact-level discrepancies are recorded there and repeated here for completeness: one package’s PowerShell build script was supplied but not executed on Windows in its preparation environment; one package’s README lists a `SHA256SUMS.txt` that is not present among its delivered files; the eighth package’s program rewrites its own delivered `data/verification.json`, computing the package root as the parent of its own directory, so the record reproduced above is the delivered one and a rerun would overwrite it; member 12’s program writes its output file, by default `verification.json` in the current directory, and its `verify` target also overwrites the text record; member 11’s build script and README redirect the program’s output onto the delivered text record; neither of those two packages carries a checksum manifest; member 12’s record labels the mixed-frequency ambient dimension 1 without computing it; and member 13’s program writes a record only when given `-output`, while its package README suggests `-output verification.txt`, and its manifest lists three package files — the article source, the 21-page PDF and the README — that are not shipped here.

Maintained review rerun. The eight earlier programs (members 01, 02 and 05–10), unmodified, were subsequently run on separate temporary copies under Python 3.13.14 and SymPy 1.14.0 (member 01 uses only the standard library). All exited successfully and reproduced the respective check totals 1,395, 115, 149, 258, 291, 150, 259 and 62, with zero failures. Their generated outputs stayed outside the repository; the delivered `code/` and `data/` files were preserved. A separate exact matrix-coefficient check confirmed the added $[t^4]$ cocycle counterexample in Example 17.3; it is not included in those historical suite counts. The later programs of members 11 and 12 were not rerun in this review. Member 13’s program, unmodified, was run once on a separate temporary copy during its merge, under Python 3.14.4 and SymPy 1.14.0: it exited successfully with 70 passed and 0 failed, and its output agreed with the delivered record line for line apart from the recorded Python version. These finite checks do not prove the general theorems.

E Building this report

The source `article.tex` is standalone: the bibliography is internal, and no external bibliography database, graphics file, font file, shell escape or network access is required. A standard TeX installation providing the packages named in the preamble suffices. From this directory,

```
latexmk -pdf -interaction=nonstopmode article.tex
```

and, if `latexmk` is unavailable, run `pdflatex` with `-interaction=nonstopmode` three times so that the table of contents, the cross-references and the `cleveref` labels settle. The directory also contains `README.md`, member 11’s review log `11-autonomous-dynamics-sources.md`, and the unmodified `code/` and `data/` directories of the eleven source packages. To re-run a source suite, run it *on a copy*: several of the delivered programs rewrite their own evidence files beside themselves, the eighth writes to a `data/` directory it creates one level *above* its own, member 12’s program writes `verification.json` into the current directory, the build scripts of members 11 and 12 redirect output onto their recorded transcripts, and member 13’s program writes whatever file its `-output` option names, all still under the packages’ original file names; so a byte-identity check performed after a run compares two equally modified copies and tells you nothing.

References

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